# Human Thought

Its Forms and Its Tasks

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# Contents

# Foreword

The life of thought, which it is the task of my book to describe, is intimately interwoven with the other sides of the life of the spirit. I have sought to show that thought cannot serve the life of the spirit as a whole unless it first and foremost becomes master in its own domain and unfolds itself according to its own laws as an independent side of the life of the spirit. That this unfolding, when it is rightly carried through, will bring with it great and far-reaching changes in personal life as a whole, is something of which I become more and more convinced. There are here oppositions and problems that will take on ever sharper forms, and I feel no admiration for those who here sense no sting, although they have sufficient opportunity to come to know the life of thought.

Just as I have on the whole stressed the personal factor in all thinking, the more this approaches the border-regions, so too I have not concealed that my own stance at thought's final station has a personal character, and I have not been able to refrain from stating how, in my experience, one feels at this station.

I still have much to learn. My work will need filling out and correction on many points. But beyond the framework I have given here, I cannot go. And it affords me a good place to work. May there be one or another who could also find work here, even if this work should lead him to alter both the framework and its content.

August 1910. *Harald Høffding.*

# I. The Psychology of Thought (Functions)

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## A. Psychic Energy

1. The task I have set myself in this book is to investigate human thought—a task that all philosophers have set themselves, whether they have been conscious of it or not. For all questions that can be raised concerning life and existence depend—both with respect to the way in which they are posed and with respect to the way in which they are answered—on the conception one has of the nature and operation of thought. It is therefore also an outline of my philosophy that I shall give here. That is to say: I shall give an account of the thoughts to which I have come back again and again, when I have occupied myself with such questions as: the presuppositions of science, the significance of the results of science for one's view of life, and the relation between scientific cognition and other sides of the human life of the spirit.

2. We know thought best as reflection. The word *Eftertanke*—"after-thought"—I use not only in its strict sense, "in one's thinking to *seek after* something or *pursue* something," but also in the sense that thought *comes after* an immediate experience. Reflection thus presupposes that we have seen, felt, experienced something. During the experience we were perhaps wholly absorbed in it; but now we seek to bring it forth anew, either to dwell once more on what was experienced, or to go through its various parts or properties or its relations to other experiences. In mere dwelling, reflection is one with a beholding, a mere representing. But such dwelling can scarcely persist for long in a wholly quiescent form. Involuntarily—perhaps precisely because of the interest that what is experienced holds—a movement of thought will be released, one that serves to bring out its inner and outer relations. Thus when the poet involuntarily reaches for similes and images in order really to let his subject come to expression. Or when perhaps

it is precisely the contrast with other experiences that serves to illuminate the peculiar character of the single experience. Both likeness and difference can thus serve to cast a clearer light on what was given in the original experience, and reflection therefore does not necessarily stand in opposition to it, but can work in its service and in its spirit, when it has itself vanished.

By the word experience we think chiefly of states in which we were essentially receptive. But reflection can also serve the involuntary life of the spirit, where this appears as urge or striving. In an instinctive or inspired manner, involuntary striving is often led to its goal without any reflection being needed, and perhaps better than could happen with the help of any reflection. When reflection comes to operate in our action, it will be because the involuntary striving met resistance. There may then be needed a deliberation upon what it really is that is being striven for, upon this goal's relation to other possible or actual endeavors, and upon the ways and means that can lead to the goal. Here too it is through the bringing-out of likenesses and differences that reflection renders its service. And here too a difference can appear between dwelling and movement. It may be that the representation of the goal is held fast without great oscillations, and that the relation to ways and means or to other possibilities plays a subordinate role, as when the helmsman stares at the compass or the seamark, and his hand involuntarily moves the rudder according to what he sees. But it may also be that the strongest movement back and forth among various possibilities takes place and must take place before the decisive concentration that is called decision sets in.

The arising of reflection is like a kind of awakening, not from sleep or from unconsciousness, but from the involuntary mental life and its experiences, from states that may provisionally be characterized as sensation, intuiting, representing, mood, emotion, urge, and striving. In its dwelling form, reflection stands closest to the involuntary mental life. In its striving to bring out likenesses and differences, it moves further away from the involuntary. Linguistic usage marks this difference rather aptly by distinguishing between *thinking of* something and *thinking over* something. It is the latter kind of reflection, the discursive, mobile kind, that we shall have most use for in what follows, since it is the one we can best observe, and likewise the one that plays the decisive role in the tasks of thought. Only for this kind of reflection do problems exist. In what follows I therefore understand by thinking this kind of reflection. And yet there is only a difference of degree between the two kinds. Only for a brief moment is a dwelling representing, a holding-fast of a memory-image, possible; precise observation shows that in greater or smaller rhythms we move toward and away from the object we are thinking of. And every return can bring with it a recognition, so that the relation of likeness comes to play a role just as in discursive reflection. Perhaps the representation was new to us at the first moment; but through repeated dwelling it becomes familiar. Apart from this too, dwelling reflection by no means excludes that manifold other representations and experiences can contribute and perhaps precisely condition the interest that makes continued holding-fast of a single representation possible. What holds for discursive reflection will therefore also hold, to a certain degree, for the dwelling kind.

It emerges from this description that the work of thought consists in comparison, in distinguishing or finding likenesses. When I think over something, I seek to determine its relation to something else, and the more relations of likeness and difference I find, the better I have thought it through. This holds of course also when it is one and the same object that I examine in different connections and under different conditions; I then compare the different ways in which the object appears in the different cases.

But comparison in turn presupposes a holding-together or combining, synthesis. I can compare two objects only when I either have them given to me simultaneously in my sensation, my memory, or my imagination, or else retain a representation of the one while the other is present to sensory observation. If the one object constantly vanished entirely when the other emerged, no comparison could take place. The work of thought consists in a combining by which likenesses and differences become determinative for the relations in which the objects are placed to one another in the connection that results from the work. Synthesis is an expression of energy, of the capacity to perform work.

The resistance that is overcome through this work is due to the objects against which reflection is directed and which are given in the involuntary mental life—in sense-sensation, memory or imagination, feelings of pleasure and displeasure, urge and striving.

The combining work must become the more difficult, the greater the multiplicity of objects that are to be brought into relation with one another. The fullness of content can be so great that the psychic energy available in the particular case cannot master it. There then threatens a dissolution of the mental life, a transition to chaos.

Yet it depends not only on the multiplicity, but also on the degree of difference that the objects present. The more they resemble one another, so that they can fuse, or so that at least the work done in the case of the one need not be repeated in the case of the other, the less work is required. Objects that stand in strong mutual opposition, or perhaps even conflict, require a great energy if they are all to be taken up into the conscious connection of the mental life. This holds both for the inquirer, who must find a point of view from which the conflicting can stand as one, and for the practically striving person, who must subordinate the conflicting impulses and interests to the thought of a great and guiding purpose.

It will, thirdly, depend on how independent a place the individual objects are to occupy in the connection that is being worked on. A schematic whole is easier to form than a whole that lets the peculiarities of the individual objects each come into their own. Here it is precisely likeness that can create a difficulty, since it easily leads to fusion and thereby to the abolition of the independence of individual objects. On the other hand, from this point of view a pronounced difference or opposition can be a relief.

Furthermore, a greater work is required the further apart the objects lie from one another in time and space. The simultaneous and the near join together involuntarily, and the connection perhaps even forces itself upon us. And if we leave the objects before us in order to immerse ourselves in the world of memory and imagination, here too the connection can often form itself with ease; "with great ease thoughts take up their abode with one another," but when it is a matter of holding together a present object with distant memories, or with ideals that have been properly to the fore in the soul only in isolated moments of enthusiasm, then special demands are made on psychic energy. To have, in the situation of the moment, the treasures of one's experiences, one's reflection, and one's ideality fully at one's disposal is perhaps the greatest testimony to mental energy.

Finally, it plays a role how close and intimate a connection the objects, despite their difference, are to enter into. There are here many degrees, from wholly external juxtaposition and grouping to the inner connection that a scientific concept, an artistically formed image, or a great decision can express.

Besides these five conditions—which might be said to form an analogy to what mass is in the case of physical energy—account must also be taken of the speed with which the synthesis proceeds. The genius and the man of will can in an instant expend a capital of energy that other people must distribute over a longer span of time. In the domain of the life of feeling, this opposition appears as an opposition between vehemence and inwardness, emotion and disposition[1], and analogous oppositions hold for the other sides of the mental life. In research, discovery often forms an analogy to emotion, the conduct of proof to disposition.

The psychic energy that expresses itself in synthesis we measure by taking all these circumstances into account. Of exact measurement there is of course no question here,

[1]See on this my *Psykologi* IV, 7; VI E, 4–5.

only of an evaluative estimate. But as soon as we seek to give grounds for why one work of thought, one intellectual orientation, one view of life can be called lower or higher than another, we must take all these circumstances into account, and we in fact do so, more or less consciously, and more or less precisely. There is less psychic energy available in the dream-state, where the changing objects each release their own train of representations, than in the waking state, where dominant purposive thoughts bring about a main direction. It is a purely psychological measure that we apply here. And even though, in forming the concept of psychic energy, we have had guidance in the analogy with the concept of physical energy, that concept is nevertheless built upon its own peculiar experiences, in which physical or physiological observations and investigations need play no decisive role. For this reason we need not fail to appreciate the significance of the attempts that have been made, from the experimental-psychological and physiological side, to measure the brain-work presupposed in certain elementary mental activities—e.g. calculating or remembering—through its relation to the muscular work that can be performed simultaneously with it. One can detect such thought-work by the diminution it produces in the muscular work performed at the same time. But this very measurement presupposes that one can, by purely psychological means, distinguish between more difficult and easier mental work; otherwise one moves in a circle. The brain-work corresponding to the mental work described above is, after all, not known directly, but only through the very mental work to which it is assumed to correspond.

3. Reflection presupposes objects on which it can exercise its combining and comparing activity. But now these objects? Does the law of synthesis hold for them as well? We stand here before a question of a peculiar difficulty, closely bound up with the whole nature of the mental life. It is only reflection that acquaints us with the objects. It is the medium through which they are accessible to us. The involuntary mental life in its immediate unfolding must not be described and judged straightway according to the way in which it shows itself to reflection. It is a life within which we are not permitted to presuppose properties and determinations that first come into being through the work of reflection. When, in isolated moments, we are able to immerse ourselves in this involuntary—but not for that reason unconscious—life in such a way that we can afterwards retain a memory of it, it stands before us as a stream, a gliding-along of manifold objects, which we designate as sensations, memories, moods, endeavors, and so on. It is not a chaos of independent objects; already in the involuntary mental life there are connections and wholes as well as differences and oppositions. Every word we use about what we can retain from the involuntary states bears witness to this. Most

striking, no doubt, is the continuity; this is why the word "stream" offers itself so readily for what we call the immediately given. But it is not a uniform stream. Differences of quality make themselves felt; otherwise we could not distinguish between the kinds of objects called above sensations, memories, and so on. And there can be eddies in the stream, or windings, so that the direction is not kept in a straight line. That is to say, strivings of different tendency, of more or less opposite direction, can make themselves felt. With this is bound up, again, that inhibitions can set in, while in other cases, where the different tendencies go in the same direction, a facilitation or an acceleration can appear.

Through the indefinite and figurative expressions that alone stand at our disposal for indicating the basis of objects that reflection presupposes, it shines through that this basis forms no absolute opposition to what emerges as a consequence of the work of reflection. There too we find both unity and multiplicity, both continuity and discontinuity, and we find many degrees and nuances of these relations. It is therefore not justified, with the older English school (*Locke*, *Hume*, *Mill*), to conceive the immediately given in the mental life in an atomistic way, as if it consisted in a multiplicity of mutually independent elements. Both in the psychic and in the physical domain, it is only reflection that leads to the formation of atomic concepts, guided by the urge to make the differences as distinct as possible, so that the connection of them can become as clear and perspicuous as possible. The unity to which reflection bears witness, and which it itself seeks to bring about, thereby becomes enigmatic. But it is also unjustified, as in our day especially *William James*[2] and *Henri Bergson*[3] have attempted, to present that involuntary life as standing in the sharpest opposition to reflection and its work. It then becomes, in the last analysis, the indescribable and inconceivable, and it would become an insoluble enigma how there can be any knowledge at all of that "immediately given" from which reflection is nonetheless constantly to draw. James and Bergson represent a significant reaction against intellectualism's misunderstanding and overvaluation of the significance of reflection, a reaction that in several respects recalls the stand taken by *Hamann*, *Herder*, and *Jacobi* against *Kant.* Not pure reason, but the immediate stream of the life of the spirit is, they maintain, the highest, and brings truth forth. The distinctions and definitions of thought are necessary in order that we may practically find our way in life. And analyzing science has carried the opposition to the immediately given still further—an

[2] *The Stream of Thought*. Mind 1884 (reprinted as Chap. 9 of *Principles of Psychology*). Later especially *A pluralistic universe* (1909), Sect. 7.

[3] *Les données immédiates de la conscience* (1888). Later especially in *Introduction à la Métaphysique*. (Revue de Métaphysique et de Morale 1903).

opposition that language itself already favors, by forming independent words for the individual objects and by taking spatial relations as the basis for our intuitive images. It is as though a Fall had taken place when reflection awoke and turned, inquiringly, toward the involuntary life from which it had itself sprung.

Without a kinship between the involuntary mental life and reflection, it would be impossible to understand how the latter could arise. But in the very descriptions that James and Bergson give of what, for them, in sharp opposition to all elaboration by thought, is the immediately given—descriptions that are masterpieces of descriptive psychology—it is nonetheless expressly stated that in this "given" there are differences and connections, qualities and continuities. James emphasizes especially the constant transitions (continuous transitions) and the difficulty of distinguishing between relation and term (relation and matter related); Bergson stresses especially the undivided and yet heterogeneous nature of the inner stream (durée hétérogène et indistincte; durée dont les moments se pénètrent). But with this an inner possibility for the arising of reflection has been indicated. The whole weight must not be placed on the external causes (practical need, the mechanics of language, and the technique of science). The motive for the awakening of reflection is given with the opposition between continuity and qualitative multiplicity, between the connection and the individual elements. Herein lies the possibility of an inhibition of the involuntary stream, an inhibition that only discursive reflection can remedy, through the clarity and definiteness it is able to bring. As a combining that is accomplished through comparison, reflection can bring about clarity concerning the relation between continuity and discontinuity, unity and multiplicity. And it is not anything wholly foreign that it brings in by its work; for already in the immediately given there appear, after all, likenesses and differences, connection and multiplicity—and likewise, what is especially significant, an involuntary striving to bring about harmony among the different tendencies. If there were not such an original striving, it would be unjustified to speak of a stream, of a continuité mouvante, a unité de direction[4]! There shows itself here, then, an analogy to what, in the case of reflection, we called synthesis—a tendency to gather and unite. And reflection awakens when there is a special urge for a fuller carrying-through of this tendency than is possible within the involuntary mental life. Already within this—not only after reflection has

[4]One of my young listeners wrote to me some years ago: "I can declare myself free of every purpose, as of every interest or belief in anything outside or inside me. I do indeed feel something willing, something pressing forward, but it lives, as it were, its own life, pushes aside everything that gets in its way, and surely never commits any act that is not to its own advantage."—Here reflection has brought no result, only dissolution; but the involuntary life has maintained itself in its unity and in its definite direction, which was all the more striking to the young psychologist the less his reflection could attain to any firm standpoint.

awakened—crises and catastrophes can arise, and chaos can threaten. A work can be required even before reflection can awaken; but the difficulties arise, are worked on, and are resolved on different terms in the involuntary mental life than where reflection has taken the helm. Perhaps some of the most significant things in our mental life take place in the regions below the domain of reflection, and it is often only the after-effects of these deep-lying processes that reflection has to deal with. It will always be impossible to demonstrate that at any point we stand before an immediately given in the absolute sense, that is, one in which absolutely no psychic work has been done. What we in each particular case call immediately given deserves this name only in relation to a certain definite psychic work, while in comparison with other objects it would perhaps have to be regarded as the result of a work.

4. By starting from reflection and seeking to become clear about its way of operating, we arrived at the concept of synthesis, and we then found that this concept could also be applied to the involuntary mental life, which forms the constant basis from which reflection springs, in order thereupon to make that basis itself an object of its activity. We must make ourselves familiar with the fact that the involuntary mental life, as it unfolds prior to the awakening of reflection, can be illuminated and designated only by way of analogy. We apprehend it only in analogy with clear reflection, and the expressions of language are throughout drawn from the world of reflection. So far is that involuntary mental life from being the thing easiest to apprehend, that it takes precisely great art to get a proper view of it in its difference from the mental life shaped by reflection. What one calls sound common sense operates, in its psychology, with concepts drawn from the world of reflection. A childlike tendency makes itself felt to regard everything in the mental life as the fruit of clearly conscious thinking, of deliberation and calculation. One personifies, so to speak, not only outer but also inner nature, placing a person behind the person. The childlike psychology explains personal life itself, in its involuntary unfolding, by supposing that behind it there is a person who consciously thinks whatever stirs in it—just as in older works on optics one can find depicted a man inside the head, looking at the image that the light-rays form on the retina!

Reflection is required in order to discover and apprehend the involuntary mental life, and yet the peculiarity of the latter consists in the fact that reflection plays no role in it. This difficulty can be resolved only by our apprehending and expressing the involuntary—insofar as any memory of it persists after the awakening of reflection—in analogy with what we know clearly from the world of reflection itself. This is what we do when we say that synthesis is the fundamental form, not only for comparing reflection, but also for the involuntary mental life, and is thus the most comprehensive psychological concept.—

But now a new difficulty arises. Synthesis means combining—but there must then be something that is combined, and this something must be a multiplicity. The objects toward which reflection directs itself in order to hold them together and compare them form a coherent multiplicity; each individual object comes into being for us through an act of distinguishing. And in a single object we can again distinguish individual sides, properties, or parts, which we may call elements. The result seems then to be that we end in an opposition between synthesis and elements, and that the multiplicity of elements is the constant presupposition for synthesis being able to operate! From this one might then draw the conclusion that what is original in the mental life was a multiplicity of elements, and that only at a later point of development did the combining activity intervene. This conception was also asserted in its day by John Locke, and it still often haunts psychology. It relies on the analogy with the formation of a mechanical whole out of elements that could be pointed out beforehand each by itself, as when a heap of sand is formed from grains of sand that previously lay each in its own place and are only now brought together. The question becomes whether this analogy is justified.

When reflection awakens, the mental life does not, however, begin with it; for then it would have no objects. The elements—that is, the parts, sides, or properties that reflection can distinguish in the object or objects, and that it can determine through their mutual likenesses and differences—are, after all, precisely the fruit of a work of thought. They are drawn out of the connection in which they involuntarily occurred, and set in a certain opposition to one another—an opposition without which they could not become objects of clear consciousness. And the characterization that each individual element, and thereby each individual object, receives is conditioned by its relation to other elements or objects. At no point is it justified to assume isolated elements in the mental life, each absolutely independent by itself—a kind of absolute psychic atoms. That is to confuse the effect of reflection with reflection's own basis—at bottom, again an example of the naive psychology that tends, after all, to confuse differences that thought first makes with originally given differences.

And insofar as we succeed in retaining a memory of the state we are in when involuntariness prevails, this state appears—and here the masters of descriptive psychology are undoubtedly right—not as a chaos of independent parts, but as a stream, a pulsation, yet not without differences, but such that there is a connection even between the lowest troughs and the highest crests of the waves. We find, as far as can be judged, unity and multiplicity so interwoven with each other that they cannot be separated in an external manner. It was, after all, precisely in the urge to gain greater clarity about their relation that we found the possibility and the motive for the awakening of reflection.

At no element can we, in the domain of the mental life (any more than in that of matter), come to a stop as absolutely isolated or absolutely simple. There constantly poses itself, for reflection, the task of finding a connection between the elements or objects that might seem to be isolated, and of finding an inner multiplicity in the elements or objects that appear as simple. It is a sign of the dullness of reflection when it comes to a stop too early in either of these respects. In other words: the law of synthesis must always be carried through anew with respect to the objects or elements themselves.

On the other hand, neither can there be any absolutely conclusive synthesis. As long as the mental life lasts, new objects are taken up into the involuntary stream and are involuntarily worked into it. They are new drops, or perhaps new brooks, which are partly reshaped by the stream and partly can also gain influence on its character and its course. To this is added the character of the new riverbeds that the stream must form for itself when the terrain changes. And as long as reflection is awake, it will constantly be able to set itself the task of bringing about a greater, more comprehensive, and more intimate synthesis than before. There is always a psychic work to be done. The synthesis that has hitherto enclosed the multiplicity of the mental life can again come to stand as object or element for a later synthesis. As soon as we become conscious of our preceding mental states and activities, such a transition to a new synthesis takes place, whether or not this signifies a progress, a more valuable state. It is a new I that becomes conscious of our earlier I's. No synthesis is definitive. Even the philosophy that spoke most of higher unity always fell into embarrassment when it had to speak of the highest unity.

It can of course be entirely justified, in a special connection, with a view to a particular investigation, to disregard these difficulties of principle and to regard elements or objects as if they were absolute and simple. Every special scientific investigation presupposes a certain abstraction. If, e.g., experimental psychology really needs to operate with a kind of psychic atoms or flashes as independent and simple, this can be just as justified as when, in natural science, one operates with material atoms. But one must avoid the dogmatism that confuses concepts we have arranged for the needs of an investigation with ultimate realities. And likewise it can be justified that in everyday speech, or in a practical—e.g. ethical or juridical—respect, we regard the individual as an absolute unity with a completed synthesis, without regard to the fact that synthesis at bottom signifies a work, an activity, not something given once and for all. There is an irrational relation between the mental life and what we call its elements. The point now is to find as many elements as possible; but it can be justified and necessary to confine oneself to a few decimal places, provided only that one does not forget that the question is then not exhausted. The irrational is here, as everywhere, both a barrier and a task.

There expresses itself here an antinomy that is closely bound up with the essence of the mental life and is peculiar to the concept of personality. Mental life and personality cannot be conceived as products of previously given elements, since the elements we know always have their properties precisely by being members of the mental life. We here come to move in a circle, unless we are clear that an inexhaustible reciprocal action prevails: the parts are determined by the whole, and the whole by the parts. In each particular case, in each particular psychological consideration, we must investigate how far the synthesis, the formation of the whole, has advanced, and on this basis observe how the emergence of a new object operates. But the very distinction between synthesis and object is always more or less artificial in relation to the constant interplay of meeting and parting, of continuity and discontinuity, of wholeness and dissolution. There is here a mental rhythmics that forms an analogy to—or rather a prototype for—the rhythms that characterize the course of organic life, a rhythmics that by no means begins with reflection, but, by all we can judge, already makes itself felt in the involuntary mental life. Indeed, this rhythmics recurs in the very relation between reflection and the involuntary course of the mental life. The difficulty of understanding the origin of the mental life rests precisely on the fact that it cannot be explained as a product of previously given elements, but from its very first moment moves in a rhythm—and that in a rhythm of the peculiar kind that none of the members in the rhythm can have absolute priority over the other.

5. In the history of the concept of synthesis this antinomy has been a stumbling-block. The concept of synthesis itself is an old idea, traceable in Plato and Aristotle, and especially in Plotinus in antiquity, and which from Leibniz's time on has always, with more or less energy, been asserted in psychology and in philosophy generally[5].

*Hume* cut the knot by dissolving the mental life into absolutely independent elements, and consistently he therefore found an insoluble enigma in all unity and connection in consciousness. The principle of unity (the uniting principle) was for him the unintelligible. *Kant* started from the opposite point of view, that it is precisely the unconnected, the

[5]On the history of the concept of synthesis in more recent times, see *Georges Dwelshauvers*: *La synthèse mentale* (Paris 1908), pp. 179–223 (an account of the concept of synthesis in Leibniz, Kant, Wundt, Høffding, and Pierre Janet).—In *S. Kierkegaard* this concept appears in an energetic and profound manner, especially in the measure he applies in his doctrine of the stages. See my book *Danske Filosofer*, pp. 153 f.; p. 165 f.

isolated and mutually independent, that is unintelligible to us, both in psychology and wherever else we meet it. But he still held fast to Locke's presupposition that, as the basis for thought's combining and ordering activity, there was given a multiplicity that was only now to be connected. Thereby he arrived at his fateful distinction between the form of cognition and its matter. This too was to cut the knot. And yet Kant did not lack the insight that already in the involuntary mental life there operate forms analogous to those that clear reflection employs. Synthesis is, he says, the effect of a blind but indispensable function of the soul, without which we would have no cognition at all, but of which we only rarely become conscious. It is the task of thought to form the synthesis into a concept. And this same blind function of the soul (Kant calls it imagination) is likewise what conditions the connection between the two extremes of our cognition, "sensation" and "understanding." It was surely only Kant's reluctance to give psychological considerations too large a place in his investigation that held him back from giving a fuller characterization of the involuntary course of the mental life. J. F. *Fries* developed Kant's doctrine further from this side and is even regarded by his disciples as the real discoverer of the involuntary basis[6]. But he then believed that in this basis—the spontaneous striving, the immediate cognition—he possessed a rule of truth exalted above all error. Unfortunately this rule is ineffable, since it can come to consciousness only in an imperfect and fragmentary way through reflection! A dogmatism comes to the fore in the firm belief in the validity of the involuntary mental life—a dogmatism that signifies a decided step backward in relation to Kant. Already when Fries assumes that "spontaneity" is not only continuous but also constant, he says more than any reflection could demonstrate. The irrational relation between immediacy and reflection is not strictly maintained.

This relation is, as already mentioned, very sharply emphasized by *Henri Bergson*. This thinker is not satisfied with the concept of synthesis, because it always presupposes an opposition between unity and multiplicity. Is it not an opposition—he asks—that recurs, whatever one thinks of or speaks about? What matters is to say *which* kind of unity and *which* kind of multiplicity we have in the mental life. The streaming life of the personality cannot be defined by means of concepts that always stand in a certain relation of opposition to one another. One cannot, through thesis and antithesis, arrive at a real unity. One must, conversely, immerse oneself in the immediate stream of the mental life in order to see the opposition emerge from it. Immediate beholding (intuition)

[6]Against this, see my review in *Göttinger Gelehrte Anzeiger* (1907) of "Abhandlungen der Frieseschen Schule. Neue Folge."

must take the place of reflection[7].

Bergson's standpoint recalls, viewed from one side, that of Fries, except that he gives us, with far greater mastery, an understanding of what is meant by "spontaneity." But whereas Fries and his disciples held that true cognition is attained by translating, as well as we can, that spontaneous basis into the forms of reflection, Bergson demands that we subdue reflection and, by an act of will, dive down again into the immediate, feeling ourselves one with it. It stands here as an enigma how the validity of a cognition can be demonstrated without reflection. One of the special problems of thought will later give us occasion to bring out this difficult point in the philosophy of the brilliant French thinker.

In response to the remark that the concepts we use in our reflection to characterize the mental life—synthesis, unity, multiplicity, and the like—find application to all objects, I will maintain that the reason why that group of concepts constantly finds, and must find, application to all objects is to be sought precisely in the fact that they express the forms in which the mental life, and cognition in particular, unfolds itself. Only through the application of these forms does the appropriation, the taking-up into the connection of our own being, in which all understanding, all cognition consists, become possible. No wonder that all products of cognition bear their stamp! Our cognition consists in a conscious or unconscious analogy with our own being.—

The relation of opposition that has been discussed in the foregoing under various expressions (object and reflection, matter and form, spontaneity and reflection, intuition and analysis), *Stumpf* designates as a relation of opposition between phenomena and psychic functions[8]. With respect to this relation he advances, with all possible reservation, the hypothesis that psychic phenomena are not altered because they are analyzed. When we become aware of something in an observation that we did not notice at the first moment, this is supposed to bring about no change in our observation itself, but only in the degree of explicit consciousness we apply to it. A sound rings the same whether or not I notice individual tones in it; a dish makes the same impression whether I can distinguish something sour or sweet in its taste, or whether the taste acts as a unified whole. In my judgment it is, from the outset, not probable that this is how matters stand. The differentiated and differentiating attention in which analysis consists must alter the object for us. The fact that we have discovered a spot in an otherwise beautiful picture—a spot we did not notice at the first moment—can give the whole picture a

[7] *Introduction à la Métaphysique.* (Revue de Métaphysique et de Morale 1903, pp. 15–17.)

[8] *Erscheinungen und psychische Funktionen.* (Abhdl. der preussischen Gesellschaft der Wiss. 1907), pp. 15–21.

different character. The timbre changes; an element has, quite simply, been added that was not present in the first glance. And the relation between the individual parts is in each case a different one after the analysis than before. It can cost an exertion of will to return to the immediate beholding in which the particulars have not yet become the object of special apprehension. There is here a difference between experience and reflection that can scarcely be effaced. There are of course many degrees, as we shall come to see in what follows, in the investigation of the relation between intuiting and judging. And individual differences certainly also make themselves felt. In elastic natures the transition from experience to reflection, and vice versa, will take place relatively easily; and yet I should think that even in them a certain contrast-effect will make itself felt between the two states.

*Stumpf*, for that matter, will only maintain the possibility of a differentiated subsistence, during the experience, of what is first differentiated with consciousness during reflection. And he concedes that there is probably no absolute independence in the relation between phenomenon and function "under the usual, complicated circumstances of the psychic process." Whether experimental investigations can, as he surmises, lead to any result here is doubtful. It is not easy to see how the difference between immediate experience and reflection could fall away. And in a psychological experiment it is difficult to obtain a wholly immediate experience.

Chemists often say that the atoms of the elements are present in compound substances—so that, e.g., carbon and oxygen are really present in carbonic acid. And Stumpf holds that the psychologist must have the same right in his domain. I think, however, that there is a difference. For the chemist can demonstrate that there are just as many parts by weight in carbonic acid as in carbon and oxygen each by itself; and at bottom that is all he means by that manner of speaking. Such an equivalence the psychologist is, for good reasons, debarred from demonstrating.—For that matter, a chemist such as *Sir William Ramsay* has expressly denied the right to take that manner of speaking literally, and *Ernst Mach* has expressed himself in the same direction. The analogy is therefore doubtful.—

We do not get beyond an antinomy between experience and reflection, and we may not, without further ado, transfer to the one what holds for the other. Experience itself we may think of neither as absolute continuity nor as a chaos of differences. The continuous and the discontinuous occur in it in all possible degrees, and it is only reflection that takes the opposition between continuity and discontinuity sharply into view, in that it—as will later be shown—receives the double task of finding differences where continuity was immediately given, and intermediate links where discontinuity was immediately given.

6. In the concept of synthesis there lies, as we have seen, an opposition between object and form, which gives rise to great difficulties that become insuperable when by object one understands an absolutely unformed matter. As already indicated, every object will pose a double task for combining thought: to determine its relation to other objects, and to find out differences and relations within the object itself. Hereby arises the constant striving to make the discontinuous continuous and the continuous discontinuous.

But when we do not separate thought from the other sides of the mental life, there is yet a third element, besides form and object, that we must bring out. When a task is posed, there is presupposed an urge to have the task solved. The work on the solution then acquires value, since we designate as valuable everything that satisfies an urge, whether this urge expressly announces itself, or whether it is only from the satisfaction we feel at the work or its result that we infer that there has been a lack that has now been filled.

Thought has biological significance, in that attentiveness to changes in the surrounding world makes it possible for dangers to be averted and advantages gained. But here too the significance of synthesis shows itself, since it is only under very elementary conditions that attentiveness to a single change is sufficient; it will be of importance that the single experience be brought into connection and reciprocal action with other, present or earlier experiences; only then can the reaction to what is experienced become entirely appropriate. Only instinct reacts to an elementary stimulus; when reflection is awakened, a more serious orientation, and thereby a richer relation to surrounding existence, is possible. An interesting example of a synthesis that is perhaps of a purely physiological kind is afforded by what *Pavlov*[9] has called conditioned reflexes. Salivation can be produced not only by direct stimulation with a substance brought into the mouth, but also by stimulation of sight, hearing, smell, or the skin-sense, on the condition that these sense-stimulations have not occurred too long beforehand together with the action of that substance. Here an associative synthesis is at work, which makes it possible for one kind of stimulation to take the place of another with which it originally had nothing to do. Already in so simple a case as this there takes place a combination of present and earlier states. What shows itself in this simple example repeats itself at the higher stages of psychic development.

Immediate value belongs from the first to the result—in the given example, salivation (which of course, as we may here disregard, itself again has its value as a member of the

[9] *The Lancet*, 6 Oct. 1906.

digestive process). The sense-sensations that indirectly produce it acquire at first only mediate value. And at more developed stages it will, similarly, be the result of reflection that has value, while reflection itself is only a means. But it is quite possible that a value can pass over from being mediate to becoming immediate. There then occurs what we in psychology call a displacement of motive.

When it is bitter need that leads to thinking, the value or motive stands in a purely external relation to thought. The value is here itself an object, since, as it cannot be immediately attained, it poses the task of finding ways and means, and thereby comes to stand as a purpose. In the definite and compelled relation between purpose and means we have man's first encounter with the necessity of thought. If he wills A (the purpose), he must also will B (the means). By this path man enters the world of reflection even where there are no more direct motives for entering it. But it can acquire an independent value, can become an object of direct interest, to investigate that necessary relation between purpose and means, which is, after all, the same as the relation between effect and cause; and the value is then bound to reflection itself and its work. It comes to stand as an activity that it is attractive to exercise, whatever result it may lead to.

When a stage has been reached where the value or motive attaches itself closely to the activity of thought itself, so that this is both end and means—when, in other words, an intellectual feeling has developed, a joy in thought itself, as it unfolds according to its own laws and according to the demands of the objects—then we need not specially recall motives and values in the case of cognition. We then desire no other satisfaction than that which is bound to the use of our faculty of thought.

It is such a stage that we presuppose in an investigation of human thought. All purposes other than that of seeing where consistent thinking leads then do not exist, and we need not constantly recall the standpoint of value; this is given once and for all together with the very urge to—observe and to infer. Even where other motives than the intellectual interest itself actually make themselves felt in a movement of thought, it will nevertheless be fortunate if such motives—in relation to which the work of thought stands only as a means—pass over, through displacement of motive, for a while into purely intellectual motives. Thereby the work is best done. And it will in any case always be, in the end, an examination undertaken from purely intellectual motives that must revise the work performed from other motives, whatever these may have been.

7. The concept of energy in the sense in which we have here taken it has its origin in natural science, and it is by way of analogy that we have attempted to transfer it to the domain of the mental life. All psychological expressions are, after all, transferred from

the material to the spiritual, from the world of space to the world of spirit and thought. But there are difficulties in carrying the analogy through. An exact measurement is possible in the psychic domain only as regards relatively elementary mental phenomena, and never in so perfect a manner as in the case of material energy. Modern physiology stands, as regards method, on the standpoint that it requires physiological states to be explained by physiological causes, insofar as physical causes from the surrounding world do not intervene. Every state of the brain or brain-activity is therefore also to be explained on the basis of the earlier states of the brain and of the organism. According to strict natural-scientific method, every material state within and outside the organism must find its explanation in being seen as having arisen through the conversion into new form of the physical energy that expressed itself in preceding material states. When two brain-states thus succeed one another, the explanation will therefore not be found until it is demonstrated that all the energy that expressed itself in the first state has, without remainder, been converted into the second state, insofar as it has not radiated out into other physiological processes. It is perhaps impossible to carry out experiments in this direction; but such experiments would mark the completion of the physiological method, which can set as its motto *Spinoza*'s words, that an explanation of a bodily state by the intervention of the soul is one with an admission that one cannot explain it. Even if now one of the experiments one might make seemed to show that, in the arising of an organic state, physical energy arose or vanished without a physical equivalent, one would, from a purely natural-scientific standpoint, sooner believe that there was an error in the experiment than rest content with having discovered a limit to the conservation of physical energy. "The principle of the conservation of energy," says *Maxwell*[10], "has acquired so great a scientific weight that no physiologist would place any confidence in an experiment that showed a considerable difference between the work performed by a living being and the sum of the energy received and expended." Since Maxwell wrote down this proposition, experiments have been carried out which—within certain limits of error—show that the exchange of matter and force during the performance of mental and bodily work presents an equivalence between what the organism took in and what it expended[11]. It will hereafter become even more difficult than before to assume that in any part of the organism something should take place that conflicts with the energy principle.

But how, under these conditions, will it be possible to maintain psychic energy as

[10] *Scientific Papers.* II, p. 759.

[11] *Atwater: Neue Versuche über Stoff- und Kraftwechsel des menschlichen Körpers* (Ergebnisse der Physiologie III, 1. 1904).

a peculiar form of energy? Is it not, then, more reasonable to assume that all energy is physical, and that the psychic energy we have spoken of in the foregoing is only a symptom of a physical energy to which we have no direct access?—For we cannot carry through the principle of equivalence within the psychic domain itself, as we can in the physical domain. Interruptions occur within one and the same individual conscious life (through sleep, faint, and so on), and between different individual consciousnesses we cannot demonstrate psychic continuity, as we can demonstrate physical continuity between different bodies—e.g. between two organisms.

Accordingly, the demand has been raised that the psychological definitions should be replaced by physiological ones. Among us, *Carl Lange* (in his "Nydelsernes Fysiologi") has asserted this demand. In strict epistemological and psychological form, *Richard Avenarius*, in his acute work *Kritik der reinen Erfahrung*, has maintained a similar view. Full scientific understanding is attained, he holds, only when the psychic states are traced back to the corresponding physiological states: hence he calls the series of psychic states the dependent vital series, the series of physiological states the independent vital series. And his justification is that only in this way can we reduce the qualitative differences that the psychic phenomena present to quantitative ones, so that an exact formulation of the relation between the changing states becomes possible. The completion of our cognition will thus consist in a reduction of the psychic to the physical[12].

In the face of the views that thus demand a reduction of psychology to physiology in order for a scientific psychology to become possible—views that thus would really abolish psychology in order to make it a science—it must be acknowledged that the actual discontinuity and the qualitative difference among the psychic phenomena will always set limits to a strict carrying-through of psychology as a science. But the question is whether the path they point out leads to the goal—whether the method they praise does not at bottom contain a palpable self-contradiction.

When it is demanded that the psychological definitions should be replaced by physiological definitions, the presupposition for this must surely be that the psychological definitions are complete and exact. But to provide these definitions is psychology's own affair, and if it cannot provide such definitions, physiology cannot know what it is that it should seek to explain in the brain. When that which is to be replaced is vague and wavering, the replacement must turn out accordingly. There will be no certainty that one has really found the brain-states corresponding to the definite psychic phenomena one has in view. The independence of psychology must therefore in any case be

[12] Cf. my characterization of *Avenarius*'s philosophy in *Moderne Filosofer*, pp. 98–106.

acknowledged, since it is to give us the symptomatology that sets physiology its tasks. Now, for every empirical science it is a long and difficult work to arrive at conclusive definitions; such are possible only when the empirical science is essentially finished; they come at the end, not at the beginning, of the investigations. All too often one operates with unfinished psychological definitions that are regarded as reliable starting-points for brain-physiological interpretation. Thus *Descartes*, and in more recent times *Lotze*, regarded the very concept "soul" as so clear and fixed that one could confidently call upon physiologists and anatomists to search for "the seat of the soul." And in the most recent times *Flechsig*[13] has regarded the concept of association of representations as so simple and clear that he could suppose he had found special tracts in the brain for processes that are supposed to bring about the connection between the physiological correlates of the individual representations. It is in reality a great abstraction that Flechsig works with when he sets the connection itself (the association) in opposition to the elements (representations) that are to be connected, so that the former could be thought of as localized in places other than the latter. He thereby quite overlooks the psychological antinomy brought out above, which warns against confusing without further ado the differences and distinctions that the investigation necessitates with oppositions in the unfolding of the mental life itself. There is no "element" in a mental life that does not stand in, and is not determined by, some connection or other, some relation or other. When a representation resists entering into a certain, seemingly natural association, the reason is to be sought in the fact that it is a member of another, more intimate association and cannot be detached from it. The conditions here are very complicated, and with a purely mechanical schema neither physiology nor psychology comes into its own.

But let us suppose that the psychological definitions are complete and perfect. Even then great, unsolved problems will remain. How can physiological states have psychological symptoms? How can qualitative differences correspond to quantitative ones, and how can discontinuous phenomena be bound to continuous processes? Far from such questions falling away through the "reduction" of psychology to physiology, they are precisely sharpened by it and appear in a more glaring form. And in any case it stands as an unsolved enigma how the qualitative differences and the discontinuities come into being. Even if one should succeed in showing that two phenomena, A and B, which one has hitherto regarded as different, are in reality identical, so that one can set A = B, one has not thereby explained how A and B could in the first place stand as different. To call this difference "subjective" does not help; it is not thereby gotten rid of out of existence. It

[13] *Gehirn und Seele*. Leipzig 1896, p. 24 f.

is, after all, precisely the fact that there is something "subjective" in the world that makes psychology both possible and necessary. It is, moreover, within this "subjective" that all cognition of material nature—physiology too, and with it the knowledge of our own body, especially of our own brain—comes into being. Cognition is itself a psychic function, both where it appears as sensation and where it appears as thinking. By regarding psychic states and activities as accidental appendages ("epiphenomena"), the natural scientist would saw off the branch on which he sits. To study this branch itself is, to be sure, not his task; but he does well, nonetheless, not to forget where he is sitting.

When *Avenarius* attempts to carry through the reduction of the doctrine of "the dependent vital series" (psychology) to the doctrine of "the independent vital series" (physiology), it is nonetheless apparent through his exposition that he constantly infers from the "dependent" to the "independent" series, not the reverse. It is on the basis of the psychological observations he has collected that he constructs the physiological schema which for him is the proper object of science, and one cannot understand what is said about the "independent" series except by means of the facts that the "dependent" series contains. In general it turns out that brain-physiology is still far from developed enough to give a thorough elucidation of the material processes to which the great, richly nuanced world of psychic processes is in one way or another bound. What one finds in this respect among physiological psychologists and experimental psychologists are mere schemata, which are supposed to "explain" what psychological experience presents. No one has more occasion to experience this than a psychiatrist who is at the same time an experienced and critical psychologist. Thus *Pierre Janet* has declared that, apart from individual observations and histological investigations, the physiological interpretations are only translations of the psychological concepts into another language, and that if one will think only anatomically, one must wholly give up thinking psychiatrically[14].

8. It is the discontinuity and the qualitative differences that cause psychology's difficulties. But greater weight is often placed on them than experience warrants. Precise reflection often leads to the discovery that even where there seemed to be a psychically empty space, there was in reality a psychic content given, although it was not grasped by attention or by recognition, or was forgotten immediately after the experience. And as regards the qualitative differences, observation will often find more and finer nuances than those one first stopped at and perhaps confidently declared to be utterly disparate. The continuity in conscious life is surely greater than is often assumed, and there can lie a scientific danger in too strongly proclaiming discontinuity and qualitative difference

[14] *Les Obsessions et la Psychasthénie.* I, p. 496; 605. Similar statements by *Wundt* (*Philos. Studien* XIII, p. 363) and by *Ramon y Cajal* (cited in *Herrlin*: *Minnet*, p. 21).

as the distinctive mark of the phenomena of consciousness. The inner connection in consciousness—the fact that the more we immerse ourselves in, and make ourselves one with, the involuntary course of the mental life, the more this appears to us as a whole, a streaming—is and remains the primal phenomenon in the psychological domain. We overlook it so easily only because in everyday speech, and on the whole in practical life, we have most use for certain single drops or waves in the gliding stream. We must perform an integration when we go back from our more or less mechanical practice to the immediately given. And even if we confine ourselves to functions that can be exercised with clear consciousness—e.g. memory and comparison—these very functions bear witness to an inner connection behind the particulars. They are forms of synthesis, of the tendency to combining and holding-together that attests the mental life's constant striving to be a whole. The unconnected is the psychologically enigmatic, that which sets psychology its problems. The task of psychology is to find connection between the elements of consciousness as well as between its different states.

*Leibniz* has the merit of having energetically maintained the principle of continuity in the psychic as well as in the physical domain, especially by pointing to the elements that are vanishingly small in relation to the clearly conscious states, and that only by being summed up and combined produce a result accessible to untrained self-observation. Already a comparison between modern and ancient or medieval poetry shows how many more mental nuances self-observation has now come to perceive. It is scarcely only because the nuancing in the psychic domain is now in itself greater than in earlier times that this greater richness is found. It is partly precisely the knowledge and understanding of the immediate and involuntary in the mental life itself that has grown. The childlike and primitive psychology is inclined to conceive everything that takes place in the soul as clearly conscious and all actions as undertaken with deliberation and intent. Now an explicit demonstration is required that reflection and deliberation have been present in the particular case. The transitions themselves between the unconscious and the conscious, and between the involuntary and the voluntary, can take place through many intermediate forms, many degrees. The more practiced self-observation and criticism are, the more difficult it is found to be to put one's finger on a definite place where the boundary should lie.

Leibniz was inclined to regard all psychic differences as differences of degree with respect to clarity and obscurity. This attempt was characteristic of the century of the Enlightenment, which Leibniz inaugurated. But there are other paths by which psychology can find a continuous connection despite the qualitative differences. There constantly take place fusions and compositions in the domain of conscious life—of sensations, representations, feelings, and endeavors. And displacements take place by which what at first stood in the foreground is supplanted by what was at first background—where what at first had only mediate value comes to stand as immediately valuable. And in integral characters, the highest and noblest objects of psychology, there appears to contemplation so intimate a dependence of everything that stirs in the soul on one purpose, one guiding thought, that we have here a causal connection that is every bit as firm and definite as any in the physical domain. Every single representation, every single mood, and every single impulse is, in such a character, clearly determined by its place within the whole to which it belongs and which it itself helps to shape.

Where, despite everything, a lack of connection shows itself in the psychic domain, we need not, however, give up the principle of continuity, any more than the physicist gives it up when energy seems to have vanished in a state of rest or equilibrium. Just as the physicist introduces the concept of potential physical energy to express the fact that what seemed to have vanished can be released again, so the psychologist will be justified in speaking of potential psychic energy, supporting himself especially on the fact that psychic elements can be reproduced after an interval in which they have not been in consciousness. Words such as "predisposition," "possibility," "disposition," "capacity," "trace," "tendency" express the same as the word potential energy. The psychologist is, however, worse placed in applying this concept than the physicist in applying his corresponding concept. In the first place, the physicist can directly point to the phenomenon in which the potential energy has its seat—e.g. the tensed spring or the stone resting on the roof. In the psychic domain, by contrast, complete equilibrium (rest) means the same as unconsciousness—hence something inaccessible to observation (whatever it might otherwise be in itself). And in the second place, in the physical domain one can carry out conversions that show that a quantum of energy can be released corresponding to the quantum that apparently vanished on the transition to the "potential" state. But in the psychic domain it cannot be demonstrated in this way that predisposition and development correspond to each other. If the concept of potential psychic energy nevertheless has its justification, it is because it stands as the expression of an unsolved but undeniable task, and because it contains a summons to go ever further in observation and analysis in order to find as much continuity as possible between the different states of the mental life. There is a long way to absolute rest—and it might be that it occurs as little in the psychic as in the physical domain. Possibility everywhere subsists in an actuality, whether or not this is accessible to us.

9. The only way in which psychology and physiology can work together without either of them coming to break with its guiding thoughts is provided by the psychic phenomena being regarded as symptoms of physiological states—and, conversely, physiological phenomena being regarded as symptoms of psychic states. We can then *infer* from the one series of phenomena to the other, although we cannot *derive* the one from the other.

It is the natural-scientific method that step by step leads to such a conception. The law of inertia requires that every change in the state of a material point (of rest or of rectilinear motion with a certain velocity) be derived from the influence of another material point. Thereby it is excluded that a non-material (that is, non-spatial) cause can acquire natural-scientific significance. Thereby the principle is established that a material phenomenon must have a material cause. *Maxwell* finds the experiential proof of the law of inertia in the fact that, every time we encounter a change in a body's state of motion, we can trace this change back to an action between this body and another body. This experiential proof natural science is constantly in the process of furnishing.—To the law of inertia is joined the proposition of the conservation of physical energy, which, as already mentioned, seems to have to be maintained also in the case of organic nature.

There are in the physics of our time intimations that the said propositions, despite their great importance for the whole conception of nature, are nevertheless derived from laws and relations that hold for the ether-vibrations in which one sees more and more the fundamental thing. Philosophically regarded, this would change nothing in the conception of the problem of the relation between the spiritual and the material. The vibrations of the imponderable ether are just as different from the psychic phenomena as the movements of the ponderable atoms. The problem remains the same, even if electrons take the place of atoms. It is the spatial character that, for philosophy, is the decisive property of matter; in this *Descartes* saw rightly. It must not here mislead us that physicists sometimes use the expression the immaterial in the sense of the imponderable.—

If now this is so, there can be found no place for the mental life in the series of physical, or respectively physiological, phenomena. The only possibility then becomes the conception that in the mental life we have the expression that self-observation gives us for what appears to our senses as organic processes. Soul and body are then not two different things or beings, but two languages in which existence speaks to us. We can perhaps translate from the one language into the other; but we cannot derive the one language from the other.

Since the mental life cannot be derived from the laws and forces of material nature, we must assume that in all material activity there cooperate, besides the properties whose presence natural science demonstrates, other properties that are not accessible to sensation, and in which the mental life would be able to find its explanation. Philosophically regarded, natural science's concept of matter cannot exhaust existence. But by the "predispositions" or "dispositions" that we presuppose in order to understand the possibility of the mental life's arising in nature, we do not in the least shake the application that natural science makes of what it has learned and seen.—

The conception to which we here adhere really returns to the childlike and practical view that regards the body as the outer expression of personality, accessible to the senses. And we thereby obtain at the same time the best and most durable working hypothesis, in that we are summoned to follow at once the series of psychological and of physiological phenomena, each by itself, as far as is at all possible, and to investigate the relation between them, the analogies and proportionalities, as exactly as possible. It is a functional relation, in the mathematical sense of the word, that this conception assumes between the mental and the material phenomena; we seek, and often we also find, that a variation of the mental states corresponds to a variation of the bodily states, especially the brain-states, and vice versa. And to this is in reality confined the knowledge we have of these things. Our hypothesis (for which the most fitting name is the identity hypothesis, since it does not assume that soul and body are two different beings) we regard provisionally only as a working hypothesis. Whether it gives a conclusive explanation of the relation between the spiritual and the material, between psychic and physical energy, is another question, which will be discussed in a later connection. Even thinkers who reject our hypothesis as a definitive solution nevertheless concede that it is a useful working hypothesis. Thus *H. Bergson* and *C. Stumpf*. Whether this concession is consistent from their standpoint is a question in itself. A hypothesis that has no ground at all in the nature of things will not in the long run be suited to serve as a working hypothesis.—

It has sometimes been thought that the enigma could be solved by assuming (or saying) that psychic energy is nothing other than nerve-energy. But when one does not assume that all nerve-energy is connected with psychic phenomena, a distinction must be made between conscious and unconscious nerve-energy—and the problem is then posed anew. Besides, we know as good as nothing about this so-called nerve-energy and about its relation to other forms of physical energy, and it is of no use to give the nerve-energy to which psychic phenomena are bound a special name. Thereby one only introduces a qualitas occulta and rests content with it. The natural-scientific concept of energy is, wherever one meets it, always formed on the basis of phenomena with spatial, geometrical properties. Natural science knows energy only as the expression of relations between phenomena in space. And we cannot, unfortunately, form a concept of energy from which the physical and the psychic concepts of energy could be derived as special forms. In any case, every attempt to form such a concept would fall outside the limits of psychology. In a later connection we shall return to the problem we here leave[15].

10. We have in the foregoing determined thought as psychic energy and sought to show how, provisionally and purely methodically, we must regard it in relation to physical energy. But there will perhaps arise in many the question, *who* it is that *has* or *exercises* this energy? One reasons somewhat thus: "Every activity must be exercised by someone or something. We must distinguish between that which acts and the activity itself. In particular, in the psychological domain we must distinguish between that which thinks, feels, and wills, and thinking, feeling, and willing themselves."

Such a conception, which occurs not only in everyday speech but also among philosophers, spiritualists and materialists in brotherly union, is due partly to the influence of language's mode of expression, partly to the influence of material analogies.

Language distinguishes between subject and predicate and as a rule designates each of them with its own word. There is therefore formed one word for the acting being, another for its activity. Even where it is difficult for intuition to make this distinction between the acting thing and the activity, language nevertheless often employs a formal subject, as in "impersonal" sentences of the form "it is raining." It is a grammatical schema that prevails here, a schema one has no right to give philosophical significance, either psychological or metaphysical. Long ago the acute philologist *Højsgaard*[16] impressed upon us the difference between grammar and philosophy. "Grammatically," he says, "a word or a discourse is considered when one looks not so much at the things that are understood by the words, as at the words themselves, their order and case, and at their mutual dependence or relations to one another. Whereas a philosopher considers only the things that the words call to mind." This warning is constantly needed.

Because language often needs two words to express what happens when something acts, it does not follow that we have two "things" before us. And how, really, should the relation between these two things be thought—the relation between consciousness and that which "has" it, between energy and that which "exercises" it?

In material phenomena, which have exerted precisely the greatest influence on

[15]With respect to many details concerning this problem I refer to my *Psykologi*, chaps. II and III.

[16]*Methodisk Forsøg til en fuldstændig dansk Syntax*. Copenhagen 1752. (The section "Register og Forklaringer.")

language's mode of expression, we distinguish with ease between the subject of the activity (the movement) and the activity (the movement) itself. We have often seen the body that is now in motion lying apparently still in its place, and we thus know it both as resting and as in motion. When we say that the body moves, it is really these two states that we compare. We therefore fall into embarrassment when we discover that no absolute rest can rightly be assumed. And language especially falls into embarrassment when one has to speak not of "ponderable" but of "imponderable" matter, since this latter is presupposed to be in constant wave-motion. "Ether" can therefore be defined only as that which undulates. It is now the analogy with ponderable matter, and with the apparent difference between equilibrium and motion, that operates when one distinguishes between the soul (the I) and its activity. But the soul we know (like the ether) only as active. What we know here is precisely psychic energy, not any resting "being." The soul's being or nature is activity. It is an intuiting, often a lazy, imagination, not a real thinking, that has the word when one makes that distinction between the thinking thing and the thinking. Spatial images or symbols are confused with psychological concepts. The urge for fixed, resting images, such as material nature seems to present, prevents one from holding fast the concept of activity or energy in its peculiarity and its consistency. Or one fears that the mental life will evaporate if it is not hung upon a "something," a "substance," as upon a secure hook—although one is not so consistent as to ask upon what this hook is fastened. The "substance" could just as well need a "bearer" as the activity. Just as one earlier asked who "has" the energy, so one might here ask who "has" the soul or the I.—Thinkers such as *Fichte*, *Sibbern*, and *Wundt* have the merit of having seen clearly on this point.

We can very well say what we mean by "ourselves" or "our I," without entering into any spiritualistic or materialistic hypothesis. Keeping to experience, we see the mental life unfold itself through different stages and under different forms, and we seek to describe these and to explain the transitions between them. At every given stage of development we then have the mental life before us with certain properties and dispositions, and the task is then to find out what, under certain conditions, can develop from them. (Cf. 8.) "We ourselves" or "our I" has at every stage its content in the properties and dispositions that the phenomena of consciousness make known, and its form in the degree and manner in which these are combined and ordered.

Now the more the continued mental development is conditioned by the given properties and dispositions in their mutual relation, the more energy we ascribe to the mental life and the more active it is said to be. The more, on the other hand, the continued mental development is conditioned by influences that make themselves felt independently of the individual mental life's past, the more passive it is said to be.

Just as the planets in our world-system tore themselves loose, with a certain initial velocity and initial direction, from the great nebular mass, so the mental life—at the point where it awakens from the night of unconsciousness—begins by acting according to the same laws that it later follows. It begins with peculiar dispositions that give it a certain independence in the face of outer influences. With this agrees the fact that the nervous system is one of the parts of the organism that first appear with distinctness, and that from the first it occupies a central position in relation to other parts of the organism. The functions of the nervous system, to which the mental activities correspond, are not mere effects of outer influence, but determined in form and direction from the individual's first coming into being.—

In what follows we investigate, for the closer elucidation of the nature of human thought, first the more immediate and involuntary way in which it expresses itself in intuiting, association, and comparison (B), then the more conscious and exact form it takes on in judgment (C).

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## B. Intuition, Association, and Comparison

### a. Sensation

11. When one often distinguishes sharply between sensation and thinking, it is because one conceives sensation as a purely passive coming-into-being of new psychic content, while in thinking one sees a working-over of such content. But one cannot at any point wholly separate passivity and activity—that is, wholly separate states whose causes lie entirely outside an individual's preceding states from such as should be wholly due to these. There always show themselves traces of psychic work, even in the simplest elements of consciousness that can be distinguished. We can be more or less active—that is, the causes of the changes that take place in us can to a greater or lesser degree lie in ourselves, in our character and our preceding experiences; but wholly passive we never are.

If we bring out the conditions for the arising of a single sense-sensation, e.g. a light-sensation, it turns out that the condition for this arising is a change that has a certain degree of strength, a certain degree of suddenness, and a certain degree of qualitative difference—all in relation to the given state of consciousness. A sensation arises only when the new, in the three respects mentioned, stands in a certain definite relation of opposition to what was previously given. Perhaps, of the three conditions, suddenness is the most important, which from a practical biological point of view is easily understandable, since it is of importance in the struggle for existence to be awakened to attention for changes in the surrounding world, even apart from the degree and character of such changes. But all three conditions point, psychologically regarded, in the same direction. The weaker the change is, the more slowly it sets in, and the more alike the new and the earlier state are, the less possibility there is for the arising of an independent sensation; hence one can rightly designate every sensation as a distinguishing.

It is, to be sure, a distinguishing of a purely elementary kind. The single sensation steps forth in consciousness with its definite degree of independence and with its definite quality, and it is only indirectly, especially by way of physical and physiological investigation, that we can here demonstrate conditions similar to—in any case analogous to—those that condition a distinguishing undertaken with reflection and intent. When we set ourselves to apprehend the difference between two things distinctly, we can do nothing further toward it than to bring these two things into immediate proximity to each other—then the difference announces itself involuntarily, springs forth as it were, since each of the two things stands involuntarily with its own peculiarity. We here produce, by express holding-together, the opposition necessary for a distinguishing; but the apprehension of the difference we call forth only indirectly. And thus too, in the arising of the simplest sensations, the given changes or oppositions are only releasing conditions. The sensation arises as if it were a result of comparison; but the two opposed terms are here not known each by itself; it is only the change in the state, released by the opposition, that appears as an element in consciousness.

It is, then, a justified analogy to call the sensation a distinguishing—and so far a thinking—and already in it to find an expression of psychic energy. Language, which has formed itself under the influence of reflection, has no expressions for functions so simple as those we here confront. Linguistic usage is inclined now to set a very great difference between the simplest psychic functions and the more developed ones, now to read "inferences" and other forms of reflection without further ado into the simplest utterances of consciousness. But by the concept of analogy both the kinship and the distance between sensation and thinking can come into their own.—How one conceives the unconscious basis from which the simplest utterances of conscious life are thus released by the power of the oppositions—whether one declares it to be merely purely physiological processes, or whether in the unconscious one will see a potential mental life—is in this connection a matter of indifference. The main thing is that we already here trace the chief form of the mental life. For that matter, it shows itself already on a purely physiological consideration how little the sense-sensation can be conceived as a passive effect of stimulations from without. The physical impression traverses, on its way from the outer sense-organ to the great brain, so many layers of nerve-cells that it can only most improperly be said to be the same impression when it reaches the brain as when it comes to the sense-organ. And in the brain itself there is released a reciprocal action of a mass of cells as a consequence of the stimulation. What emerges in consciousness as a sensation corresponds, therefore, physiologically regarded, to a multiplicity of processes; there is gathered into unity in it what physiologically stands as a plurality.

12. That the sensations are thus not simple, passive effects of stimulations from without, but can be regarded as elementary forms of psychic energy, becomes of essential importance with respect to the validity that can be ascribed to them as means to cognition. We build up our world-picture on the basis of them, and yet one became very early aware that they could not be used without further ado to form a coherent and intelligible world-picture. Reflection must at once use them and disregard them if it is to reach its goal. The reason for this is that they are not only the expression of something different from our consciousness, but also of the very nature and mode of operation of our consciousness.

In a characteristic way, reflection's two-sided relation to sensation appears already in one of *Democritus*'s fragments, in which the senses speak thus to the understanding: "You poor understanding, you fetch your grounds from us and yet want to overturn us! Thereby you prepare your own downfall."—At the founding of modern natural science this paradox appears again, in that *Galileo*, *Descartes*, and *Hobbes*, at about the same time and on the basis of the same considerations, set up the doctrine of the purely subjective validity of the sense-qualities, all while the new science was nonetheless grounded upon sense-observations. In *Kant* we meet it again in his distinction between matter and form, in that he counts all sense-sensations to the matter and regards only the form—which, however, never occurs separated from the matter—as that with which science has to do.

If sensation were purely passive, it could be assumed simply to reproduce what calls it forth from without. The subjectivity of the sense-qualities is grounded precisely on the fact that they come about as a consequence of an activity on the part of consciousness. That there exists a problem of cognition rests—as will later show itself—precisely on the fact that we are self-active in our cognition. How can we, by our own activity, cognize something different from ourselves? By our very activity we transform it, after all, according to the laws of our consciousness. Cognition is our work. And in this respect there is, according to the foregoing, no difference of principle between sensation and thinking. The paradox then disappears when one gives up the absolute difference between sensation and thinking, matter and form. The work of reflection then stands as a continuation of the psychic work that sensation has already begun.

Insofar as reflection does the work of sensation over again, it is thus not because with it a wholly new principle comes into force. It is only psychic energy operating in another manner, more appropriate for the solution of the task of cognition. The understanding can thus quite well at once use and overturn sensation. And the “overturning” takes place partly with sensation’s own help, in that the one sense controls the other. To the relation between the qualities within the one sense there can correspond relations between qualities within the other sense, and it is through the analogies that can here be demonstrated that reflection reaches its result. The senses by which the forms of synthesis most clearly come to light, and by which the highest degree of continuity can be attained, are taken as basis. And it is especially the senses that permit clear apprehension of movement. One might call them the rational senses. To these belong sight, hearing, the sense of touch, and especially sensations of movement. The sensation of movement takes place distinctly by a synthesis, since it presupposes the holding-together of the successive positions and stages of the moving thing; only by such a holding-together does movement become an object of consciousness. The special qualities—color, hardness or softness, sound—that correspond to the different stages of the movement could in themselves be thought of as exchanged for others, provided only that the relation between the qualities among themselves continued to be the same. One can then disregard them, and the sensations then become, when they stand in the service of cognition, only symbols for certain stages or places in continuous processes of movement. In a sense-sensation, with its degree of independence and its peculiar quality, a concentration has taken place, a combining of what, physically and physiologically regarded, is a multiplicity. Modern physical theories interpret what for us is only the object of a single act of sense as an aggregate of small parts in constant movement. Our sensation appears here as analogous to a statistical result of a mass of particulars that are each by itself inaccessible to us and that stand in the most complicated mutual connection. The kinetic theory of gases is especially an example of this. According to it, every gas is an elastic fluid whose elements are in constant movement, with numerous collisions against one another and against the barriers, and consequent changes of direction and velocity. For our sensation it nonetheless appears as a whole, within which we distinguish nothing. And the same holds, as already remarked earlier, of the relation of sensations to the brain-processes. What in our consciousness appears as a single sensation corresponds to a very composite reciprocal action between the millions of cells of the great brain and between these and the other parts of the brain.

What takes place in the nervous system and in the brain does not resemble what stimulates the sense-organ from without. And what takes place in the brain probably does not resemble what takes place in the sense-organ, the ingoing nerve, and the lower centers that the impressions pass through on their way. The sense-sensation resembles neither any of these physiological processes nor the outer stimulations that release them.

Innumerable enigmas conceal themselves here. And yet it is a fact that sense-sensations can guide living beings in their struggle for existence. Our states can very well be decisively determined by our own nature, provided only that they lead to the release of activities by which we can maintain ourselves against the surrounding world. In instinct, which is released by definite sense-stimulations, this appears especially clearly. The answer is given in a quite different way than the question is put. In instinct, urge and capacity are so united that they are both released and operate together when a certain definite change sets in in the state of the living being. Thus when the pregnant bird builds a nest, or when the newly hatched chick scratches if it feels gravel under its feet.

Already the first thinkers who became clear about the subjective nature of the sense-qualities emphasized, in connection therewith, the practical significance of sensations as the original, the theoretical significance as the derived. Thus especially *Descartes*, *Malebranche*, and *Berkeley*. Modern evolutionary theory has again brought out this point of view. Sensation is first and foremost a means in the struggle for life, only secondarily a means to come to know the nature of existence. The connection between its practical and its theoretical significance lies in the concept of sign or symbol. The sign can be a signal for action; but it can also become a signal that announces the setting-in of a certain change, apart from the action necessitated by this change. There are many intermediate degrees between these two significations. But in neither of the two significations is there needed any likeness between the sign and what is to be signified. And the sign perhaps operates best by being as simple as possible—by presenting as a unity what is in itself a multiplicity. The main thing is that it can serve as an introduction either to an action or to a train of thought that can attest their validity—the action by leading to the satisfaction of an urge, the train of thought by leading to understanding and to agreement with new sensations. In this last case too it is a matter of the satisfaction of an urge—namely of the very urge to understanding, to the agreement of our reflection with as many sensations as possible. Motive (value), form, and object thus always operate together. (Cf. 6.)

The special qualities with which the different sense-sensations arise nevertheless contain a great, perhaps insoluble, problem. For from a biological as well as from an epistemological point of view, whatever quality a sensation appears with, sensation can perform its function in the service of life and of reflection, provided only that it is able to release activities or trains of thought that can satisfy the practical or theoretical need. Why precisely these definite colors arise for us when certain electromagnetic stimulations have propagated themselves through space to our eye, or why we hear precisely these definite tones when certain air-waves have propagated themselves to our ear, stands as a constant enigma. We stand here at mere facts and must for the present remain standing at them.

Each single one of our senses has its series of qualities, more or less continuous and systematic. Even where the continuity is greatest, as in the domain of the sense of sight, it is nonetheless interrupted at definite places, and the series of qualities is always limited. To infra-red rays the retina is not receptive, and ultra-violet rays affect it only at a particularly great degree of strength. This limitation, which probably has its ground in certain physical or physiological conditions, and perhaps has its appropriateness, entails a limitation of the possibilities of knowledge.

Each sense-organ has its own peculiar advantage. The ear possesses the fine power of discrimination by which partial tones can be recognized within the sound in which they occur. The eye's power of discrimination shows itself especially as the capacity to localize the different visual impressions. In the different living beings, according to their conditions of life, different senses show themselves especially developed. All such special degrees of development must, just like the limitation, be assumed to have arisen through reciprocal action between the inner and the outer conditions of life.

When we here speak of living beings and of a surrounding world, we make use of a knowledge that, like all other knowledge, is itself won on the basis of sensations and by the work of reflection on this basis. It is then presupposed that the psychic energy that already expresses itself in sensation has developed in the form of reflection. The biological knowledge we make use of for understanding the development of the psychic activities is itself a product of these activities.

13. It has already shown itself above that a single sensation is an abstraction. The character of every sensation rests on the relation between the conditions under which it arises and the conditions under which simultaneous or preceding sensations arise. And

there showed itself here an analogy between its arising and the discovery of a difference effected by reflection. It was for this reason that we could call it a distinguishing.

The psychic equilibrium must be abolished for a sensation to arise. But there always expresses itself a tendency to equalize this disturbance. And this tendency appears in two forms, a more elementary one, which consists in fusion, and a higher one, which consists in the formation of a sense-intuition.

Fusion sets in, as already indicated, when the change produced by the impression is slight in relation to the state we are already in, or when changes follow very rapidly upon one another. Examples of fusion we have in the timbre of tones and in the cooperation of colored and uncolored light-impressions in the coming-into-being of light-sensations. In a visual image due to seeing with one eye, it is not noticed that the blind spot on the retina gives no contribution to the filling-out of the image. When a regular alternation takes place of light-stimulation in the time a and complete non-stimulation in the time b, there is nonetheless obtained, when the time a + b is very short, a continuous sensation. Even changes that take place simultaneously in different sense-domains fuse insofar as they mutually inhibit or reinforce one another.

It would scarcely be right to see, in such a fusion—in which the individual elements cannot make themselves independently felt—the original form of conscious life, so that an "undifferentiated continuum" should be the starting-point for mental development[17]. All conscious life is scarcely ever fused; there will in any case probably stand different fused complexes over against one another. A complete state of fusedness would be a state of rest, which would be one with a state of unconsciousness. A certain breaking of oppositions, or at least of differences, is necessary for the subsistence of conscious life.

When fusion is not possible, and there nonetheless expresses itself a tendency to equalization, a sense-intuition can form itself, in that the sensations are ordered in such a way that each single one can make itself felt without swallowing up or supplanting the others. With the least possible change, the new content that a sensation brings is then incorporated into the whole connection of consciousness. The sense-intuition solves a double task, in that it at once lets the novelty and peculiarity of the single sensation and the connection of conscious life come into their own. It presents at once multiplicity and unity, while fusion abolished the multiplicity. The sense-intuition is clearest when the differences have previously appeared each by itself and only thereupon are connected in such a way that a single act of attention can survey them. The clearest form of such a surveying is space-intuition, which we therefore have a pronounced tendency to use as a

[17] *Einar Buch*: *Om Fornemmelsers Sammensmeltning, særlig ved Klangindtryk.* Copenhagen 1898, pp. 31–35; 60 f.

schema for everything we wish to make clear to ourselves. In particular, a time-intuition first becomes possible through the symbolic use of space-intuition, although we can of course quite well apprehend change and succession—e.g. of tones or of our own inner states—without using a spatial schema.

Sense-intuition denotes a higher stage of psychic energy than fusion. The multiplicity is preserved through an involuntary ordering. There will always be one or more single elements that determine the ordering, and around which the others group themselves—namely those that correspond to the dominant urge or interest. Therefore every involuntary sense-intuition will be articulated, pointed in a definite direction, that in which attention is concentrated. This determinacy of direction forms an analogy to what later stands as purpose for striving and object for reflection. Out of this concentration the other members in the sense-intuition fade off into the indefinite and obscure.—

In this description we must constantly remember what was above called the psychic antinomy. When it is said that elements fuse or become members in a sense-intuition, the meaning is not that the elements, apart from these kinds of synthesis, have the same character as they have within the syntheses. It is only in our abstract language that we can let the elements go in and out of the different connections like a kind of atoms.

## b. Recognition

14. A pure sense-intuition would be one in which a multiplicity is involuntarily ordered without earlier lived experiences or experiences contributing. Apart from the most primitive states, such a pure sensation scarcely occurs. As soon as, in the multiplicity that is ordered, there are found elements that have previously been present, there is the possibility of a new form of psychic energy: recognition. The simplest example of recognition we have where a sensation, by reason of repetition, immediately appears with a peculiar quality, different from that with which it appeared when it first appeared. I use the expression quality to designate the wholly immediate difference that can obtain between the known and the new—a difference that can be as simple and indescribable as the difference between yellow and blue. The quality that a sensation can acquire through repetition I call the familiarity-quality.

While in the simple sense-intuition it is essentially the difference between the elements that operates, in recognition the relation of likeness makes itself felt. What conditions the transition from new to known is not itself something new, but is, on the contrary, precisely a repetition. How the familiarity-quality is to be explained physiologically is not decisive for its psychological and epistemological character. It is, however, most natural here to point to the influence that repetition and practice have in altering the course of processes and functions. Through repetition one and the same function, probably also the brain's function, can take place with greater ease, and recognition might then be the psychic correlate of the altered character of the functions. However this may be, the circumstance that the relation of likeness now makes itself felt marks a significant advance. One can already here speak of an understanding: the present here stands immediately as one with something earlier experienced. There is a foothold, a home for it in consciousness. *Plato* called the dog a philosophical animal, because its behavior toward men is determined by whether it knows them or not. In this sense newly hatched chicks too are philosophical animals, since they scratch only when they feel gravel under their feet, not on a carpet. The remarkable thing about instinctive actions is precisely that they are released by stimulations that are at first new to the individual and yet operate as if they were known.

In everyday speech we often use the word "understand" in the sense of recognizing. It is an understanding of the most elementary kind. I understand a sound when I know what kind of sound it is, that is, recognize it. I can in this respect understand a word when I know its sound, even if I do not know its meaning. Understanding has many degrees. I understand what kind of figure it is that approaches when I recognize the human figure in it; but there can still be much to understand, namely which person it is, and so on[18]—

Just as in the simplest apprehension of difference, so in the simplest recognition language lacks apt designations. But there is a significant analogy between such recognition and the comparison undertaken with reflection. In genuine, free comparison two independently given sensations or representations are presupposed, whereas immediate recognition takes place without the present and the earlier experienced needing to appear each by itself in consciousness. Just as a sensation can arise immediately when there is a certain opposition between the influence of a stimulation and the state at hand, so immediate recognition arises on the presupposition of a relation of likeness between the influence of a stimulation and the dispositions or after-effects that earlier stimulations have left behind. In both cases it is only the results, not the presuppositions, that appear in consciousness. The elementary distinguishing and the immediate identifying are the first intimations of what at higher stages appears as thinking in the narrower sense of

[18]It is therefore not an apt objection that *Husserl* (*Logische Untersuchungen*, II, p. 73 f.) makes against me, that unintelligible words too can sound familiar. In the wholly immediate "understanding" of a word, it is only the sound or the sign, not the meaning, that is known.

the word.

15. Distinguishing and recognizing denote two poles in our cognition, which under different forms are found again at all its stages of development, and which correspond to two necessities of life: receptivity for the new, the changing, and the capacity to take it up into oneself in as intimate a way as possible. The two capacities can stand in sharp opposition to each other. They are two different kinds of psychic energy, a centrifugal and a centripetal one. But there is a natural reciprocal action between them. Recognition makes possible a saving of force. If, in the face of changes that repeat themselves, we had to go through anew each time the experiences that the first change called forth, it would lay claim to all our energy, and we would not get anywhere. By reason of recognition, on the other hand, we can turn with fresh force toward the new, and the very opposition between the known and the new lets the new, by a contrast-effect, step forth more vividly and characteristically.

Yet recognition itself presupposes energy to perform the synthesis of the present and the earlier. In that weakening of psychic energy which *Pierre Janet* calls psychasthenia, and which essentially expresses itself as a lack of will (aboulia), the capacity to recognize (understand) the present situation is often lacking. The bond that should connect the present with the earlier is broken. On the occasion of one of his aboulic patients, Janet says: "The attention to the present, the connection (synthèse) of the new sensations with the old images, which constitutes recognition, has here, like the will, vanished on account of the mental weakening"[19]. Yet such weakening can also lead to false recognition or memory-illusion; for energy is also required to distinguish the new from the known.—

It is only an analogy that obtains between immediate recognition and reflection. One has no right to identify it without further ado with acts of thought performed with clear consciousness. It would thus be incorrect to call it a judgment, as several authors have done[20]. For a judgment there is required a transition of thought from a subject to a predicate; but in immediate recognition no transition takes place between two terms; the familiarity-quality appears as if at a single stroke. Immediate recognition can form the basis and presupposition for a judgment, but the judgment first comes into being when a distinction is made between the given, present thing and the earlier thing with which it is identified. The so-called exclamatory judgments, which we shall later come to discuss, are the first judgments that become possible on the basis of recognition; in them there is a predicate, even if there is no subject. Thus when the gypsy boy in *Blicher*

[19] *Les Obsessions et la Psychasthénie.* II, p. 29.

[20] *Meinong*: *Beiträge zur Theorie der psychologischen Analyse.* p. 374.—*Hans Cornelius*: *Versuch einer Theorie der Existenzurteile.* p. 56 f.

cries: “A dog! A hunter!” These are immediate expressions of recognitions.

16. When it is a whole, not a single property or a composite phenomenon, that is recognized, it will not always be necessary to recognize expressly all the members (the parts or properties). Recognition of a single member can be sufficient, and the familiarity-quality is then involuntarily transferred to the other members, so that the whole stands as familiar, or an expectation is awakened that the other members too are present—an expectation that can be so secure that one does not seek confirmation of it. The recognition is then partial, supporting itself merely on a part of the whole that serves as a mark of recognition.

If the expectation leads to a continued going-through of the members of the whole, and these, when they are sensed, correspond to the expectation, a mediate recognition of them takes place. A person’s gait and bearing seem familiar to us, and we then expect that other properties will correspond to them; we address him, and his expression and speech are as we expected; they are then recognized mediately, in that the representation contained in the expectation covers what we see and hear.

In immediate, in partial, and in mediate recognition alike, an involuntary interpretation is woven into apparently simple observations. It is a surplus of energy that makes us at once interpret the given and, in virtue of the fundamental law of synthesis, bring it into connection with previously given content. Later, renewed observation may perhaps correct our first interpretations and expectations. But it is only by the contribution we ourselves give that we really come to know things. If we sat still and waited, the circle of experiences would become narrow. Now, on the contrary, the urge to use our energy leads us to an involuntary experimenting, which can then confirm or correct our expectations and representations. We have here, in purely elementary form, already an intimation of the method that developed science applies: an observation (recognition) leads to a conjecture (expectation) or a hypothesis, which can be confirmed or corrected by new observations. In the simplest mediate recognition there is already described that arc from observation through reflection to new observation in which the peculiarity of all scientific method consists. This kind of recognition testifies to greater psychic energy than immediate recognition: the latter is bound to the given and present, whereas the former goes out beyond this in expectation and representation. It is from the first the urge that drives out beyond the given—partly the urge for an outlet for energy, partly the urge for the attainment of something that does not involuntarily fall to our lot.—

Some psychologists have held that there is no immediate recognition, but that all recognition takes place mediately, that is, through previously awakened expectations and representations, and perhaps even through express comparison. After the discussions conducted on this, it can nonetheless be regarded as settled that there is an immediate recognition, and that it is an interesting transitional form between sensation and reflection[21].

## c. Memory- and Imagination-Intuition

17. The longer the interval becomes between the arising of the expectation and its confirmation or correction, the more independently the representations appear—the more liberated they become, or can become, from the impressions that first awakened them. There can then—on the presupposition of sufficient energy—form itself a content of consciousness with a certain independence of the supply from without, a circle of free representations.

Just as little as the sensations do the representations subsist independently of one another. They are members in a train of representations. They call one another forth and connect themselves with one another, and by reason of their mutual connection (association) a new kind of intuition becomes possible, *memory- and imagination-intuition.* In many cases one can fairly distinctly point out the process of association by which such an intuition forms itself; but the formation can also take place so rapidly that a total image appears in consciousness at a single stroke. This holds especially in inspired imagining, where nature operates "as if it were art," performs deeds that in other cases would require conscious deliberation, if they could be performed at all. Only by roundabout ways can one, in inspired production, gain an insight into how imagination-intuition has come about. And the matter becomes still more complicated by the fact that urge and interest cooperate in all memory and imagination, and that this cooperation here takes place more hidden than in sensation and recognition. The whole past of conscious life operates along with it. The psychic energy is therefore of a higher degree than in the more elementary processes.

Memory- and imagination-intuition is, like sense-intuition, a form in which the energy of consciousness solves the task of combining a multiplicity of elements. Both the unity and the multiplicity are to come into their own. In memory and imagination a more or less rich content of experience and representation is worked together into a

[21]Cf. my *Psykologiske Undersøgelser* (Vid. Selsk. Skr. 6. Række), pp. 8–26. *Katz* in *Zeitschr. für Psychologie und Physiologie der Sinnesorgane* XL., 6. 1906. p. 432. *Herrlin*: *Minnet.* pp. 128–218.—*H. Bergson* nonetheless presupposes, in his critique of the theory of an immediate recognition (*Matière et Mémoire.* p. 91 f.), that all recognition presupposes a free representation.—Besides the kinds of recognition discussed above (immediate, partial, and mediate) there are several kinds. In *Psykologiske Undersøgelser*, p. 35 f., I have attempted a systematic survey.

whole, often in such a way that a quite different ordering comes about than the original one. Such total images have their definite directions, determined by the interests or purposes that cooperate in their formation. Every time we lay a plan, we make use of the past to prepare the future and transform, often without ourselves knowing it, the given representations in such a way that they can enter as members into the new whole. When it is intellectual or aesthetic interest that determines the ordering, it is not an outer end that is aimed at, but the involuntary endeavor is directed at the intuition's becoming as clear or as characteristic as possible.

Once an intuition has been formed, there will be a striving to hold it fast as long as possible. Under the influence of new experiences, incitements, or associations of representations, individual members are perhaps shed or taken up, or else the mutual relation of the members is altered. The direction of the intuition can also be changed. Just as in quite simple organic beings, which have no proper limbs, displacements of the organic mass in certain directions can take place, so too the intuition can be articulated functionally without being replaced by analyzing and judgment. A simple example from the domain of sense-intuition is offered by the so-called Schröder staircase figure, which by changing accessory sensations and associations of representations, or by altered attention, is seen now as a staircase, now as an overhanging wall. When we stare long down into the water from a bridge, it can come to look as if we ourselves and the bridge were moving in the direction opposite to the water's real movement. According to changing moods or outer occasions, an image that stands in our memory or our imagination can thus shift without our noticing it. It can be one of our earlier experiences that is thus altered, or the image we have formed of a poetic figure. Our memories and imagination-images have their own history, whose individual stages it is not always easy to trace, any more than it is easy to find the grounds for their shifts. Their history takes place in the involuntary mental life, right down in the ground of nature within us, and reflection can here often only ascertain that changes have taken place, when it holds together two mutually divergent intuitive images of the same object from different points of time in our life. In other cases reflection can in time become aware of the change that threatens to set in, and work to prevent it. The astronomical world-pictures, which from the first formed themselves involuntarily, are an example of this. Scientific hypotheses often have their advantage in their intuitability and in the support they have in sensation's apparently immediate testimony, and there is therefore often a hard struggle before they are let go. To a still higher degree this holds of religious intuitions, in which a whole host of life-experiences—experiences of the course and value of existence—have formed themselves into great images. Here too the most important thing—both as regards the intuition's original coming-into-being and its alterations—takes place right down in the involuntary mental life, although reflection can enter the lists for the right to hold fast, despite everything, the images that have been laid down. In the relation between intuition and reflection many individual differences make themselves felt. In some it is feeling that holds back, and reflection is then forced to fight for the images in which it has found rest. In others it is the urge for intuitability that opposes reflection's intervention[22]. In others again there is such spiritual elasticity that emotional interest and the urge for intuiting relatively easily give way to reflection.

In animals the capacity to help oneself with functionally articulated sense-intuition probably has preponderance over the capacity for reflection, if the latter occurs at all except in the most highly developed animals. The dog that runs toward a man whom it "believes" to be its master will, during the running, constantly alter its intuiting, according as the man's features become more distinct, and when these features finally show themselves to be quite other than the master's, a disappointment will arise; but whether the free memory-image of the master thereby appears in its opposition to the sense-image of the stranger is impossible to decide. There is no reason to assume a proper judgment or inference.

What I call articulated intuiting, or functional articulation of the intuition, is nonetheless regarded by several authors as a judgment. But it is then in any case a judgment of a quite different kind than the proper judgments that rest on reflection. When *Gassendi* thus ascribes to the dog that "believes" it meets its master a capacity to judge, he adds: "An affirmative judgment is, after all, nothing other than the apprehension of a thing with an addition or a property." The dog sees, Gassendi holds, the man immediately as its master, and this seeing or representing (imaginari hominem herum) is—according to Gassendi—the same as forming the judgment that the man is the master (enunciare hominem esse herum); yet this judgment is passed tacitly or potentially (tacite vel virtute)[23]. In a similar way several psychologists express themselves. *Lloyd Morgan* thus distinguishes between perceptive and conceptive judgments. The perceptive judgment (perceptual judgment) is what I call articulated intuiting, in that it consists in a single feature's becoming predominant in a previously formed image. A conceptive judgment, on the other hand, presupposes a conscious distinguishing between individual features; only this, therefore, is a proper judgment. What Morgan calls a perceptive judgment, *Ribot* calls the logic of images (logique des images). This expression is more fortunate,

[22]Cf. on the opposition between intuitive and reflective natures my *Religionsfilosofi* § 95.

[23]*Gassendi*: *Opera*. Lugd. 1658. II. p. 411.

since the image-forming, intuiting process is decidedly the essential thing in the phenomena here in question. They belong under intuition, not under judgment[24]. The expression articulated intuition seems to fit best, since the intuitive image need not be dissolved because a single element is strongly brought out. If a new articulation within the intuition is not possible, a new intuition can form itself, within which the changes come into their own. Possibly the earlier intuition then vanishes entirely from consciousness. If, on the other hand, it is held fast or reproduced, a comparison and an analysis can get going, which may perhaps lead to a judgment.

The articulation stands as a preparation for, or a presage of, an analyzing and judging. Thus what at later stages of development becomes a special organ is, in embryonic development, intimated in bulging parts of the still uniform and continuous tissue.

18. In itself one might suppose that the intuition-formations, with their shifts and articulations, could be continued to infinity, without any judging conditioned by reflection being necessary. It must be taken as a simple fact that this is not so, and that—at least in human beings—a necessity for reflection and judgment sets in. The equalization and equilibrium that can be attained through intuiting does not always show itself to be sufficient. Conditions can arise that make the combining have to take place with clearer consciousness than happens in the involuntary and half or wholly unconscious process by which the intuition comes into being. In sense-intuition, recognition, memory- and imagination-intuition, it is really only the result that appears clearly in consciousness, while the formation-process itself takes place predominantly below the threshold of consciousness.

Reflection is in a certain sense not productive. It only brings to clear consciousness what is already contained in the involuntarily formed intuitions. It compares and tests, chooses and rejects, connects and separates, fills out deficiencies and determines exactly the mutual relation of the elements. The new that reflection brings and expresses in concepts, judgments, and inferences is the clear consciousness of the individual parts of the content of consciousness and of their mutual relation. Through the purgatory of reflection it can soon show itself that we know more than we thought we knew, in that there can lie in the content of consciousness unsuspected presuppositions for orderings and inferences—soon that we know less than we thought we knew, in that incompatible elements or invalid connections are discovered.

[24] *Lloyd Morgan*: *Animal Life and Intelligence.* p. 328 ff.—*Th. Ribot*: *L'évolution des idées générales.* p. 32 ff.—*Hobhouse* (*Mind in Evolution.* p. 200) ascribes to higher animals a capacity for "practical judging." But he adds cautiously that if animals have no proper [free] representations, they must have something that can perform a corresponding function.

Judgment is the form in which the content of consciousness can come into equilibrium anew, when reflection has caused a dissolution of the intuition. In a judgment different elements of consciousness are brought into definite and conscious connection with one another. Insofar as the appearance of judgment is a sign that the wholeness and equilibrium of the intuition no longer subsists, it can be regarded as a sign of an imperfection, especially since the single judgment cannot express the whole fullness that can lie in an intuition. Perhaps an infinity of judgments is required for the whole content of consciousness, with all its nuances, to come into its own. *Goethe* thus sets "das zersplitternde Urteil" in opposition to "die stille Fruchtbarkeit" in immediate intuiting, and *Wundt* even defines judgment as the dissolution of a composite representation. But judgment in reality signifies a new formation of a whole, which has become necessary because the tension within an involuntarily formed intuitive whole has become so strong that a dissolution must set in. It has a similar advantage over intuition as intuition has over fusion: both the differences and the likenesses in the content step forth more distinctly than before. The formation of judgment is therefore a higher expression of psychic energy.

There must of course be a close connection between the judgment and the intuition it builds on. One often expresses this by saying that the judgment "lies in" the intuition, or that the intuition is a potential judgment, as Gassendi said. The difference between intuiting and judgment is that the latter lets the elements step forth independently, so that a distinction can be made between "subject" and "predicate," while in the intuition a more continuous connection prevails. When I believe I know a person who approaches, I see him immediately before me as the one I take him to be; but if I feel an urge for a closer testing, I institute a comparison that distinguishes between the figure that approaches and the memory-image of my acquaintance, and thereafter there is the possibility of a positive or a negative judgment.

But the way from intuition to judgment can be long and toilsome, and it is not said that the "possible" judgments can become actual. There can be a logical possibility for a judgment without a psychological possibility being present. Thus it is said to Peer Gynt: "We are thoughts; you should have thought us!"—The logical possibility is present when the intuition contains elements that can be separated out, and whose mutual relation can receive a closer determination. But this separating-out and determination is undertaken only when there are sufficient conditions for an abolition of the whole equilibrium of consciousness that the intuition denotes. Here, as at the first arising of consciousness, there must take place an interruption of the uniform state, in order for a further-reaching mental work to get going. A special interest must attach itself to individual elements, so that an urge arises to become clearly conscious of their mutual relation. Thus, e.g., if an intuitive image contains something that could become a means to the attainment of a purpose, or serve for a better understanding of another intuitive image, or form a contrast to it.

The judgment must correspond to the intuition, since it is to express its content in another way, more appropriate under the given conditions. But this does not exclude that it is an essential change that takes place in the transition from intuition to judgment. When one does not embrace the psychological atomism that I have combated above, it must be clear that the elements of consciousness undergo essential changes by passing into new connections, and the connections of judgment can be quite other than those of intuition. The elements are not indifferent to the connection in which they stand. In the intuition an element stands in a single special connection; the judgment brings it into connection partly with elements that lie within the same intuition, but perhaps at great distance, partly with elements from other intuitions. In the intuition I dwell, e.g., on a man's action as a total image and am absorbed in the mood this awakens in me; in the judgment "he has acted nobly!" I make use of a standard with which I more or less consciously compare the action. The validity of the judgment presupposes only that from the relation established between subject and predicate one can infer back to the intuition that lies at the basis, so that the judgment can again lead to a construction of a similar intuition in those who hear the judgment without having had the intuition at first hand. There must be analogy, but not necessarily identity, between judgment and intuition.

## d. Association of Ideas

19. The content to which judgment corresponds does not always appear in the resting form of intuition, and as a surveyable whole. Judgment can be formed out of a series of associations of representations, in which intuitive images of greater or lesser extent replace one another more or less rapidly. The difference between association and intuiting is only a difference of degree. The more the successive gains the upper hand over the simultaneous, the more association—which in its higher degrees becomes a flight of thought—steps in place of intuition. What intuition is in static form, association is in dynamic form. Already in the formation of memory- and thought-intuitions associations cooperate, and no less is this the case in shifts and articulations, as well as in the dissolution of the intuition.—But we now regard association as an independent form of

the involuntary conscious life.

Just as involuntarily as an intuitive image can emerge in consciousness, just as involuntarily can one representation draw another after it, this a third, and so on. In both cases laws operate that do not themselves come to consciousness. The articulating intuition stands closest to association; but in it too the dynamic is still subordinate to the static, since the shift quickly leads to a resting image.

The transition from association to judgment takes place through the relation between the members of the association-series becoming the object of distinct consciousness. The urge for this will already arise from practical interest, as when the preceding members stand as means in relation to the following ones. Where merely practical interest prevails, one will attend to the preceding members only insofar as they indicate means to the attainment of an end, and the whole order in which the members are brought will be determined no longer by their involuntary sequence, but by their relation to the end. It is, as *Hobbes* remarks, the thought of a purpose (cogitatio finis) that makes us, in the waking state, order our representations in a definite and coherent way, while in dreams, where the firm thought of purpose is lacking, they can unfold more chaotically. For a thoroughgoing ordering there will be required holding-together and comparison with other association-series and intuitions than those involuntarily setting in, and the more such series and images thus interact, the greater the psychic energy that comes to light.

Every attempt to order, by way of reflection, the members in the involuntarily arisen association-series consists in a testing of the members' relations of likeness and difference. Even if the objects a and b have occurred together ever so often in my experience, so that a has an irresistible tendency to call forth the representation of b, the justification of expecting b when a has set in will nonetheless depend on whether the a that earlier occurred as predecessor of b is identical with the a that has now set in, and further on whether it can be shown that a cannot set in without being followed by b. The greater the continuity between a and b, so that they really form a whole, the more justified the expectation will be. If it is an association by likeness that we have before us, the transition from the one member ($a_1$) to the other ($a_2$) will be the more justified the more the likeness approaches identity; it is then merely a transition from one example to another. Thus association approaches judgment the more the transition from object to object is determined by continuity and identity. In the genius, the formation of judgment can proceed as if it were an involuntary association-process, since the psychic energy is released with great ease and in great fullness. Reflection is here limited to a minimum. In everyone, practice in a definite train of thought can operate in the same direction.

Where reflection can be spared, it is spared.

20. Association of representations is, just as much as reflection, an expression of psychic energy, and it is wrongly that one has regarded it as a purely passive process and set it in absolute opposition to "thinking." What happens in every association is really that a whole, a larger connection, is built up, constructed out of a single given object that occurs in a certain isolation. There are indeed always several objects or elements that cooperate at the springing-forth of an association-process; but a single one will be the decisive one, and for simplicity's sake we can keep to it. Consciousness has both an urge and a capacity to go out beyond the single object or element and connect it with others. In this instinctive tendency the psychic energy expresses itself. All association can be traced back to a transition from part to whole; what we call association by likeness and by contiguity are only limiting cases. In every association relations of likeness and of contiguity operate together, in such a way that now the one, now the other has preponderance[25]. The urge for the release of energy is, in the domain of the life of representation, so strong that a minimal occasion can operate as a release. There is here an analogy to the urge for movement. Now the incompletely given is filled out (by predominant association by contiguity), now one circles around it through dwelling with kindred representations (by predominant association by likeness). In both ways the single element can be incorporated into the whole psychic world, and its relative isolation can be abolished. Which of the different paths is taken in the particular case—and which special relations of likeness or contiguity come to operate—nonetheless does not depend on purely intellectual conditions. Like a hidden weight, interest can press the scale-pan down for a single definite relation of likeness or contiguity that in itself would not be decisive. It is ultimately man's striving that determines his associations of representations, even though these, once they have got going, can in turn alter the strength and direction of the original striving.

When now association is replaced by judgment, the relation is reversed. It is no longer a transition from part to whole that takes place, but a transition from whole to part, that is, an analysis. Association is, like intuition, an integration, but judgment presupposes a differentiation. In the association-process it holds that a little tussock can upset a great load; in the formation of judgment a needle is found in the great load. The associations can bring forth the whole content of consciousness, but the formation of judgment draws out a single thread from the rich web. Association is productive, the formation of judgment is critical.

[25] See more closely my *Psykologi* V B., 8.

Yet neither is this opposition complete. The associations are—as already remarked—not determined exclusively by a single element. This is seen already from the fact that one and the same representation does not, under all conditions, awaken the same associations in one and the same individual. And on the other hand the formation of judgment presupposes that attention is directed toward a single element's relation to other elements (in or outside one and the same association-series), and so far it too can be said to begin at a single point, with the little tussock.

Yet another opposition has been found between association and judgment, in that the former is supposed always to be able to be continued, since there is no inner limitation for it, while the latter is supposed to denote a definite conclusion. "Association," says *W. Jerusalem*[26], "is never concluded in itself. Ever new members attach themselves, ever new deviations from the original direction take place, and nowhere is there found a natural conclusion lying in the very nature of association ... In judgment a process is released from its connection with the rest of the content of consciousness; it is isolated and, as it were, concluded for consciousness. We are for the present done with it, and precisely this concluding moment in the act of judgment forbids regarding it as an association." Against this it must be objected that an association-series can very well find its natural conclusion, namely when the motive that has called it forth is fully satisfied. One can also reach a representation that perhaps lies far from the starting-point, but that offers an independent interest and therefore takes attention captive. When one takes into account the interest, which is never lacking in the association of representations, an inner conclusion of this can very well be given, not to mention that the relations of likeness and contiguity that are at one's disposal in the particular case will always be limited. On the other hand, judgment does indeed denote a settling of accounts, a decision conditioned by reflection, but only a provisional decision, since new reflection can correct it or else bring it into connection with other judgments and thereby unfold what "lies in" it.

## e. Comparison

21. The transition from intuition or association to judgment takes place through reflection, which consists in an analysis, a successive turning of attention to the individual objects and elements in the intuition or in the association-series. Thereby free comparison becomes possible, in distinction from the elementary comparison in sensation and the bound comparison in recognition.

[26] *Die Urteilsfunktion.* p. 79; 82.

There lie two objects before me on the table, which I can see at a single glance. But although they are both within my field of vision, I can fix my attention now on the one, now on the other, move back and forth between them without necessarily moving my eyes. The one object does not fall out of the field of vision because attention is directed toward the other. By such alternation of attention an experiment is made, whose result is that the two objects stand either as similar or as different. The likeness or the difference springs forth for us as a peculiar quality, an element that was not present as long as we regarded the objects each by itself without holding them together. It is this holding-together that brings it about that the new property we call likeness or difference comes forth. The older English psychology (especially *James Mill* and *Stuart Mill*), which laid the whole weight on the individual representations each by itself, could not explain how comparison was possible. For them, relation-representations differed from other representations only in that they presupposed two "things" instead of one.

The active element in comparison—apart from the energy that holds together the elements that are compared—is one with attention. This is a successive concentration toward the different members of the intuition or the association, a concentration that precisely by reason of its alternation lets the content between the elements step forth distinctly. Experience shows that we, all else being equal, apprehend a successive relation more distinctly than a simultaneous one.

If the objects we compare are not given simultaneously, the comparison comes about through the one member being reproduced while the other is directly observed. Memory thus provides the simultaneity within which attention can then swing.

If, at the passage of attention from the one object to the other, no difference at all appears—apart from the exertion involved in the alternation of attention itself—the relation between the two objects stands as a *relation of identity*. Identity in the strictest sense is the likeness an object has with itself. When different objects are said to be identical, the meaning is that it is as if attention returned to the same object; and conversely, that an object is the same means that attention can return to it without any difference appearing.

If it is emphasized that the two objects, despite their lack of difference, nonetheless occur at different times or in different places or both, their relation is called *Sameness* (*Congruence*).

If some of the two objects' parts or properties appear as identical or the same, others as different, we get a composite *Likeness*. It has sometimes rightly been held that all likeness—apart from identity—was of this kind.

If a likeness is present without there being identity or sameness at any point, we can call the relation *Qualitative Likeness*. It is a peculiar kind of qualitative difference. The remotest degree of likeness is *Relational Likeness* (*Analogy*), where the likeness does not concern any single property or part in the two objects, but the relation between properties or parts in the two objects.

22. Comparison can take place involuntarily or voluntarily. Comparison is involuntary when no intention, no preceding representation, no wish to find out likeness or difference makes itself felt. Voluntary comparison presupposes that one has set oneself as a task, as a purpose, to find whether there is likeness or difference between two objects. The transition from involuntary to voluntary comparison is formed by such cases in which the one object is reproduced, so that it is merely held together with the other. It is a transition reminiscent of the transition from immediate to mediate recognition. In mediate recognition a representation is released, which thereafter covers a following sensation. An analogy in the domain of will is formed by the transition from urge or instinct to drive (when by drive we understand an urge connected with a representation of its goal). And here there is really more than analogy; for there is an urge to compare just as to act, and in both cases it can be of importance that one becomes conscious of what one feels an urge for.

Now it is not always the case that both the objects to be compared are given or can without further ado be called forth in memory. Often only the one object is at one's disposal, and the task is to find another object that stands in a definite relation of likeness or difference to it. The more clearly and distinctly this relation can be represented, the greater the possibility that the new object can be found. The most exact indication of that relation we have where the task can be put into an equation. We then have a definite standard to decide whether the sought object appears or not, and by a methodical procedure we can perhaps force out the object that satisfies the demand. In a more indefinite way something similar happens when we wish to call forth something in our memory and know of it only that it stands in a certain connection with a representation already present in us; by concentration of attention on this, or, as it has been called, by a boring, the sought thing can then spring forth. The psychic energy appears quite especially in such cases. The point then is not merely to connect given members, but to find a member that is perhaps quite new, but that stands in a familiar relation to a given member. Where we can indeed determine the relation to the given member, but on account of the limitation of our experience cannot find the new member, there we end in a problem. Thus we are placed, e.g., when we assume that in the psychic domain there is something corresponding to the difference between potential and actual energy in the physical domain; for of potential psychic energy we can form no positive representation; we can only say that it is related to actual psychic energy as potential energy is related to actual energy in the physical domain. (Cf. above 8–9.)

In all comparison an urge makes itself felt, even if the finding-out of relations of likeness and difference is not directly set up as a purpose. It is an urge for clarity and distinctness, for the use of the combining capacity that constitutes the innermost essence of consciousness. Often, and especially under the most elementary spiritual conditions, it is a downright urge for self-preservation that leads to thinking. When something that has value for the subsistence of life does not immediately fall to our lot, we must make it a purpose and strive to find the means for its realization. In the relation between end and means, man meets for the first time with necessity: if he wills the end, he must also will the means. The urge for the end begets the urge for the means. Will and thought here go together. This meeting has not merely practical significance. Practically regarded, it is a roundabout way that man here comes to make; but this roundabout way points to a definite unavoidable connection that can become the object of an independent interest, a joy of cognition. In this way the intellectual interest can, through displacement, develop out of the practical one (Cf. 6).

When it is a matter of determining the relation between two objects or of finding an object that stands in a definite relation to a given object, several possibilities will often make themselves felt. Several different possible relations or objects emerge, and a choice must then be made, which is determined by which of them best satisfies the definite conditions. The choice is determined by a comparison between what is required and what offers itself. If the two objects are given, the point is the fixing of the relation that takes place between them; if only the one object and the relation to the sought object are given, the point is the fixing of which object stands in the definite relation to the given object.

As far as possible, comparison leads to an Either—Or, so that there is a choice between two possibilities. The other possibilities are rejected by a preliminary comparison, or they show themselves able to be subsumed under one of the two main possibilities. We stand here before a fundamental peculiarity of our thinking. This moves forward through two-membered relations. A predicate must either be ascribed or not ascribed to a subject. No concept can at once contain and exclude a certain mark. A grounding of this peculiarity of our thinking cannot be given. It is evidently bound up with the psychological fact that attention, during reflection, can be concentrated only on one element at a time, and that comparison takes place through the passage from one element to another. *George Boole* has called the logical principle springing from this necessity *the Law of Duality.* Before it even the most pronounced practical interest must bow when it wishes to take reflection into its service.

Boole shows in an interesting way the purely factual character of the said feature of our thinking. Starting from the fact that the combination of a concept with itself has no logical significance, he finds the nature of our thinking expressed in the equation $x^2 = x$, which is satisfied only by $x = 1$ or $x = 0$. Here 1 means All, 0 Nothing. The derived equation $x(1 - x) = 0$ expresses the Law of Duality. A combination that should at once contain a concept and its negation is invalid. But if now the fundamental equation had not been of the second, but of the third degree, our thought-activity would be otherwise constituted than it in fact is. We would then not work our way forward through two-membered relations, holding together two elements at a time and affirming or rejecting their likeness (or difference), but through three-membered relations, trichotomies instead of dichotomies—a procedure of which we can form no real representation. Boole by no means wishes, by this presentation, to teach that the fundamental method of our thinking is accidental or arbitrary, but only that it must be taken as a simple fact[27].

Already earlier *Salomon Maimon* had shown that, on account of the unity of our consciousness, what is to be connected by our thought in a definite way can immediately consist only of two members. Every immediate comparison that leads to a definite connection concerns two immediately consecutive members. A mediate connection can take place only after a series of immediate connections; thus a mediate connection comes about between a and d when there has successively taken place a connection of a and b, b and c, c and d[28].

By Maimon's and Boole's remarks, the law of the excluded middle between two mutually negating assumptions—already set up by *Aristotle*—is illuminated with respect to its connection with the whole nature of our reflection.

23. Besides the object there are, as already said several times, in all reflection two moments that stand in a constant relation of reciprocal action: the form of the thought-work and the motive that awakens reflection and drives it forward.

A motive or an interest that could not be satisfied by the laws and forms that human thinking must by its nature follow will in the long run be without intellectual significance. Such a motive can perhaps become fruitful for imagination, for the capacity to form images and symbols, but it would be a misunderstanding to attempt to force thought

[27] *Boole*: *An investigation of the laws of thought.* London 1854. pp. 48–50.
[28] S. *Maimon*: *Versuch einer neuen Logik oder Theorie des Denkens.* Berlin 1794. p. 73 f.

onto paths it cannot by its nature walk. Fortunately, "reason" cannot be taken captive forever; it will with inner necessity burst its prison, and its captivity will not be of benefit to the interests that were to be served.

The laws and forms of reflection are the same, whether the thought-work takes place involuntarily or voluntarily, half unconsciously or with full clarity about goal and ways. These laws and forms operate, as we have seen, in analogous forms already in sensation, in memory and imagination, and in association of representations. That the thought-work, the comparison, takes place "with will" entails merely greater definiteness and firmness in the synthesis and greater clarity in the transitions from member to member. However significant the transition from the involuntary to the voluntary can also be, it nonetheless does not mean that a wholly new "capacity" should come into force.

One of the older psychologists who has laid most weight on the will-element in thinking, *Charles Bonnet*, has stated the proposition: L'attention est la mère du génie. But however decisive concentration in a single direction is, it nonetheless gives only the introduction, the impulse. No less important is it that, as a consequence of this impulse, a series of intuitions and associations is involuntarily released, which can then become objects for reflection. In the content of consciousness thus emerging, new elements and new connections can come forth that show themselves able to pass the purgatory of reflection, and genius often consists precisely in the richness and depth of the involuntary intuitions and associations. *Kepler* and *Fechner* are examples of inquirers whose guiding thoughts first came forth in the form of imaginative combinations, which through critical testing led to significant results. Such minds unite the freedom and freshness of the involuntary with the exactness of attention and comparison. Genius can finally also express itself in the indefatigability and carefulness with which the working-over of the objects at hand takes place. Regarded from this side, *Buffon* was right when he defined genius as great patience.

We easily come, in our psychological theories, to emphasize the difference between the involuntary and the voluntary more strongly than close observation warrants. The transition itself from involuntary to voluntary comparison of course takes place involuntarily: the representation of a goal that is the condition for voluntary striving—the one that *Dante* called "the will's first thought"—must itself first arise by reason of involuntary association. Here can lie a significant turning-point in the life of thought and in the whole development of the mental life; but it does not justify assuming a break in the continuity.

# C. Judging

## a. The Formation of Judgment

24. It has already been mentioned several times, provisionally, how intuition and association can be replaced by judgment. Judgment is a higher solution of the task that is solved in more elementary forms through fusion, sense-intuition, memory- and imagination-intuition, and association. Such a higher form becomes necessary when neither the likeness nor the difference between the objects fully comes into its own in those forms.

Judgment is a synthesis, a connection of objects that, prior to the judgment, have occurred in one or more of those simpler forms. The connection can only be clear and definite when the objects are each by itself clear and definite. When we become clearly conscious of an object (or a group of objects), so that our representation of it does not vary in the different cases in which it appears, we have formed a concept. So far it might seem that concept-formation would have to precede the formation of judgment. But a closer consideration will show that there is a constant reciprocal relation between the formation of judgment and the formation of concepts, or rather, that these two processes are at bottom not different.

Concepts as well as judgments are formed by analysis of intuitions or associations. When we become clearly conscious of an object that is given in a single state, through attention's successive turning to its different elements (all that can be distinguished in the object), we form a concrete individual-concept; when we become clearly conscious of an object that has appeared in a series of different states, we form a typical individual-concept, in that a single state comes to stand as an example of the whole series; when we hold together and compare a whole series of objects and can regard a single object as an example of the whole series, we have formed a general concept. Between the different examples of one and the same typical or general concept there need not be identity, only analogy. It is only in certain definite respects (indicated in the concept's definition) that the examples can take the place of one another. In place of the example there can step a schema, a symbol, or a word. They are directions for finding examples, when one wishes to save the time that the more detailed representation of such an example takes. The main thing is that one keeps, in one's attention, to the elements that can approximately recur in the different objects or in the different states of the single object, even though such elements by themselves alone would not be sufficient to constitute a content of representation. But it is then really by a series of judgments that concept-formation takes place. In order to form concepts we must connect elements in one or more objects in a clear and definite way. It is by judgments that we establish that an object presents certain elements—that the same object occurs in different states—and that different objects present the same (analogous) elements.

We seem then to move in a circle: judgment presupposes concepts, and concepts are formed by judgments! But the matter is that concept-formation and the formation of judgment are really one and the same process—reflection's analysis of what is given in intuition or association. The difference lies merely in the purpose and direction of the analysis. Now it is a chief interest to find out the elements that the objects contain and gather them into a whole, and then we speak of concept-formation; now our interest is essentially directed at becoming conscious of the relation between different objects or states of the same object, and then we speak of the formation of judgment.

Insofar as a difference can be made between the two processes, they stand in constant reciprocal action. The clearer the concepts can become, the clearer too can their connection become in the judgment; and the clearer a connection can be shown through judgments between a concept and other concepts, the more perfect the concept itself becomes.

The difference that can nonetheless always be made between concept and judgment is a new example of the antinomy that—as earlier shown—lies in the nature of consciousness: synthesis presupposes elements, but these are only what they are by being comprehended by the synthesis. Hereby is explained the constant unrest in the life of thought, and in the mental life as a whole; ever new tasks are set: now that of expanding the content of consciousness, now that of bringing about a more intimate connection within the content.

25. The connection that judgment brings about is due to a recognition. Two concepts can be connected into a judgment only when the one can be found in the other. The one concept then denotes a point of view from which the other can be seen, a closer determination of it. Here again it is seen that thinking is comparison. When I say "A is B," I must have found B in A. This re-finding can have different degrees or forms. It can show itself that A and B have wholly the same content, so that it is only due to limited or inexact apprehension when they earlier stood as different concepts. Or B can be a single one of A's marks, hence be a means of recognizing A (by a kind of partial recognition). Or the connection of A and B presupposes that account is taken of a larger connection within which A and B belong, so that only under certain conditions, which cannot be derived from A or B alone, can one find a relation of identity; it is then not A and B that

are identified, but $A_x$ and $B_x$[29]. The two first-named relations might seem to belong to Kant's "analytic" judgments, the third to his "synthetic" judgments. But the critique of this Kantian division of judgments has shown that no real boundary subsists between them. Even in absolute identity (A = B) there will always be certain presuppositions and limits for the identity. The difference between the three named kinds of relation will in particular cases depend on how far back one goes in the series of presuppositions.

When, with *Leibniz*, one conceives judgment as the expression of a relation of identity, its kinship shows itself clearly with the forms and processes that form its psychological basis. Like these, it is a form in which consciousness seeks unity and harmony in its content, and it is, like them, a stage in the constant breaking of likenesses and differences. It lets both the relation of likeness and that of difference come into their own to a greater degree than the more elementary forms were able to.

Judgment itself can enter as a member in reflection's continued work. Inference is related to judgment as judgment is to the concept. Judgments can be connected into inferences when identical concepts can be inserted from the one judgment into the other. The relation between premises and conclusion is analogous to that between subject and predicate in the judgment, and that between intuition and concept. At all stages reflection can lead only to a higher degree of consciousness of what is contained in its objects. In the concept it becomes conscious of the intuition's content, in the predicate of the subject's, in the conclusion of the premises'.

## b. Obstacles to the Formation of Judgment

26. No judgment is formed when two intuitions or two members in an association-series replace one another in such a way that the first object has vanished when the second steps up. If the dog mentioned earlier, which "believed" it was going to meet its master, had wholly forgotten this "belief" when it turned out that it was a stranger who came, it would feel no disappointment and would have no occasion for any judgment. And even when the conflicting objects, the representation of the master and the sight of the stranger, meet and suddenly replace one another, so that a disappointment is felt, it is not said that this leads to a judgment. (It would have been the negating judgment: "it was not my master.") The contrast, the collision, can constitute the whole experience. Experience shows that one must to a certain degree have freed oneself from an experience in order to be able to convert it into a judgment. It is more than rhetoric when *Shakespeare*, both in a comedy and in a tragedy, puts utterances in this direction into the mouths of his

[29] This tripartite division is due to *Jevons*, *Principles of Science* (1874), Chap. 3.

characters. "Silence is the perfectest herald of joy: I were but little happy, if I could say how much," says Claudio (Much Ado About Nothing II, 1). "The worst is not so long as we can say, 'This is the worst,"' exclaims Edgar (King Lear IV, 1). What in memory, or in the mind of the outside spectator, forms itself into a judgment need not, in the experience itself, be formed so. It is a frequent psychological error that one reads judgments into where the immediate experience laid claim to the whole energy of the mind. We have met this error already in immediate recognition, which one has also wished to conceive as a judgment. Disappointment can, like every relation of contrast, be formulated as a judgment, but need not entail one.

Far sooner than entailing a judgment that expresses what has happened, disappointment will—when it does not lead to complete slackening, but when there is energy still to spare—lead to a more or less impatient demand, a claim that the expected shall come. The collision between expectation's rich image and impoverished reality begets the demand. In children the demand appears not only before the judgment, but also before the question. Or rather, the question is at first a demand. The purely theoretical question always presupposes a certain development—a cooling-down.

The question can be a determination-question, in that a whole stands before consciousness within which there is a part that is not filled in. There is lacking, e.g., a time- or place-determination for an event. Here the answer consists in a factual indication, the bringing forward of an object by which the deficiency is filled. In the decision-question, which presupposes a higher stage of development than the determination-question, the opposition between two mutually exclusive total images makes itself felt, and the answer must be either Yes or No. (Is he dead? Is it ten o'clock?) Yet the difference between the two kinds of question is not absolute; for in the decision-question too it is a matter of the filling-out of a whole, namely a whole that comprises both of the two mutually exclusive wholes. In the question "Is he dead?" it is a matter of whether we are to think of the world of the living with him or without him. Only, this larger horizon here stands more in the background, and that comprehensive world can in turn have different extent.

Negation too appears first in practical form; it has its first germ in aversion, anger, and fear, and appears as rejection. A dim urge lies at the basis both of negation and of the demand. A transition between practical and theoretical negation is formed by the triumphant or despairing ascertainment of what is lacking.

Negation itself, like disappointment, presupposes a transition from an object with a manifold content to another object in which something of this content is lacking, and also a holding-fast of the first object after the second has set in. AB is replaced, e.g., by

A, and AB and A are held together. The negating judgment “B is not here” ascertains the difference. It springs immediately from the opposition between the two objects. In simplest form the negative judgment, like the positive, can be expressed by a mere exclamation: “Gone!”

Probably positive judgments arise before negative ones. Consciousness will at first be more occupied with an increase of its content than with a diminution of it. The transition from A to AB brings a fullness that can be enjoyed or become a motive for action. For a judgment to appear, it is not necessary that the preceding object (A alone) be preserved in memory; consciousness revels in B’s entrance, the new positive thing. If, on the other hand, the transition from AB to A is to beget a negating judgment, AB must be held fast in memory, since it is B’s absence that is to be ascertained. All else being equal, a negative judgment therefore requires greater energy than a positive one. Energy is required to miss and to grieve, and to demand and to wish. There is lacking here the help that the positive B provides when A is replaced by AB.—The relation becomes of course more complicated when, after the transition from AB to A, A gets a new companion, so that AX appears. What the consequence will be will depend on what relation X stands in to the earlier content of consciousness—whether AX can become an equivalent for AB or not.

We have here an example of a small series in which the order of the members is not indifferent. The transition from A to AB is of another kind than the transition from AB to A. The two processes that lead to positive and to negative judgment can be equally simple, but the psychological motives for positive judging can be present earlier than those that lead to negative judging.

The state or experience that gets its expression in a negative judgment can very well in itself have a positive character. Lack is a positive state; it is only its cause that we express by a negation. There are no negative states at all. The negative numbers denote, after all, merely a quantifying in the opposite direction from the quantifying that the positive numbers denote. The Hindus denoted what we call the positive numbers by a word meaning fortune, the negative numbers by a word meaning debt. The representation of fortune must, from the nature of the case, precede the representation of debt. Life leads to positing before it leads to negating. A logical parallel to this psychological fact is offered by the circumstance that a negative judgment, as regards content, presupposes a corresponding positive judgment, just as a negative concept presupposes a corresponding positive concept. “Alle Begriffe der Negation sind abgeleitet,” says *Kant.* The person born blind does not know what darkness is, the ignorant person does not know what ignorance is[30]. A negating judgment gives us no new content, but rejects a positive content, which is thus presupposed known. A negative judgment can therefore, logically regarded, not be placed on a par with the corresponding positive judgment, although it can, as we have seen, build on just as simple a relation of opposition as the latter. It stands, on the other hand, on a level with that kind of positive judgment which rejects a possible negative judgment, and which is often expressed by means of more or less solemnly asseverating words ("truly," "really") or by strong emphasis on the copula. Such affirmative judgments are a peculiar kind of positive judgments. *Hegel*[31] wrongly held that narrative sentences which do not reject a doubt about the correctness of what is narrated do not deserve the name of judgments. A narrative judgment will, when it is more than idle repetition, be the fruit of an analysis of events or reports, hence of a work of thought, even though it does not stand in opposition to a doubt.

Therefore *Sigwart* and *Lotze* (and in antiquity *Menedemus of Eretria*) have rightly denied the negative judgments a place among the logically simple or primary judgments. Sigwart calls the negative judgment a judgment about a judgment, namely a judgment whose content is that another judgment does not hold[32]. When *Hermann Cohen* does not agree with this[33], it is because he overlooks the difference between positive judgments in their simplest form and the special kind that can be called affirmative judgments; with these, not with the former, the negative judgments must, logically regarded, be paralleled.

27. The formation of judgment can, as we have seen, be hindered by the memory of one of the objects falling away, or by not all the objects necessary for the completion of the judgment being given, so that only demand or question are possible. It can further happen that a judgment is formed only to be rejected, so that negation prevails. There is yet another way in which a judgment can be hindered from asserting itself in consciousness, namely when there also appears another judgment with which it shows itself incompatible. A problem then arises. The word problem really means "that which is laid before," "task." In this general sense a problem can already be said to be set when closer determinations are lacking. The demand and the question thus already set problems, in that they set the task of finding determination and decision. In a narrower and sharper sense the problem arises when two judgments—either two positive judgments, or a positive and a negative, or two negative—stand as incompatible.

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[30] *Kritik der reinen Vernunft*[2], p. 603.

[31] *Wissenschaft der Logik*. II. p. 76 f.

[32] *Sigwart*: *Logik*[1] I. p. 119; 260.—*Lotze*: *Logik*. p. 61.—Cf. also *Bergson*: *L'évolution créatrice*. p. 312.—On Menedemus see *Diog. Laërt.* II., 135.

[33] *Logik der reinen Erkenntnis*. Berlin 1902. p. 89.

One might call the one kind of problems completion-problems, the other kind liberation-problems. In the first kind the sting of the problem lies in the incompleteness of our world of thought; in the second kind the sting lies in a discord within our world of thought. Common to both kinds is the relation of tension (vital difference) described by *Avenarius* in his natural history of problems[34], which arises through a psychic equilibrium or harmony being abolished.

Psychic energy is required to have problems. Both lack and discord subsist only when there is a capacity to hold oppositions together. If one lets go the one opposed term, neither the incompleteness nor the contradiction is felt. There are problems in all spiritual domains, and it is through their purgatory that spirits are tested.

In this connection we keep to the second kind of problems, that due to the collision between judgments. Where a simple positive and the corresponding simple negative judgment stand over against each other, the rejection of the one will, according to the Law of Duality, be one with the acceptance of the other, and vice versa. B must either be a mark of A or not be one. Where two positive or two negative judgments stand against each other, the relation will not always be so simple. The relation can be many-membered, so that both judgments can be false and it is a third that holds. The rejection of a hypothesis thus does not necessarily lead to the acceptance of a definite other hypothesis, when more than two hypotheses are possible. The task must in such cases be to reduce the different possibilities to two, between which the choice must then be made.

Often the choice is made on negative grounds. One becomes clear that one cannot go the one way, and only therefore does one go the other. Necessity is then expressed negatively. In the history of logic this shows itself in that the negative logical principles, the principle of contradiction and the principle of the excluded middle, were set up before the positive one (the principle of identity). Our criterion of reality often gets a negative expression: we cannot but believe this or that! The primitive ethical commandments have a predominantly negative form: "Thou shalt not—." It has been the task to set up barriers within which life could unfold. The commandments are the tribe's reaction against those who deviate from the paths that have shown themselves salutary. Indignation and disapproval express themselves before recognition and approval. It was above all a matter of preventing encroachments. The prohibitions were of course motivated by positive purposes that were to be promoted; but the negative measures stood as most necessary.

[34]See my book *Moderne Filosofer.* p. 100–104.

Often one uses, out of politeness, considerateness, or melancholy, negative expressions to cover over the crass and sharp character with which a positive reality can announce itself. One then denies the opposite of what one means. It is the so-called litotes. "It is no joke." "I cannot praise him." "It is not going well."[35] When *Luther* at Worms said: "I can do no other," he thereby not only indicated that he had tried all possibilities, but at the same time expressed his resolve in the most considerate form. His conviction received at once the strongest and the gentlest expression.

28. Only those negative judgments are fruitful which are necessary intermediate links for arriving at a positive judgment. But it follows precisely from the conditions of our cognition that we can often reach a positive determination only through a series of negative determinations. "Empty" space was at first a purely negative concept; one could ascribe no physical properties to it. Later such properties, especially electrical and magnetic ones, have been demonstrated. When one says that an organ dies away through "non-use," it is because one stops at a preliminary observation. What in reality takes place is that when the function falls away, the chemical processes in the tissues stop, and the organ then becomes an inactive mass that is absorbed by the neighboring tissues.

The history of classification shows us interesting examples of how we often have to content ourselves with staking off a domain, in that we can say nothing else about it than that it is different from another, definitely and positively characterized domain. It then becomes the task of continued inquiry to find positive marks for the domain thus staked off. *Aristotle* divided animals into rational and irrational, the irrational again into those that had (red) blood and those that did not have (red) blood[36]. *Linnaeus*'s classification of animals seems to support itself on positive marks. He has six classes: Mammals, Birds, Amphibians, Fish, Insects, and Worms. But *Cuvier* pointed out that the sixth class is characterized only negatively. "It is well known," he says, "that Linnaeus united under the designation *Worms* extraordinarily numerous and differently formed animals, for which it was difficult to indicate any common mark. While I was working on my first treatises on comparative anatomy, I came to face the impossibility of saying anything universally valid about the worms' nervous system, or about their circulation, or about their organs of respiration and reproduction, indeed not even about their organs of digestion. Thereby it became clear to me that this class was not, like the others, founded

[35] *Kr. Nyrop*: *Ordenes Liv.* p. 47.

[36] Yet Aristotle did not base his classification on a single mark. He precisely combats dichotomy. *Gomperz*: *Griechische Denker.* III. p. 117.

on positive marks."[37]

The inductive method often moves, like classification, forward through negations, in that different possibilities or hypotheses are set up and tested, and only those that are not rejected under the testing are held fast. *Kepler* arrived at the assumption of the elliptical form of the planetary orbits first after he had rejected various other forms. It was the "method of exclusion" that led to his result.

Between the significance of negation under induction and under classification there is the difference that the negative paths under induction mean paths that are barred, while under classification they mean paths that are left for later inquiry to travel. Perhaps our cognition ends finally at a crossroads where it can get no further, neither as classification nor as induction. We would then stand at a conclusion of the process that began with the first, primitive distinguishing. Through the forward-driving force of differences and oppositions, lacks and contradictions, ever higher forms of psychic energy have been developed, until the limit is reached where perhaps pure speculation or poetic analogizing still develops thoughts about the nature and worth of existence, but where experiential confirmation of our judgments is no longer possible.

That negation plays so great a role during this whole development comes from the fact that we have to orient ourselves in existence from a definite standpoint in time and space, and from definite presuppositions that are given with the very nature of thought. The richness of existence is too great for it to be surveyed at a glance or by a single progressing series of intuitions. On the other hand, thought can see more possibilities than experience shows us realized in the existence we know. Now it is our all too limited points of view, now our all too comprehensive ideas, that must be negated or at least corrected, before true cognition can be attained.

## c. Subject and Predicate

29. A wholly definite point where judgment replaces intuition or association cannot be demonstrated. There is a multiplicity of transitions in the degree of attention that is directed toward the relation between the elements in an intuition or in an association-series. The boundary is formed on the one side by the immediate and involuntary streaming-along, in which neither objects nor processes are distinguished, and on the other side by hypnotic or ecstatic absorption in a single element. Between these two boundaries lie a series of states that contain more or less favorable conditions for the arising of a judgment. The decisive thing is that, however much attention is directed

[37] *Cuvier*: *Sur un nouveau rapprochement à établir entre les classes qui composent le regne animal* (1812).

toward a single member, the connection with the whole intuition or the whole series in which it occurs is nonetheless preserved, so that its relation to the whole can stand clear. Continuity is the constant presupposition for a fruitful application of attention in a single definite direction. It does not go, as older presentations of logic might mislead one to believe, that one first knows the individual members each by itself and then knows the relation in which they must stand to one another when they are to constitute a whole. If there were not beforehand a whole that enclosed the members, it would be inexplicable how thought could come to connect them. What thinking can do is only to bring the relation between the members of a given whole to clear consciousness, and this may perhaps require a rearrangement of the immediately given relation between them.

One has often thought to find a definite mark of distinction between judgment and the simpler mental processes we call intuition and association in this, that judgment is expressed in words. But the significance of the word for thinking consists essentially in the help it provides toward holding fast the typical and general in its difference from the concrete and the individual. Herein it is presupposed that a comparison of the objects in question has beforehand been instituted. And such a comparison is also necessary in order to decide the right to transfer a word from one object to another. The judgment is expressed by the word, but does not owe its coming-into-being to the word.

The word is, in its simplest form, an exclamation in which the state of mind called forth by an object gets air. An exclamation common to several individuals can become a means of communication. And this means of communication can in turn become a means of thought, a means of holding fast a thought in its limitation, in particular of holding fast the difference between points of likeness and of difference within a circle of objects. Hereby the word comes into so close a connection with thinking that it is no wonder it has been difficult to keep the grammatical, the psychological, and the logical side of the nature of judgment apart from one another. The psychology of cognition and logic have developed under the constant influence of grammar and its forms. Especially as regards judgment, one has transferred relations and forms from the nature of the grammatical sentence, determined by the laws of language, to hold immediately for the thought-activity itself. This has especially had influence on the conception of the relation between subject and predicate in judgment, a relation I shall now attempt to illuminate.

30. The logical predicate is the most important member in the judgment. It denotes the new that comes to be added and that is pointed out. The subject is always presupposed. All words therefore begin as predicates, and it is only later that they are used to express

subjects. It is at first the regard for the hearers that leads the speaker to express the subject in words: for the speaker himself his object, the subject, the thing at hand, the situation, stands as given; he sees in each case the object that is the starting-point for his thought and speech before him in spirit; but this is not always the case with the hearers, and therefore he must indicate that about which he wishes to say something.[38] There are, however, sentences in which the initial representation, the logical subject, is not expressed at all. This does not mean that the subject is lacking in the speaker's consciousness, but only that there is no urge or time to express it. The interest rests on the predicate. The logical subject is often hidden, and yet is that about which everything turns. It is the pivot on which the door turns;—but it is the turning, not the pivot in itself, that is the new thing.

Such sentences, in which the logical subject is not expressed, we can call predicate-judgments. Different kinds of them can be distinguished. The exclamation is the shortest and most forceful linguistic expression for the fact that something new in relation to the thing at hand or preceding has set in. No special initial representation (terminus a quo) is formed at all, if consciousness is immediately occupied by the final representation (terminus ad quem). Small children are especially occupied with what moves. A little boy of a year and a half said, when he saw an animal coming, "Bow" (from bow-wow), and the same word he used when he saw something move, e.g. when the tall grass on the lawn where he played moved in the wind. It was clearly enough the movement that laid claim to his attention, in contrast to the preceding, relatively resting state. The resting and unchanged called forth no outburst[39]. The exclamation "A star!" when a star is suddenly discovered in the bright evening sky is a predicate-judgment. Likewise the exclamation "A good idea!" when a happy inspiration puts an end to an embarrassment. In naive narration the initial representation is often presupposed known, in that the narrator, absorbed in his subject, hastens straight to the predicate and forgets to acquaint readers or hearers with the situation. In *Herodotus* several delightful examples of this are found (I, 144; V, 35; VIII, 21).

Often what one feels an urge to express is only the value of what happens. It can

[38]Cf. on this *Ph. Wegener*: *Untersuchungen über die Grundfragen des Sprachbaus.* Halle 1883. p. 54.

[39]W. *Stern* (*Beiträge zur Psychologie der Aussage. Erste Folge.* Leipzig 1903. p. 52 f; 124 ff), by showing children of seven years pictures, has found that what first interests them are persons and things, later activities and actions, last relations and properties. These experiments have, like so many psychological experiments, the defect that they do not show us the purely involuntary. Furthermore, they concern children who already have a certain development and command many memories. And finally the striking opposition between rest and movement does not come forth. "Activity and action" cannot be presented as strikingly in a picture as they are in the experience. "Substance" is not, as Stern holds, the child's first "category."

be done by a single word ("Splendid!" "Bravo!"). What is splendid stands as given and presupposed, and one has neither desire nor time to express it. In such exclamations, which contain the germ of the proper value-judgments, is expressed an object's intervention in the whole inner state of the speaker, and the logical predicate is here the property ascribed to the object as a consequence of this intervention; the object itself is not expressed.

Different from the mere exclamations are the so-called impersonal sentences. Here too only the logical predicate is expressed. There are two kinds of such.

There are impersonal, subjectless sentences in which the subject is either named beforehand or emerges from the connection in such a way that one at least knows of what kind it is, and can construct it in more or less definite form. In *Kaalund*'s poem "Nissernes Vandring" it is described how the nisses leave a region and in great throngs swarm over a bridge; ever new flocks come: "it was as if it would never come to an end." What seemed as if it would never come to an end was the swarming of the nisses. The indefinite impression of the endless is emphasized by the sentence's having no definite grammatical subject. In *Schiller*'s poem "Die Erwartung" the lover waits for his beloved; every sound and every sight can announce her coming; it is only a question of whether the interpretation of what is heard and seen is correct. "Hör' ich nicht Tritte erschallen? Rauscht's nicht den Laubgang daher? . . . Glänzt's nicht wie seidnes Gewand?" By the indefiniteness of the grammatical subject, the phenomenon comes to step forth in all its immediacy. And yet there is an initial representation; for only on the basis of the definite expectation do the rustling of the leaves and the bright gleam have interest. In other cases the initial representation is more indefinite. Only its kind can be represented. E.g. "There is dancing," "Someone is knocking." Often one here uses the indefinite grammatical subjects "one" or "they." "One says" = "it is said."

In another class of subjectless sentences it is not grammatically possible to indicate any definite subject. A wholly indefinite representation lies at the basis. These are the impersonal or subjectless sentences in the narrower sense. They are used to express natural phenomena, indefinite states, the course of fate. E.g. "It is cold." "It is spring." "Things are going backward." "It goes up and down." The chaotic and total character of the initial representation shows itself in that one can often, instead of "it," put "everything." E.g. "It is so still" = "Everything is so still." The word "it" is, in the opinion of the philologists, used in impersonal sentences only to fill an empty place in analogy with complete sentences—a grammatical counterpart to the way in which one often reads the forms and functions of developed consciousness into the elementary psychic phenomena.

Psychologically regarded, an initial element is never lacking (even if it does not become a distinct initial representation), and logically regarded one will always be able to construct such a terminus a quo and thereby translate the most primitive exclamations into logical judgments. But the initial element can be vanishing in relation to the final element. Hypnotic and ecstatic states denote the culmination in this respect.

If one defines a judgment as a conscious and definite connection of concepts, this definition does not fit predicate-judgments, in which the subject-element lies wholly in the background of consciousness and is not represented distinctly according to its whole content, as we require of a concept. But that definition can hold as an ideal. Logic can say that it really has to do only with the fully developed judgment, in which both subject and predicate are completely determined. The predicate-judgment and other judgments in which this is not the case then become only approximations to judgments. And of such approximations there can be extraordinarily many, since there can be many degrees in distinctness and definiteness.—

Predicate-judgments express acts of recognition. The application of the predicate contains a referring of the object to other objects, a kind of classification. The simplest cases of recognition permit no special designation. The object can stand as "known"—but this is also the only predicate at one's disposal. When immediate recognition is translated into the form of judgment, the classification consists in the object's being reckoned to the known in contrast to the new. In contrast to this there can be predicate-judgments where "unknown" or "new" becomes the only predicate, in that a closer description would already presuppose recognition of certain features in the new. In the very simplest cases one can only point.—

The more many-sided a relation the predicate-content is brought into with the subject-content, the greater the role played not only by acts of attention and recognition, but also by further-reaching associations and comparisons. If the predicate's relation to the subject is to be determined exhaustively, the consideration must not be limited to the immediately given, but this must be held together and compared with as many other objects as possible. Greater and greater demands are then made on the psychic energy.

31. Also in judgments in which both subject and predicate are expressed, the logical predicate continues to be the most prominent element, while the subject can stand in different degrees of obscurity. It is on this relation between the obscure or indefinite and the clear or definite that one must fix attention when it is to be decided what is logical subject and what is logical predicate. One cannot always see it from the single sentence, but must often infer it from the whole connection.

Often the logical predicate is placed first in a sentence for emphasis' sake, while the subject follows toneless afterward: "Great is Diana of the Ephesians!" "Now the time is up!" "Felix qui potuit rerum cognoscere causas!"—The logical predicate is often the grammatical subject or an adjective belonging to it: "*Thou* art the man!" "*All* the guests have come."—One will always be able to recognize the logical predicate by the emphasis, whatever place it otherwise occupies in a grammatical respect: "The king is *not* coming!" "He *has* gone."—In questions—both determination- and decision-questions—it is always a logical predicate that is lacking and sought. One has the terminus a quo and seeks the terminus ad quem. When we go back to the chief questions that lie at the basis of all other questions, we come to the fundamental predicates, the so-called categories. *Kant* aptly called the categories, the fundamental concepts of thinking, "predicates of possible judgments." To think, he says, is cognition by concepts, and concepts determine, as predicates for possible judgments, something that hitherto stood undetermined[40].

One and the same subject can be successively determined by a series of predicates. Indeed, in sentences of a describing or summarizing kind almost every single word can express a logical predicate. E.g. "A poor peasant went in to his neighbor to ask him to stand godfather to his newborn son" (the beginning of *Tolstoy*'s tale "Gudsønnen"). We here get a whole series of new things to know at once.—

32. If now a predicate is attached to a subject as a closer determination of it, then a new, more complete concept has been formed than the subject-concept one began with. The earlier predicate has now become an attribute. The attributes of existence (fundamental properties or fundamental forms) are found by analysis of experience and of the objects it presents. It is also this path that *Spinoza* has gone in his doctrine of attributes, despite his system's apparently deductive character. He finds in experience objects of spiritual nature and objects of material (spatial) nature, and since these two characteristic marks are irreducible, he regards them both as attributes or predicates of that which lies at the basis of all existence ("Substance")[41]. And so it goes in each single case; we constantly make a transition from a predicate found by analysis to an attribute. If I discover that a man is just, then I can for the future operate with the concept "this just man." The new concept thus formed can then in turn become the initial representation for a new formation of judgment, and so on.

Of special interest here are such sentences in which the grammatical subject appears with a quantitative determination attached, whether this determination is expressed by a number or in words (all, some, one). If the emphasis lies on this determination, it is in

[40] *Kritik der reinen Vernunft*² p. 94; 105 ff.

[41] *Ethica.* II. Ax. 5.

reality the logical predicate. In the sentence already used above, "*All* the guests have come," the new thing, hence the logical predicate, is "all"; the formally logical formulation would thus be: "the guests who have come are all who are expected." So it goes with all quantitative judgments, where the quantity is the main point. *Maimon*, *Herbart*, and *Sigwart* have therefore rightly maintained that there is no ground for setting up such judgments as a separate kind of judgments.—Something corresponding holds for the hypothetical judgments, just as we have already seen it for the negating judgments.

*Benno Erdmann* polemicizes against this and maintains that in a judgment such as the one cited the quantitative attribute belongs to the logical subject and is not predicate; one could otherwise, he holds, with the same right make all attributes predicates, which would be a forced reinterpretation of the linguistic mode of expression[42]. But precisely the linguistic mode of expression distinguishes between two different cases here. Where the tone lies on the quantity-determination, this is terminus ad quem, the final representation. And where the quantity-determination is not emphasized, it must have been won by an earlier judgment, so that we in any case have a secondary judgment before us.

*Husserl* concedes that all attributive designations mediately or immediately have sprung from judgments. But he holds that with these earlier judgments we have nothing more to do[43]. This is now in a certain sense correct. But the main thing is whether a certain kind of attributes (e.g. the quantitative) entail acts of judgment of a certain peculiar kind. And it is this that must be denied. When Husserl praises Benno Erdmann because he has "the courage to follow the natural meaning of the words" (Wortsinn), —and when he adds: "Ich halte es mit ihm für gewiss, dass 'alle S sind P' ein rein affirmatives Urteil ist ... das 'alle' gehört zum Subjekt,"—then it must be answered that one cannot at all pronounce on the sentence "all S are P" without knowing the emphasis; for this too belongs to the "word-meaning" (Wortsinn)—in any case it belongs to the meaning (Sinn)!

It is remarkable that two such acute thinkers as Erdmann and Husserl can still discuss the logical significance of the sentence without taking into account either the emphasis or the whole connection in which the single sentence occurs, and to which one must go back in order to see where the tone is to be laid, that is, where the logical predicate lies. Every skilled reader-aloud knows this. And it was long ago stated in the old *Art de penser* (Port-Royal's Logic) that only by attending to the connection in which a sentence occurs can one decide what is the logical subject and what is the logical predicate: L'unique et véritable règle est de *regarder par le sens* ce dont on affirme, et ce qu'on affirme. Car le premier est toujours le sujet, et le dernier l'attribut [i.e. le prédicat], en quelque ordre

[42] *Logik* I. p. 325–329.
[43] *Archiv für systematische Philosophie.* 1903. p. 114.—*Logische Untersuchungen.* II. p. 438.

qu'ils se trouvent (II, 9).

It makes no difference in this connection that judgments with quantitatively determined subjects acquire a quite special methodical significance on account of the great importance that counting, measuring, and weighing have for research. This has nothing to do with the judgments' purely logical and psychological character. Here it is a matter of not letting oneself be misled by differences in the linguistic expressions into assuming differences of kind with respect to the logical significance of acts of judgment as well as of judgments.

33. It will now be time to bring out the difference between the psychological and the logical consideration of judgment. Logically regarded, it is really only the relation between the judgment's two members that matters, while psychologically regarded it is the process by which thought reaches from the initial representation to the final representation that matters. Logically regarded, it is ultimately indifferent with which of the two members we begin, provided only that the relation between them stands clear and definite. But psychologically regarded, we in each particular case start from a certain representation and pass from it to another; the former becomes in the judgment the logical subject, the latter the logical predicate, however the judgment may be expressed purely grammatically.

No object can become logical subject without an interest attaching to it. If there are wholly interestless representations, they glide without effect through consciousness. The interest attached to an object makes the representation of it into an initial representation and operates as a spur to finding a closer determination of it. A certain tension arises before the final representation is reached, and this tension is replaced by the interest in the predicate found, which is therefore emphasized in the linguistic utterance. The interest in the subject begets the interest in the predicate. The whole formation of judgment would be superfluous if the interest in the subject were not inhibited by its indefiniteness. The mystic, who revels in his one, constant thought, forms no judgment; and his object would—on account of its unity and lack of difference—only be violated by any predicate whatever. And he who is absorbed in intuiting or in associative imagining will also seek to keep judgment at bay. *Goethe* thus says in his “Italienische Reise”: “Ich halte die Augen nur immer offen und drücke mir die Gegenstände recht ein. Urteilen möchte ich gar nicht, wenn es nur möglich wäre.”

In the completed logical judgment subject and predicate appear clearly and definitely connected. Yet even under the psychological process in which the formation of judgment consists, they do not at any point fall wholly outside each other. If that were the case, it would be unintelligible how they could be connected into a judgment. While the judgment successively develops out of the intuition or the association, the whole from which the judgment's members are fetched stands constantly before consciousness. It is within this whole that attention's transition takes place from the initial representation, which becomes the judgment's subject, to the final representation, which becomes its predicate. And the movement will be repeated several times back and forth. Two currents will form, a forward-going and a backward-going. What was at first initial representation becomes, by the backward current, final representation: in the judgment the subject is therefore just as much determined by the predicate as the latter by the former.

In logical theories one has now had more regard for the one determination-relation, now more for the other. The most common conception has been that the subject is, in the judgment, subordinated (subsumed) under the predicate, whose extension was thus greater than the former's. "The man is good" will here mean: "The man belongs to the good beings." The predicate is continens, the subject contentum. Here only the subject gets a determination, the predicate none—beyond this, that it contains, among other things, the subject in itself. In contrast to this, *Benno Erdmann*[44] has maintained that it is the subject that determines the predicate, in that the latter's content is subordinated to the former. "The man is good" will thus here mean: "The property goodness is a part of the man's being." These apparently conflicting conceptions are easily reconciled when one remembers attention's double movement during the formation of judgment. When Galileo, according to the legend, came to think of the motion of falling by seeing the church lamps swing, his attention—before the judgment "to swing is to fall"—moved back and forth between the two representations; he saw the swinging as a kind of falling-motion, but at the same time saw a falling-motion contained in the swinging. The intuition of the swinging lamps became clearer through the final representation found, of falling. The judgment's clarity is attained only by a going-back from terminus ad quem to terminus a quo.

In the impersonal sentences the initial representation still stands in indefiniteness. One notices a rustling in the foliage and a gleam (Schiller's "Erwartung"); but it is still uncertain who causes these sensations. If it turns out to be the awaited one, the impersonal sentence can be replaced by a new judgment;—but here the subject then becomes what in the impersonal sentence was predicate, and the predicate becomes the subject that in the impersonal sentence stood indefinite: "the one whose steps make the foliage rustle, and whose white gown shines through the leaves—it is *she*!"—In an

[44] *Logik* I. p. 251 f; 261 f.

example like this the two currents are clearly seen, before the proper judgment is formed.

Logically, the time it takes before the relation between subject and predicate can be determined plays no role, and the stages the judgment-process must run through have also more psychological than logical interest. Discovery and its winding ways interest the psychologist and the historian; but the logician as such asks for grounding and proof. In the last instance, therefore, it really does not concern logic what, during the formation of judgment, is initial representation and what is final representation. The difference between subject and predicate loses its significance when one formulates the judgment as exactly as possible, which is attained by expressing it in the form of an equation, as the designation of a relation of identity of a certain kind. This path Leibniz entered upon in some treatises that were first printed in 1840. Later, by various paths, William Hamilton, Morgan, Boole, Jevons, and their successors have worked in the same direction.

If the concept A shows itself by analysis to contain the concept B, one can with Leibniz express this by: $A = A + B$. For one can everywhere, where A is found, put $A + B$, since B is given when A is given. B has here, during the formation of judgment, been the final representation, terminus ad quem, and in the judgment it is predicate (A is B). But when that logical equation is formed, there is no ground to distinguish between start and conclusion. We can now keep to the result won, which can be read from right to left as well as from left to right. That distinction is only a memory from the preceding psychological process, insofar as it is not due to an influence from grammar.

Leibniz's mode of expression at the same time reminds us aptly that we do not let go of the subject when we pass over to the predicate; A occurs, after all, on both sides of the equals sign. Yet every mode of expression will more or less distinctly show the mutual determination of subject and predicate. Even according to the usual manner of presentation, according to which the judgment means that the subject's extension belongs under the predicate's, so that the judgment can be expressed by $A < B$—even there not only is A determined by B, but also, conversely, B by A, insofar as it can serve to characterize B that it comprises A in itself. When I, e.g., learn that Amphioxus belongs to the vertebrates, I learn not only something about Amphioxus, but also about the vertebrates, e.g. that a brain is not a necessary organ in them. In the languages in which the predicate-word conforms, in gender, number, and case, to the subject, the mutual connection between the two members is also seen; this mode of expression reminds one of Leibniz's.

In a logical equation the relation between the judgment's members stands before us at a single stroke, wholly clear and precise. Here has been attained, in a perfect way, what intuition attained at earlier stages of development: the abolition of equilibrium that took place at the dissolution of the intuition is now replaced by a new form of unity. The judgment expresses a higher form of psychic energy, insofar as the whole is here attained through a more composite work, and insofar as the whole that is attained is more exactly determined. But if the whole fullness of objects that an intuition (or an association-series) can contain is to be expressed in the form of judgment, a great mass of judgments will be needed for it, and it will always be only a matter of an approximation.—

What holds of the relation between subject and predicate in the judgment—namely that the time it takes to get from the former to the latter plays no role when the judgment is finished—holds analogously of the relation between premises and conclusion: there is no time-relation between them. The conclusion "lies in" the premises, just as the predicate "lies in" the subject, and just as the judgment as a whole "lies in" the intuition (or the association). Inference is really only a continuation of judgment: we infer when we discover that the predicate we have ascribed to a subject is itself subject to another predicate (*Aristotle*'s first figure), or that a subject has two different predicates (his third figure). Thus inferences grow out of judgments. But the more judgments that must be connected for an inference to be drawn, and especially when they must be fetched from highly different connections, the higher degree of psychic energy the inference-process presupposes.

Just as little as all intuitions can be exhausted in judgments, just as little can all judgments enter as members in inferences. And even where judgments and inferences can be formed, the original forms of unity—intuiting and associating—do not lose their significance. They form the involuntary basis that progressing reflection cannot do without. And perhaps, with the help of the judgments and inferences that have been formed, a higher intuition can be built, an intuition in *Spinoza*'s sense, a new comprehension of the whole won through reflection, in which the single thing could be intuited in its connection with the larger whole to which it has shown itself to belong. What value such an intuiting can have will be discussed at a later place in our investigation.

34. Historically there appears, right from *Aristotle* on, a tendency to lay the chief weight on the logical subject, although this is always given prior to the judgment, and although reflection's interest is attached to the closer determination that can be won by the predicate. According to Aristotle[45] there are concepts that can be used only as subjects, never as predicates: these are concepts of individual beings ("substances"). In Aristotle subject and substance are denoted by the same name (ὑποκείμενον); the predicate is in

[45] *Analyt. post.* 1. 22.—*De categ.* c. 5.

the judgment added to the subject, just as in reality actions, states, and properties exist only in the beings that perform the actions and live in the states. The judgment's subject thereby gets a higher dignity than the predicate: the subject corresponds, after all, to the one who performs the action, has the state, or is the bearer of the property—to "that which lies at the basis." This conception has, under various forms, maintained itself to the most recent times. One did not heed the lesson that language itself gave through emphasis, perhaps because one studied the written more than the living language.

The inclination to lay the chief weight on the judgment's subject expressed itself in a characteristic way in the first attempts to form a Danish terminology in logic. *Eilschow*[46] translated "Subject" by "Hoved-Sag" (principal matter) and "Predicate" by "Bi-Sag" (subordinate matter), and *Jens Kraft*[47] used in his logic the expressions "Sag" (thing) and "Tillæg" (addition). Even *Højsgaard*[48], who otherwise sharply distinguishes between the grammatical and the philosophical significance of the sentence, does not let his fine linguistic apprehension gain sufficient influence on the philosophical conception. He aptly emphasizes the predicate's predominant significance by translating "Verbum" by "Hoved-Ord" (head-word) and adds: "There is no kind of word that has so important a post in any discourse as the verb has. Every verb is in its proposition (sentence) like a captain for his company, and is not unlike the head, which enlivens all the limbs, and by whose nod they must all direct themselves." But he nonetheless starts from the assumption that, philosophically regarded, the judgment's subject expresses "the efficient cause," which quite conflicts with the fact that this subject first gets its determination by the predicate, whether or not this determination is to be a cause.

In more recent times *Wundt*[49] finds in the relation between subject and predicate an expression for the relation between thought-activity and its special content; *Benno Erdmann*[50] finds in that relation an analogy to the inherence of properties in the thing; *W. Jerusalem*[51] sees in every judgment an expression for the relation between will and action. But all this is analogies, due to language's tendency to personify—analogies that can have their significance with respect to making things intuitable, but are not necessary for understanding the logical nature of judgment. The analogy between the subject's relation to the predicate and the thing's to its properties, or the will's to its actions, can do a similar service to that which other logicians have found in the use of geometrical

[46] *Cogitationes de scientiis vernacula lingva discendis.* Hafniæ 1747. p. 36.
[47] *Logik.* København 1761. p. 141.
[48] *Methodisk Forsøg til en fuldstændig dansk Syntax.* Kbhvn. 1752. p. 256.
[49] *Logik*³ I. p. 149 f.
[50] *Logik* I. p. 261.
[51] *Die Urteilsfunktion.* Wien 1895. p. 92–96.

figures, especially for making the doctrine of inferences intuitable. The proof never lies in the analogies or the schemata. One must first show one's logical right to connect two concepts as subject and predicate before one can apply those analogies. Even though we perhaps always involuntarily use certain analogies, schemata, or symbols when we think, there are nonetheless great individual differences both with respect to which intuitive forms one uses, and with respect to the more or less predominant role they play in individuals. Logic must therefore distinguish definitely between the thought-relation itself and its being made intuitable. Signs, as well as words, are always only invitations or directions to carry out a certain act of thought. We must therefore always subject our images, analogies, and schemata, as well as our words, to a thorough critique, by testing whether we can draw all the consequences of them, carry them through in all details. Where this is not possible, there is a limit to the validity of the sign in question. We can perhaps not think at all without images or signs; but thought shows its independence of them in the constant critique to which it subjects them. *Leibniz* has clearly illuminated this relation, especially in the following utterance: "Even though the signs are arbitrary, there is nonetheless in the use of them and in the manner in which they are connected something that is not arbitrary"[52].

In the use of examples the relation between thought and image appears clearly. We draw only those inferences from the example that we could just as well draw from any other example. And if we do not keep to this rule, we misuse the example, just as when we extend a simile beyond the tertium comparationis.

35. Until the judgment is finished, the subject lies, as we have seen, in a certain indefiniteness. And even when the judgment lies formed, it is a question whether the whole content of the initial representation has been taken up into the predicate (or the predicates). This content must always undergo a limitation in order to enter into a judgment. So far every determination is a negation. To define is to delimit. Beneath our clear judgments there therefore always lies an obscure basis that is not exhausted, and for which predicates may perhaps later be sought. The limitation that is necessary for the formation of the judgment may perhaps later be abolished. There is an irrational relation between judgment and intuiting (or association). There is always the possibility of more decimals.

A first subject can therefore not be found. But just as little can a last, concluding predicate be found. Every predicate can in turn become subject, or the subject determined by a predicate can in turn become the basis for a new formation of judgment. Our thought

[52] *Dialogus de connexione inter res et verba.* (Opera philos. ed. Erdmann) p. 77.—Cf. my treatise on *Analogien og dens filosofiske Betydning.* (Mindre Arbejder. Anden Række). p. 35.

moves constantly from a more or less indefinite basis toward a complete determination, which nonetheless constantly sets new tasks and stands as an ideal. Existence, which announces itself in our experience, cannot be exhausted in any series of predicates, or in ever so many series of predicates. New initial representations can always emerge, and new final representations must be sought.

## d. The Validity of Judgment

36. Since judgment denotes a new state of wholeness or equilibrium, it will be connected with an assurance of its validity, even if this assurance does not specially come to expression. But this is not to be understood as if it were first the judgment that brought the assurance or the presupposition of validity at all. Consciousness operates from the first with an involuntary, instinctive trust in everything that emerges in it and for it, both in its individual elements and in their connections. We are all born in faith; doubt comes only afterward—if it comes. And that faith is there is seen from the involuntary way in which it shows itself in action. A sense-impression releases a movement just as quickly as it releases a sensation, and perhaps quite as quickly, as often in reflexes and instincts. The single impression operates like the spark that falls in the powder; it releases an energy that can seem wholly disproportionate to the stimulation. And this energy can get full outlet only through movement and action. Faith shows itself in action before it appears as a special state of consciousness. It expresses itself in the immediate absorption in, or devotion to, that with which it enters into relation. Drive and expectation precede the deliberating thought, the calm contemplation.

English psychologists have strongly emphasized this original and involuntary faith. *Hume* laid the weight especially on the expansion by which an arousal, a lively movement from a single point in consciousness, can spread to the whole state; under the influence of a strong sensation or feeling we are inclined to place trust in all representations that hang together with them; they get a share in the latter's life and strength. *Bain* has laid weight especially on the original motor tendency, the urge for movement, that makes us often act on representations before we have tested them. The single representation's connection with the urge for movement is from the first closer than its connection with other representations that could determine and limit, perhaps even inhibit, it. It is the urge for movement that makes us involuntarily test our representations by applying them in action and thus apply a kind of experimental method.

This original faith or expectation makes itself felt, as we have seen above (16), already in mediate recognition. It operates in simplest form already before proper, free representations have really developed. It is by no means the case that we first develop our representations completely and clearly, and then afterward ask about their validity.

But apart from the influence of expansion and the urge for movement, it follows from what one might call psychic inertia that sensations and representations win full trust and assent as long as no conflicting elements appear. We have no ground to doubt their validity as long as nothing inhibits them. Just as a movement changes speed and direction only when an outer force intervenes, so the development of our consciousness in a certain direction changes only when motives arise that lead in another direction. An urge or a drive that has once been awakened can (when it does not die from lack of nourishment) be inhibited only by another drive. And thus men's first assurances and expectations, whether they are due to their own observations or to traditions, change only when opposing experiences or assertions exert their influence. Doubt is always secondary.

Every intuiting, every association, and every judgment therefore has an existential character. And the same holds for the inferences we draw from our judgments. Dogma precedes critique. The history of thought bears witness to this by the tendency that expresses itself to regard points of view, concepts, and forms of intuition as absolute realities. The Pythagoreans' numbers, Plato's Ideas, Spinoza's Substance, Newton's absolute space, materialism's atoms, Kant's "Ding an sich"—are examples of this. There stirs here in the thinkers the same tendency that gives the ordinary consciousness so secure an assurance of the reality of the sensory surrounding world, despite all subjectivist objections and despite all the disappointments produced by sense-illusions.

This primitive assurance, which in practice bears our whole life and stirs in our cognition from its first germ, has its ground in the instinct of self-preservation and is of great significance in the struggle for life. But it does not have quite the same character as the conviction that can be attained through the purgatory of doubt and contradictions (cf. above 26–27). It does not appear as an independent element different from the intuition, the association, or the judgment itself, and does not itself become the content of a predicate. Precisely because it is wholly involuntary, it does not become the object of a seeking and does not come to stand as final representation, terminus ad quem. It determines the whole character of the state, but is not itself object or subject-matter. This wholly immediate character of the assurance one could express by ascribing to our intuitions, associations, and judgments an original existence-quality, which first becomes the object of explicit consciousness when an opposite quality, non-existence, can appear over against it. This happens first through disappointment and lack. Then the sharp difference between being and not-being is experienced. It goes with the existence-quality as with innocence; it is first brought forth before consciousness when it comes to stand over against its opposite—is first experienced when it is threatened or abolished.

I here use the word quality, as where the talk was of the simplest kind of recognition, to designate the wholly immediate and involuntary, ultimately indescribable feature of a state or an object, and in order to exclude the tendency that, both as regards recognition and existence-assurance, makes itself felt to read relations that first come forth at more developed stages of consciousness into the purely elementary phenomena of consciousness. Just as one has held that all recognition is due to a judgment, so one has also held that the difference that at later stages can be found between existence-conviction and mere representation must be present from the first. But what one can find out by analysis of the grown European's consciousness one has no right to presuppose in the child or the Patagonian. Analytic and genetic psychology here enter into decided opposition to each other. When one, with *F. Brentano* and his disciples, starts from the assumption that a judgment must be presupposed for there to be existence-assurance, one naturally comes to read judgments into the most involuntary utterances of the mental life. It becomes scholasticism, neither psychology nor epistemology[53].

37. The involuntary experimenting that the original presence of the existence-quality calls forth can lead to experiences by whose help critique can be applied to the immediate existence-quality. If the urge to act in a certain way or in a certain direction meets disproportionate resistance or entails unbearable suffering, it will fall away by degrees, if such an energy cannot be mustered—perhaps by the very resistance or suffering—that the inhibitions are overcome. Or it can happen that the original secure expectation, by reason of the opposition to what actual experience shows, is formed into a more distinct intuition or into a judgment, so that a clear comparison between the expected and the occurred can take place. Here one of two things can then happen. Either the content of the expectation can lose the existential character and pass over to becoming "mere" representational content, a representation of something "merely" possible, indeed perhaps of something impossible, or at least of something whose time is past, if it has ever been. Or the original expectation can be altered in a similar way to that in which we have seen the immediate intuition can be altered. A lesson is then drawn from the disappointment met, and the expectation is limited or reshaped in such a way that it is

[53] Cf. my critique of *Meinong*, one of Brentano's most eminent disciples, in *Göttinger Gelehrter Anzeiger* 1896. p. 299 f (where I have for the first time used the expression existence-quality).—*Stumpf*, who in his *Tonpsychologie* (1883) joined Brentano's conception and regarded every observing and noticing as a judgment, now distinguishes definitely between these more elementary functions and judgment. *Erscheinungen und psychische Funktionen.* (Abhdl. der preuss. Academie der Wissenschaften. 1906). p. 16.

no longer shaken by conflicting experiences.

The chief task for the psychology of cognition becomes, after the whole consideration we have here made, not to explain how our sensations and representations can come to stand as the expression of something real; but the task becomes, on the contrary, to show how representations of something merely possible and of something unreal or impossible can arise. And only when this has happened can there be a question of how we distinguish between the real and the unreal—what criterion of reality we have—and what reality really is.

The criterion of reality now shows itself to be merely a use, clarified through reflection, of the involuntary experimenting to which the existence-quality and the original expectation already lead. The expectation must be formulated as a definite question. Definite inferences must be drawn from the objects at hand and the involuntary assumptions they have led to, and it must then be investigated how far these inferences are confirmed by continued lived experiences and experiences. The expectation is thus formulated as a hypothesis that seeks confirmation. The criterion of reality consists in agreement and law-governed connection between as many sensations and representations as possible. The testimonies of the different senses, the different memories, the different inferences, and the different individuals must agree and show themselves compatible, preferably in such a way that a law-governed connection shows itself between them, so that one can infer from the one to the other. All that exists must ultimately constitute a great totality whose members join together in a natural, or even necessary, way. Illusions and hallucinations, dreams and fantasies, speculations and intimations must, even where they appear with the most vivid existence-quality, give way to the great, consistent connection that is our last refuge in cases of doubt, the firm fortress from which we judge what is to be called real and what is to be called mere representation. If the world of waking and sober life could just as well be subsumed under and understood from the world of dreams as the latter from the former, it would be impossible to draw any boundary between reality and unreality.

Reality is real truth. Formal truth a circle of representations can be said to have when it is won by consistent reflection from a definite point of view. And since the firm, consistent connection is an essential element in the concept of reality, the concept of reality presupposes the concept of formal truth, which it is logic's affair to develop. The concept of reality comprises, besides formal truth, also the law-governed connection and agreement between as many different points of view as possible. It requires a working-together of all the points of view from which we can orient ourselves in existence[54].

That the criterion of reality cannot consist in anything wholly simple was already impressed in antiquity by *Carneades*[55], in that he pointed out that one cannot rely on any single and momentary sensation or representation. It is not the immediately persuasive force of a representation that is decisive; what matters is whether it can be held fast and carried through by exhaustive testing. (The representation must not only be πιθανή, but above all ἀπερίστατος and διεξωδευμένη.) One cannot rely on a single mark, but on the agreement (συνδρομή) of all marks.

The law-governed connection is the criterion of reality of the more recent science. Thus *Galileo* says: "The mark of what is natural and true (la proprietà e condizione delle cose naturale e vere) is not merely that no self-contradiction is present; nor merely that an easy transition can be made from the one to the other; but it consists in the transition's taking place with necessity, and in its being impossible that it can proceed in another way"[56]. And the same thought recurs in all the philosophers of the more recent time from Descartes *(sixth Meditation)* to *Kant* (*Kritik der reinen Vernunft*, p. 236–247). In the period of Romanticism it became a chief thought that truth or reality must be a totality, outside of which nothing can be understood or have reality. And the fundamental features of this totality one held one could construct by way of pure speculation.—In a work that on many points seeks to hold fast Hegel's fundamental thoughts, F. H. *Bradley*'s *Appearance and Reality* (1893), the idea of reality as a totality of all possible experience has taken on the character of a standard: a reality is the higher the greater the multiplicity it encloses, and the greater the unity that makes itself felt in this multiplicity. But Bradley has given up the Romantics' hope of being able to construct the absolute totality (the one absorbing experience)[57].

38. How great the totality of experiences must be, into which the single phenomenon must fit in order to be able to be regarded as real—that cannot be decided once and for all. The world of experience expands constantly and cannot be concluded. Therefore the concept of reality or existence is, however strange it sounds, an ideal concept. For an absolutely secured assumption of reality it would really be required that the single object had been held together with all possible observations, and this is of course impossible. We can only strive to work with as large a horizon as possible. Only a provisional assurance can ever be won. The concept of reality is constantly in its becoming. But this

[54]Cf. with respect to this whole consideration my *Psykologi* V D (The Apprehension of the Real).
[55]See *Sextus Empiricus.* ed. Bekker. p. 229–232.
[56]*Dialogo sopra i duo massimi systemi.* Giornata quarta, (Ed. Fiorenza 1710. p. 417.
[57]Cf. on Bradley's philosophy my book *Moderne Filosofer.* p. 44–54.

does not exclude that this concept, having emerged from the application of the criterion of reality that is possible at a given time, can now become a predicate-concept, in that reflection investigates how far it can find application to the single object at hand. Only now do proper existence-judgments become possible. The concept existence is formed in a way analogous to that in which the concept likeness is formed. Sensations and representations must often have been placed together on account of their likeness before the concept likeness could be formed and become predicate in judgments. The association by likeness operates long before the concept likeness is set up. Thus the existence-quality can long have been present without reflection having turned against it and investigated its conditions and its justification. Existential judgments, such judgments in which the concept existence is predicate, cannot be primary, but presuppose a whole development. Behind the immediate assurance they express, e.g. "There *are* good people!" "There *are* flying mammals!"—there really lies the presupposition that the judgment's subject belongs within the great connection of our experience as a single member. The presupposition for the existential judgment is a combining and comparing process, whose possibility is maintained, even if the process is not carried through explicitly.

It is by way of reflection, not by way of mere sensation or the involuntary train of representations, that we reach reality. It is so far of ideal nature, is an object of thought. In many ways, as the history of science shows, thought must reshape the sensuously and involuntarily given in order to be able to ground a conviction of reality. Naive, secure realism has another concept of reality than the constructive realism that seeks to carry through a justified concept of reality, and that it is the affair of the cognition of nature, of historical research, and of philosophy to separate out. Slowly but surely human thought moves in this direction.

Yet it will always show itself that there are certain definite objects that form starting-points for reflection in its striving to apply the criterion of reality. All objects do not stand on the same line; reflection must begin by presupposing the reality of certain objects and testing the others in relation to them. Our thought must have a place to stand, and only at a later stage of reflection can this place itself be drawn into the discussion, namely when another starting-point has been won. The beginning-point is in a certain sense accidental; it is perhaps historically and psychologically explicable; but history and psychology are themselves part of the cognition whose validity is to be tested, become themselves objects for epistemology. From this side too it shows itself that the concept of reality is an ideal. It is not only the content of our apprehension of reality, it is also the very standpoint from which we have found this content, that must pass through the critical purgatory. And without a standpoint no reflection can, as said, be exercised.

39. Value-judgments offer an analogy to existential judgments. In their simplest, most involuntary form they appear as mere exclamatory judgments (see 30). When doubt arises about the validity of the involuntary valuation—when, e.g., the one cries "Abominable!" where the other cries "Splendid!"—a criterion is needed here too. And here too the only criterion will show itself to be agreement between the individual valuations. The two conflicting parties must seek out something about whose valuation they agree, and then investigate whether one of them did not commit an inconsistency in the valuation that became the occasion of the dispute. If they can find no such point at all, the dispute is hopeless. Each one must then seek to preserve agreement with himself in his valuations. But whether or not agreement can now be reached, it will show itself that, just as the criterion of reality in the individual cases presupposes the reality of certain objects as given, so there is in fact, in the value-judgments that are passed, presupposed one or more values. Their validity is not—at least provisionally—drawn into the discussion, but the relation to them conditions the validity of all other values. Such values we can call fundamental values. A fundamental value denotes, for each individual personality and for each individual historically given group of men, the presupposition on which valuation in the particular case, the fixing of the individual values each by itself, rests. Only in reflection's further development can the fundamental value itself be drawn into the discussion—but this will presuppose that a new fundamental value has developed and consolidated itself. Reflection can here, just as little as in the purely intellectual domain, hover in the air; but it can successively investigate its earlier standpoints, and it can become conscious of the very standpoint from which it makes its concluding valuation.

The two considerations we have here distinguished between, that which concerns reality and that which concerns value, stand in a very close relation to each other. For on the one hand the cognition of reality itself has its value, a value that for many minds stands as the fundamental value, and on the other hand the values, especially the fundamental values, must be so constituted that they can be held fast and developed within the world whose reality is shown by the greatest possible connection between the greatest possible number of experiences. This possibility belongs to the conditions of the value's validity. Finally, the reality that a purely intellectual interest leads to discover gets a peculiar value if it shows itself to afford possibilities for the development of a world of values. In the special discussion of the tasks of thought we shall return to what is here intimated.

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# II. The History of Thought

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## A. Animism, Platonism, and Positivism

40. The psychology of thought presupposes its history. The functions arising step by step, which have been described in the foregoing, have developed under definite conditions of life, physical and social, and their character is determined by a constant reciprocal action between inner and outer conditions. It falls to sociology and comparative psychology to unravel more closely this relation, which in the foregoing could only be intimated. In *Hobhouse*'s book *Mind in Evolution* (1901) a comparative-psychological treatment of the development of the life of thought is given. And it is here shown at the same time how social conditions enter into reciprocal action with intellectual development, in that the life of thought is developed through the living-together of individuals, while on the other hand social instincts and feelings presuppose a certain intellectual development, without which recognition of other individuals and understanding of their conditions would not be possible. Already language, which is the work of the race, not of the single individual, shows the life of thought's dependence on social development. The individuals do indeed give their contribution to the growth of language, more or less significant and more or less conscious contributions. But essentially language is nonetheless a social inheritance into which the single individual lives himself. In it many representations and points of view are already arranged. It has, as has already shown itself in the foregoing, its own metaphysics, which is a result of development that now promotes, now inhibits the individual life of thought. And language, as well as the thought-content deposited in it and the thought-forms developed through it, has itself developed in connection with the whole social life in family, tribe, religion, and state. The biological-psychological consideration thus leads over into the sociological, which shows the development of the life of thought under the influence of social institutions. It has even been maintained from the sociological side that, just as little as obligation can be

understood from the standpoint of the individual will, just as little can the necessity with which certain fundamental concepts (categories) make themselves felt be understood from the individual's intellectual development; in both places the explanation lies in social and historical development. The categories are of social origin, are tools of thought that human societies have laboriously formed through centuries, and in which they have gathered the best of their intellectual capital. *Émile Durkheim*, who has especially asserted this pronounced sociological point of view, does not, however, thereby wish to dispute the significance of a direct psychological investigation of the life of thought. To a complete investigation of the human fundamental concepts belongs, according to him, also an investigation of the germs of thought-life that can be found by direct analysis of the individual consciousness[58]—such an investigation, then, as I have attempted in the foregoing. It must at the same time, in the face of the strong emphasis on the sociological point of view, be definitely maintained that the overwhelming social influence on the life of thought is easiest to demonstrate at more primitive stages of human development. During the development of culture the individual thought frees itself more and more. Already comparative inquiry in the domain of psychology and sociology shows this. Sociology itself would not be possible if such a liberation had not taken place, which of course can never be absolute, any more than, for that matter, the social influence can ever be absolute. When it is so easy to demonstrate the social influence on spiritual development in earlier periods, this has for no small part its simple explanation in the fact that the individual influences are forgotten. One has the products left, but the factors have vanished, and one then lets "society" or "the group" bring forth itself in a mystical way. At every given stage "society" consists of definite individuals, each with his dispositions, with his urge and his circle of experiences. "Society" easily becomes an asylum ignorantiæ. It is a convenient word to use where one does not know or cannot survey the operative individual forces in their many differences and in their infinite reciprocal action. The historical point of view must not be made without further ado one with the social point of view. Every individual lives himself into a historical connection, also as regards the life of thought; but this historical connection is itself the result of manifold individual forces' cooperation through the ages, of giving and receiving between individuals in constant reciprocal action. The social is what makes the reciprocal action possible; but the reciprocal action itself presupposes the individual forces.

41. That the individual life of thought always unfolds on a historical basis was

[58] *É. Durckheim*: *Sociologie religieuse et théorie de la connaissance.* (Revue de Métaphysique et de Morale. 1909) p. 750.

strongly emphasized by both the two great main directions in the domain of philosophy in the nineteenth century, Romanticism and Positivism. Both directions had—in common opposition to the eighteenth century—a clear understanding of the social and historical atmosphere within which the spiritual development of individuals takes place. They could think of no spiritual development on bare ground. By *Hegel* and *Comte*, two eminent representatives of the two main directions, the history of thought has therefore also been written in broad strokes. In his *Phänomenologie des Geistes* (1807) Hegel showed how cognition begins with the single and isolated thing, but is from there driven forward step by step through the contradictions in which it becomes entangled. Through step-by-step development thought unfolds its essence and shows itself to be an ideal totality within which everything single can find its place, at once as a peculiar member and yet "sublated" in the "higher unity" that encloses all oppositions in itself. The individual stages in this development are illuminated by examples and types, partly of individual, partly of social kind. But it is Hegel's view that only the thought that can unfold on the basis of what the history of spirit has brought about will have the conditions for exercising a significant work. The "Spirit" whose history of manifestation (Phenomenology) he describes in his work is the spirit of the human race, in which the individual spirit is only a single moment. It is from the "higher unity"—won at the given stage—of formerly struggling oppositions that the individual thought-work is to be exercised.

*Comte* too distinguishes (*Plan des Travaux.* 1822) between different stages in the development of the life of thought. But with him it is not the spurring force of oppositions and contradictions that leads forward. His chief thought is that at all stages a relation makes itself felt between what is given in experience (the objects) and the representations by which it is interpreted. The difference between the stages rests partly on how large a circle of objects is at one's disposal, partly on which representations prevail in man, so that they can more or less involuntarily be taken as basis in the interpretation. The narrower the circle of experience is, the greater the role the ruling representations will play; the more comprehensive it is, the more the thought-work will come merely to aim at bringing about connection within the given, so that this interprets itself.

The points of view that the two philosophers take up supplement each other. For when experience necessitates a new mode of interpretation, or when the difficulty of carrying through an interpretation leads to seeking out new experiences, it is evidently because it discovers contradictions or oppositions in the hitherto given, or between the prevailing interpretation and the given. Thought is shaken in its accustomed interpretation by new experiences, or it hopes that its difficulties will fall away through new experiences[59]. Whether by these paths "higher unities," more comprehensive unified intuitions, can always be reached, as Hegel held, is a matter in itself, which will be discussed later. And whether Comte is right in the sharp difference he sets up between three stages in cognition, the theological, the metaphysical, and the positive, we will here also not discuss more closely[60]. But with some change in the points of view one will rightly be able to preserve Comte's tripartite division, as I shall now attempt to show. The change entails a corresponding change of the designation; one will suitably be able to distinguish between Animism, Platonism, and Positivism as three main stages, in that these three words are each taken in an extended sense.

42. In its simplest form the difference between object and interpretation appears in the opposition between distinguishing and recognition, the two poles in our cognition, which correspond to the two great conditions of life: openness for the new and application of the old to the new (Cf. 14). But only when judgments are formed, and especially when questions and problems are set (26–27), is there a matter of interpretation proper. The presupposition for this is that one does not immediately proceed to action. For when the emerging cognition-elements immediately release reactions in a reflex and instinctive manner, there is neither urge, time, nor force for interpretation. Interpretation presupposes a certain degree of intellectual interest and, in connection therewith, experiences that there are limits to our capacity to intervene in the course of things, so that we must come to know the essence and laws of things in order to be able to place them in the service of our purposes.

*Frazer*'s hypothesis about the relation between magic and religion[61] is of interest in this connection. Prior to religion, which always presupposes experience of dependence, there goes, according to Frazer, a period in which man has trust in his will's power to such a degree that he believes he can bend nature under himself, not only where he really intervenes actively, but also by mere wishing, made known by word or ceremonies. Or rather, the difference between action, word, and ceremony does not exist for man at this stage. It is experience of the will's limitation that leads beyond this stage, and it is likewise this that leads into the world of cognition. The magical art does not always lead to the goal. Events often take place independently of or in defiance of the human will. Through disappointment and the feeling of impotence, man learns that in existence there

[59]These two sides were emphasized by Danish philosophers in the eighteenth century, the first by *Holberg*, the second by *J. S. Schneedorff*. (*Danske Filosofer*. p. 5; 18.)

[60]See on this *Den nyere Filosofis Historie*[2] II. p. 334–338 and a treatise by *Hobhouse* in *Sociological Review*. 1908. p. 262–279.

[61]*The Golden Bough*[2] I. p. 61–78; III. 458 f.—*Adonis—Attis—Osiris*. p. 3–4.

prevail powers that are stronger than he. These powers he conceives as personal beings, and this conception—however it has arisen—afforded the first means of interpretation, the first "categories." Man did not give up magic; but it was now exercised toward the overmighty beings whose will it was a matter of bending, or whose power it was a matter of supporting by the performance of certain ritual actions. Already the magician at the pre-religious stage performs his ceremonies in a certain definite way, since he otherwise has no trust in their operation, and he so far recognizes a certain order of things; but this recognition becomes more penetrating and comprehensive when magic is replaced by, or at least subordinated to, religion.

I here disregard the question whether the arising of religion can be explained merely by the disappointments to which autonomous magic led. There must, after all, have been many other paths by which man could experience his dependence. What in this connection is of interest is the indisputable opposition between a standpoint where man has no occasion to recognize an order of things, and another standpoint where such a recognition is alive. From the one standpoint to the other man can only have come by way of experience, a way that has probably often led through hard disappointments. Only when this way has been traversed is there a matter of interpretation. And history teaches that the first attempts at interpretation have the character of *Animism*, in that the explanation of natural phenomena, at least the most striking and far-reaching ones, was sought in the intervention of personal, man-like beings.

Animism appears in a series of forms—from momentary and special gods to polytheism's individualized god-figures, and from these to a more or less thoroughgoing monotheism[62]. These transitions are not only of significance for the development of the religious life, but also for the development of the life of thought. The mythological imagination's individualization of a god-figure—which corresponds to the psychological transition from concrete to typical individual-representations (24)—leads naturally to a more definite distinction than before between the natural phenomena that depend on the god's intervention and the god himself. This distance-relation makes a more thorough investigation of the natural phenomena themselves possible; the god withdraws and hides himself more behind his work than before. Human experience gets time and space to spread out, since it does not everywhere or immediately strike upon the god. And the god-figure itself now presents an inner connection, a more or less consistent characterization, so that the representation of it denotes an advance in psychological insight. A typical individual-representation presupposes that one regards all the individual

[62]See my *Religionsfilosofi* § 46–53.

utterances and situations in which a being lays itself bare as examples of its character. It indicates, where it is distinct, the law for the development of the being in question, la loi du changement (Leibniz). Both physically and psychologically, then, polytheism denotes a great advance. Comte has even called its development the most important advance within the "theological stage." The transition from polytheism to monotheism too has of course its great significance. An absolute monotheism is scarcely ever carried through. But the fewer gods there are, the more the divine intervention in the course of natural phenomena can approach a minimum. On the other hand, the strictest monotheism can get a likeness to the belief in momentary gods, when the one god does all too many miracles.

During this whole process there is a reciprocal action between religious and intellectual elements, or, if one will, between religious experiences and other experiences. When by religious experience one understands the way in which the human life of feeling is attuned through what is experienced with respect to the fate of human purposes and ideals under the course of things, then this experience will contribute much to the way in which the powers of existence are conceived; religious experience has a spurring and forward-driving influence on the whole world-view. One need only compare Homer's world-view with Hesiod's, Plato's with Marcus Aurelius's, the Vedic poems' with Buddha's, older Judaism's with that of the prophets. But by degrees experience and understanding of the natural connection of things, hence physical insight, is also gathered. And the more social life and the personal life in individuals develops, the more connection the human life of personality presents, the less can one think of the gods' operation as guided by momentary and changing motives. Psychological and physical experience thus meet with religious experience.

43. Animism (in the widest sense) has not yet disappeared as a principle of interpretation because strict science develops. It is limited to "the creation of the world," or to certain extraordinary phenomena strongly intervening in human destinies, or one finds in it the ultimate explanation of the fact that there prevails law-governedness in nature at all. In some of the greatest inquirers of the more recent time the animistic principle of interpretation is found in one or more of these forms. *Kepler* and *Descartes* ground the conviction of the universe's natural law-governedness on God's perfection. According to *Leibniz*, all monads, existence's ultimate elements, "radiate" from a primal monad, by whose choice the best of all possible worlds has come to be, a world whose course is determined by strict natural laws, but in such a way that these natural laws themselves are determined by regard for the greatest possible "perfection." The solar system's har-

monious structure and other purposive natural arrangements are, according to *Newton*, due to God's intervention and are not subjected to natural-scientific explanation; the infinite space itself that mathematics presupposes is God's sense-organ, the organ for his omnipresence. *Linnaeus* derived the different species of organic beings from just as many different divine acts of creation, and *Kant* found in the concept of time the only possible solution of the great problem set by the opposition between moral ideality and nature's actual course.

Yet during the development of the life of thought there takes place a constant elimination of animism. This is used less and less for the interpretation of the objects at hand. It is by this indirect way, by non-use, that animism disappears, not by direct disproof. It really cannot be disproved. *Kepler* saw a disproof of the assumption that the planets were moved by souls in the fact that the moving force decreases with distance[63]. But how could he really know whether a soul's operation did not also decrease with distance?

Animism has, despite everything, exercised a significant function during the development of the life of thought. It was the first coherent train of thought in which man could set his scattered experiences and lived experiences. And if it had been possible to develop and carry through the concept of personal beings in such a way that it could afford frame and basis for all understanding of material and spiritual nature, and at the same time guide toward setting research new tasks, then its time would not be past. Theology would then be both the highest and the fundamental science. The history of the life of thought shows that it has not been possible.

To this is added that the personal life itself shows itself to be subject to definite laws and is itself a part of the objects that are to be interpreted. Animism consists really in a part of our experience, namely our experience of the mental life, being taken as basis in the understanding of all other parts of our experience. One cannot, with Comte, distinguish absolutely between the given and its interpretation, since every interpretation ultimately builds on something that is itself given. Interpretation consists at all stages in the one object's coming to illuminate the other. There is thus a constant reciprocal action between experience and interpretation. In each single case there are some objects or experiences that are to be interpreted, and others that serve for interpretation. This was, in Comte's view, to be the characteristic of the "positive" stage, but it is in reality characteristic of all stages. The difference is only the choice that is made of interpreting objects, and the exactness with which the interpretation is undertaken; and both these things hang closely together, when it shows itself that not all sides or parts of our

[63]Cf. *Den nyere Filosofis Historie*[2] I. 165.

experience are equally well suited as basis for the understanding of the others.

44. Animism always had to undertake a certain ordering, in any case have a capacity for recognition, in order to be able to know under which divine being's domain of power the single object belonged. In "special gods" (departmental gods) this shows itself especially clearly; but it is a relation that is found again under one form or another in all religions. But the ordering or classification that here took place was very external; there was no inner relation between frame and content. And yet an ordering survey of the objects that are to be interpreted is the first condition for their understanding. The point is then to order them according to likenesses and differences. Reflection's essence as comparison expresses itself here. It soon showed itself that there are many degrees of likeness (and difference), and it dawned on thought that it is possible to bring about an ordering of the content of our experience from this point of view. One had then to try to discover as many and as essential likenesses between the objects as possible, and order them accordingly. The significance of Greek philosophy lies for a great part in its having impressed the importance of this task and made the first attempts at its treatment, and *Plato* can here be regarded as its typical representative. For Plato thought's essence laid itself bare in its capacity to form concepts that contained the points of likeness between large groups of objects and thereby comprehended these in themselves. The discussion of objects that led to sharp determination of their likenesses and differences Plato called dialectic, and the dialectician had to have a capacity for comprehending survey (ὁ μὲν γὰρ συνοπτικὸς διαλεκτικός, ὁ δὲ μὴ οὔ. Republic 536 C). There could by this way be formed a world of Ideas, in which the nature of the objects found its highest expression. The individual things and events arose, changed, and vanished, but this changed nothing in the concepts in which thought had comprehended their essence. All that is beautiful, e.g., is perishable, but the Idea of the beautiful is imperishable.

The ascent from the world of differences and changes to that of Ideas, in which thinking for Plato essentially consisted, was now for him not merely a formal systematic ordering, a classification, but in the Ideas or general concepts that held for a group of objects he found at the same time the ground that such objects existed at all. The doctrine of Ideas is not merely a doctrine of method, an attempt at a conceptual ordering of all thought-content. For Plato's aesthetic-religious conception the Ideas also came to stand as plastic figures, as constituting the true reality, of which the individual special objects are only imperfect copies. The value of the copies rests on their nearer or remoter relation to the prototypes, and of these prototypes again the Idea of the Good is the highest[64].

[64] *Natorp*, in his work *Plato's Ideenlehre* (1903), keeps only to the methodological significance of the doctrine of Ideas, and interprets Plato rather too much from modern points of view. Thus especially when

Interesting here is Platonism's relation to Animism. They meet in the presupposition that what lies at the basis of the phenomena must also be something that can be the object of men's urge and longing, contain values or be the cause of values. For Animism it was the god that was both the ground of things and the source of values. Plato gives up the personification, and the highest value he found in the very relation to the Ideas, in the happiness of beholding them. And this holds for gods as for men. The advantage the gods have is only that they stand nearer to the world of Ideas. In the great procession that in the "Phaedrus" sets out to admire the Ideas, the eternal stars in the heaven of thought, the gods drive first. Herewith is denoted a decisive turning-point in the relation of religious representations to scientific thoughts.

By the projection that let the Ideas stand not merely as thoughts, but as things, there came forth in Platonism a great problem: how do the Ideas relate to the phenomena? It was a reflection of Animism's problem, how the gods relate to the phenomena. A critique that possibly derives from *Aristotle* emphasized that consistently there would have to be set up an Idea under which both the individual Ideas and the phenomena belonging under them were to be comprehended. The justification of this critique lies in the fact that reflection cannot end in any principle that should "have its ground in itself," but must always anew discuss the relation of the Ideas or principles won to experience—and that (what Aristotle did not see) not merely to the experiences hitherto won, the objects hitherto at hand, but also to the new ones that can emerge or perhaps even be called forth.

At another point too Platonism shows its insufficiency. In that it makes classification, tracing-back to Ideas, one with causal explanation, it comes to assume just as many causes as there are kinds of objects to explain. That the Ideas are again to form a systematic connection, and finally be subordinated to the Idea of the Good, makes no difference in this connection. Here too Platonism reminds one of Animism; just as the latter has a god for each group of things or events it is able to distinguish, so Platonism has an Idea for each such group. Thereby a way was paved for the kind of pseudo-explanations such as that life is derived from a life-force, gravity from a force of gravity, and so on. It was this peculiarity that Comte especially laid weight on for the characterization of the second of his three stages, the one he called the "metaphysical."

Yet it denotes a great advance that the "Idea" or the force stands as something that operates constantly, always in the same way under the same conditions, something that

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he interprets the phenomena's "participation" in the Idea by the relation of single events to the law (p. 151).—J. A. *Stewart* (*Plato's doctrine of Ideas* 1908) distinguishes sharply between the methodological and the contemplative significance, but concedes that Plato constantly confuses these two significations.

could not be said of Animism's gods. But the advance lies especially in Platonism's weight on the ordering thought-work that systematizes according to likenesses and differences. This will always continue to be the first step toward gaining penetrating understanding. From the mere catalogue with its external ordering to the "natural" systems resting on deeper-lying principles, such work appears in its great significance. Platonism's time will therefore never be past, as long as series-formations, type-arrangements, and overview (synopsis) continue to be of necessity. And there are domains, such as the psychological and the historical, where research perhaps never gets beyond this kind of work, at least as regards the most significant questions.

45. *Plato*'s Ideas express relations of likeness between different objects. When it is asked why the individual objects appear precisely with their definite properties and at a definite time and place, Platonism gives—just as little as Animism—any answer to this. Its method can only lead to an ordering that is grounded on comparison between the objects with the properties they happen to have, but cannot say anything about the origin of the objects and their properties. It sees existence in equilibrium; the comparing and ordering reflection stops the stream of time in order to fix the types that the objects present. But since the objects and their properties arise each by itself at definite stages of the stream, the task naturally comes up of investigating the relation between the coming-into-being of the different objects and properties. Does the one object arise independently of all others? Can the single property appear without other properties appearing or vanishing? Answers to these questions are not obtained by a merely resting ordering, but by finding out an ordering of the succession in time according to definite rules, so that the ground for the arising of the following objects and properties can be found in the presence of the preceding ones.

We can use the word Positivism to denote the direction in the life of thought that sets new tasks in the way just intimated. Here again the point is to understand objects through comparison. The comparison concerns the definite conditions under which an object comes to be or a property appears, the conditions for its localization in time or in space or in both respects, and the connection is to give the unavoidable historical succession as determined by a law. Every event is thus attached to another event, and it is at this that the endeavor aims, not merely at finding the likeness and difference between the different objects and properties. And the explanation is now sought not outside the series of events, in a being or an Idea, but in the very law according to which the members of the series follow one another. What is called cause is thus itself to be demonstrated as object, as member in experience, just as well as the object that was to be explained. It is the standpoint of the more recent empirical science, as it was indicated by *Kepler* and *Galileo* through the demand for a cause demonstrable in experience (vera causa).

The differences stood unexplained in Animism and Platonism. How could very different, perhaps mutually conflicting, objects and properties have their ground in the same god or in the same Idea? The more rich in content and comprehensive experience became, the more the sting of this question had to be sharpened, and attention had to turn toward the mutual connection of the differences. The urge for recognition can nonetheless constantly find satisfaction, if it succeeds in re-finding the same events when the same conditions are present. The earlier experiences comprised in a certain sense the new ones, if these are due to the same law as the former.

Yet it is not merely the urge to overcome in thought the multiplicity and change of objects that operates at the transition to Positivism: it is also the wish for power—the wish that in magic expressed itself in immediate acting, in Animism more mediately and on the presupposition of a certain resignation, and in Platonism faded into intellectual contemplation. Power over nature is now won by understanding of the laws according to which nature operates and by attempts, in the single case, to bring about the law-determined conditions.—

Is now this form the definitive one, or will there be able to come new stages in thought's great history of development? To this, from the nature of the case, no answer can be given. But we can see—and it is an insight of the very greatest significance—that the positivistic tendency itself, the standpoint of strict empirical science itself, rests on certain presuppositions concerning thought's relation to existence, so that, even if this standpoint should become the definitive one, problems remain with respect to the ultimate grounding of it. It is at this point that philosophy in the more recent time has taken up its work. The seventeenth century's great systems sought to find out what world-view resulted when the new standpoint, which I have here for brevity's sake called the positivistic, was strictly carried through. The eighteenth century's philosophy then went back to the presuppositions for the whole standpoint and subjected them to a critical testing. The nineteenth century's philosophy made partly an attempt to return to Platonism, partly to stop the thought-work there where the positivistic principle did not lead us further. It will be a chief task for the continued thought-work to carry further the treatment of the problems that through the purgatory of the great discussion have shown themselves unavoidable for every waking reflection. A contribution to this it will be attempted to give through a closer investigation of thought's modes of operation and tasks. But first there is yet another opposition to illuminate within the history of thought, an opposition that indeed reminds one of the opposition between Platonism and Positivism, but does not quite coincide with it and in any case illuminates it from a new side.

## B. Ancient and Modern Thinking

46. The opposition between the second and the third stage, as it has been described in the foregoing, coincides in part with the opposition between ancient and modern thinking. But it will serve for the closer illumination of the nature of human thought if we give a more thorough characterization of this opposition. It can be summed up under two points of view. Ancient thinking has a predominantly formal character; on this rests in part its great and lasting significance. And next, and in connection therewith, it is characterized by the presupposition that the unchangeable stands, not only in a logical but also in an ethical respect, in value above the changeable.

47. The opposition between formal and real knowledge appears already in *Parmenides* of Elea, and so sharply that he recognizes only formal knowledge as real knowledge. Only those propositions that can be formed as pure identities can be the object of true conviction, and only such propositions can express what truly is. Being must be unborn, imperishable, unchangeable, indivisible, concluded in itself. Of the manifold, changeable, and perishable no knowledge can be had, only "opinions of mortals."

Here appears, with the thought-energy that distinguished the Greeks, the principle of identity, the demand for thought's unity with itself, the strict recognition, as the first principle of all thinking, as the formal ideal to which thought seeks to bring all cognition as near as possible. Parmenides would not even concede that differences and changes could set serious problems, whether such problems can be solved or not. Only conjectures, not true science, are here possible[65].

Two other Greek thinkers, the one Parmenides' predecessor, the other his successor, made, on the contrary, attempts to bring identity, the highest form and norm, into closer connection with the objects' difference and change. *Heraclitus* maintained that in all the constant streaming and under all the conflicting oppositions there is nonetheless something that is constant, namely the law, existence's eternal fundamental thought (λόγος), which subsists under the constant change of all phenomena. And *Democritus* set up the proposition that one reaches a true understanding only when one disregards

[65] Cf. my treatise on Parmenides in *Mindre Arbejder*. Anden Række, p. 141–149.

all the qualitative differences that the senses show us, and explains everything by the reciprocal action of homogeneous parts. His atomic theory is Parmenides' doctrine of identity applied to each single one of existence's smallest parts.

While Parmenides denied the reality of what falls outside pure identity, Heraclitus and Democritus set themselves the task of tracing the given differences and changes back to identity, partly the law's identity under different conditions, partly the elements' identity in different connections.

In *Plato* appears the same conviction as in Parmenides of the world of identity as the true world, only that he finds the identical in the general concepts or Ideas that hold for the sense-world's multiplicity and change, and that for him are not merely human thought-products, but the expression and content of what truly is. The logical point of view, which more hiddenly made itself felt in his predecessors, and which he had in Socrates' school been trained to use, came in him to clear consciousness. What filled his mind with ever new enthusiasm was the fact that concepts can be formed which, in their unchangeableness, their identity with themselves, hold for large groups of differences. The victory of likeness over the differences in human thought was for him a testimony that the world of Ideas was the true world. All cognition was recognition of the Ideas in the sensory world, and the highest spiritual capacity was that by which, from the points of likeness within the sense-world, a world of thought is formed, so that an ascent to the true reality becomes possible.

The Greek concept of science is predominantly formal. True science was for the Greeks logic and mathematics; their empirical science for the most part did not get beyond the collection of more or less exact observations. Democritus and Plato are the classical Greek thinkers, and against them both there was in antiquity directed the objection that they set aside and rejected what the senses teach (τὰ αἰσθητά), and kept merely to what pure thought grasps (μόνοις ἑπόμενοι τοῖς νοητοῖς)[66]. Both Democritus' atoms and Plato's Ideas were pure objects of thought, constructed from the demand for identity. Both for the founder of the atomic theory and for the founder of the doctrine of Ideas the sense-qualities conflicted with thought's pure identity.

*Aristotle* maintains—in opposition to Plato's doctrine of Ideas and Democritus' atomic theory—the reality of quality and of individual existences. It was thereby established (without Aristotle's fault, for millennia) that the science of empirical objects could consist only in classification. Aristotle himself did not fail to appreciate formal science; he is, after all, himself the organizer of formal logic, probably influenced by the development

[66] *Sextus Empiricus* ed. Bekker. p. 299.—Cf. my treatise on *Forholdet mellem Platon og Demokritos* in Nordisk Tidsskrift for Filologi. 1909.

that mathematics had gained; but logical thinking and qualitative comparison continue in him to stand side by side. He saw well that movement, change of place, lay at the basis of all other change in nature—of qualitative change, of coming-to-be and passing-away, and of increase and decrease; but he made no attempt to carry through this thought, which would have led him to the point where modern natural science begins.

It was two different, but both deep-lying, presuppositions that hindered the Greeks from letting formal and real thinking cooperate in a fruitful way.

From the Orient the Greeks had inherited the presupposition that the star-world's movements took place with absolute regularity. It was this unchangeableness, in opposition to the earthly (sublunary) world's mutability, that first opened the view for the possibility of science. Here one found the self-identical being realized. And one found then, in thought's firm and constant forms, expressed in the general concepts, a counterpart to it. Both the visible and the invisible unchangeableness were embraced with religious awe, as a refuge under the change of things. A peculiar religious type makes itself felt here[67]—but in remarkable connection with a physical and a logical ideal. The presupposition that the unchangeable is better or more perfect than the changeable the Greeks did not give up, any more than the Indians gave it up.

In a peculiar opposition to this tendency in Greek spiritual life stands the Greek sense for the concrete, the individual, and the intuitable, the living urge for plastic formation of the spiritual content. Not only mythology's gods and art's figures, but also thought's Ideas stood as clear images, whose self-resting firmness neither thought's analysis nor experience's increasing fullness could dissolve. The single observation and the single individual stood as something apart, both in opposition to conceptual thinking. In Aristotle this opposition culminates; for just as convinced as he is that only the universal is object for thought and science, just as convinced is he that only the individual has existence or being. The individual is that which is the object of all assertions, but which can never itself be used as predicate. (Cf. 34.)

48. The Renaissance's thinking led beyond the opposition at which Greek science had stopped. The multiplicity and mutability that life and experience present were now conceived precisely as means to come to know nature rightly. *Nicolaus Cusanus* and *Giordano Bruno* showed that thought itself works its way forward through oppositions, operates by constantly setting limits and overstepping them in order to find new limits, so that no opposition, no barrier, can be established once and for all. And experience itself now led to the discovery that one had drawn in far too external a way the boundary

[67] Cf. my *Religionsfilosofi*[2] p. 51–55; 125–128.

between the world of unchangeableness and that of changeableness. *Tycho Brahe*'s "new star," which so suddenly emerged in the sphere of unchangeableness, moreover presented changes both in brightness and color, and his investigations of the comets showed that these apparently so capricious beings did not move about in the sublunary world, but on the contrary traveled through the ether-world higher up. And Galileo stated outright that the earth appeared to him precisely "most noble and admirable" (nobilissima e ammirabile) on account of the many changes that take place on it. He cannot in itself find any perfection in the unchangeable. The standard for value becomes another. It is a matter not merely of ascending to the unchangeable world of eternal Ideas, but of finding the laws according to which the changes in the world of experience take place. As one had found changes in the world of "unchangeableness," so one also sought to find unchangeable laws in the world of change. In antiquity *Archimedes* had made the doctrine of equilibrium an exact science. Galileo began to subject the doctrine of movement to an equally exact investigation. Ancient science had essentially been statics and morphology; now dynamics gradually came into its own in the different domains.

If one was to find an unchangeable law through the multiplicity of changes, one had to keep to the properties of things and events that make quantitative determination possible. Counting, measuring, and weighing had now to become means of thought to a far higher degree than before. One had then to keep to properties such as magnitude, figure, and movement, and try to determine the appearance and changes of the qualitative properties through their relation to them. The opposition between formal and real thinking now became an opposition between quantitative and qualitative knowledge, and in this form a more intimate cooperation between the two kinds of thinking was possible. Speculation and observation, deduction and induction, could now work into each other's hands, in that the inferences drawn from general propositions could, by reason of the quantitative formulation, be tested in an exact way by their relation to what experience showed, while conversely, from experience, ever more exact general propositions could be formulated. Thinking could now describe an arc from experience through inference to new experience. Ancient thinking really knew only the first part of this arc. When it had found a starting-point for its abstraction, it ascended to the world of Ideas, the world of logical and mathematical forms, and shrank from descending again to apply and test the thought-results won in the world of experience. It believed in apotheoses, but not in incarnations.

It was now no longer the ideal to order qualities under general concepts of different degrees and thus form classificatory systems, but to find relations of dependence between changes in the world and express them in definite laws. Explanation steps in place of mere classification, and quantitative description forms the transition from the latter to the former. Logic and mathematics enter as tools into the work of real cognition; they no longer form a world subsisting by itself, but their principles and laws are woven into the world of thought that builds on the ground of experience. Platonism and Positivism strive to enter into as close a connection as possible.

49. Purely logically, the change that now takes place in thought's task and mode of working shows itself in the fact that judgment, not the concept, becomes the most important thought-form. As we have earlier seen, the opposition between judgment and concept means only a different weight on two sides of the same thought-process (24). But the opposition between ancient and modern thinking can very well be expressed thus, that the former uses the formation of judgment to win concepts, the latter uses concept-formation to reach judgments. The necessity of forming concepts was Socrates' great discovery, which gave Plato enough to think about throughout his whole life, and which led to the formation of a world of Ideas as the expression of true reality. But for modern thinking the point is to find judgments that express definite relations between definite changes. Through such judgments—by virtue of the relation of reciprocal action between concept and judgment—concept-formation will in turn be able to attain greater perfection.

In a characteristic way this tendency appears in the fact that two of the Renaissance's thinkers interpret Plato's Ideas as the laws of things. *Bruno* teaches that the task is not merely to form general concepts that express what is common to several phenomena, but to find expression for the real connection between different phenomena (e.g. the parts of an organism) among themselves, since only by such a connection can one understand the nature of the individual phenomena. *Bacon* expresses himself in a similar direction when, for the understanding of a phenomenon (e.g. heat or organic life), he demands that one must not remain standing at abstract concepts, but shall try to find the continuous, often hidden process (latens processus—per minima) by which things get their definite constitution[68].

With this there now also follows a change in the conception of inference. As long as one had merely to do with a system of super- and subordinate concepts, it was natural to build the theory of inference on the principle that what belongs under a concept also belongs under the concepts under which this concept belongs. Inference then stands as

[68] *Den nyere Filosofis Historie*[2]. p. 126; 194–196.

a process of subsumption, culminating in a highest concept. But when weight is laid on the fact that concepts as well as judgments are to enter as members in the ever-continued thought-movement, it will be natural—as has also (see 33) shown itself to accord best with the essence of judgment—to conceive judgment as the expression of a relation of identity, so that inference can take place by different judgments being combined by means of substitution. The purely logical thought-movement thereby approaches the mathematical, which proceeds through equations and substitutions. *Leibniz*'s identity-logic is thus a characteristic expression of the more recent time's thinking—even though a later time's logic has in several respects had to strike out on other paths[69]. It must especially be emphasized that Leibniz did not bring his logic into connection with a doctrine of categories, and thereby came to overlook that there are many other relations than identity that can be of great significance for the thought-work (even though they can all be purely formally expressed by predicates in identical judgments). In a later section we shall have occasion to discuss the relation of identity in comparison with other relations. But it is extraordinarily characteristic that identity for Leibniz does not stand as an object of mystical beholding, like the "Idea of Likeness" for *Plato* (*Phaedo* p. 74), but as a tool of thought. This shows itself already in his definition of identity as the possibility of substitution: identical are the concepts that can be put in place of one another without the validity being violated (salva veritate). There lies in this a constant summons to test identity experimentally, to see whether it can be maintained in the individual special connections in which the concepts occur. It is in the world of multiplicity that the thought-work is to be exercised, and the question is how far relations of identity can there be demonstrated.—

While Platonism was inclined to confuse conceptual determination with causal explanation and could thereby lead to assuming just as many "forces" as there were kinds of objects, in modern thinking the concept law stepped more and more in place of the concept force, or rather this concept is subordinated to the former. In the Aristotelian doctrine of nature, nature formed a series of stages, within which every stage contained the possibility or the force (δύναμις) for the following one. But about the possibility or the force one then learned nothing other than precisely this, that it was—possibility or force for the following one. There were then just as many possibilities as there were realities. The qualitative differences were explained by just as many qualitates occultæ. In scholasticism, which in many domains maintained itself far beyond the Middle Ages, one lived high on that kind of word-explanation. When it is first and foremost a matter of

[69]Cf. on this *Couturat*: *La Logique de Leibniz*. p. 386 f; 433.—Couturat overlooks, however, that all relations can be expressed as predicates.

finding a law, that is, a judgment that states one phenomenon's dependence on another, one need not indeed give up the concept force, but this gets the definite meaning of denoting that in the preceding phenomenon which contains the ground for the following phenomenon's setting-in—the peculiarity of a phenomenon that from it one can infer another's setting-in. Leibniz is clear that the concept force contains no explanation, but only a summons to seek why the transition to a new phenomenon takes place precisely under these definite conditions. The concept of force is constructed in each single case on the basis of the law that has shown itself to hold for such a transition[70].

50. The new concept of science could not immediately be applied in all domains, and it became of decisive significance for the philosophical treatment of problems in the centuries immediately following the Renaissance that there were some domains that were especially suited to be treated according to its demands. These domains then naturally came to stand in the foreground in the scientific conception of the world. It was the doctrine of movement and mechanics that determined the conception of nature in the seventeenth century's great systems. In these domains the differences of quality and individuality stepped back before the universal laws. *Leibniz* here forms the transition to a new period, in that he so strongly brings out the concept of individuality, and demands not only a continuous connection and law-governedness demonstrated between all existing beings, but also within the little world that every single individual being constitutes. Within the individual beings each by itself too there take place changes, and if we could find the law for these changes, we would thereby have determined the individual being's peculiarity: La loi du changement fait l'individualité de chaque substance particulière (Lettre à Basnage 1698).

By degrees, as we in the series of the sciences pass over from the formal to the real, the more difficult the cognition of law becomes, the more we stop at mere experience without being able to carry through the reciprocal action of deduction and induction. The logical law of the inverse relation between concepts' content and extension shows itself here in its great significance: the simpler the objects we face, the easier it is to form concepts and laws; the more concrete and composite the objects we have before us, the more we stop at mere descriptions and at characterizations of individual peculiarities. Biology, psychology, and sociology present in increasing degree difficulties for the development of a cognition of law.

Modern thinking has not disowned Platonism. As the Greeks sought universal Ideas,

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[70] *Den nyere Filosofis Historie*[2]. I. p. 346.—In a fragment brought forward by *Couturat* (see Bulletin de la Société Française de Philosophie. II. p. 88 f) Leibniz says: Vis activa nihil aliud dicit quam id, quo sequitur transitio, sed non explicat in quo consistat.

so modern science seeks universal laws. And herein shows itself the unity of human thought under all forms. Yet it would not be right, with *Henri Bergson*, to assert that the modern concept of science is only different in degree from the ancient. Bergson grounds this conception on the fact that, just as the ancient "Ideas" meant general concepts about things, so the modern "laws" mean general concepts about changes. Modern thinking has, therein one finds the advance, taken up time as an element. But the main thing for it is nonetheless always to be the universal and abstract, not the single and individual[71]. To this it must be remarked that modern science, the more it becomes conscious of its proper goal, sees it as its task to use the general concepts and laws as means and ways to understand the individual and historical totalities. It is the individual totalities whose understanding stands as science's last goal, and the poorer-in-content, but at the same time more comprehensive, thoughts are to serve to lead in the direction of this goal. The "law of change that constitutes the individuality of the single being" is the highest ideal of modern thinking. Therein lies a difference of principle between the ancient and the modern conception of science, at the same time that the world of problems becomes far greater than it could be for ancient thinking. The difference thus lies not only in the way, but also in the goal.—The doctrine of categories will give occasion to take up this question again.

[71] *Bergson*: *L'évolution créatrice.* Paris 1907. p. 359 f.

# III. The Forms of Thought (The Categories)

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## A. The History and Method of the Doctrine of Categories

51. Through the preceding psychological and historical presentation we have come to know the different stages by which human thought works its way forward. Through the work we here, as everywhere, come to know the energy. Sensation, intuiting (with association), and judging stood, psychologically regarded, as the most important stages, and the rise from stage to stage meant an ever clearer consciousness of objects and elements and their mutual relation. In a historical respect the relation between the objects and their interpretation was the main thing, and the ideal here was that interpretation takes place through the objects' mutual illumination of one another, an ideal that, however, in turn pointed back to definite presuppositions.

It will now be the task to give a survey of the fundamental forms with which thought works, and which condition presuppositions to which it is by its nature bound. Such fundamental forms have from old times been called categories.

The word category denotes, according to a usage established by *Aristotle*, assertion, predicate. There are therefore, in the widest sense of the word, infinitely many categories. But in philosophy one understands by category a fundamental predicate, a fundamental concept, and the doctrine of categories sets itself the task of gathering and ordering the predicates that with greater or lesser degree of consciousness are applied by thinking in the different domains where it works, and tracing them back to certain mutually kindred and yet each individually peculiar fundamental forms.

The doctrine of categories occupies a central position in philosophy. It points on the one side back to the study of the cognition won, as the psychology of cognition on the one hand and the history of science on the other present that development, on the other side forward to the philosophical treatment of problems, which investigates how far the assertions we have at our disposal can lead us.

The categories come into being involuntarily during thinking's work at satisfying the objects' demands. They arise sporadically, to meet individual special demands or answer momentary questions, and only later can the question be set whether their significance is limited to a single domain or a single situation, or can extend further. The point is then to gather them, to demonstrate their connection among themselves and with the nature and tasks of human thinking, and to trace them back to as few and simple forms as possible.

The most significant attempts in this direction have, at intervals of millennia, been made by *Aristotle* and *Kant*. All other attempts that exist, also the attempt to be made here, relate to these models as modifications that strive to give closer grounding and more exact connection. Both masters of the doctrine of categories started from an investigation of judgments—quite naturally, since predicates appear only in judgments. The fundamental predicates must be the determinations that constantly recur in the judgments that objects give us occasion to pass. An analysis of the different assertions a judgment can contain, or of the different kinds of judgment that can be passed, was necessary in order to find the fundamental predicates. Aristotle goes the one way, Kant the other.

52. Judgment is, as we have seen, not the first form in which objects appear to us. Judgment presupposes intuiting—sensation, memory, or imagination. In intuition the objects are involuntarily ordered in us and for us, and already at the coming-into-being of intuition there operates the psychic energy that can later appear in the formation of judgment. In intuition, whether it is sensation, memory, or imagination, a multiplicity is involuntarily ordered; a form of wholeness is brought about. The arising of every new sensation is an interruption of consciousness's given state, and work is now done involuntarily at ordering the new object or element (the new distinguishable thing) into the whole connection. As long as possible consciousness dwells in intuiting—in the three named forms, which, for that matter, can only at a later stage be distinctly distinguished from one another. But when there appear changes that can no longer be equalized in the form of resting intuition, there begins a work at a new kind of connection. Reflection is awakened, attention moves from object to object, from element to element, brings them out each by itself in its peculiarity, and seeks through such analysis to find a new ordering of the objects. By this way judgment can step in place of intuiting.

During the formation of judgment, as we have seen, the thought-movement goes from a given and presupposed, more or less indefinite basis, the subject, over to a more definite determination, the predicate. We think in predicates, or rather by predicating (asserting). The formation of judgment begins with the one leg of thought's compass standing in or being set on a single point, and a place then being sought where the other leg can be set. A doctrine of categories becomes possible through all special determinations or predicates being able to be subordinated under a few fundamental predicates that recur again and again. The categories are the chief forms in which decision- and determination-questions (cf. 26 and 31) are preserved.

When an object emerges in my sensation, in my memory, or in my imagination, I will involuntarily be able to raise the following questions:

Is it one with something I know?

Does it resemble something I know?

If it is different from something I know, is it then a difference in kind, in time, in number, in degree, or in place?

Does it contradict something I know?

Can it be derived from something I know, or can something I know be derived from it?

Is it cause of something I know, or is something I know cause of it?

Is it part of something I know, or is something I know comprised in it as part in whole?

Through what different intermediate forms has it become what it is?

What value does it have? Is it a means for something valuable that I know, or does it perhaps denote a higher purpose than the valuable I have hitherto known?

In the preceding psychological and historical investigations we have in various places come upon objects that can give occasion to such questions. But whether the list of questions gives us a direction to all categories must remain open. It will on the whole be impossible to demonstrate that any list of categories is complete, although both Aristotle and Kant and most of their successors have considered it possible to set up concluded systems of categories. We form our judgments on the basis of intuiting, and within intuition new objects and elements can always appear, which can require new judgments and give occasion to new predicates. The involuntary, often obscure, basis for the analyzing thought-work can shift, without our being able to say in advance how far this shifting can go—and wholly new questions can be called forth thereby. And even if the basis does not shift, proof can never be given that it is exhausted through the judgments we have with full consciousness and exactness formulated. Reflection constantly draws from a source whose spring is hidden[72]. History shows that categories can arise and pass away by reason of the shifting of the basis, or because hitherto unheeded objects or elements draw attention to themselves. A presentation of the doctrine of categories can then have validity only for a certain stage in the development of cognition. It has its essential significance in being a stocktaking that can lead to a clear understanding of the problems.

There must (cf. 3–4) indeed be a distinction made between the basis and the reflection that is directed toward it. Reflection can be compelled to make distinctions and establish differences that cannot without further ado, or in the same way as they lie before reflection, be said to be present in or "lie in" the basis. Of course there must in the latter be occasion and justification for setting up those distinctions; but they can there make themselves felt in another form. There is analogy, not identity, between intuition and judgment. They are both forms of the psychic energy that at all stages strives for connection and equilibrium; but they are two different forms of this energy.

Only in the clearer conscious life can there, e.g., be distinguished, as *Kant* does it, between matter and form, or, as *Stumpf* in the most recent time does it, between phenomena and functions. (Cf. 5–6.) In the involuntary mental life this difference does not come forth. One will perhaps say that the functions are present, even if they are not specially noticed and not set in opposition to the phenomena. But it must in any case make an essential difference whether a state appears as an undivided whole, or whether analysis has brought out differences—whether it (to use Stumpf's expression) appears "unzergliedert" or "zergliedert." There is a difference of quality between these two forms, even though we can understand them as forms of one and the same energy at different stages.

It is this difference that makes it that the doctrine of categories, however important its connection with psychology (as well as with the history of science) is, cannot without further ado be called a part of psychology. It stands between this and epistemology proper.—In sharp opposition to the psychologism that effaces the difference between the two disciplines, there appears in the history of thinking a constant tendency to give the doctrine of categories without further ado a metaphysical significance. One then sees in the fundamental concepts of the human spirit an immediate revelation of existence's innermost essence. The Platonic doctrine of Ideas is the greatest example of this assumption, which later appears under various forms, not only in philosophers like Spinoza and Hegel, but also in natural scientists like Newton (especially in his concept

[72]Cf. my treatise *Filosofien og Livet.* (Mindre Arbejder. Anden Række.)

of spatium absolutum as sensorium dei).—The doctrine of categories can be called a descriptive epistemology, an introduction to the special discussion of the problem of cognition, and it takes up as such other points of view than psychology and metaphysics.

53. Some features of the history of the doctrine of categories will serve for the illumination of its significance. I begin with the two great masters of Greek thinking. The opposition between Plato and Aristotle appears here in a characteristic and instructive way.

For *Plato* it was the greatest fact of thought and life that it was possible to form concepts that hold for multiplicities of objects. His philosophizing was a constant ascent to the world of general concepts, of Ideas. (Cf. 44.) Starting from the objects' obscure, chaotic basis, he formed a world of predicates. In the predicates, not in the subjects, the world of objects, his thought found its home. There expresses itself here the correct insight that the predicates are thought's proper work and property. Only, Plato overlooks that the objects are the constant presuppositions, and that during the formation of judgment a constant reciprocal action between subject and predicate takes place, so that in the judgment passed they steadily determine each other. (See 33–35.)

Not all Ideas stand for Plato on the same line. Repeatedly he comes back, in his dialogues, to a class of Ideas that show themselves to be inseparable from thought's activity, whatever special objects this concerns. He is, as we have earlier seen, one of the predecessors of the synthesis-theory. The soul must be able to exercise a combining, as surely as it can have sensations of different kinds and compare them among themselves. And by this very combining and comparing certain predicates are applied, which express the relations that are ascertained: Being and Not-being, likeness and unlikeness, identity and difference, number-concepts. It is by the application of the forms expressed in these concepts that thought ascends from the objects to the Ideas, and the named concepts are therefore themselves the most fundamental Ideas. (*Theaetetus* p. 185–187.) And he especially emphasizes judgment as a connection (συμπλοκή) by which the Ideas are determined in their mutual relation (*Sophist* p. 259).

In his last, Pythagoreanizing period he occupied himself with ordering the fundamental concepts in a series analogous to the number-series. But of this attempt little certain is known.—A doctrine of categories proper is first found in the thinker who introduced the very word category.

Here and there in *Aristotle*'s different writings some concepts are applied that denote the most general points of view that can be used in more special investigations, no matter what these concern. Nowhere, however, has he—as Plato did—brought them into inner connection with thought-activity itself, and in the writing (*Categories*) that (whether it directly derives from Aristotle's hand, or only indirectly) especially treats them, they are placed—ten in number—in a very external way, in a purely catalogue-like manner side by side. Probably, however, he came to them through analysis of a number of judgments.

First in the series of categories stands the concept thing (essence, substance: οὐσία; τὸ ὑποκείμενον). And yet it is really no category—for it cannot be used as predicate, cannot be asserted of anything. It denotes the basis, whose closer determination is sought through the other nine categories. In itself it can only be denoted negatively, namely as that which is not asserted of anything—and then one must, for that matter, point back to intuition. Aristotle has here aptly characterized thought's constant basis, that which is first to be determined through the thought-work itself and therefore prior to this cannot have any determination. Every positive determination would already be a predicate, and the "essence" was to be precisely that which was to be determined by predicates. In opposition to Plato, Aristotle lays great weight on this constant basis, while Plato forgets it in his enthusiasm for the Ideas, which were yet really its predicates. Here shows itself the realistic character of Aristotle's philosophy. He has a sharp eye for the fact that thought always works on objects at hand.

The object, the given, is for Aristotle always an individual (ἄτομον καὶ ἓν ἀριθμῷ. *Categ.* c. 5). Only the individual exists. Thought has here one of its limits; it needs a starting-point that cannot itself be used to determine anything else! Further back we cannot come.—Analogies to this concept appear in the absolute atomic theory, in Spinoza's Substance, in Leibniz's Monads, in Kant's "Ding an sich," as well as in the higher religions' concepts of God. (Cf. 34–35.) Aristotle demands absolute conclusion of thinking. The one limit thus lies in "Substance"; the other lies in the principles (ἀρχαί) that condition all judging and inferring and therefore cannot themselves be proved (*Analytica* post. I, 22).

The (other) nine categories are: Quantity, Quality, Relation (πρός τι), Acting, Being-acted-upon, Place, Time, Condition, and Having (the possession of a property). This list Aristotle considers complete, although he does not demonstrate the necessity that precisely these, neither more nor fewer, exist. He considers them irreducible, independent points of view that cannot in turn be brought under a common point of view, but only stand in a relation of analogy among themselves. More fully he discusses only the first three (Quantity, Quality, and Relation).

Of special interest, especially in comparison with Plato's doctrine of categories, is the category of Relation (relational determinacy). As examples he names equality and inequality of magnitude, numerical relations, cause and effect, cognition and the cognized. These are relations that have their root precisely in the combining and comparing thought-activity taken as basis by Plato[73]. Both Quality, Quantity, and the other nine categories could, on closer consideration, be brought under the category of relational determinacy, and it will later show itself that in the more recent time attempts have been made to set up Relation as the first, most comprehensive of all categories. Such an attempt Aristotle did not make, however much most of his examples point in that direction. And the ground for this must surely be sought in the fact that he did not (at least not in the writings we have) attempt to bring the individual categories into natural connection with thought's mode of operation as a whole.

The changes that the doctrine of categories underwent in antiquity after Aristotle it is not necessary for our purpose to enter into. For spiritual development it was, however, a characteristic sign of the times that the last eminent thinker of antiquity, *Plotinus*, limited the Aristotelian categories to holding only for the sensory world. In the supersensory world the differences and oppositions that the categories denote fell away, as well as the very difference between the essence and its predicates. All is there in inner unity and harmonious fullness. Since Plotinus nonetheless wishes to think about this supersensory world, he is compelled to use some of the categories, but they then get a more or less mystical significance[74]. In a similar way Christian theology placed itself. *Augustine* reproaches himself in his *Confessiones* (IV, 16) for having in his youth been so occupied with Aristotle's doctrine of the categories (decem prædicamenta), since they cannot, after all, serve for thinking God, in whom no distinction can be made between essence and properties. God cannot be subordinate (subjectum) to his own properties; one cannot, e.g., distinguish between him and his goodness. The difficulty that all theological thinking has to struggle with here came clearly forth. And it was the Neoplatonists who had first felt it.—

Apart from this, the Aristotelian doctrine of categories prevailed for millennia, and its after-effects can still be traced.

54. The next great and epoch-making attempt at a doctrine of categories was made by *Kant*. For details of his theory I must refer to my treatise *Om Kontinuiteten i Kant's filosofiske Udviklingsgang* (Vid. Selsk. Skr. VI) and to my work *Den nyere Filosofis Historie*. Here I shall only bring out some chief points.

With full consciousness Kant derives the fundamental predicates or categories from

[73]See more closely on the concept of relation in Aristotle: *Trendelenburg*: *Geschichte der Kategorienlehre*. p. 117–129.—G. *Grote* (*Aristotle*. I. p. 115–122) shows that relation lies at the basis of all nine categories.

[74]Cf. *Arthur Drews*: *Plotin und der Untergang der antiken Weltanschauung*. (1907). p. 158–173.

the judgment-function. And in that he conceives judgment as arising from a connecting operation, a synthesis, and furthermore sees in synthesis an operation that lies at the basis of all consciousness, whether we notice it or not, he has brought the doctrine of categories into inner connection with the whole nature of human consciousness.

Therefore Kant is also clear that the categories are provisionally only forms of our thought, while Aristotle immediately conceived them as predicates of existence (κατηγορίαι τοῦ ὄντος. *Categ.* c. 2). After having set up the categories, Kant then passes over to investigating how, with what right, and to what extent they can be said to lead to valid cognition. He thus distinguishes definitely between the fact that we work with certain forms of thought, and the validity of the cognition that can be won by their help.

Kant was convinced that there was a definite number of categories, which follow with necessity from the nature of thought. He supports this conviction on the fact that there is a definite number of mutually kind-different classes of judgments, and that to each such class a category must correspond. Thus, e.g., the negating judgments were to form a class of their own, likewise the quantitative judgments. According to what we have earlier developed (26; 32), there can here be no question of a difference of kind. As regards the other kinds of judgment (e.g. the problematic and hypothetical) too, it can be shown that it is linguistic, not purely logical, grounds that lead to distinguishing between them. Kant's number twelve is therefore just as little grounded as Aristotle's number ten[75].

Of great interest, on the other hand, is that Kant himself demonstrates that there is a common characteristic mark for all categories, namely the concept of continuity, which in turn appears in two forms, in the concept of magnitude and in the concept of cause. Under the concept of magnitude belong again concepts concerning extension in time and space, and the concept of intensive magnitude (degree), while the concept of cause, besides the cause-relation itself, also comprises the concepts of possibility, actuality, and necessity.

Kant's doctrine of categories can accordingly be schematized in the following way:

Synthesis (Continuity)

| Magnitude | Causality. |
|---|---|
| a) Unity, Plurality, Totality. (Extension). | a) Subsistence and Change. Cause and Effect. Reciprocity. |
| b) Reality, Negation, Limitation. (Intensity). | b) Possibility, Actuality, Necessity. |

[75]Cf. already G. E. *Schulze*'s critique in *Kritik der Theoretischen Philosophie* (1801). II. p. 291–333.

The concept of continuity hangs closely together with the concept of synthesis. For the closer the connection between objects or elements, the more intermediate links can be found between them, the more complete and exact the synthesis becomes. From his doctrine of categories Kant therefore derives the following principle: “To allow nothing in the empirical combining that could hinder the continuous connection of all phenomena; for only thereby does that unity of experience become possible within which all observations have their place.”[76] Experience is for Kant the connection within which every single object is a member determined by all other members. And experience in this sense becomes possible only by reason of consciousness’s original combining energy. The fruitful thing in his way of considering lies in the summons it contains to ever-renewed inquiry in order to find new members in the connection of experience and to determine this connection in an ever more exact way.

Besides continuity there is yet a concept that consistently ought to have had a place in Kant’s doctrine of categories, namely the concept of Relation (relational determinacy). It is seen from Kant’s notes to the “Kritik der reinen Vernunft” (the “Reflexionen” published by Benno Erdmann) that it was by occupying himself with the concept of relation and its significance for thinking that he became clear about synthesis as the fundamental form of all consciousness and all thinking. He states thus not only: “Our reason contains nothing other than relations,” but also: “The category of relation (of the unity of consciousness) is the most important of all. For unity really concerns only relation.”[77] He has thus assumed a very close connection between the concepts synthesis and relation. When he has not entered them among his categories, the ground has been that he regarded them as presuppositions for all the special categories. Therefore the significant utterance cited has not gained influence on his systematic presentation.

An essential objection to Kant’s doctrine of categories is that he narrows the concept of category too much.—In the first place, space and time are supposed to be not categories, but “forms of intuition.” But in our cognition we do not stop at the mere staring at what spreads out in space and time; it is the definite relations of space and time that we have to do with, relations of a peculiar kind that do not coincide with any of the relations that Kant’s categories denote, but that must nonetheless be reckoned among the fundamental relations we work with in our thinking. Kant on this point makes too sharp a distinction between “intuition” and “understanding.”—In the second place, he does not reckon the purely logical relations identity, difference, and likeness among the categories, although they, as Plato already clearly saw, are the simplest and most necessary of the forms with

[76] *Kritik der reinen Vernunft*$^2$. p. 282.

[77] See *Om Kontinuiteten i Kants filosofiske Udvikling*. (Vid. Selsk. Skr. 6. Række. p. 53. kfr. p. 27).

which our thought works, and although we must ask about the validity of these forms just as much as about the validity of concepts like magnitude and cause.—In the third place, he distinguishes sharply between categories, which are always applied to finite, limited relations, and "Ideas" (partly in another sense than the Platonic), which demand absolute conclusion of the thought-work in a comprehension of the whole. He overlooks that the categories too really make ideal demands on our thought-work. When are the concepts of magnitude and causality completely satisfied? Do they not also, just as much as, e.g., the "Idea" of the world, require a comprehension of the whole, if they are to be carried through completely? Continuity, indeed experience itself (in the sense in which Kant takes this word), set up great ideals for our cognition. Already above (37–38) it showed itself that the concept of reality is an ideal concept.

55. While Plato and Kant started from an analysis of thought-activity, as it expresses itself in judgments, *Hegel* begins his logical system directly with the simplest, most abstract concepts, relying on the fact that every concept, when only it is thought through rightly, must lead over to other concepts. For a contradiction will come forth when a single concept is set as the sole, all-comprising expression of existence. And on the forward-driving force of self-contradiction rests the train of thought in Hegel's logic. Self-contradiction is only a testimony that no thought can stand isolated—that the world of thought constitutes a whole against which one sins by holding fast a single point of view. But precisely for this reason one could, Hegel maintains, safely begin with a single thought, even the poorest of all; it will with inner necessity lead over to the others, so that ever richer thoughts are won.

Hegel has, however, in fact conceded that the doctrine of categories presupposes a psychological-historical preparation. In his *Phänomenologie des Geistes* he has presented the stages by which the human spirit—and the "world-spirit" through it—raises itself to a standpoint on which the thoughts can unfold according to their own law, constantly through contradictions working their way forward to higher unities encompassing the conflicting oppositions.

The category that is all-determining in Hegel is one that in Aristotle was only one among many, but that will show itself to be the one that most immediately hangs together with the nature of thought: relational determinacy, relation. Contradiction comes forth in Hegel only through one's treating a concept's content without regard to the relations that determine and limit it. It is a contradiction to remain standing at the abstract concept Being, since it can get no content at all and is so far a Being of Nothing—a Not-being. It is a contradiction to remain standing at the concept Identity, for it presupposes members that are identical, and these members must be different, if there can be any distinguishing between them at all; the concept Identity then points over to the concept Difference, and is moreover itself different from it!—It is Relation's law, the law of relation, or perhaps better the law of relational determinacy, that Hegelian dialectic impresses.

What Hegel overlooks is that every one of the concepts that can thus be taken up and tested as examples of our thoughts' "dialectical" character have themselves grown up in a definite, given connection. That they, when they have been tested and determined, can be ordered in series, proceeding from the simplest and most abstract to the most concrete and individual, is a matter in itself; the question in this respect will only be whether Hegel's ordering by triads—position, opposition, and higher unity of both—is really the natural ordering. The main thing is that what in Hegel stands as thought's self-development, a kind of logical generatio æquivoca, is in reality a fruit of a constant reciprocal action between thought's forms and its objects. Our thought is a reflection—both in the sense that there is always something it seeks after, and in the sense that at every stage it presupposes objects and so far comes after these (cf. 2–4). Hegel constantly draws from the memory of experiences, the race's and his own; from there he has the objects that his reflection presupposes.

Reflection itself (and the concepts or Ideas in which it deposits its work) comes in Hegel's world of thought not first but last—precisely because it is the richest, most concrete category, the one that in its law of relational determinacy contains all other categories in itself. His logical system looks, by virtue of the law of the triad, thus:

I. Being. a. Quality. b. Quantity. c. Degree.
II. Essence. a. Ground. b. Phenomenon. c. Actuality.
III. Concept. a. Subjective Concept. b. Objective Concept. c. Idea.

From the simplest thoughts (I) one proceeds to thoughts that are distinctly relationally determined (II), and from these to thoughts that express wholes (III). By Idea Hegel understands, like Kant, a concept of totality. The Idea is liberated from all the limitations and contradictions under which all preceding members in the system more or less suffer.

But Hegel definitely rejects Kant's limitation of the categories and Ideas to being human fundamental thoughts whose applicability was first to be tested. His doctrine of categories is, like Plato's and Aristotle's, a metaphysics. The categories are laws of existence, not merely human laws of thought, and what drives forward from stage to stage, what operates in the dialectical method, is existence's own innermost essence. Hegel could say with Goethe: "Ist nicht der Kern der Natur Menschen im Herzen?" It is existence's essence as ideal and absolute whole that, because it cannot be exhausted in any single thought, compels thought to proceed to ever higher unities. Thought's law is one with existence's own law.

Even if one would accept this conception, there would be no ground to assume that contradiction should be the sole motive that drives thought forward. New objects can arise that demand their right, even if they do not stand in contradiction with other objects, and that can bring about the downfall of some categories and the transformation of others, or demand wholly new categories. And reflection can also be driven forward to conjectures or new seeking by a tendency to expansion, to extend won insight beyond the hitherto given, a tendency that is of such great significance for the development of the life of the spirit.—

The thinkers who in our day stand or think they stand nearest to Hegel in the doctrine of categories have, however, modified his philosophy more or less in the direction of Kant's conception. The dogmatic certainty is lacking. What animates them is the conviction that no thought can have validity when it is held fast isolated and absolute. Truth must constitute a totality, an infinite connection. "The Absolute" would be the completed, all-comprising experience, but such a one stands as a constant ideal. We work constantly at supplementing and integrating the fragmentary in our cognition. While Hegel in his logic lets the concepts develop out of one another with inner necessity and prefers to dwell up in this ethereal world of thought, his modern disciples are most occupied with ascending from the world of finite and limited experiences to that standpoint of completion. Their works remind one more of Hegel's "Phänomenologie des Geistes" than of his "System der Logik." Their thinking is really only an attempt to carry through the criterion of reality, and it is characteristic that F. H. *Bradley*, who can be regarded as the type for this group of thinkers, uses the word total experience (experience entire) for the ideal of cognition. (Cf. above 37–38.)

56. An interesting attempt at a doctrine of categories with the emphasis on Relation (relational determinacy) as the chief concept was made, in essential agreement with Kant, by *William Hamilton.* Every act of thought establishes a condition and a limitation: to think is to condition. All the fundamental concepts our thought works with indicate relations and conditions. Subject and object, thing and property, cause and effect, relations of time, space, and degree are examples of this. And with the law of condition or of relation it accords that comparison and judging cooperate in every apprehension, in every function of cognition. What is peculiar to Hamilton's conception, as to that of other Kantians, is that the law of relation is conceived especially as entailing a limitation—that it is the expression of our finitude. He does not emphasize that every

positive cognition and understanding, every advance in our knowledge, rests on more relations being demonstrated and drawn into consideration, and on each single relation being determined with greater exactness[78].

In *Comte*, the founder of the so-called positivist school, we find a doctrine of categories intimated that in its chief features reminds one of Kant's. For Comte connection is thought's most general form: tout se réduit toujours à lier. Cognition springs from an urge for unity. And connection takes place in two ways, partly by reason of the objects' mutual likeness, partly by reason of the order in which they follow one another in time. Herewith relation is already given as an essential determination. Besides the relations between the objects among themselves, Comte also emphasizes—and in express agreement with Kant—the relation between subject and object.

The difference between likeness and succession lies at the basis of the difference between relations of equilibrium and relations of movement, statics and dynamics, which Comte seeks to demonstrate in the different scientific domains. Where equilibrium and rest prevail, there are only relations of likeness to demonstrate; where there is succession and movement, the task becomes to demonstrate the dependence of the following on the preceding. Where the relation of likeness prevails, general propositions can more easily be set up and deduction applied; where the relation of succession prevails, so that the objects get a historical character, we do not so easily get beyond the special, and induction becomes predominant. According to this principle (généralité décroissante—complication croissante) Comte has laid out his ordering of the special sciences—from mathematics through astronomy, physics, chemistry, and biology to sociology. In this last, most concrete domain the relation between statics and dynamics appears as a relation between order and progress[79].

In *Charles Renouvier*, the most important representative of critical philosophy in France, both Comte's and Kant's influence is traced in the doctrine of categories. In his list of categories Relation stands first. It makes itself known in thought's distinguishing, identifying, and determining activity, an activity that is also decisive for all other categories, although Renouvier sets them up alongside the first. A definite principle for the ordering of the categories cannot be seen. The ordering is the following: 1) relation; 2) nombre; 3) position; 4) succession; 5) qualité; 6) devenir; 7) causalité; 8) finalité; 9) personnalité. Within each of them Renouvier seeks to demonstrate a triad of thesis, antithesis, and synthesis, which Kant had already intimated, but which cannot be carried through without artifice. A closer critique of Renouvier's arrangement I shall

[78] See *Den nyere Filosofis Historie*$^2$. II, p. 388 f.
[79] *Den nyere Filosofis Historie*$^2$. II. p. 333 f; 352–355.

not give; such a one will indirectly emerge from the ordering I shall later attempt to demonstrate among the categories I find. Renouvier is clear that a deduction of the categories such as Kant had attempted is impossible, and that only experience can decide whether our system of categories is the right one. Experience itself is a synthesis, and the categories are the constant relations that make themselves felt in experience—relations that presuppose just as many special forms of synthesis[80].

In *Eduard v. Hartmann* too Relation is a chief category (Urkategorie), because our conscious thinking is a reflecting, a positing of relations; it brings out the relations to which everything that becomes an object of consciousness is subject. All Being is relational Being. And this holds also for all sensation, although this has its special categories (quality, quantity, extension in time and in space). Within thinking Hartmann distinguishes between the reflecting thinking, which expresses itself in comparison, distinguishing and connecting, measuring, inferring, and the speculative thinking, whose categories are causality, finality, and substantiality[81].

Here too I cannot enter into a closer critique. It was in this survey especially a matter for me of bringing out the chief points, at which there also shows itself a certain convergence between the different catalogues. The convergence shows itself especially in the weight that is laid on consciousness's, especially thinking's, combining or connecting character and, in close connection therewith, on relational determinacy, relation as necessary consequence thereof and moreover as common characteristic feature for the more special concepts. It is of no slight interest that in a chief point of the doctrine of categories, this central part of philosophy, there is found such an agreement. The more special categories do not hang together so closely with the general nature of thought that they could be derived from this without further ado. They get a more historical character, and the order in which they are placed is conditioned in many ways by more special philosophical problems and views, something that is especially clearly seen in Renouvier and Hartmann.—

When some authors (Lotze, Sigwart, Wundt), following *Herbart*, place the categories object and property before the category relation, the objection must be made that a thing's concept contains only relations within itself and to other things, and that properties only express such relations. And to this is added, as the most essential objection, that the object is known only as thought's object; it cannot then itself be the first category. Aristotle indeed placed it first; but he added that it was really no category, since it was

[80]Cf. besides my book *Moderne Filosofer.* p. 72 f.: *G. Séailles*: *La philosophie de Charles Renouvier.* (1905). p. 84–133.—*L. Dauriac* in Revue de Métaphysique et de Morale. 1909. p. 487–496.

[81]Cf. *Arthur Drews*: *Hartmanns philosophisches System.* (1902). p. 766–846.

precisely that which was to have its determination through all categories.—

57. As regards the method of the doctrine of categories, it has already been said that the categories must be found by reflection's seeking to separate out the forms in which the life of thought involuntarily moves in reciprocal action with objects and tasks. The individual categories arise at different times and in different connections. But the presentation of them can be laid out in such a way that we begin with the simplest and proceed to the more rich in content. The simplest will probably be those that hang together most closely with the nature of thinking and therefore express the most necessary presuppositions that the objects must satisfy; they must therefore also hold in the greatest extent. They can merely indicate the demands that must be made at all on every object that is to be able to be thought. The more rich in content a category is, the less comprehensive a circle of phenomena it will be able to be applied to. It is a rule that holds for all our concepts, that content and extension—in concepts that lie in the same ascending series—stand in inverse relation to each other. From this springs, as will later show itself, one of the greatest problems for our cognition.

The transition from the one category to the other, that is, from the simpler to the more rich in content, is necessitated by the appearance of objects or elements that are not satisfied by the simpler, more comprehensive categories. Existence is then richer than thought. There must then be sought more special categories, forms in which the new objects or elements can come into their own. The more special categories will not be able to be derived from the more general; but it will perhaps be possible to show that they are forms in which the more general categories must appear under definite given conditions, or to find intermediate links between the old categories and the new. At every point thought's own nature stands in reciprocal action with its objects and its tasks. There will be a constant breaking between the two sides, a breaking that in our day has called forth a dispute between Criticism and Pragmatism. The doctrine of categories need not enter into this dispute, which belongs under the discussion of the problem of cognition itself. But it will perhaps be able to give a contribution to the illumination of the disputed question.

## B. Fundamental Categories

58. What is given as an object for reflection is always a whole, whether this be loose and changing, or firm and lasting. If the object were an absolute single, thought would become one with a staring and lead to ecstasy. Even in the moment consciousness awakens from

the unconscious by the abolition of the equilibrium that the unconscious presupposes, the state is determined by the opposition to the preceding state and constitutes a whole with this. Phenomena of consciousness are always determined by a relation between background and foreground, a relation that can be successive or simultaneous, or both. In the simplest case it is only the background that determines the foreground; but when the background does not immediately vanish, but is preserved or recalled, it can itself in turn undergo change, become foreground in relation to what was before foreground. In all sensation, memory, and imagination this reciprocal relation lays itself bare. Especially clearly, however, it comes forth, as we have seen (33), in judgment, where the predicate not only determines the subject, but is also itself determined by it. In all judging, as well as in sensation, intuiting, and association, the psychic energy makes itself known as a combining, and the law of synthesis therefore becomes the fundamental law of cognition. *Synthesis* becomes the first category. It is applied involuntarily as long as consciousness subsists, and it is in analogy with it that we apprehend everything. The type for all cognition, and therefore also for everything cognized, is connection of a multiplicity into unity. Neither a wholly connectionless multiplicity nor a differenceless unity (singleness) can be cognized. Already purely psychologically, states of these kinds are impossible, although there can be strong approximations to them.

In this first category now lies at once contained the second. When a multiplicity of objects or elements is connected or combined, they are brought into relation with one another. This happens in every sensation, every memory, and every imagination, as well as in every judgment. When a definite connection is dissolved, as when intuiting is replaced by judging, this dissolution is only a conversion into other relations. The wholly relationless can be neither sensed, remembered, represented, nor judged. To think is, as *Fichte* has rightly said, to set something in relation to another.

*Relation*, relational determinacy, then becomes the second category. It denotes both something that appears involuntarily and an ideal. Involuntarily, in a consciousness all objects and elements are set in a certain relation to one another. But our reflection seeks to determine this relation as exactly as possible and to reshape it in such a way that all objects, earlier emerged and possible new ones, can come into their own. Wherein this "own" consists is decided by the more special categories.

The often misunderstood proposition that everything is relative does not mean that everything is equally right or equally wrong, but precisely the opposite, that everything that is cognized stands and must stand in definite relations, and is known the better the more exactly these objects are determined. Within the definite relation, cognition is absolute.

59. There are, as said, two limits for consciousness and therefore also for thinking, absolute unity and connectionless multiplicity. At neither of these limits does any relation hold, and therefore no synthesis either. Between both limits lie relations that can be designated as *Continuity and Discontinuity.*

These two categories must be named together, since they mutually presuppose each other. An abolition of equilibrium is required for consciousness to replace unconsciousness—for an intuition to be able to be articulated (17)—for judging to replace intuition and association. An interruption is required, both for the simplest sensation to come to be, and for reflection to be awakened. So far *Søren Kierkegaard* is right that thought begins with a leap, and that a special interest is required to make this leap; in any case the presupposition is an involuntary interest in preserving or restoring the equilibrium and the connection. This leap expresses a discontinuity; but it reveals at the same time that there was previously a continuity, and it may perhaps itself later show itself to be a member in a deeper-lying continuity.

It might perhaps be disputed that continuity is in itself a relation. For insofar as it subsists, no distinction can be made between members that could stand in relation to one another. When it prevails, consciousness glides imperceptibly forward, seems constantly identical with itself, and it is only the reflection awakened by discontinuity that becomes clear that changes have uninterruptedly taken place, even if these were so small that they could be set as near to unchangeableness as one likes. One can say with *Poincaré* that in all continuity a contradiction is contained. If a continuous series of the members A B C subsists, then $A = B$ and $B = C$, but $A \gtrless C$[82]. But this contradiction is first discovered when experience or memory by a leap leads to an immediate juxtaposition of A and C. Then comes the awakening. One can then ask: "whither are we gliding?" As long as one is absorbed in the gliding, one does not ask; the contradiction does not exist as long as continuity prevails. Continuity is a relation that is not discovered until it is past. It is only when discontinuity has come into force that one can apprehend continuity as a connection of objects or elements that lie infinitely near one another.

We stand here again at the earlier-mentioned psychological antinomy. When we say that in continuity, in the continuous states of consciousness, a multiplicity of members is contained, this can hold only for reflection, which looks back and makes distinctions. In the states themselves the multiplicity did not come forth as such. Both in our apprehension of sensation, of intuiting, and of thinking, this must indeed be noted, so that

[82] *La Science et l'Hypothèse.* p. 35.

one does not read something into the more immediate and involuntary states that first comes forth in reflection. In any case one has no right to identify without further ado the differences that reflection sets up with those that can already be said to "lie in" the immediate processes. (Cf. 5.) Or, as one can also express it, the rational continuum, the continuum formed by reflection, is different from the empirical continuum. When reflection is awakened, a closer determination is sought of the empirical continuum that the involuntary mental life constitutes, and this can only take place by apprehending it as discontinuous multiplicity. Discontinuity is thus not merely a condition for the awakening of reflection; but we seek, when reflection is awakened, to apprehend continuity in analogy with discontinuity. And only now can one define continuity positively; at first it stands as a negative concept, as absence of multiplicity as well as of unity. But one now assumes within continuity differences of a lower degree than the clear oppositions that discontinuity presupposes. Often great problems are hereby set. Thus when a new mental peculiarity, a new psychic tendency, suddenly reveals itself: can there previously have been such a continuum as one formerly assumed? In the material domain the discovery of radioactivity, in the biological domain the demonstration of mutations, leads to analogous questions.

There is, however, no ground to deny that discontinuity is just as original a property of conscious life as continuity. It would be incorrect, with some philosophers in our day (cf. 3), to conceive discontinuity as a more or less artificial result of attention and reflection, set in motion by practical motives. Consciousness subsists under a constant breaking of continuity and discontinuity, and this holds especially for the life of thought. The experience of difference, opposition, and interruption is just as elementary as the experience of the gliding transitions. And the experience of discontinuity is indeed, biologically-practically regarded, even a more significant experience, since it is only by an awakening that energy can be released. Experience shows us strife and tension between different tendencies in conscious life, no less than harmony and continuity, and it is only thereby that the psychic energy gets its tasks, its resistance to overcome. The differences belong to life, not merely to reflection. And since they present the clearest relations, reflection uses them to make the continuous states and objects intelligible to itself. There expresses itself here a feature that constantly recurs in our cognition—that a part or side in our experience is used to illuminate other parts or sides by way of analogy.

Conversely, continuity too is then used to illuminate discontinuity. Given differences, oppositions, and relations of tension are made distinct by the finding-out or assumption of intermediate links and transitions in as great a number as possible, so that the differences finally become infinitely small. Objects or elements that stand in pronounced opposition to one another are then sought to be apprehended as extreme points of connected series. The very impulse or leap with which both sensation and thinking begin is then sought to be brought into a relation of continuity to other objects and elements in conscious life or outside it. The apparently accidental thing in the starting-point thereby falls away. In this way the principle of continuity has constantly shown itself, in the history of science, as a great principle of discovery, in that one tries to give a once-discovered relation as great an extension as possible. The scientific imagination works according to this principle until reflection anew demonstrates discontinuity. Thus Newton asked whether the force that lets the stone fall to the earth should not also operate on the moon, and further extended this question to hold for all bodies in relation to one another. In all representations and inferences concerning the human past, life's past, nature's past as a whole, we infer from the forces and laws we now know and of whose validity we have convinced ourselves. The principle lying in this, the principle of actuality, is only a peculiar form of the principle of continuity. There makes itself known here an expansion that psychology demonstrates not only for the life of thought, but also for the life of feeling. Thus a mood called forth by a special experience will have a tendency to spread over the whole feeling-state, as long as it is not met by a mood called forth from another side.

Thinking thus gets a task, both when there are too many intermediate links, and when there are too few.

It seeks to make continuity discontinuous and discontinuity continuous, and both these apparently opposite processes stand in the closest connection with each other. In both of them thought moves forward through combining in definite relations.

60. If a closer determination of Relation is sought—besides its being able to be continuous or discontinuous—it shows itself that in the individual cases it is either a relation of likeness or a relation of difference. This is already presupposed in the foregoing. For continuity is a relation of likeness within a multiplicity that is not apprehended as multiplicity, because the difference within it is not noticed, and discontinuity presupposes a certain difference within a multiplicity.

That all thinking is a positing of relations can then be more closely expressed thus, that all thinking is a comparison. (See already 2.) It can also be said to hold for all sensation and all intuiting that it is relations of difference and likeness that are determinative for the individual objects' or elements' appearance and for the way in which they are connected. (See 11; 14; 17.) There is an analogy between these more elementary functions and judgment. But we keep from now on especially to this latter, because the relations are more easily accessible here, and because judgment is of decisive significance for the further development of cognition. Every judgment expresses both a relation of likeness and a relation of difference. Despite the difference between subject and predicate, we nonetheless have in the predicate the subject itself from a new side or in a new form. It is the relation of difference that makes the judgment possible and gives it a content; but the judgment is at the same time precisely a victory over the difference, in that this is either wholly abolished or gets a limitation.

We have earlier had occasion to mention the different kinds or degrees of likeness: identity, congruence (sameness), composite likeness, qualitative likeness, and relational likeness (analogy). (21.) The different kinds or degrees of difference we shall in what immediately follows have occasion to mention, since it is they that give occasion to the appearance of the more special categories.—

That every relation is either a relation of likeness or of difference cannot be proved in any other way than by one's finding in each single case, by analysis, that the said relations lie at the basis. It is not, however, conceded from all sides that all combining and positing of relations is a comparison. Thus Th. *Lipps*[83] maintains that a distinction must be made between combination-judgments and comparison-judgments: in the former a multiplicity is connected into a whole; in the latter different objects are compared among themselves. A comparison-judgment we have when it is said that the one of two tones is higher than the other. A combination-judgment is, e.g., the judgment: "this is a tree." In this judgment, says Lipps, "the given manifold—'this'—is by the name tree combined into an objectively necessary unity: this naming-judgment includes in itself a consciousness that, when this designation is to hold, I must think that manifold as one."—To this it must in the first place be remarked that only the different can be called manifold and connected into a whole. If the elements in the given object were absolutely identical in quality, time, and place, they would not be manifold: the object would then be an absolute unity. The relation of difference is thus presupposed. Next, there lies in the naming itself a presupposition of likeness between the object ("this") and other objects, and it is this likeness that the predicate expresses. Finally, the object, in order for its elements to be able to be gathered under a common predicate, must already be distinguished from other objects as something apart (cf. 30 ff.). Here again the relation of difference lies at the basis.

The breaking between unity and multiplicity will always show itself to be a breaking

[83] *Grundzüge der Logik.* (1893). p. 93–95.

between likenesses and differences. In this breaking there constantly intervenes the interest or the purpose that guides thought. It cooperates in the individual cases in determining which group is formed—which object is separated out to be determined—hence which difference is made; and it cooperates also in the point of view that is taken toward the object. The separated-out object (the group) corresponds to the judgment's subject, the point of view to its predicate. Instead of saying "this is a tree," I could under other conditions, toward the same "this," come to say "this is hard," or "this is something I must go around." And likewise I could under other conditions have formed another group, so that "this," e.g., became the tree with the earth it grows in, with the birds that build a nest in it, the insects in its bark, and so on. The object then became a whole little world. That already the judgment's subject presupposes a distinguishing is easily overlooked, since it can stand in a certain indefiniteness—since it is presupposed known and therefore not always expressed—and since the distinguishing act is undertaken beforehand and perhaps wholly involuntarily.

61. We involuntarily order given multiplicities according to likenesses and differences, and this naturally becomes the first step in our cognition. The question is what a complete ordering would be, and whether such is always possible. Here, in the doctrine of categories as descriptive epistemology, we have to do only with the possible orderings; the question of validity belongs under the problem of cognition.

Instead of the word ordering I shall in what follows for simplicity's sake use the word series, although this word is as a rule used in logic in a narrower sense, namely of such orderings as make inference and calculation possible (cf. 33). Epistemology must investigate whether there are not other possibilities than those that make science possible. It is no matter of course that everywhere, at all times, and in all domains, orderings can be formed that make inference and thereby science possible. We have no right to establish in advance the rationality of objects. It is, as we have seen (35; 52), not certain that thought can exhaust the whole content of sensation, intuition, and association in the clear form of judgment. And there is also no certainty that our sensations and our memories are exhaustive. There is every ground to believe that they are always fragmentary. Our objects are always limited. Already for this reason orderings or series of an imperfect kind could be thought of, at which thought must nonetheless stop. The doctrine of categories then needs an investigation, an orientation with respect to the in-themselves-possible series of difference and likeness.

When I try to order a multiplicity of objects or elements according to likeness and difference, it shows itself that the members in the series that is formed can stand in very different relations to one another. There are here especially three relations that become of significance. There can possibly be formed a series of such a constitution that any member whatever can be pushed out, and that the found relation of likeness or difference nonetheless continues to hold between the remaining members. Such a series *Augustus de Morgan*[84] called a *transitive* series. *William James* has later called it a series that makes mediate comparison possible[85]. And there can possibly be formed a series of such a constitution that any member whatever in it can be exchanged with its neighbor-member without the relation of likeness or difference thereby being changed. Such a series I will call *convertible*. In a transitive series ABCD, B can fall away, and the relation between A and C is then the same as between A and B, and as between B and C. In a convertible series ABCD … B and C, e.g., can be exchanged, and the relation between A and C, C and B, B and D is then the same as between A and B, B and C, C and D. A convertible series (in the indicated sense of the word) must at the same time be transitive. The exchange operates in part as a pushing-out. Finally, a series can be formed in which the relation continues to be the same, whether one goes through the series from the front or from behind. Such a series can be called *symmetric*. The series ABCD… is symmetric when the relation between D and C is the same as between C and D, likewise the relation between C and B the same as between B and C, and so on. Such a series can be intransitive, namely when neighbor-members cannot be exchanged. Because the relation D–C is the same as the relation C–D, it is not said that I can put D in place of C, since it depends on how the relation between B and D is. In an intransitive series every member is determined by its relation to other members, and exchange is therefore not possible. While a symmetric series can be both transitive and intransitive, a convertible series can only be transitive.

De Morgan did not distinguish between convertible and symmetric series. And neither later has that distinction, as far as I know, been made. One has as a rule used the expressions synonymously and thought only of whether the whole series could be reversed, not of whether the members in the series could be reversed among themselves.

Finally, it will show itself that a series can often be read from behind, when one constantly reverses the relation. It is an inversely symmetric series.—

I now go through these possibilities in detail and seek to find examples of them. We shall thereby come upon more special kinds of relations of likeness and difference and

[84] *Cambridge Philosophical Transactions* IX. (1860). p. 104.

[85] *Principles of Psychology*. II. p. 646: The principle of mediate comparison is only one form of a law which holds in many series of homogeneously related terms, the law that skipping intermediary terms leaves relation the same. Cf. Vol. I. p. 190.

thus be able to make the transition from the fundamental categories to the more special ones.

62. It could be thought that absolute difference prevailed between the objects among themselves, so that it is only an occasion accidental with respect to their mutual relation that has led to placing them together. And the differences could further be thought to be the same between any objects whatever, so that the order is indifferent and members can be pushed out. Such a series, which can be called a *chaotic difference-series*, would be at once transitive, convertible, and symmetric. It can be expressed thus (the sign $\wedge$ expressing a wholly indefinite relation of difference):

$$A \wedge B \wedge C \wedge D \;\ldots\ldots\; (1).$$

The difference between A and C is the same as between A and B and between A and D, and so on.

Where such a series can be formed, we are at the limit of our cognition—either at its beginning or at its end. If everything given, all objects and elements, were of such a kind that such series could be formed, cognition would be impossible. Of course there must be an occasion to form the series. The series can indicate what shows itself to us at a certain point of time or at a definite place, in our sensation, in our imagination, or in our dreams. But if no other ground for the placing-together shows itself, the members can be exchanged in any way whatever, or pushed out, or read in any order whatever. The only bond that for a while connects them has nothing to do with their mutual relation.

Immediately given such a series can really never be. Only by a mass of thorough comparisons can one reach the conviction that all points of likeness or definite relations of difference are lacking. An approximation to a chaotic difference-series we have in the mere collection of observations and pieces of knowledge (experientia vaga). That between our objects there are no other definite relations than precisely this, that they are gathered at the same time and place, can only be shown by thorough investigation. Absolute unreason (irrationality) can only be shown by the use of reason (ratio); only this can know its limit, and if it does not know it, there is no other force or capacity that knows it. The limit of cognition conditions, when it is demonstrated in this way, a docta ignorantia.

If a chaotic difference-series held for all objects in all possible experiences, an absolute pluralism would follow from it. In the assumption of absolute atoms, in Hume's doctrine of connectionless "perceptions," in Kant's "matter" as well as in his "things in themselves," in Herbart's "reals" there are greater or lesser approximations to such a conception. But pluralism is maintained by an inquiry that goes beyond the immediately given. One cannot here appeal to immediate experience. Pluralism must always be rationalistic. Only reflection would be able to demonstrate an absolute chaos, and the demonstration would be difficult.

63. The chaotic begins to disappear when it is possible to form a series in which the differences vary from member to member, so that the members can neither be pushed out nor exchanged, and in which, moreover, no definite direction in the variation can be demonstrated. Such a series can be called an *indefinitely varying difference-series.* It is inconvertible, intransitive, and asymmetric, and can be expressed thus:

$$A \underset{\mu}{\wedge} B \underset{\gamma}{\wedge} C \underset{\pi}{\wedge} D \;\ldots\ldots\; (2)$$

An example of such a series we have when the ordering takes place according to a memory-schema, or according to our different interests for the different objects, or purely externally chronologically. In distinction from the chaotic series there is here a principle of ordering. It is not merely the placing-together in general, but also the ordering, that takes place from a definite point of view, only that this is just as inessential for the objects' mutual relation as, in the former case, the principle of placing-together.

64. A step further we come when the differences indeed constantly vary from object to object, but the variation goes in a definite direction. The ordering can then take place according to decreasing likeness and increasing difference, or conversely. Such a series can be called a *regularly varying difference-series* and is, like the preceding, inconvertible, intransitive, and asymmetric. It can be expressed thus:

$$A \underset{\alpha}{\wedge} B \underset{\beta}{\wedge} C \underset{\gamma}{\wedge} D \;\ldots\ldots\; (3)$$

An example we have in the color-series. There is a qualitative difference in the transition from color to neighboring color; but yellow is more different from red than orange is, and the difference between orange and yellow is another than that between red and orange. In all comparative science (comparative biology, psychology, and sociology) the objects are ordered in series according to as essential likenesses or differences as possible. If one lays weight on the fact that even after such an ordering many questions—especially about the causes of the likenesses and differences—still stand unsolved, one comes back to the indefinitely varying difference-series. It is the point of view that determines how the individual series is characterized.

65. An *identically varying difference-series* we have when the difference between

any two neighbor-members whatever always varies in the same way, each time one goes from member to member. Of this kind is the relation between the numbers in the number-series, when this is regarded as a series of ordinal numbers (serial numbers), whose distance from the beginning constantly grows in the same way. Between events among themselves and between locations among themselves the same relation can obtain. Such an ordering is more primitive than counting and measuring proper.

The identically varying difference-series is inconvertible, intransitive, and asymmetric, and can be expressed in the following way:

$$A \underset{=}{\wedge} B \underset{=}{\wedge} C \underset{=}{\wedge} D \ \ldots\ldots \ (4)$$

The series (2)–(4) denote an advance in relation to series (1) in that the members lie fast and are determined by the place they occupy in the series. The differences are now bound, and chaos is abolished. It is the first condition for getting further.—A further advance is made when it succeeds in forming a transitive series, even if it continues to be inconvertible and asymmetric.

66. Such a series is *the progressive difference-series*, in which every single member relates to the preceding as the following relates to it—e.g. as whole to part, as the later to the earlier, as right to left, as up to down, as the greater to the smaller. Here members can be pushed out, and inference can therefore take place: when B lies to the right of A and C to the right of B, C lies to the right of A, and so on.—This series can be expressed thus:

$$A < B < C < D \ \ldots\ldots \ (5).$$

The sign < then here does not merely mean "less than," but denotes any inconvertible, transitive, and asymmetric relation.

The relation of time and of place are here not (as in series (2) and (4)) mere outer points of view or principles of ordering; reflection is here directed toward the very relation of time and of space as such.

67. As soon as a difference-series is to be formed, likeness of course plays a part along with it, since decreasing difference is increasing likeness, and conversely. Even in the chaotic (1) and the indefinitely varying (2) difference-series, there is at each individual ordering a common point of view, although it can shift, as in the chaotic, or be wholly external, as in the indefinitely varying. In the regularly varying (3) and the identically varying (4) likeness makes itself felt within the series itself, in the former between the members among themselves (two and two), in the latter between their relations. And in

the progressive relation-series (5) it steps forth still more definitely, in that here quite the same relation holds between any members whatever in the series, which shows itself by pushing-out.

Still more definitely, however, the relation of likeness appears when every following member can by analysis be found in the preceding, hence be said to be contained therein. Such is the relation when the following member is predicate in relation to the preceding as subject, or consequence in relation to the preceding as ground. Such a series can be called a *partial identity-series* and is expressed in the following way:

$$A \rightarrow B \rightarrow C \rightarrow D \;\ldots\ldots\; (6).$$

The arrow means “subject to” or “ground of.” In this series the members cannot be exchanged, since one cannot without further ado let predicate and subject, consequence and ground, change places. But members can be pushed out, so that inference is possible. The series is thus, like (5), inconvertible, transitive, and asymmetric.

When one conceives judgment as an act by which the subject is subordinated (subsumed) under the predicate’s extension as part under whole (cf. 33), a chain of judgments can be brought under (5) as well as under (6). And series (5) can sometimes be formulated as a partial identity-series, since the part is, after all, identical with a part of the whole, the smaller with a part of the greater. But it then becomes a series that runs in the opposite direction from (6), for in (6) it is conversely the following member that is comprised in the preceding. In the transition from part to whole something is added; in the transition from subject to predicate a closer determination and thereby a limitation takes place.

68. There can be series that are symmetric, but neither convertible nor transitive. Such a series one can just as well read from behind as from the front. It can be called a *reciprocity-series*. As examples de Morgan names: A associates with B, B with C, and so on, or: A is in conflict with B, B with C, and so on, or: A is cousin to B, B to C, and so on. Here one naturally cannot push out members or exchange them among themselves; but the series can be read in reverse order. It can be expressed thus:

$$A \smile B \smile C \smile D \;\ldots\ldots\; (7).$$

The sign $\smile$ means that the relation is the same between any neighbor-members whatever, but that no member can be pushed out. A is not in conflict with C because he is in conflict with B, and because B is in conflict with C. Rather he will become friends with C.—From (3) and (4) this series is distinguished by the fact that the relation does not vary, from (5) and (6) by the fact that it is indifferent whether the series is read forward or backward.

The series (4)–(6) can also become symmetric, hence be read from behind, when one reverses the relations, hence goes from the remotest location to the nearer, from the greater to the smaller, from predicate to subject. But in (6) the justification of this will require a special investigation in the individual cases. One could call such series inversely symmetric series or *absolute correlate-series.*—A "war of all against all" would belong under the chaotic difference-series.

69. If the differences in one respect or another, or in all respects, fall away on closer investigation, we get an *absolute identity-series*:

$$A = B = C = D \;\ldots\ldots\; (8).$$

Where such a series can be formed, it shows itself that the given differences were disguised identities. Such a series is at once symmetric, transitive, and convertible. It can be read from behind as well as from the front; the members can be exchanged; and members can be pushed out. These properties this series has in common with its complete opposite, the chaotic difference-series. Between these two limits our cognition moves. The two limits have this in common, that they can only be demonstrated after careful investigation. An absolute identity-series can only be set up when one can everywhere put one member in place of any other whatever, without any change taking place in the whole connection. This is aptly expressed in *Leibniz*'s definition of identity (49): "Identical are the things that can be put in each other's place without breach of validity." Difference, on the other hand, is present where the one cannot everywhere be put in place of the other (in quibus substitutio aliquando non succedit)[86]. It is clear that whether the substitution succeeds everywhere can be very difficult to decide, especially in special domains where many conditions operate together. One will, strictly speaking, never get beyond the hypothetical—any more than at the opposite limit, the chaotic difference-series.

Absolute identity often holds only in a single respect. All are, in the more recent civil law, "equal before the law," however different they otherwise are in station and circumstances. Regarded as "siblings," all children of the same marriage stand alike. In general an absolute identity-series can be set up when different objects fall under the same concept. It lies precisely in the concept of example that different objects can be placed in place of one another in a definite respect, however different they otherwise are.

[86] *Non inelegans specimen demonstrandi in abstractis.* Opera phil. ed. Erdmann. p. 94.

Regarded as horses, a yellow horse is identical with a red one, a Norwegian pony with a Norman cart-horse, and so on. Analogies can on the whole—in the definite respect in which the analogy holds—be regarded as identities.

In common with (1) and (7), (8) has that it can be read from behind as well as from the front. But it distinguishes itself from both by the absolute equivalence of the members, e.g. of the examples of the same general concept, or of ground and consequence, when the inference can be reversed. From (7) it distinguishes itself at the same time by the fact that the members can be exchanged or pushed out.—

Since the absolute identity-series, where it can be set up, denotes the completion of the thought-work, it is made the standard of thinking, and one gladly places it (since *Leibniz*) at the head of the logical principles in the form $A = A$. It then expresses the fundamental demand that thought shall be identical with itself, and that therefore every concept, in whatever connection it occurs, shall everywhere have the same content. This principle is really presupposed in all series, since every member in these, whatever its relation to the other members, must be identical with itself, and since every member can be defined by its place in the series (even if this place, as in the chaotic difference-series, can shift and holds only momentarily).

If all series could be reshaped and traced back to a single absolute identity-series, an absolute monism would be the result. Despite the great weight *Leibniz* laid on individuality and difference, as his doctrine of monads shows, it was nonetheless for him the ideal to trace all factual truths back to identical judgments, and it was in his view only human limitation that hindered the carrying-through of this. One has even found in this the original motive for *Leibniz*'s system[87]. But it is certain that the monads, which according to *Leibniz*'s conception form what has above been called a regularly varying difference-series, cannot be reduced to disguised identities. *Leibniz*'s logic and his doctrine of monads stand in sharp opposition to each other. Still more pronounced and dogmatic does identity appear as ideal in *Parmenides*, *Plato*, and *Spinoza*.

But even if all objects could conceptually be traced back to being members in an absolute identity-series, the question would still stand unanswered whence the differences derive with which the thought-work begins, and which set its tasks, and how it is to be explained that identity did not immediately reveal itself. It is of no use to declare the differences merely "subjective," or "empirical," or "illusory," as one has done with sense- and value-qualities. The qualitative and the historical will always make themselves felt anew, however often, when an identity is reached, one tries to cast off the bridge behind

[87] *Couturat*: *La Logique de Leibniz.* See also Bulletin de la Société française de philosophie. II. p. 65–89.—Cf. already *Kant*: *Kritik der reinen Vernunft*[2]. p. 337.

oneself. We must therefore constantly work with the different series, ascending upward from chaos in the direction of identity.

70. The purpose of setting up these series has only been to give a schematic presentation of the thought-work, as it takes place under constant breaking of relations of difference and likeness. There may perhaps be found still more series. But the weight does not lie on completeness. The significance that the categories of likeness and difference have for our thinking, when synthesis and relation are to be applied in the particular, shines forth sufficiently from the end-points (chaos—identity) and the intermediate forms.

I shall only mention further that the series set up end indefinitely. It remains open how far they can be concluded. In such cases where a conclusion is given or lies in the nature of the matter, we can get series-forms that do not occur in the list set up. If a series merely consists of two members, A and B, and if the relation between A and B is the same as between B and A, we have a little series that is both convertible and symmetric (like (1) and (8)); while there can of course be no question of pushing-out; it is thus neither transitive nor intransitive. As one of my auditors, *Naphtali Cohen*, has pointed out, one can, in closed series of several members, have both convertibility and intransitivity. Every member is then determined by its relation to *all* other members; none can be pushed out, but all can be exchanged. Thus, e.g., the series formed by the three vertices in an equilateral triangle; each has the same distance from both the others. The series of the Decemvirs is closed; none can be pushed out; but the members can be exchanged arbitrarily, provided only that one says of each single Decemvir what can also be said of the others.—Such series are at the same time symmetric, since it is indifferent whether one begins the series from the one or the other end. And such series can moreover also be formulated as absolute identity-series, in that one regards the individual members as examples of one and the same concept (vertex in an equilateral triangle; Decemvir), only that in this case the concept's extension is definitely delimited.

71. As already remarked (61), one has especially been interested in the series ((5), (6), and (8)) that make inference possible. They are the properly rational series. Inference is conditioned by pushing-out (transitivity). But an inference always presupposes a limitation of the series: one wishes to prove a definite judgment and then seeks the premises, or one has a judgment given and then seeks what follows from it. There must be a goal set, a task given. Purely objectively regarded, it is arbitrary where the beginning and the end of the series are set (apart from the just-mentioned conceptually closed series). A series within which pushing-out can take place makes the formation of a logical totality possible, in that *only* such members can come into consideration as

can serve as beginning-, intermediate-, and end-members in an inference-process. It is a thought of purpose, a wish, that is the driving force for the formation of such a totality. Yet this wish can itself be an effect of the relation between the series-members at hand; one wishes clarity about this relation, wishes to see the consequence.

But we must not, in the doctrine of categories, overlook the fact that there are also other series than the rational ones. It is precisely our task to orient ourselves with respect to the question whether and how far it is possible to form firm series of difference and likeness. We must then also take into account other possibilities than those that make rational thinking possible. A chaotic difference-series would even make it impossible.

The series that make inference possible we naturally try to build wherever the objects permit it. And when we are now to attempt a survey of the special categories, conceived as peculiar forms of likeness and difference, we must, among the examples with which we have been able to support the different series-formations, keep to those that offer the greatest significance of principle, because they are suited to be guiding thoughts for research. It is a matter of a struggle between likeness and difference; the task is to reduce the differences to the least possible and yet fully to let them come into their own. We begin then with the thought-forms that are most intimately attached to the very thought-activity, as this is known psychologically and historically. Here there is the greatest independence of the objects; for of all objects it holds that thought must apply to them the forms without which it cannot work. Already here—in *the formal categories*—there appears a breaking between likeness and difference. This becomes still more pronounced when the difference between objects and thought-forms is especially emphasized. It will show itself that this especially happens when the relation of time between the objects' appearance becomes of significance. Even if all other differences could in one way or another be conceived as disguised identities, the relation of time nonetheless always remains as irreducible, as that which constantly anew sets thought in tension. In thought's attempt to overcome this tension it applies *the real categories*. When the interest that is always attached to the thought-work is especially emphasized, the concept of value presses forward. And since there now in fact are other interests than the one that animates the thought-work itself, the different values naturally become objects for thought. The question becomes whether there is also, for such objects, a possibility of a rational thinking, that is, of a thinking that makes use of such series as make inference possible. Since every value can be said to contain a demand, an ideal, namely either attainment, preservation, or development of the valuable, we can call the fundamental concepts applicable to this class of objects *the ideal categories*.

It is time that leads us from the formal to the real categories; it is value that leads us from the formal and real categories to the ideal. Everywhere thought's nature must be the same; but the thought-work takes place under different conditions. It is the triad Form—Object—Interest that entails the tripartite division of the special doctrine of categories.

## C. Formal Categories

72. Whatever it is that we think over, it is a comparison that we institute. In every experience there is an involuntary comparison; the new is referred to what was previously experienced, and if this is not possible, the novelty appears, by reason of the opposition, so much the more strongly. Every description takes place by the indication of points of likeness and difference between an object and other objects; these points of likeness and difference are what I call *Elements*. And it is here indifferent whether an object is apprehended as "real" or not. Memory and imagination are in this respect treated just like sensation.

Psychologically regarded, the relation of difference appears first, since it is the condition for an object's being able to come to be for us at all. To sense is to distinguish. An object must raise itself up in relation to other objects in order to be able to be apprehended. There must be relief. The same holds in the world of memory and of imagination. When the thought-work replaces the immediate intuiting, one strives to bring out the likenesses. We have distinguished a series of stages by which thought can ascend from the world of differences to that of identity (62–69). Identity is thought's ideal and standard, the presupposition for all concept-formation, judging, and inferring. Different objects must in one respect or another be able to be regarded as identical, in order for us to be able to form a common concept for them. And without a certain identity between initial representation and final representation no judgment can be formed. Inference, finally, is possible only by substitution, by identity of concepts that occur in different judgments.

While psychologically and historically identity is an abolition of difference, logically regarded difference is an abolition of the identity that is thought's standard; the different is that which cannot be substituted—the irrational (since all rationality rests on the possibility of substitution). *Identity* is a relation between $a$ and $a$, ultimately an object's relation to itself. In opposition to it stand the quality-relations, partly qualitative likeness ($a_1$ and $a_2$), partly qualitative difference ($a$ and $b$), which both exclude substitution. The thought-work, in the face of *the qualities*, aims at making substitution possible, or at least at reshaping qualitative likenesses and qualitative differences in such a way that they approach identity to as high a degree as possible. The more continuous the series become, the more such an approximation is reached; the difference then steps forth only when one—by a leap—compares more remotely lying members in the series. There is thereby attained an overview and thereby already a certain mastery over the objects, even if the purely rational power over them should not be attainable. If no such series can be formed, we must content ourselves with a separating-out of that which cannot be placed in a series with certain objects. Thereby we apply *the category of Negation* (*a* and non-*a*), leaving it to the future to find a positive relation.—It is in the quality-category that the series (2)–(4) come to application, while Negation stops at (1). If it succeeds in demonstrating relations corresponding to the series (5), (6), and (8), which are transitive, in such a way that either the relations or the members or both are always identical, wherever in the series we are, then inference is possible, and *Rationality* comes into force.—As for series (7), it too gets its significance, insofar as the relations between the members two and two can be reversed and yet are identical.—Rationality's schema is $a \xrightarrow[=]{\angle} b \xrightarrow[=]{\angle} a \ldots$, corresponding to the progressive difference-series, the partial and the absolute identity-series. Even the chaotic difference-series can be interpreted as a kind of (5): if A, B, C, and D form a chaotic difference-series, I know that since A is different from C, it is also different from D. It is a ray of rationality down into the chaos of differences.

## a. Identity

73. Identity is a degree or kind of likeness. It is through a series of degrees of likeness that thought ascends to identity. (Cf. 21, besides the series (1) to (8).)

It has been held that the relation is the reverse, so that the concept of identity would lie at the basis of the concept of likeness. Identity, it has been said, cannot be defined, but likeness can be defined by means of identity, namely as the respect in which the identity takes place; likeness is the relation that subsists between things that fall under the *same* species[88]. But it must then be asked how we come to form species-concepts. Does the species-concept not express that there are certain points of view from which the things in question can be regarded as identical, so that they can be put in place of one another? And this is precisely the definition of identity (see 49; 69), which can therefore very well be defined. The possibility of substitution is now the highest degree of likeness. Likeness is thus the most comprehensive concept, identity a special case of it. What likeness is

[88] *Husserl*: *Logische Untersuchungen.* II. p. 112 f.

in general cannot be defined. That two things resemble each other is something we apprehend immediately, just as that a thing is red or blue. (See 21.) But if I presuppose likeness as immediately known, I can define identity as the degree of likeness that makes substitution (in some or all respects) possible. This definition has the great significance that it reminds us to distinguish well between the different degrees or kinds of likeness, since not all such can be used for the function so important for the thought-work that substitution is. It also shows the way by which we reach the concept of identity. It is an analytic definition, in that we go from a larger whole (likeness in general) to a part or kind (identity), just as when in geometry we define the point by the line, the line by the surface, the surface by the solid[89].

One can therefore also not define likeness as identity + difference. This is a single kind of likeness (see 21), but not the only one. When two colors resemble each other, one cannot "divide" this likeness into identity and difference. Likeness here stands for us as an immediate quality that appears involuntarily when we place two colors like yellow and orange side by side, at least when we compare this relation, e.g., with the relation between yellow and violet. There is, as said, a kind of likeness to which that definition can fit, namely the composite likeness that is present where two objects have in some of their parts wholly the same constitution, in others a different constitution. We can often substitute a part of one thing for the corresponding part of another thing, e.g. put a leg of one chair on another chair whose seat and back are different from the first's. But this composite likeness is always derived. It presupposes the separating-out of the concept of identity from the general and indefinite concept of likeness[90].

We define identity psychologically-genetically as the highest degree of likeness, and logically, in agreement with *Leibniz*, as possibility of substitution. This logical definition is experimental. In each single case differences will make themselves felt, and we must test our way forward, how far substitution can be carried through. It depends in the individual cases on the task and the point of view whether objects are to be regarded as identical or not. It is the purpose that decides. If I have a purpose that for me is one and all, everything that does not serve this purpose stands as identical, as indifferent. For the fanatic, all belief that does not coincide with his own (positive or negative) belief is indifferent. Intellectually regarded, identity is a mighty weapon of thought against the confusion of multiplicity.

In concept-formation whole groups of objects come to stand for us as examples, both

[89]Cf. *Zeuthen*: *Matematikens Historie* I. p. 102 f and *E. Mach*: *Erkenntnis und Irrtum.* p. 354 on the advantages of analytic definitions.

[90]On the fateful influence this definition has had in psychology, cf. my *Psykologi*[5] p. 208, 221 f.

in the sense of equally valid representatives and in the sense of prototypes for future ordering. *Socrates* demanded of Euthyphro that he should determine for him the concept of piety, so that he could then use the definition as an example or prototype (the Greek παράδειγμα means both), and then call what accords with the concept pious, and not what does not (Plato's Euthyphro p. 6 E). But actions that accord with the concept of piety can very well be different among themselves and not everywhere be placed in place of one another.

By reason of the law of the conservation of energy, the different natural forces can be regarded as equivalent and substituted for one another, where it is merely a matter of the energy-standpoint itself, not of the special form of the energy or its practical significance. Potential energy is thus, from that point of view, identical with actual energy. In the psychic domain we can within certain limits apply a corresponding point of view. We can here too speak of equivalents. Thus self-mastery is only possible by reason of the equivalence of different drives or interests. A drive can (when it does not die a slow death from lack of nourishment) not be supplanted except by another drive's drawing energy to itself, as when, e.g., the interest in art supplants the drive to sensory enjoyment. The identity-standpoint is here: something that fills the mind and lays claim to its energy.

A great example of the purification of the quality of likeness into identity is the principle of the absolute uniformity of all parts of space and time. This thought works its way forward in the thinkers of the Renaissance (*Telesio*, *Bruno*)[91] and lies at the basis of the more recent time's exact natural science. It is formulated in the following way by *Maxwell*: "The difference between two events does not depend merely on the mere difference between the times and places where they occur, but only on differences in the nature, form, or motion of the bodies concerned"[92]. Antiquity, on the other hand, spoke of "natural places."

To take a very external example: things of the same weight—a human being, a block of ice, a lump of gold, and so on—can be regarded as identical when weight is the only point of view.—The word "thing" itself expresses an identity of highly different objects, when by "thing" one means something distinguishable that is to a certain degree concluded as a whole in itself. Everything is a thing.—Here belongs also the beautiful explanation that Adam Homo's mother gives her little son of what it is "to be."—

Naturally there are different points of view from which identity is asserted, not equally significant. In its struggle with multiplicity, thought ascends from external and accidental points of view to essential points of view that can be held fast under the

[91] *Den nyere Filosofis Historie*[2] I. p. 89; 119 f.
[92] *Matter and Motion.* § 19.

change of objects. The world of points of view thus built up is the true world of Ideas. It does not hover over experience, but is precisely the condition for experiences being able to be made in the strict sense of the word, not merely isolated observations that cannot even be distinguished from illusions. The objects must, in order for true experience to be possible, be so constituted that they permit the application of firm points of view. It is this that *Plato* in his language expresses by saying that all phenomena "strive after" the Idea, especially the Idea of Likeness (*Phaedo* p. 74–75), pure identity. We would rather say that it is we who strive to form thoughts that can lead us through the multiplicity of objects. And while identity ("the One") for Plato denoted the absolute conclusion of the thought-work, it is for us indeed the conclusion of a thought-work, but at the same time the beginning of a new one, since it will be a matter of bringing the identity-standpoint won into relation with other points of view.

## b. Quality-Relations

74. Where substitution is not possible, the multiplicity and the differences remain standing. All differences are immediately qualitative differences, since the word quality denotes the immediate peculiarity with which an object (something distinguishable) appears in relation to other objects, a peculiarity that can only be described through likenesses and differences toward other objects. The very difference between likeness and difference is a qualitative difference. Other examples are: the differences between our different senses and between qualities or nuances within the same sense; relations of magnitude such as fullness and lack, whole and part, as they immediately act on us; the difference between the psychic and the material, between the known and the unknown, between the real and the unreal, between the valuable and the valueless, between values among themselves.

When thought stands before such qualitatively different objects, and substitution is not possible, at least provisionally—its first task is to order them according to decreasing likeness (or increasing difference). Qualitative likeness and qualitative difference would then mutually bring out each other. Herein lies the intellectual significance of ordering and classification. It brings about a certain continuity, which in turn can lead to the discovery of finer discontinuities than the originally given ones. And conversely, many immediately appearing discontinuities can be diminished or abolished by an ordering of the mentioned kind. This work succeeds in different degrees. The inconvertible and transitive series (5) and (6) here become of special significance. Not only does each member here have its definite place, but transition can constantly take place to new members according to the same law as that by which the preceding members were reached, and intermediate members can be pushed out, so that new relations can be formed between members that previously lay far from one another.

There are now some in-themselves indescribable and indefinable qualities toward which attention has early been directed, because they are quite especially suited for series-formations. These are time, number, degree, and place. The series that these four qualities each form can then be used to cast light on other qualities that do not so easily permit clear series-formations. We then place these more imperfect quality-series together with those more perfect ones and can thereby, by way of analogy, reach greater clarity about them and perhaps even discover new members in them, hence lengthen them or make them more continuous.

We stand here before the concept of analogy and its extraordinary significance for thinking in all domains[93].

75. For *Aristotle* and *Kant* analogy was no category. It stood for them as the last means of connecting what could not go under a common concept. They did not see that it is already analogy that prevails between different objects that go under the same concept. Apart from the identity-standpoint (73), the differences prevail, e.g. differences in magnitude, form, and color in different horses. It is really only the relation between the different parts or properties that can be said to be the same in different objects for which the same concept holds. It is thus relational likeness, analogy. (Cf. 69; 73.)

The history of science has demonstrated the significance of analogy as a means of thought in the most diverse domains.

Already quite involuntarily it plays a role in man's interpretation of his observations, as well as in all mythology, religion, and metaphysical speculation. It is by an involuntary analogizing that man applies the representations that stand for him with special clarity and order to as many other domains as possible. It is an expansion that here operates, a spreading of a tendency that has provisionally shown itself appropriate to objects other than the original ones. There can be dispute about which analogy is the right one, or, if several analogies are applicable, how the relation between them among themselves is to be conceived. Religious and metaphysical disputes often concern such questions. In the domain of strict experience such disputes can often be decided by exact carrying-through of the analogy in all its consequences and by holding these consequences together with the testimony of experience. Thus the question was decided whether the movement of light was to be conceived in analogy with a wave-movement (undulation) or in

[93]This point I have already discussed in my treatise *Om Begrebet Analogi og dets filosofiske Betydning*. (Mindre Arbejder. Anden Række.)

analogy with a hurling-out (emission). In the religious and the metaphysical domain it is both difficult, if not impossible, to carry through the analogies applicable there with full consistency, and to institute strict comparison between the consequences and experience; the justification of the analogies can therefore not be verified.

*Leibniz* first discovered[94] that all idealistic metaphysics, and thereby also all religious representations, rests on analogy, in that man conceives existence's innermost forces in likeness with what stirs in his own soul.—A materialistic metaphysics too, for that matter, builds on analogy, only that it fetches this from another side of experience.

An extraordinarily important use of analogy was made by *Kant*[95], in that he taught that the causal proposition only expresses an analogy between the changes in nature and the transition from ground to consequence in thinking. The series of changes and events are thus interpreted in analogy with a series of premises and conclusions in logic. This too was a discovery. Earlier one identified the concepts ground and cause (thus Spinoza and Leibniz), and when this identity showed itself untenable, it stood as the only way out to deny the causal proposition (Hume).

*Maxwell* and *Cournot* conceived the application of the concept of number to real objects as grounded on analogy between the number-series and the series of objects. "By physical analogy," says *Maxwell*[96], "I understand the partial likeness between the laws of two domains of experience that brings it about that the one illuminates the other. Thus all applications of mathematics in science are grounded on relations between the laws of the physical magnitudes and the laws of the whole numbers, so that the endeavor of the exact sciences aims at tracing nature's problems back to the determination of magnitudes by number-operations." And *Cournot*[97] has stated that the concept of number and the concept of grouping and ordering hang closely together; quantity is a kind of quality that can vary continuously and thereby serve for regular determination in a way that no other quality is able to.

Within logic, now the analogy with geometry, now the analogy with algebra has played a role. One has made the concepts' relations to one another intuitable by the relation between smaller and larger circles or lines, or one has formulated the judgments as logical equations. And sometimes one has found the strict logical proof in the possibility of such formulations, although the validity of the analogy itself is, after all, the constant

[94] *Den nyere Filosofis Historie*[2] I. p. 350 f.

[95] See the section on "The Analogies of Experience" in "Kritik der reinen Vernunft."

[96] *Om Faradays Kraftlinier*. 1855. (German trans. by Boltzmann in Ostwald's "Klassiker der Naturwissenschaft," p. 4.) Cf. my book "Moderne Filosofer" p. 83 f.—E. Mach and H. Hertz have carried Maxwell's idea further. (Moderne Filosofer. p. 90 f; 93.)

[97] *L'enchainement des idées fondamentales.* (1861.) I. p. 13 f; 22.

presupposition.

Within physics the so-called mechanical conception of nature offers a great example of a carried-through analogy, namely the analogy between movements in space and qualitative changes. Light, sound, electricity, and magnetism were conceived as expressions of wave-movements in media of different kinds. The analogy brought one domain after another under its dominion. And during the great work here undertaken, analogy between those different phenomena among themselves has again played a great role. Thus the analogy between sound and light in *Young*, the analogy between electricity and magnetism in *Faraday*, and so on[98].—*Leibniz* conceived rest in analogy with movement, while Greek thinking was inclined to apply the reverse analogy and precisely therefore found such great difficulties in thinking movement.

New analogies will be to develop, if the modern attempts to lay electromagnetism at the basis of mechanics succeed. It has shown itself that from the theory of electricity propositions can be derived that are analogous to mechanics' propositions about work and energy[99].

Within biology the analogy between different organs, by reason of the likeness in the significance they have in different species, contributes to a clearer understanding of such organs' forms and changes. One and the same function is performed by organs that differ to a high degree in structure and degree of development. There is an analogy between lungs, gills, and tracheae, and there is at the same time an analogy between such special organs' function and the skin-respiration that also takes place in animals that have no special breathing-apparatus, but that can breathe with the whole surface of the body.

Between the nervous system's structure and function on the one side, and the mental life's peculiarity and mode of operation on the other side, there subsists a significant analogy. In both places an activity is exercised that has the character of a combining, a concentration. However one conceives the relation between mental life and nervous system, this analogy must play an essential role in the formation of the hypothesis.

Analogous problems can arise in very different scientific domains. A problem similar to the one that radioactivity sets physics is set in biology by the mutations and in psychology by mental new-formations and sudden changes of character. And thinking's stance toward such questions must also be analogous.

Sometimes within a natural science there can arise a dispute similar to that between *Luther* and *Zwingli*: between "it is" and "it means." Thus when some chemists (in

[98] *Th. Merz*: *History of European Thought in the 19th. Century.* II. p. 16–89, has a whole series of examples.
[99] *Chr. Christiansen*: *Om det elektromagnetiske Grundlag for Mekaniken.* (Tidsskrift for Fysik. V.) p. 17.

more recent times, e.g., *Arrhenius*) hold that the atoms of the elements continue to subsist as such in the compound substances, while others (like *Ostwald*, *Mach*, *Ramsay*) maintain that it is only figurative speech when one says that the atoms subsist after the composition has taken place. What one can ascertain is only that the weight of the compound substance is equal to the weight of the elements of which it is formed. There is otherwise no qualitative likeness present.—Of an analogous question in the psychological domain there has been spoken at an earlier place (5).

However it may now be in the individual case with respect to the question "identity or analogy?", it is certain that it is a sign of dogmatism to confuse them. In philosophy a rationalistic dogmatism came forth by the identification of ground and cause, whereby substitution and succession were confounded. One overlooked the great difference that the relation of time entails toward all purely logical and mathematical determinations. In natural science, and in the philosophy that believed it had in natural science's analogies the absolute truth, a materialistic dogmatism came forth in that the mechanical conception of nature is regarded as proved by the possibility of its applicability to the different domains, in that one overlooked that the mere analogy just as well as identity can make such applicability possible.—

After having demonstrated by examples the significance of analogy for thinking, where substitution is not possible, we now return to the earlier-mentioned four qualities (time, degree, number, and place), which can afford a basis for fruitful analogies and thereby cooperate in the rationalization of other qualities.

### 1) Time

76. No thinking, no consciousness is possible without a change of qualities. It is change that maintains consciousness and sets thought in activity. But it is at the same time change that sets us all our problems. Change, alteration—time in the most elementary sense of the word—is itself a quality, an immediate experience that cannot be described. That B replaces A is as simple an experience as is at all possible. It does not even necessarily presuppose that we have a distinct consciousness of A and B each by itself; the change can be felt as an indefinite impulse. The simplest experience of time is the sensation of change. A representation of time, time as a category, first becomes possible through expectation and memory. But even where expectation and memory make themselves felt, qualitative differences still prevail within the representation of time. It will at first be especially practical motives that entail this representation. If every urge, every wish, were immediately fulfilled, it would scarcely arise. But when there

is an interval between the appearance of the urge or purpose and the aimed-at state, we get three members: the now, where the urge is felt—the intervening time, where the fulfillment is awaited or aimed at—and the attainment, or the disappointment. These members can be qualitatively different to a high degree. Think of the difference between urge—lack and longing—and enjoyment. Most pronounced the difference between the members becomes when in the intervening time we are compelled to hold ourselves passively waiting; there arises then a state of inhibition or a subduedness that stands in sharp opposition both to the flaring-up of the urge and to the dwelling of enjoyment. The more active, on the other hand, we can be in the intervening time, the greater the continuity between the three members. With the highest degree of activity, time can as good as vanish.

A fourth member can be added when there is such a surplus of energy that thoughts can be spent on what lies before the now, hence before the urge or the thought of purpose. A passionate urge will look only forward, not backward. Memories get practical significance only by entering as members into wishes and purposes. Dwelling on the past, the time-before-the-now, is a sign of a certain liberation from the practical motives. Memory itself can get immediate value, even if what is remembered belongs absolutely to the past.

When the representation of this four-membered rhythm is formed, and when experience shows that this rhythm recurs in highly different experiences—when at the same time many such rhythms have been experienced and in memory join together into a chain—then the representation of time as a general form or frame for highly different qualitative content can form itself. And it then shows itself that the frame can constantly be expanded by virtue of the same rule according to which it is formed. The attainment can beget new urge, and so on, and the past that was remembered can awaken memory of a still more remote time, and so on. Step by step there can then take place a purging of all qualitative and practical elements. This very purging can be of practical significance through the greater overview and the more definite measuring-out that thereby becomes possible. It can be practical to disregard the practical.

The relation between the members of the time-series then becomes pure succession, the pure relation of "before" and "after," a relation that is not in itself assumed to have any influence on the qualitative content of time. The Maxwellian principle (73) then comes into force. Every moment is then assumed qualitatively the same as the other, and the relation between them conditions a progressive difference-series (66). Time has then become "a pure form of intuition," or rather a typical individual-representation: the different times are conceived as belonging to one and the same time, which advances without being able to be concluded.

77. But this whole process of purification cannot, as experience shows, be accomplished except under the influence of two other categories that we have not yet discussed, namely space and number. The independence of the representation of time is given by the fact that we have an immediate sensation of change, and that we can recognize (estimate) very small time-series. In longer time-series, however, we are compelled to use space as schema or symbol. Here, then, analogy intervenes. And in the measuring-out of time the concept of number finds application, just as in the measuring-out of space. But the application of both these two categories to the apprehension and valuation of time presupposes that the purging of time's primitive qualitative nature has already advanced to a certain point. The indifference of the moments must first stand firm.

A definition of time is, as already remarked, not possible. Mechanics too does not define time, but only what it understands by two times being equal. When two bodies of quite the same constitution, under quite the same conditions, starting from the same place and in the same moment, traverse the same length of path, the times they have needed are said to be equal. But the presupposition here is that all moments are the same—in a qualitative and in a quantitative respect. There must then have taken place an elaboration by thought of the representation of time before the mathematical determinations can come into force. But on account of the tendency to expansion and projection that is peculiar to consciousness, this does not see that time in its fully developed form is in a certain sense its own work, conditioned by its own forms; it conceives time as something existing in itself, realized in the movements of the heavenly bodies (Aristotle) or standing fast as an absolute order of things (Newton's "true, absolute, mathematical, equably flowing time"). In opposition thereto one has then wished to limit time's validity to being merely subjective or empirical, in that one kept to the fact that the relation of time, through the analogy with the relation of space, could be expressed as a purely mathematical relation of dependence. Or one has regarded the relation of time as conditioned by a falling-away from the rest of unchangeable eternity, whereby the restless striving after the future stepped in place of the concentrated and concluded rest[100].

But there is and always remains a historical element in our cognition—despite exact

[100]Cf. the remarkable utterance of *Plotinus*: *Ennead.* III, 7, II, where time is derived from the world-soul's unrest (δύναμις οὐχ ἥσυχος).—*Augustine*, in his acute theological-psychological discussion of the concept of time, lets time first set in at the creation and can thereby reject the question what God did *before* the creation. *Confessiones* XI, 13.

natural science and speculative metaphysics and theology. In individual great moments existence can perhaps, insofar as our experiences and lived experiences have reached a certain rounding-off, stand for us as a logic or as a system of equations, or we can see "the still world at our feet in eternal evening light." But again and again we must dive down again into the stream of time and of changes. Time can never be "abolished" once and for all.

### 2) Number

78. When I go from one object to another, or from one moment to another, a series of acts of consciousness or of thought is performed, each of which corresponds to an object, no matter what this is, or to a moment, no matter how it is filled—no matter whether the objects or the moments are homogeneous or heterogeneous, single or composite. Such acts can be regarded as identical, apart from their content or objects. Herein lies the possibility of the concept of number.

But here too the purging of the qualitative differences takes place step by step. Provisionally, each act of consciousness or of thought has a quality determined by its content and by its place in the time-series. From the content it is easiest to abstract. There then remains the peculiar constitution that the act of thought gets by the place in the successive series of acts of thought at which it takes place. This constitution can be designated as a number-quality. For psychological reasons a difference makes itself felt, e.g., between the seventh and the eighth act of thought, when I go through a series of objects. I feel somewhat different at the one than at the other—this is due now to fatigue or to other causes. And the difference will be the greater, the greater the distance between two members within the series, when they are compared.

The most primitive form of the concept of number is the serial number that the single object gets as the object of an act of thought that follows after or precedes other acts of thought concerning the same series of objects. The condition for such a serial number, or for the number-quality, is that the objects do not arise absolutely continuously, but can be distinguished in time or in quality or in both respects. There can then be formed an identically varying difference-series (65).—It is more easily formed by the successive appearance of the objects than when these appear simultaneously. In the latter case we ourselves must take the initiative, while the succession itself contains an incitement to connect the following with the preceding.

The proper concept of number, the concept of amount, presupposes the discovery that one can by many different ways come to the same place in the number-series. I can, e.g., come to No. 8 from any other number whatever, however great the distance is, and on whichever side it lies. The difference is only that more or fewer acts are required in the different cases, and that the direction can be different. These acts are of the same kind as those that led from No. 1 to No. 8. Thereby the concept 8 is defined in several different ways, not merely as 7 + 1, but also, e.g., as 100 ÷ 92. It shows itself hereby distinctly that the different acts of thought are really only repetitions of one and the same act of thought—that they are identical among themselves, independent of the place in the identically varying series as well as of the constitution of the object. The concept of number can now get its purely formal character; the quality-element is separated out.

*Helmholtz* and *Kronecker*, in their psychological-genetic development of the concept of number, take their starting-point from the ordinal number as a mark that indicates the single element's place as a member in a series. They then seek to show how the concept of number is gradually developed to its pure form.[101] Against this *Georg Cantor* has protested. For him the concept of number is a general concept that holds for different sets (whatever the constitution and ordering of the elements within these sets are). Ordinal numbers—in the sense in which the two named authors take this word—are for Cantor "the last and most inessential thing in the scientific theory of numbers"; they are marks for indicating order (notæ ordinales), not proper numbers (not numeri ordinarii).[102]—It seems to me that Cantor overlooks that the starting-point for a presentation of a concept's course of development can very well later, when the development has reached a certain conclusion, come to stand as something "inessential" and derived. A genetic presentation follows another order than the systematic. The concept of number as a general concept has had a long history of development, which it can be of great epistemological interest to condense into an overview, as Helmholtz and Kronecker have done.

It is the positive significance of repetition that separates the theory of numbers, and mathematics as a whole, from logic. Purely logically, as *Leibniz* showed, $A + A = A$, or as *Boole* expressed it: $x^2 = x$. A concept's content or extension cannot be changed by repetition. A horse is a horse, however many horses there may be. But mathematically $A + A = 2A$ (or $A < A^2$). The concept of number comes into being by a synthesis of identical differences, that is, differences that are objects of one and the same repeated act of thought—or else by a synthesis of different identities, that is, of objects that are only different by being the object of acts of thought that fall in different moments. Repetition

[101] *Helmholtz*: *Zahlen und Messen, erkenntnis-theoretisch betrachtet.*—*Kronecker*: *Ueber den Zahlbegriff.*—(Both treatises are found in the Festschrift for E. Zeller 1887).

[102] *Cantor*: *Mitteilungen zur Lehre vom Transfiniten.* (Zeitschrift für Philosophie und philosophische Kritik. Neue Folge. 91. Bd. 1887.) p. 86.

presupposes at once identity and difference. Logic keeps to the fact that repetition presupposes identity. But mathematics points out that even identical objects can be the object of different acts of thought. Two concepts cannot be united by synthesis into a new concept (concept-combination) except when they are mutually different. But two numbers can by synthesis produce a third number, whether they are identical or different, because in both cases it is really the acts of thought that are combined. Logic is powerless in the face of absolute differences. (Cf. 62 and 69.) But mathematics' first beginning is given with the thought that even in identities (A A A…) different (time-different) acts of thought are applied, and even in differences (A B C…) identical (only temporally different) acts of thought can be applied. Mathematics becomes, by taking this point of view, a more concrete science than logic.

The process by which the concept of number comes into being is continued through the history of the theory of numbers. Already above there appeared two directions in which one can count: from 1 to 8 or from 100 to 92. By reason of these two directions in the numbering, the number-series can come to comprise both the positive and the negative numbers; they mutually form the continuation of each other. And now between the numbers in the number-series (the amounts) one can continue numbering and counting—by fractions and by the approximate number-values to which attempts to determine irrational relations lead. If the number-formation goes in another direction than the double direction denoted by the series of the positive and the negative numbers—sideways, instead of forward or backward—one gets the series of imaginary numbers (symbolized by a line perpendicular to the line of the usual number-series). Ever greater continuity is reached within the series—by differentials of different order—as well as beyond the once-formed series—by infinitely growing numbers. The concept of infinity has its basis in the fact that the same rule and law can hold, however short the distance between a series' members is thought, or however far continued the series as a whole is thought to be. As *Leibniz*[103] has expressed it: "La considération de l'infini vient de celle de la similitude ou de la même raison."

The very possibility of forming such series, which by virtue of their law can constantly be continued, is of interest for the characterization of human thought. It shows us the constant relation of breaking of identity and difference in pure form and in the domain where it can get its sharpest determination. Thereby a free and yet point-by-point law-governed expansion becomes possible. It is of purely formal nature, since the question always remains whether there are objects that can correspond to the different members

[103] *Nouveaux Essais.* II, 17. (Op. phil. ed. Erdmann p. 244.)

in the series. But it has its value in itself as satisfaction for an intellectual urge that is grounded in the general nature of human thought—an urge that gets air in the pure world of forms, just as it leads to working in the world of real objects and in the world of values. And through the satisfaction of this urge an important means of thought is brought about, since, as we have already seen, the number-series by way of analogy makes ordering and inference in other domains possible. Quantity, which immediately is only a quality, appears, through the development of the concept of number, as a peculiar category.

**3) Degree**

79. The difference between qualitative differences and differences of degree is not original, but is the fruit of an interpretation in which the analogy with the number-series plays a decisive role.

From a purely psychological point of view no sharp difference can be drawn between a sensation's degree of strength (intensity) and its quality. The opposition between light and dark, between white, gray, and black, is for immediate sensation just as much qualitative as that between yellow and blue and red. Just as there are innumerable nuances and transitions between white and black, so there are also successive transitions between the color-qualities among themselves. We place gray between white and black, just as we place, e.g., orange between red and yellow. It is not from a psychological, but from a purely physical standpoint that one sets only light- and dark-sensations in opposition to color-sensations; but here more is presupposed than the immediate sensations, namely knowledge of a series of physical differences, or rather of two physical series, the one consisting of differences in the energy (amplitude) of the ether-vibrations, the other of differences in their wavelength. To the first series corresponds the series of sensations from light to dark, to the second corresponds the series of sensations from red to violet. It is the practical or theoretical acquaintance with physical optics that leads to distinguishing between the strength and the quality of light-sensations. One and the same color can vary with different physical light-strength without losing its quality, unless the variation leads to the highest degree of strength, or to a point where no light-stimulation at all takes place. And in a similar way it is with our other senses. The difference between a weaker and a stronger tone, a weaker and a stronger pressure, a lower or a higher degree of warmth, and so on, is for our sensation of the same kind as that between two colors, but it is interpreted as a difference of strength when we have knowledge of differences in the strength of the outer stimulations. The strength of the outer stimulations can be measured by the physical work that is performed—the degree of warmth, e.g., by the height to which the mercury rises in a glass tube. It is from such experiences that the concept of degree of strength stems, and it is by way of analogy that it is applied to the objects' originally qualitative differences. Energy is, physically regarded, measured by the length of path over which a mass is moved. A number-series is then formed, corresponding to the different lengths, and the analogizing can now run its course, in that the number-series won is paralleled with the difference-series of our qualitative states. If we had no standard at all different from these states, the concept of degree could not be formed. The concept of degree presupposes here not only the concept of number, but also a space that can be measured.

*Kant*, in his otherwise excellent presentation of the category of degree (intensity), lays the weight on the fact that quality-series can be formed which end with a complete cessation of the sensation.[104] But no such single series can be decisive. There is needed a comparison with another series that offers a more perfect and clearer determination of the members and of their mutual relation. And here the number-series and the series of places in space stand as prototypes. Kant has here not applied the concept of analogy, although there was here just as good ground for it as in the case of causality (cf. 75). It is only in reliance on the applicability and validity of analogy that we can, as Kant aptly says, anticipate experience by setting up the proposition that all experiences are intensive magnitudes.

When the concept of degree is applied to conscious activity itself, as, e.g., when we say that the psychic energy can have different strength, it is, as we have already stated (see 2; 7), the analogy with the physical concept of energy that guides us. It is, psychically regarded, the multiplicity and heterogeneity that is combined, and the intimacy with which the combining takes place, that can have many degrees. Every state of psychic work has its peculiar timbre, and only by roundabout ways, through reflection, can the synthesis that has operated in it be traced. And just as our sensation of warmth often seems to us to have a far greater "strength" than the degree of warmth that has in fact been present, so there can also sometimes (and perhaps very often) arise a feeling of exertion and effort that in reality does not correspond to the psychic work that reflection is able to demonstrate. It belongs to the many possible illusions in the domain of the inner life.—When we speak of spiritual "progress" or "regress," the symbolic application of number and of spatial length shows itself distinctly. Although the measuring does not get beyond an estimate, it is nonetheless of significance to become conscious of what

[104] *Kritik der reinen Vernunft*[2] p. 217 f.

the estimate is built on.

80. A quality-series stops at a definite point. Orange passes over into yellow or red, lukewarmness passes over into cold or into warmth, water passes over into ice or into vapor, and so on. Such experiences lie at the basis of what has been called the law of definite number, which in more recent times has especially been impressed by *Renouvier* and *Dühring*. But there need not, as these two thinkers hold, be brought about by it a break with the law of continuity. When a quality-series has reached the zero-point (whether the zero-point means beginning or cessation), the task must be to seek a new analogy. One must seek out series that extend further than the one hitherto used for comparison, so that the beginning- and end-members of the quality-series can correspond to members within the new comparison-series. The whole earlier quality-series can then be regarded as a member or stretch within a more comprehensive series. Thus, since it shows itself that our color-series corresponds only to a fragment of the series that the degrees of refraction of the light-rays form, we must go back to the structure and mode of operation of the organs of sight and to their significance in the struggle for existence—hence to a more comprehensive series—in order to understand why color-sensations correspond only to a limited part of the series of degrees of refraction. Even here enigmas remain, which are a part of the general problem of the relation between the psychic and the physical. The disappearance of light in the evening and its return in the morning is ordered into a continuity when we take account not merely of what is immediately accessible to our observation, but of the earth's rotation and its position relative to the sun. It can often look as if nature made leaps; but every such leap contains a summons to procure for ourselves a more comprehensive field of view. Thus we conceive rest as minimal movement, as a relation of equilibrium between forces, and thereby abolish the leap between rest and movement.

The law of definite number is therefore not the expression of an independent or definitive truth. It denotes a provisional ascertainment of an enigma, points toward a new problem.

### 4) Place

81. The concept of place too, and the concept of space that belongs together with it, has developed from being a concept of qualities to being a concept of a pure form-relation that could be determined quantitatively, that is, permit the application of the concept of number.

Several different senses—especially sight, the sense of touch, and the sensation of movement—condition our apprehension of space. And in the development of this, as *Berkeley* has shown, analogy plays an essential role. It was a triumph for Berkeley that one cannot form a representation or an intuitive image in which only the properties "common" to the different senses were contained. He rightly took this in favor of his critique of the current doctrine of general representations. But he did not take account of another side of the matter, namely that we can just as well refer the sight-space, the touch-space, and the movement-space to the same concept as we can refer horses of different form, color, and size to the same concept. There is analogy between the different examples of a kind. One now takes the analogy for granted, and that to such a degree that one is inclined to conceive it as identity. But there must have taken place a great activity in the domain of sense-intuition and of elementary thinking before those analogies became as ingrained as they are.

Even after it had come to stand clear that it was the "same" space that we apprehend by sight, by touch, and by movement, space and its places nonetheless continued to appear with pronounced qualitative differences, fetched now from the one, now from the other of the cooperating senses, though mostly from the sense of movement. Up and down, right and left, forward and backward—these representations originally denoted qualitatively different relations, and they still do so involuntarily in all of us, when we are not precisely thinking mathematically or epistemologically. The different movements and positions to which those expressions practically point bring about the qualitative differences. In the ancient representation of "natural places" the qualitative peculiarity of the places in space was held fast. Each of the "four elements" had thus, according to ancient physics, its natural place. Practical motives played a role along with it, as in the apprehension of time. Space is first and foremost interspace—the space that separates us from what we wish to attain, and that we must therefore traverse or leap over. The practical apprehension of space is like the jumper's, when he estimates the distance before he takes off. Practical interest leads to comparing the one space with the other, the relation between new places with earlier-traversed spaces between known places, and with this the measuring-out begins, at first in the service of the will. The purely theoretical interest here, as elsewhere, grows out of the practical by reason of displacement of motive. Measuring presupposes an identification, in that a part of the new stretch, if not the whole, must coincide with the earlier-traversed stretch, insofar as this can be repeated in representation.

The more often one can now get different space-experiences wholly or partly to cover each other, so that they can be apprehended as identical, the more there will take place a purging of the qualitative differences of the places. It then shows itself that the process, the successive measuring-out or construction, can take place in quite the same way, from wherever I fetch the stretches that are to be measured, or the places whose location is to be determined. The identity, or rather the uniformity, of places and stretches begins to make itself felt. It was, as already mentioned (73), in the Renaissance period that this purging took place, in the domain of space as in that of time. The thought of the indifference of nature (indifferenza della natura) was carried through. It is of great significance for research, which thereby got a direction to seek explanations of what happens in nature in another way than by referring to the peculiarities of places and times.

But when the places in space are only different by their location, and the one place's location is in turn determined by its relation to the other place, and so on, always by virtue of the same law, it follows with necessity that no absolute locations, absolute places, can exist. Only slowly did the recognition of the complete relativity of place-determinations penetrate. *Copernicus* still assumed the sphere of the fixed stars to be communis universorum locus. *Kepler* refused to draw Bruno's bold consequence that the fixed stars could not be assumed to be equally far from us. And even *Newton* still spoke of loca primaria, which at once determine their own and other places' location. The difficulties are, however, here too no greater than in typical and general representations in general: we think in examples, and therefore we must, every time we speak of places' location, of movements' direction, of inertia, and so on, presuppose a definite place in relation to which all such determinations hold. In the concept of identity itself, too, it showed itself necessary to indicate a definite point of view in each single case (see 73).

Only now does the concept of pure space, as constructed by combination of homogeneous elements, become possible. During its development it has, for practical and theoretical reasons, been the interest in measuring-out, hence quantitative determination, that has been guiding. Metric geometry has therefore developed before projective or descriptive geometry, which directly investigates the likeness and unlikeness of spatial relations, and so far, logically regarded, forms the presupposition of the former, since quantitative comparison presupposes qualitative. Both metric and descriptive geometry have their interest beyond orientation in the surrounding world. For we use the intuition of space to symbolize all relations of difference, because space, as one has rightly said, is the most exact form of difference[105].

82. The combination of the homogeneous that is presupposed in all metric and

[105] *B. Russell*: *The Foundations of Geometry.* (1897) p. 186.

descriptive investigation of space can constantly be continued further by virtue of the same law as that according to which it takes place. So far space is infinite. Every limit we set is arbitrary, is due to special motives of a practical or theoretical kind. We must, as we have seen, at every place-determination, at every determination of a direction, set a limit, take our standpoint, and provisionally regard this as "absolute," as lying on a world-axis. But every such standpoint must in turn be determined by another, and so on. In principle, repetition must here always be possible, since we have to do only with our own thoughts, here acts of construction and measurement. Another matter is whether the objects that appear, without our ourselves producing them, will also constantly give us occasion to make constructions and measurements, indeed whether they will make such possible at all. It goes with space as it goes with all other categories: it is a predicate-concept, and it depends on whether we will constantly get objects that make the application of this predicate possible or necessary. From its great significance for our cognition, especially as regards exactness, nothing can in this respect be inferred with certainty.

From another side too the problem of infinity emerges, as regards space. Between two places whose location is determined, there can always be found and determined intermediate places. This process too can in principle constantly be continued. While before it was the relativity of places that led over to the concept of infinity, here it is the continuity of space. Here too we must distinguish between the formal and the real consideration.

The "absolute" space, whose places are qualitatively the same and mutually determine each other to infinity, and which not only can have no limits for its extension, but is a continuum in all its smallest parts—this space is, as *Kant* has said, an Idea[106], that is, a rule according to which all movements are to be investigated. It is a rule, then, according to which our measurements and descriptions of space in the individual cases are again and again to be undertaken.

Just as little as we can without further ado decide whether space is "really" infinite and continuous, just as little can we decide whether the kind of space we know—the measurements and constructions we can apply to our objects—are the only possible ones. The space that our experience seems to force upon us has three dimensions, and within it two straight lines cannot intersect each other when they form right angles with one and the same third line. The question has within mathematics been discussed whether this space (the Euclidean) is the only possible space. If one defines space as "being-

[106] *Metaphysische Anfangsgründe der Naturwissenschaft*. p. 149. Kant hereby in fact acknowledges that the sharp distinction he has made between "forms of intuition," "categories," and "Ideas" cannot be maintained.

outside-one-another" or as "extended multiplicity," those peculiarities do not follow with necessity from this definition, which could then give a genus-concept comprising several different species. It has shown itself possible to develop consistent doctrinal structures from the idea of space with other properties than those named. Which of the different possible geometries is the "right" one we can just as little decide as whether there "really" exists an infinite and continuous space. In both respects the decisive thing must ultimately be whether there are objects that make the one or the other answer necessary. Thought has here more possibilities than reality can be assumed to be willing to offer.

## c. Negation

83. Qualitative differences can, as we have seen, by the help of analogy be interpreted as identities. From another side, there is contained in every difference a negation. When two objects are different, the one is in one or more respects not what the other is. And in the identity-relation itself negation can be traced. For even if A is identical with B, there must nonetheless be or have been something that made it that they could need an identification, and in that respect there must then also be something that could be asserted of A but had to be denied of B. (Cf. 69.)

But negation is only one side of the relation of difference. Every determination and characterization necessarily excludes many properties and is so far a negation. If one lays, with *Spinoza*, the representation of an infinite existence comprising all realities at the basis of one's thinking, it is clear that every special characterization must stand as a negation: omnis determinatio est negatio. (Ep. 50.) And there is the truth in this proposition of Spinoza's, that in every sensation and in every act of thought we take an object out of the great stream, the comprehensive connection, in which it belongs. (See 2–3.) But in the relation of difference we nonetheless always have also the mutual relation between the objects among themselves in view from the positive side, and here the real contrast will precisely contribute to bringing out each object's peculiarity, so that negation does not become the only point of view.

Negation means in itself only the absence of an element (property or part) that is or has been present in another connection. Therefore negation first comes forth when we are about to leave intuition in its immediacy. (26.) The concept No is, as *Herbart* has aptly remarked, the clearest example of a concept that in our judging springs from experience, although it has no object in experience. The intuitive basis of negation is the opposition between AB and A; this opposition makes it that I can call A, when it appears alone, A (non B). Negation is due, like every determination, to a combining.

As with time and space and other categories, here too the concept's practical motive precedes the theoretical. Through disappointment and lack, men have learned that things and events in the world not only each have their peculiarities, but that the one is not what the other is. There must be choice, and there must be resignation. But in this school it was also learned what an important means of thought negation is. Classification and induction become possible only by it. (22; 26–28.) Negation operates now as a reference to a later closer investigation, now as a sign of a way that cannot be traveled.

84. Purely logically, negation is absence, exclusion, or abolition. Mathematically, negation means a summons to series-formation (e.g. counting) in the opposite direction. Negative numbers mean continuation of the number-series in the opposite direction from the series of the "positive" numbers. It depends on the point of view which direction one calls positive and which negative. Likewise with directions in space and in time. If I have my essential life in the world of memory, the continued course of the time-series becomes for me of negative significance. Negation as a direction in opposition to another direction gets a definiteness that purely logical negation does not possess.

There are no negative objects; negation always presupposes two members, each of which is in itself positive. *Plato* already determines negation as a relation of opposition between two beings (ὄντος πρὸς ὂν ἀντίθεσις. Sophist p. 257 E). How one regards position and negation depends on the point of view. When the zoologist in his presentation passes over from articulates to mollusks, it is natural for him in the definition of the latter to begin with the proposition: "The body is unjointed, without jointed limbs." When the theologian has maintained his faith as the only true view of life, he naturally comes to describe other views as negative, perhaps even "purely negative."

Often we must of course stop at a negation, without our having any content for a positive characterization of the one member in the relation. Negation will then, for every living thought, lead to seeking such a content. If I have sympathy or psychological interest for people who do not agree with me, I will try to enter into their state. Significant as a motive of thought negation becomes especially when it is two judgments that mutually negate each other. A problem is then set (27).

Alongside identity and analogy, negation is the most important means of thought. *Hegel*, who has so strongly emphasized its significance in this respect, often confused it with mere opposition, in that he overlooked that not all opposition is negation. He saw in negation the possibility of an extension of cognition without the help of experience; in fact he of course always filled the empty place denoted by the negation with a loan from experience. At the same time Hegel failed to appreciate the great significance of analogy for the development of cognition, and especially for all metaphysical speculation. His own system rests on a great and bold analogy and is by no means merely constructed by a dialectic spurred by contradiction.

## d. Rationality

85. When the purging of practical and qualitative elements described in the foregoing is completed, especially by tracing-back to differences in time, number, degree, and place, it becomes possible to form series of a transitive kind, so that inference can take place. Even the chaotic difference-series permits inference, since it is transitive; but thereby only a new absolute and indefinite relation of difference is put in place of another. Of importance, on the other hand, is especially series (6), the partial identity-series, which affords the schema for the relation between ground and consequence. From $A \to B \to C$ follows $A \to C$.

The relation that appears here is analogous to the one that holds between intuition or association and judgment. The judgment is only to express what already lies in the intuition or in the association-series. Analogous is the relation between predicate and subject. But transition can in these cases take place involuntarily, by an estimate or an intuition. We become, or believe we become, immediately conscious of the identity of intuition and judgment, subject and predicate. Closer investigation can, however, demonstrate that several different intuitions and associations have cooperated in the formation of judgment and the determination of the predicate. Estimate and intuition can be pathbreaking. They can lead to discoveries that perhaps must and can be grounded by wholly other ways than the one that has led to them. In the grounding it is definitely decided which presuppositions cooperate, and how far their significance reaches.

The conversion from estimate or intuition to inference (grounding) is a new example of the purging of qualities. A conviction that has formed itself involuntarily by the cooperation of observations, memories, moods, and incitements can appear with the stamp of simplicity and immediacy, just like color- or smell-sensations. One then rests securely in it, as long as no conflicting experiences awaken unrest and doubt. When this happens, the question becomes whether it can be converted into inference. Reflection on the objects and elements that condition the conviction and can be traced in it, directly or indirectly, is the presupposition for this. In the new form, as inference, the conviction gets a somewhat different character, but there is no psychological ground to believe that it must necessarily lose its energy, if only the conversion into the new form fully succeeds.

It must indeed be held fast that it is only a new form that is brought about by the transition from estimate, intuition, or conviction to inference. We still move within the world of the formal categories, and inference is a pure form of thought, just like time, number, degree, and place. The presupposition is still whether the objects, which we do not produce, permit the application of our forms, even the most compelling of them for us. We have no right, with Aristotle and Leibniz, without further ado to make rationality, the relation between premises and conclusion, into a metaphysical principle, the expression of a law of existence. By emphasizing this difference between the forms we have at our disposal and their real significance, we approach the transition to the real categories, and through them to the world of problems. But before we make this transition, there are still some remarks to be made about the category of rationality.

86. Just as there will for the most part be more in intuition than judgment can express, and in the subject more than the predicate contains, so conclusions will not always contain all that the premises contain. Precisely the middle-concepts that bring about the connection sought fall away. It is, as we have seen (71), a wish, a goal, that determines which objects it is whose content we wish determined; the determination takes place by the pushing-out of members in a transitive series, and the conclusion thus comes to contain fewer concepts than the premises. In the series $A \overset{\angle}{\underset{=}{\rightarrow}} B \overset{\angle}{\underset{=}{\rightarrow}} C \overset{\angle}{\underset{=}{\rightarrow}} D$ I must push out B and C in order to reach $A \overset{\angle}{\underset{=}{\rightarrow}} D$. The middle members can be forgotten, since there is no more use for them. The middle members had the double significance in the series of connecting and separating A and D. Now one wishes only the connection; it is this that one takes with one to the continued thought-work, and one looks back perhaps on the pushed-out members as "accidental points of view" (cf. *Herbart*'s "zufällige Ansichten").

The conclusion thus here forms an equivalent for the premises in a certain definite respect, which is determined by the whole direction and task of the thought-course. It cannot in all respects replace the premises. Perhaps these were—seen from other points of view—more significant than the former, just as the hunt can be better than the quarry, and just as it can be better to seek than to find, as *Lessing* has stated it in a well-known saying. Meanwhile there will always make itself felt an urge to get beyond this one-sided relation and attain a mutual equivalence, so that ground and consequence, premises and conclusion, come into as intimate a relation to each other as possible. This happens when ground and consequence can change places, so that from the conclusion one can always find the premises on which it rests or which it contains. The accidental thing in the way by which the result is reached then falls away; we have a little world, a totality before us, whose members cannot be separated from one another. We are then beyond the mere series-formations, which could be continued into the indefinite. We then do not forget the origin of the truth, once it has come into the world, and truth then does not so easily become a dead treasure.

How far and in how great an extent such thought-worlds, logical systems, can be formed is a matter in itself. And a special question becomes whether our whole life of thought can be concluded or rounded off into such a closed whole. The possibility of chaotic difference-series can scarcely ever be excluded; there will, as long as new objects emerge, constantly be a work to do in order to reach up to the higher series-formations and through them to the development of a world of thought.

## D. Real Categories

87. Even if all other qualities could forever be converted into, or rather interpreted as, identities or quantities, the difference of time would nonetheless always announce itself anew. One could also think of the qualities as eternal types in an absolute relation of identity to one another: $A = B = C = D$. Thus *Spinoza* conceived the relation between existence's infinitely many, mutually irreducible fundamental properties or "attributes," of which only two, Spirit and Matter, were accessible in our experience. Existence was thus expressed in infinitely many forms that were mutually logically equivalent or analogous. It would be a "still world"; the relation of time did not hold. But in the world of experience there is a constant wave-motion. Every intuiting gives only a momentary image, and the moments change.

The last great opposition in our cognition, which stands fast even if all others are abolished, is the opposition between rationality and time: there a relation between ground and consequence, here a relation between before and after. There an eternal system, here a surging sea. *Spinoza* chose the first point of view, and for him time became in the end an illusion. *Hume* chose the second point of view, and for him the rationality of existence therefore became an illusion. But our life of thought stands always before a breaking between the two points of view and has the task of finding its way through this breaking. It is therefore that we must make a difference between the formal and the real categories. This difference is determined precisely by the significance of time, which was especially impressed by the transition from ancient to modern thinking (46–50). It necessitates the setting-up of categories that are to hold for what is distinguishable in time. It is a demand that is made from the objects to the objects (71).

## a. Causality

88. The formal categories are indifferent toward the difference between the three kinds of immediate intuiting that we call observation, memory, and imagination. On the basis of all three kinds of intuiting, comparison and inference can take place, identities and qualities be ascertained, differences in time, number, degree, and place. But when the question arises how we are to conduct ourselves when there comes conflict between the three kinds of intuiting, we get beyond the purely formal, though we take the forms with us. When it is a matter of distinguishing sensation from memory and imagination, we rely provisionally on an immediate impression, the existence-quality (36), a certain distinctness and reliability that as a rule is characteristic of the observed in contrast to the merely remembered or imagined, and likewise of the remembered in contrast to what is produced by imagination. But since there can be not only many degrees in distinctness and reliability, but also downright conflict between different objects' existence-qualities as well as between observations and memories, a criterion of reality is sought, and this consists, as we have seen (37–38), in the firm and law-governed connection between the objects. The concept of reality presupposes that on the basis of given objects an expectation can be built that is confirmed by an ever-growing circle of new objects.

The objects must then not form a chaotic difference-series. Even then an expectation could arise—namely that there will constantly appear something wholly new and different. But it could not be an expectation of something wholly definite, namely of the same thing that had constantly appeared under the same conditions. That chaos will continue to be chaos is indeed a knowledge, but a wholly contentless one; it denotes the lower limit of all real operation. Could we not get beyond it, our knowledge would be merely a knowledge of the irrationality of existence. The thought-work constantly aims at a rationalization of the objects through series-formations that give identity more and more application. We work our way forward through ever more definite expectations or hypotheses—and thereby approach the ideal we call reality. Only for thought does a reality exist. Truth consists purely formally in the mutual agreement of our thoughts; and the more immediately given objects, not produced by thought itself, we can bring into a connection corresponding to the purely logical connection between our thoughts (hence between subject and predicate, between ground and consequence), the more we reach real truth or reality.

One became early clear that the criterion of reality could not be anything single, but had to consist in the firm and consistent connection of all objects. But *Kant* was the first who was clear that there was only analogy, not identity, between the logical connection of thoughts and the real connection of the successively appearing objects. It is the relation of time that has compelled the distinction between analogy and identity, and thereby in turn to make a definite difference between causality and rationality. The former contains the latter; but the latter can be present without the former: reality is truth; but not all truth is reality.

Just as time and space first appear practically as interspaces or distances between our purposes and their attainment, so causality too is first of a practical character, in that it makes itself felt, indeed forces itself upon us, through the relation between means and goal. The goal is given for our representation, in our drive or our wish, and we now seek the conditions without which it cannot be reached. Here we in a way experience causality before we think about it. Here is, as I have said elsewhere, our first encounter with necessity. The unavoidable order in the succession of objects expresses our first concept of cause.

89. The objects can often form series without there being members in such a series that can be connected as cause and effect. In the series A B C D the individual members can appear at definite times and in definite places without standing in a mutual causal relation. (Cf. 62.) We have, however, already applied the concept of cause in setting up the series as a series of real events, for we must then have applied the criterion of reality, which comprises the concept of cause in itself, at every single one of the members. When we wish to go further and find an explanation of the fact that it is precisely these objects that arise, and in this definite order, we must go further back and find the cause of the whole series. When lantern-slides are shown, the spectators have the light-source and the glass photographs behind them. What shows itself to them in each moment seems accidental, when one does not know what takes place behind. It is a similar situation that *Plato* has imagined in his famous cave-image (at the beginning of the seventh book of the "Republic").

Our experience has in many domains a sporadic character, begins from different starting-points and is continued in leaps. This sporadic character cannot always be overcome through the demonstration of a direct connection between the starting-points, but only by extending the consideration to a connection that lies further back and is of a more comprehensive kind. The connection between a series of lightning-flashes is found by going back to the thundercloud's electrical state and the changing tensions and discharges it brings with it.

Historically one must often begin merely by establishing the chronology. This can then become the basis for a further-reaching investigation. Historical criticism has first and foremost the task of investigating the reality of the individual events each by itself, as well as their chronological relation. Herein it makes use of the concept of causality, as this forms itself at the given stage of development of cognition. Whether Hannibal has lived, and when he has lived, is essentially a task of the same kind as that whether there has been a primal nebula out of which our solar system has developed. In both cases one infers back from given objects. One constructs the past out of the present and its content. It makes no difference that Hannibal has existed only once; the primal nebula too has existed only once. In every application of the criterion of reality, hence of the concept of cause, it is a single, individual fact that I wish to have certainty of. I wish to know whether it is a real person who passes my window, whether it really is a chimney I see on the house down the street, and so on. Afterward the question then comes forth why the person went past, and how the chimney has come to be. The one task may perhaps be solved even if the other cannot be solved.

Historical criticism presupposes law-governedness in the world of historical events, even if historiography should not succeed in finding the causal connection between the events among themselves, and even if comparative historical research should not succeed in setting up general historical laws.—The main thing in this connection is that we are placed before the past as before present and future. Also with respect to the past there can be a matter of expectations that are fulfilled. *Zeuthen*'s conjecture that the Greeks must have known the infinitesimal method was confirmed by *Heiberg*'s discovery of an Archimedes manuscript. A criterion of reality for the correctness of a childhood memory we can get in a letter that is found.

90. The concept of cause expresses the way in which we react to a successive difference, a change that has brought about an abolition of our equilibrium. There is equilibrium in our involuntary striving after a goal, until we meet a hindrance; such a hindrance is the distance between the goal and our present state; it is equalized, and the equilibrium is restored, when we find means and ways by which our striving can be continued in the same direction as before, though by a roundabout way. But also where no practical motives make themselves felt, the equilibrium can be abolished. The appearance of new objects or changes of those at hand more or less abolish the equilibrium in our thoughts; a previously won connection is loosened or burst, and a new one is sought. The intellectual seeking forms an analogy to the practical and develops out of it.

The relation of difference continues provisionally to stand in the foreground. Two objects replace each other and are more or less qualitatively different. What we first seek to decide is whether the second always sets in when the first has set in, and never otherwise. In its most elementary form causality expresses an unavoidable sequence. It is then that the thought-work begins which we have in the foregoing described from various sides—the work of reducing the qualitative differences to the least possible. At the same time a work will be set going to find intermediate members, as regards time, so that the transition, both as regards time and as regards quality, can take place as continuously as possible. Historically it is for a great part this interest in continuity in the causal relation that has led to the development of the formal categories. The causal synthesis has driven forward the formal syntheses. The history of science shows that all explanations aim at continuity. And the more this can be brought about, the more difficult it becomes to point to a point where the cause stops and where the effect begins. They pass over into each other gradually in quality and successively in time, no longer stand as two different "things." We discover that by our distinction between cause and effect we have intervened in nature's continuous course—that the "leap" must be set to the account of observation and of our thought, not of nature. It goes here as with logical inference, where we too grasp a single member in a whole series in order to become clear about its relation to another single member. (71; 86.)

The change that gives the impulse to our thinking, our causal thinking too, singles out individual objects in relation to others. There is so far from the first a leap, not only between two objects that replace each other, but also between these objects together and other objects, which nonetheless perhaps lie in the same series and in continuity with them. It is our interest that isolates the cause from what goes before it, and the effect from what follows after it. Both leaps are abolished the more the causal relation can develop from its elementary form, in which it is a successive relation between two qualitatively different objects, to its ideal form, in which it is a relation of continuity both as regards time and as regards quality.

91. Is the ideal now reached when the causal relation takes on the form of continuity? Must one not go a step further?—The analogy with the relation between ground and consequence (88) seems to require it. For in this relation it holds that there must be nothing in the consequence (the conclusion) that was not in the ground (the premises), and the most perfect grounding-relation even requires that ground and consequence can be exchanged, that the equivalence holds in both directions (86). This analogy led the seventeenth century's philosophers (for whom there was here more than analogy) to

formulate the causal proposition in such a way that it required that there must be no more in the effect than in the cause, that there must be absolute uniformity between cause and effect. In *Spinoza* this presupposition appears most sharply pronounced. It can still be traced in *Kant*, and especially in some of his successors (*Hamilton* and *Spencer*), who take the step fully and find a mutual equivalence contained in the concept of causality. It is of great epistemological interest that this conception played an essential role in the discovery of the law of the conservation of energy. *Robert Mayer*, *Colding*, and *Joule* all started from the assumption that the conservation of energy was a necessary presupposition contained in the very concept of cause. Mayer never tired of repeating the proposition causa æquat effectum, which for him was a consequence of "the law of logical ground"; it was for him a self-evident proposition. Colding found the opposite assumption "contrary to reason," and Joule called it "absurd." Nor did *Helmholtz* have the consciousness of setting up a new proposition[107]. Fortunately these inquirers did not stop at this consideration, but subjected their view to an experimental testing.

Against this way of considering it must be maintained that the concept of cause would not lose all significance because it did not succeed in demonstrating equivalence between cause and effect. A causal relation could hold between the objects even if a constant variation took place, provided only that the future was analogous to the past in such a way that from the fact that the event $B_1$ follows after the event $A_1$, we could infer that the event $B_2$ will follow after $A_2$—when $A_1$ $A_2$ $B_1$ $B_2$ denote heterogeneous events. Even if the qualitative differences were irreducible, the causal proposition could hold. So much the more could it hold even if no quantitative equivalence could be demonstrated between cause and effect, so that, e.g., A was cause of $B = A + \times$, and B cause of $C = B \div y$. Law-governedness does not necessarily mean equivalence; this does not stand as self-evident.

The quantitative relation between cause and effect, which for many objects it has succeeded in demonstrating, gets its significance, epistemologically regarded, especially from the fact that it is only by its help possible to apply the found causal relations to future objects. For purely qualitatively the identity of the relations cannot be determined with exactness. The degree of likeness that must be present between $A_1$ and $A_2$ in order for it to be justified to expect $B_2$ after $A_2$, just as $B_1$ came after $A_1$, can only be fixed by an estimate. Only by counting, measuring, and weighing can the identity of the relations be demonstrated, so that certain recognition becomes possible, and even then only approximately. Although no absolute repetition can be demonstrated, either in nature or

[107]Cf. *Den nyere Filosofis Historie*[2] II. p. 497–499; 597.

in history, it is nonetheless natural that thought seeks to get as far in ascertaining the identity of the relations as is in any way possible. It is then also on this that attention is first and foremost directed within the special branches of research, and we must take it as a fact that the more exactly the identity of the relations can be demonstrated, the more it shows itself that there is equivalence between what goes before and what follows after.

But even if the equivalence-relation were carried through everywhere, there would nonetheless continue to be a historical element in our cognition, which is given with the direction in which the conversion from the one object to the other takes place. There can be causal equivalence without there being value-equivalence. It is not only human history that shows that we cannot go back as well as forward in the series of causes; nature too has its history, which is characterized, physically regarded, especially by the circumstance that the conversion from heat to movement does not take place with the same ease as the conversion from movement to heat. The purely causal consideration here shows itself insufficient, and there is need for other real categories than the causal category, which is indifferent to direction. This holds whether the causal relation can be conceived as equivalence or not. And this makes it that the standpoint of continuity in reality becomes the highest standpoint from which the causal relation can be seen.

92. A judgment in which an unavoidable relation between an antecedent and a consequent is expressed, so that from the former one can infer to the latter, is called a law. But that antecedent shows itself, on closer reflection, never to be a single object, as the popular conception of the concept of cause is inclined to assume. What we call cause is always an aggregate of several cooperating objects, which we can call conditions. There is not a single causal series, but a mass of series crossing one another, and the one series cannot be understood without the other. In practice and in everyday speech we keep to the most decisive or to the most conspicuous condition. From a single object, however, just as little effect can be derived as from a single premise any conclusion can be drawn.

The contribution that a single object makes to the fulfillment of the law is called its force. Force is, to use *Leibniz*'s definition, that in a present state which contains the ground, or a part of the ground, of a new state. Force contains in itself no explanation; but the scientific significance of the concept of force lies in its pointing to the fact that it is not only the new that we understand by the old, but conversely also the old that is understood by the new. We learn in the effects the possibilities of the old. The new that happens casts light on the conditions in the preceding time. Words like possibility and force (which are of kindred meaning, since "possibility" [Mulighed] is kindred with "fortune" [Formue], which in the older language means capacity or force) have their justification by containing a direction to seek the conditions in the preceding objects that ground the emergence of a later object. And on the other hand these words contain a direction to the future; for possibilities, capacities, and forces attest their reality through the effects that proceed from them. Against *Heinrich Hertz*, who wished to abolish the concept of force in mechanics, *Ernst Mach* remarks: "Every mechanics contains a regard to the future, since every mechanics must apply concepts like time and velocity"[108]. This accords with Leibniz's definition mentioned above. It gets significance especially when the state preceding a change seems to be a state of rest or a state of equilibrium. From the fact that change can set in, we infer that the rest was a stopped movement, the equilibrium a relation of tension between forces operating in opposite directions. We understand the simultaneous by the help of the successive, the static by the help of the dynamic. It is this consideration that leads to the formation of a concept like potential energy. It is likewise this that leads the founder of the theory of mutations, *Hugo de Vries*, to state that prior to the sudden appearance of a new type there must go the formation of an inner predisposition—prior to the mutation a premutation. This latter is a process that takes place quite in the hidden (sich völlig im latenten Zustande abspielt)[109].

The greater a contribution to the fulfillment of a law an object gives, the more active it is said to be. Since all activity, all force is measured by the resistance that is overcome, it is always differences of force between objects, relative differences of activity, that we have to do with in our thinking. "We cannot," says *Maxwell*, "even say what force acts on us, but only indicate the difference between the force that acts on one thing and that which acts on another thing"[110]. Thus the law of inertia really contains a summons to find the forces that change the rectilinear movement, not the forces that in themselves give the bodies the movement they have at the moment.[111] The same holds in the case of psychic energy. If a certain state of mind is given, we can ask what is required to change it—how strongly the oppositions already found in it must be sharpened in order for a transition to a new state to set in. The mental life's changes take place just as little "of themselves" as those of physical objects; it possesses no absolute force, but we come to know its force through the oppositions it can encompass. A mental life that was not placed in the breaking of oppositions we could not ascribe energy to.

The concept of force presses—in the strict sense indicated—more and more into the

[108] *E. Mach*: *Die Mechanik in ihrer Entwickelung.*[6] p. 281.
[109] *Die Mutationstheorie.* I. p. 353. Cf. II. p. 637; 662.
[110] *Matter and Motion.* § 103.
[111] *Zeuthen*: *Matematikens Historie.* II. p. 340.

foreground since Leibniz's time, in opposition to such concepts as rest, state, matter, substance, essence. According to Leibniz, only that which acts can be called substance, and every substance is in activity. Hereby the concept of substance is eliminated, and the law for the activity steps in its place. The single individuality finds its expression in the law for the changes that take place in, with, and by it. The former dominance of the concept of substance was bound up with the old presupposition that the unchangeable and resting was the highest (48; 77), especially with the urge for rest for thought in an Idea that did not again set reflection in motion. Substance is an example of a dying, if not already dead, category. *Kant*'s "Ding an sich" was one of its last forms. As regards the mental life, *Hume* and *Fichte* have demonstrated the invalidity of the concept of substance: immediate observation does not show us the soul-substance, Hume maintained; and the more we come to know ourselves by way of reflection, the more we discover, Fichte maintained, that our soul consists in acting.

## b. Totality

93. Even where we can, in the causal series, go both forward and backward, by mutual equivalence, the direction is, as already remarked, not indifferent. Equivalence with respect to energy is not the only point of view. It shows itself psychologically in the fact that a mental state is not only described by the combining energy it has at its disposal, but is also characterized by the qualitative constitution of the psychic objects (thoughts, interests, purposes) that constitute the center of gravity, the real self. Thereby the mental life's direction at any time is determined, the striving in whose service the synthesis ultimately stands. For the combining operation does not place all elements on an equal footing; there are elements that occupy a central place, our constant thoughts, our innermost urge and longing—and there are others that occupy a more peripheral place. An exchange of these places can be one with a mental revolution[112]. And an analogy to this is found in the material domain. It is not enough to investigate the relation between cause and effect among themselves. This is best done when one can isolate a circle of objects from all others. The proposition of the conservation of energy can thus only be experimentally demonstrated when one can form isolated wholes and keep influences from without away. But this is then always ultimately an arbitrary limitation, an abstraction. And the question becomes, when one returns to the real conditions, always in which direction the causal series moves during its continued course—which members have the possibility of here anew making themselves felt. Direction belongs

[112]Cf. my treatise on *Begrebet Villie* in the first volume of the journal "Psyche."

just as essentially to a movement as mass and velocity.

*Leibniz* therefore set up, besides the law of the conservation of force, a law of the conservation of direction. According to the law of inertia, movement does not change its direction, any more than its velocity, except by the influence of outer forces. But even when the direction is changed, the earlier direction is nonetheless cooperative in the new direction, which is the resultant of the earlier direction and the outer force's direction. There is thus continuity in the direction as in the energy. Direction is not something that first emerges at a definite point of the movement, but is present as far back as we go, and the earlier directions determine the following ones, just as in the psychic domain our earlier striving and willing determine the later.

Neither in the psychic nor in the material domain is there now a single causal series, but innumerable ones. In the mental life there is indeed constantly a certain centralization, but all elements in it strive after a central position and influence, and there is therefore always a mass of crossing directions. It then depends on whether they can be harmonized, whether they can constantly anew be united in one stream. And in likeness with this, directions of movement go out from every material point. We need not even keep to the causal series; every single member in such a series, every single "cause," is really a system of conditions, each of which has its direction, and the "cause's" direction is the resultant of all these directions, each of which gives its contribution to it.

Every single cause is a totality, as well as every single causal series, and as well as several different causal series that are connected into operation in the same direction. There are here innumerable degrees and stages of composition, but the principle is the same. There is the possibility that directions diverge; but it must be taken as a fact that this is not always so; directions can be connected, so that wholes, more or less harmonious systems of causal series or directions of activity, form themselves. In how great an extent this is the case, experience must decide. But experience demonstrates that it can happen. And it is therefore that we, through the concept of direction, form the transition from the category of causality to the category of totality. By reason of the originality of direction, the totality is never an accidental result. That directions are united has its ground in the different processes' earlier directions, as far as they can be traced back. It is arbitrary here to stop at a definite point.

As examples of such totalities may be named: solar systems, planets, organisms, personalities, societies. From a general point of view they can all be conceived as totalities or systems of directions. Common to them is that they have properties different from the properties that appear in the more elementary activities in whose cooperation they consist.

Only when one forgets the necessity of direction can one think of a chaos as an absolute initial state. In such a chaos the elements of existence would then be present, and when at later stages of development wholes arise, this would be due to accidental collisions and the activities of the elements. Especially in popular presentations it often looks as if existence proceeded like Euclid's geometry from the simplest elements, corresponding to definitions and axioms, to totalities, corresponding to the most special propositions. And yet already *Hesiod* was clear that the chaos out of which he imagined the dynasties of gods to have developed had itself *come to be.* Apart from the fact that an absolute beginning cannot be thought at all, and still less proved, every attempt to establish an initial state must ascribe to it certain dispositions, definite tendencies of direction. Everywhere one must presuppose a totality, however different this must perhaps be thought from the totalities that appear at later stages.

Totality is a more special category than causality. One can think the latter without the former, since there can be thought, and perhaps are found, causal series that do not converge, do not lead to more or less harmonious totalities. Totality, on the other hand, presupposes causality. The relation between the categories is, after all, that the more special, more concrete ones have a richer content, but a smaller extension. We therefore unfortunately have no right to generalize the fact that totalities arise. The great question, which in this connection we do not, however, have to do with, is whether this fact is rule or exception.

Within the formal categories there is, in rationality, an analogy to the concept we here stand at. Also in our life of thought totalities can form themselves, in that premises and conclusions form a kind of logical wholes or organisms.—Psychologically regarded, consciousness itself forms a whole by virtue of the law of synthesis. And here too it showed itself that we have no right to ascribe to the elements our reflection can distinguish an existence outside the psychic whole of the same kind as within it, so that the whole would merely be their product. In the psychological antinomy (4) we have the example lying nearest to us of the irrational relation between whole and parts that meets us at all stages of existence.

94. The category of totality stands in the way of every attempt to explain wholes as accidental products of elements that are presented either as directionless or with wholly indefinite direction. It denotes a point of view we are compelled to take, since definite relations between the objects require it. When one often sets teleology or finality in opposition to mechanism (cf. 56: Renouvier and Hartmann), it is either an

unfortunate mode of expression or else an incorrect conception. If the formation of the whole is to be explained by a purpose, an intention, a preceding plan, then—quite apart from the anthropomorphism—the relation between whole and elements becomes just as external as in the mechanical conception, which lets the whole arise from preexisting, independent elements. Whether one lets the whole go before (as plan) or follow after (as result), one embraces an equally external conception. One overlooks how the whole in a certain sense preexists in the very earlier directions of the elements. It is only for our accidental apprehension that it comes to be all at once. It was on the way beforehand—in the different converging directions.

And when one applies an analogy with human volitional activity, in which the whole that is aimed at can be given beforehand in a thought of purpose and a plan, then it must first be remembered that voluntary willing develops out of the involuntary and in an involuntary way (see 23). And secondly it must indeed be noted that the life of thought and will itself is an example of totality—so that the problem here recurs. For here too there are elements—sensations, memories, urge and drive, and so on—whose directions' convergence and harmony is a condition for a life of thought and will to subsist. The very thought of purpose by which one would explain the formation of totality is itself a totality. One must then in any case be conscious that this analogy does not solve the problem. To be sure, we see in our highest volitional life, in a decision taken after all-sided deliberation, the most intimate connection of manifold tendencies of direction into concentrated operation in a definite direction. But it is nonetheless still only an example of what was precisely to be explained. One does not get beyond the irrational relation between whole and elements. The epistemological significance of this is that there are constantly new tasks to solve, while both teleology and the chaos-theory close off, each in its own way.

By the help of the concept of direction[113] we get the possibility of a connection between the two categories, causality and totality, a connection that is not constructed, but supports itself on definite characteristic marks in our objects, to whose number here thought itself also belongs.

95. If the different totalities are compared among themselves, they will be able to be ordered in series according to points of view that lie in the very concept of totality. A relation between unity and multiplicity will make itself felt in them all. It is, after all, the cooperation between different causal series that is the condition for totality. Thereby

[113]The significance of this concept in this connection I have already emphasized in my lecture on *Vitalisme* (1898, reprinted in "Mindre Arbejder" I), in my program *Filosofiske Problemer* (1902) p. 66, and in my treatise *Om Kategorier* (Vid. Selsk. Overs. 1908) p. 32.

there is given a relation between these series' multiplicity on the one side and the degree of unity and concentration that the totality presents on the other side. The richer a multiplicity that is gathered into unity, and the more pronounced this unity is, the higher the totality can be said to be. Herewith at the same time the greater or lesser degree of activity will be given. For the richer a system of causal series that is formed, and the more consistent it is, the greater the totality's independence and self-activity can be—the more independently it stands in relation to its surrounding world. An object is said to be the more active the more it has the ground and cause of its subsistence and operation in itself.

According to this standard we will then be able to order the totalities that experience presents. And it is also such a standard that is applied within the special domains, e.g. in the comparison between planets among themselves, between organisms, between personalities, between societies. More difficult the comparison becomes when it is to concern the different kinds of totalities among themselves. Here the point of view of degree (intensity) comes into opposition with the point of view of extent (extensity). According to the first point of view the organism stands above the planet, the personality above society; conversely from the second point of view. This consideration we cannot here continue; it leads over to a category we shall later discuss. *Pascal*'s famous utterance about man as the "thinking reed" that is nonetheless "nobler" than the great masses of the universe intimates the great problem lying here. Psychic energy presents the most intimate union of unity and multiplicity that we know, and it is from it and through it that we know and judge all other totalities.

But the higher a totality stands in the named respects, the greater enigmas it presents to research. Already in *Aristotle* this appears, in that he laid strong weight on the fact that all knowledge concerns general relations, but at the same time impressed that everything really existing is a single and individual thing, something that can always appear only as subject, never as predicate. (Cf. 53.) The individual (total) thus stands as a barrier for cognition. According to the conception we have asserted of the concept of totality, we must find that there is something correct in the Aristotelian conception, only that the opposition in him appears as a dualism, which it is not. The general propositions that science finds or sets up are not its last goal. As we have seen in the investigation of the formation of judgment, we bring the judgment into its most exact form, as an equation, only for the reason that it thus becomes easier to use it in new connections. And all our judgments are just as many attempts to determine and exhaust our objects, the basis for all thinking (25; 35). There can here only be a matter of an approximation. But the more predicates and judgments we find, the more exactly our initial representations become determined. The insight into the general laws and the use of the universal points of view is a means to the understanding of the objects that appear with the stamp of totality and individuality. Through the concept of direction, as we have seen, these concepts hang together with the other categories. Individualities and totalities we seek to understand as a "samfoldighed," an interplay of laws[114]. Even if there continues here to subsist an irrational relation, we nonetheless seek to find more and more decimals.

In the most recent time one has on this point found a decisive opposition between natural science and history (cultural science). The former—one holds—seeks only the general, laws that hold for phenomena and processes that constantly repeat themselves. Nature is the home of the general laws. But in history there is no repetition. Every event and every personality brings something new, exists only once[115]. But this opposition is only a difference of degree. Nature too presents wholes whose history is limited, and which it is the task of natural science to understand, a task in whose service the general laws ultimately step. For *Newton* it was thus not merely a task to find the laws for the movements of the planets—and ultimately of all material elements; the solar system as a whole too, this compages mirabilis, was a problem to him, even though he declared it insoluble without the help of theology. *Kant* and *Laplace* later took up the problem by a purely natural-scientific way, in that they applied the general natural laws to understand the coming-to-be of the solar system as a whole. The biological hypothesis of development seeks in an analogous way to understand the species. Historical science builds, as criticism, on general scientific laws in order to establish the reality and chronology of the individual facts; but in continuation of this it seeks to work the facts together in such a way that individual-concepts about events and personalities can be formed. In a similar way, in psychological characterization the general psychological laws (for representations, feelings, and so on) are applied to the understanding of personal life in its individual forms. Often the rounding-off must here take place by a certain artistic capacity. But it is one thing whether the goal can be fully reached, another whether it does not constantly stand in the background of all research. Provisionally science is most occupied with finding general laws and establishing individual facts; but its last goal is to write the biography of existence in the great and in the small.

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[114]The concept of "samfoldighed" (con-foldedness) was set up by H. C. *Ørsted.* See my book *Danske Filosofer* p. 51.

[115]*H. Rickert*: *Ueber die Grenzen der Naturwissenschaftlichen Begriffsbildung.* (1901–1902).—See also his treatise on "Geschichtsphilosophie" in the Festschrift for Kuno Fischer (1904 f.).—Cf. my critical remarks on the occasion of this work in "Göttingische gelehrte Anzeigen" 1906. p. 5–11, and my lecture *Tænkning og Tro i vore Dage* (Tilskueren 1907).

It is, moreover, not only in life, especially in human life, that we meet the individual, that which occurs only once (das Einmalige). Every state in nature occurs, when we make sufficiently strict demands on identity, only once (91). And the general laws are always due to sections made from a great complexity. Thus the law of inertia always presupposes the relation to a definite system within which the movement takes place. The application of the law of gravity in a single case always presupposes a system of larger and smaller bodies, and changes in it will entail changes in the gravitational relations. It is always a factual, historical element that gives every application of a general law its character.

96. Not only do different wholes stand over against one another, whose mutual relation can present great problems; but a single whole can in turn comprise several wholes in itself and itself be part of a more comprehensive whole. Here merely the point of view of extent is taken, and the relativity of the concept of whole then shows itself clearly here. In *Ardigò*'s doctrine of the undivided wholes (indistinto) and the divided-out parts (distinti)[116] this relation is well expressed, especially since the divided-out and determinate stands, as regards the point of view of degree, higher than the undivided.

There can always be found more comprehensive wholes than those we stop at in the individual case. *Kant* therefore calls the concept of an absolute totality an "Idea," that is, a concept that cannot be concluded. Thus "the world" is a concept that cannot be concluded, like "reality" (38). However high a totality stands, and however predominantly its states and changes are determined by inner conditions, it will nonetheless always present dependence on conditions that do not lie in itself. It will have its open side. Reciprocal action with something different from it then stands as a necessity. And if we ascribe to it energy as a whole, we thereby already presuppose that there are other objects than it. An absolute system would have to be unchangeable; if there were changes within it, they would nonetheless have to run their course to the end, and equilibrium set in.

One often uses the word "world" in such a way that one does not distinguish between the part of existence tolerably accessible to us and existence thought as an absolute, concluded whole. In speculations about the future of the world—especially in connection with the question whether there is impending such a distribution of energy that a complete state of equilibrium must set in—the two significations ought indeed to be kept apart from each other. For thought, "the world" can only denote a relative whole, can never lie ready-made. This follows already from the inconclusibility of the synthesis. Thought can only end with a question.

[116] *Moderne Filosofer*. p. 35–37.

And as every whole from one point of view or another will always be able to be regarded as part, so every part can in turn be regarded as whole. The series of parts and wholes (series (5)) is inversely symmetric (an absolute correlate-series, see 68). Already herein lies the fact that absolute atoms cannot be thought. There is just as little any part that cannot be whole as there is any whole that cannot be part. It has then also shown itself that such an assumption leads to contradiction; already *Huygens* and *Leibniz* saw that it conflicted with the conservation of energy[117], and in the most recent time it has become more and more clear that the atom itself stands as a whole little world within which new problems emerge. Nor in the psychic domain can we stop at last, wholly uncomposed elements; even the simplest sensations show themselves, when we take account of the conditions at their arising, to presuppose a composite process (11).

## c. Development

97. The concept of direction led us from causality to totality. A closer reflection on the relation between totalities and their presuppositions leads to the concept of development, which for that matter can also be reached immediately from the concept of cause.

The concept of cause directs us, for the understanding of an object's arising, partly to inner, partly to outer conditions. If we now keep to the inner conditions, we are led to a comparison of the object, as it now lies before us, with its own earlier states. The point is then to order these in such a way that the following states—under the constant presupposition of the cooperation of outer conditions—can be understood from the preceding. The more connection we can thus find, the more we can apply the concept of development, in that the series of the different states of the same object appears with a stamp of unity. In other words, the more we can conceive the causal relation as continuity, the more it steps forth as development. This is already intimated by *Kant*, but was especially emphasized by *Salomon Maimon* in the following utterance: "To ask about a phenomenon's cause is to ask about the continuous in it and to fill out the gaps in our observations in order thereby to make them into experiences. For what does one understand in natural science by the word cause other than a phenomenon's development and dissolution, so that one finds the sought continuity (Stetigkeit) between it and the preceding?" "The sufficient ground of a thing is the complete concept of the way in which it has arisen"[118].

There showed themselves, in the investigation of the concept of cause, two points

[117] *Den nyere Filosofis Historie.*[2] I. p. 346.
[118] *Versuch über die Transscendentalphilosophie.* (1790). p. 140; 392.

of view that led thought in different directions: equivalence and continuity. If we are right that this latter concept is the characteristic thing for the concrete causal relation, while the former concept always gets a certain abstract, methodical character, then it is also justified to maintain the inner connection between causality and development. We can then not make the causal series into an absolute identity-series, but each member in it gets, the more we keep to experience, its definite place (as in the progressive difference-series).

Neither *Aristotle* nor *Kant* have given the concept of development a place in their lists of categories. But both masters have nonetheless, each in his own way, in another connection acknowledged the significance of this concept.—Aristotle remains, in the doctrine of categories, standing at the "essence," the object of which the nine categories can be asserted (53). But in his metaphysics he discusses more closely the concept of essence and finds then that it appears in two stages, as possibility and as actuality, no matter under which of the nine categories it is to be considered. A certain property (quality) a being can have as possibility or predisposition; the knower, e.g., does not always have his knowledge present, actual;—but the property can also be there in full actuality or "energy." The transition from possibility to actuality Aristotle calls movement (κίνησις), which is thus an analogy to the modern transition from potential to actual energy. Only as actual (ἐνεργείᾳ) does a being reach its full, pronounced form. Aristotle seeks to describe existence as a series of transitions from possibility (matter) to actuality (form). What hindered Aristotle from giving a doctrine of development was his presupposition that the fully developed form was the truly real; the stages by which it is reached do not become the object of serious investigation, but were dismissed with the remark that they contained the "possibilities" of the completed form. The concept of possibility, which, as we have seen, has its significance as the marking of a problem, a future task (92), did not, in Aristotle, and still less in his successors, get this methodical significance, but all too often became a mere word that closed off further questioning.—As for Kant, it would have lain near for him to give the concept of development a place among his categories, since he conceives continuity as a chief point of view that holds for them all (54). It is only due to the artificial thing in his arrangement that this has not happened. In special questions the concept of development played a great role for him, as is seen from his cosmogonic hypothesis, his anthropology, and his historical conception.

The concept of development is a form of thought that experience has forced through. The urge to dwell on the intuition of the objects as they appear, and to lay weight on their mutual difference, long hindered its full acknowledgment. The opposition between ancient and modern thinking expresses itself quite especially on this point. The ancient tendency to hold fast the subsistence of forms, so that the transition from possibility to actuality was more a kind of disclosure or unwinding than a proper development (in the sense in which we now take the word), maintained itself very long, even in a domain where experience could seem to speak sufficiently clearly. The so-called preformation-doctrine (Leibniz, Haller, Bonnet) assumed that the organic individual was already present in the germ (whether one, like *Leibniz*, laid most weight on the seed, or, with *Haller* and *Bonnet*, on the egg), so that growth was only partly an increase in size, partly an uncovering of the forms[119]. No organs thus really come to be during the individual development. In opposition to the preformation-doctrine, C. F. *Wolff* set the so-called epigenesis-theory (1759), in that he demonstrated how an organ can develop out of a predisposition that has another form and structure than itself. The intestinal canal, e.g., appears first as a leaf, passes then over into a groove, and becomes finally a canal with openings. The difference between the two theories has great methodological significance. It is only the doctrine of epigenesis, the doctrine of the real, successive arising from a simple predisposition through changing forms to the perfect state, that contains a penetrating summons to go into the particular and trace the different transitions. When it is only a matter of an expansion of what already is, the occasion for research is not so great[120]. The concept of development here appears as an important means of thought, a forward-driving point of view. Thereby—and at the same time by its natural connection with the other categories—it is understood that its application has penetrated in all domains of research.

Remarkably enough, at about the same time as the appearance of the doctrine of epigenesis, an analogous point of view was asserted for the human personality. *Jean Jacques Rousseau* laid, in his pedagogical theory (1762), great weight on the characteristic difference between the different ages of life and protested especially against the child's being regarded merely as a little adult. The qualitative difference between the ages of life does not exclude continuity, but necessitates different psychological and pedagogical points of view. Just as the grown organism is more composite and heterogeneous in structure and mode of operation than the embryo, so it is also as regards the later stages of the mental life in relation to the earlier. And the same opposition Rousseau found in

[119]*Leibniz*: *Système nouveau de la nature.* (Op. ed. Erdmann) p. 125b: La génération apparente de l'animal n'est qu'un développement [an unwinding] et une espèce d'augmentation.

[120]Cf. *Kölliker*: *Entwickelungsgeschichte des Menschen und der höheren Tiere.* (1861). Historische Einleitung.

the domain of social life in the relation between culture, with its ever more prominent division of labor, and the uniform and involuntary form of life of the state of nature.

Already from these first forms of the modern concept of development it is seen that there are certain analogous features that recur wherever this concept is applied. There is no ground to say that it is *only* due to analogy when we speak of development both in the organic, the psychological, and the social domain. There is everywhere "only" analogy between examples of one and the same concept. (See 69; 73; 75.) The analogy is a relation of likeness that attests its significance through fruitful application. And the concept of development affords guiding thoughts that lead to raising questions where one would otherwise close off. It brings with it the unrest that a category ought to bring. Already C. F. *Wolff* pointed out the analogy that exists between the different systems within the organism (nervous system, muscle-mass, vascular system, intestinal canal), as regards development, and it was also by way of analogy that he discovered the metamorphosis of plants, whereby he stands as *Goethe*'s predecessor. The struggle for the application of the concept of development to the individual and social domain, as well as to the coming-to-be of our world-system, belongs to the eighteenth century; in the nineteenth century the struggle was waged for its application to the arising of the organic species, as well as to the coming-to-be of the earth's present constitution.—For the human race as a whole the thought of development had already come forth within great popular religions like Parsism and Christianity. In Parsism the idea of a world-history as a development toward a conclusion seems to have arisen, and has from there gained influence on other popular religions[121]—and has thereby in turn become an element in the general human circle of representations. It is in the religious form decidedly the epigenesis-type, not the preformation-type, that prevails in the representation of development; for it is under a mass of changing forms and fates that the race, according to the religious representations, appears in the great drama from origin to world-judgment.

98. When the concept of development can find application in so many different domains, the question becomes which form of it one is to lay at the basis when it is to be set up as a category. It will depend on which form is best suited as a basis for the illumination of the other domains through analogy.

Several philosophers in the nineteenth century have formulated concepts of development. *Hegel* used the analogy (which he would not admit was analogy) with logical thinking. One could, in his opinion, build series in which the first member led over to the

[121]Cf. my *Religionsfilosofi* § 14 and 43.—*Goldziher* has especially laid weight on Parsism's influence on Mohammedanism, in eschatological and other respects. (Die orientalischen Religionen. Kultur der Gegenwart. I, 3, 1. p. 108 f.)

second, and in which the third member took the two first up into itself as in a higher unity, which could then in turn become the first member in a new trinity (55). It is now, on closer investigation, clear that Hegel in reality combines other analogies with the purely logical. He thinks of psychological contrast-effects, cases where the one opposition replaces or releases the other, and he thinks of the organic and personal development, where the earlier stages' properties, however conflicting they are, can all give their contribution to the stage of completion[122]. From a biological point of view *Herbert Spencer* started, in that v. *Baer*'s theory of organic development as a progressive differentiation, a successive transition from uniformity in structure and mode of operation to heterogeneity, formed the starting-point for his systematic philosophy of development[123]. A psychological starting-point *Sibbern* seized, who in the different domains of experience, the physical, the biological, and the social, found again the type of development that psychological experience and observation had first taught him: processes that begin from different starting-points, hence sporadically, in order then, through breaking and fermentation, to lead to a harmony[124]. *Roberto Ardigò* too took a psychological starting-point, namely the natural development of human thought-life from the indefinite to the definite[125].—Under different turns of phrase certain chief features recur in these presentations, chief features that can be formulated in the following way, most nearly in agreement with Spencer's presentation.

When an object comes to be, it must happen by elements that were formerly in a more scattered state being gathered into a whole that rounds itself off with a certain independence in relation to other objects. There must take place a concentration; the sporadic must to a certain degree be overcome, in order for an object to appear as something apart. The convergence of different causal series was, as we saw (93), the necessary presupposition for the arising of a whole.

The simplest objects do not get beyond this elementary form. A cloud in the sky, a heap of sand on the beach are examples of this. And both planets, organisms, personalities, and societies have, at the simplest stages at which we know them, the character of a uniform whole with greater or lesser concentration. If development advances further, differences form themselves within the whole; differentiation replaces mere concentration. It was this peculiarity that C. F. *Wolff* brought out in his doctrine of epigenesis. It is the stage of differences, of breakings and crises.

[122] *Den nyere Filosofis Historie*$^2$ II. p. 175 f.
[123] *Den nyere Filosofis Historie*$^2$ II. p. 457; 471–473.
[124] *Danske Filosofer*. p. 99–101.
[125] *Moderne Filosofer*. p. 33–37.

As the third chief stage, which denotes the culmination of development, appears what Hegel called the higher unity (an expression that ought not to disappear from philosophical usage because abuse has been made of it). Here there has been formed a rich fullness of differences in essence and mode of operation, without, however, the concentration, the character of wholeness, suffering under it. The properties that set the two first stages in opposition to each other are here united in a harmonious way.

We have, by this schematic presentation, which is a definition of the concept of development, formed a series that cannot be reversed. Each stage is conditioned by the preceding and conditions the following; from this side it recalls the progressive difference-series (5);—but since the relation between the first and second stage is different from that between the second and third, it must be called a regularly varying difference-series (3).

How the schema is to be applied in the individual case can of course only be decided by a special investigation. But the concept of development has here gotten a definite significance, and it contains a direction to the direction in which the investigation of special objects is to be undertaken, when they appear as subjected to a series of changes. It is no necessity that there shall be development; already earlier we saw the possibility that the directions of the causal series did not converge. Of all real categories it holds that it is experience that forces them through. And within the real categories the category of development is in turn more special than causality and totality. The inverse relation between content and extension announces itself here again in all its fateful significance. Of the schematic development's three stages, the third is in turn richer in content and less comprehensive than the second, and this than the first: there will be the greatest probability of finding development-processes of the most elementary kind.

99. One could ask whether dissolution is not just as much a category as development. And one would not be able to deny it. It is experience that has led to combining our general forms of thought in such a way that a development-schema can be formed. And it is also experience that impresses the rhythm of development and dissolution. Formally we could very well use the word development in such a way that, besides the three stages discussed above, the stage of dissolution, or the stages of dissolution, formed a series (of kind (3)). Dissolution presents stages that nonetheless do not quite correspond to those of development: the harmony of concentration and differentiation is abolished;—there sets in either a stable concentration or a state of scatteredness. Whether these states are in turn suited to be initial stages for new development, so that a rhythmic series can form itself, is the last great question that the doctrine of categories cannot discuss. Only two considerations are here to be asserted.

In the first place, we can connect no meaning with applying the concepts development and dissolution to existence as a whole, for the good reason that such a concluded whole cannot be thought (96). All development and all dissolution takes place within a more comprehensive whole. *A* world can arise and pass away, but of "the world" these words could not be used if they are to have meaning.

In the second place, one will never be able to demonstrate that dissolution ends with absolutely different elements, in a chaotic difference-series. It would take an infinitely comprehensive and infinitely exact experience to convince us that we stood at such a series (62). The elements in which dissolution ends must be small worlds. One speaks then also, in the most recent time, of intra-atomic processes, and the electrons that are now thought of as the material world's last parts are by no means absolutely single. And sources of energy can emerge where one least expects it. Radium was found in a heap of refuse.

100. Theoretically regarded—and we are here, after all, moving within pure theory—dissolution is, as said, just as justified an object for reflection as development is. It is also, in civil life, just as honorable to be employed at the burial service as at the maternity hospital. When most thinkers have involuntarily occupied themselves most with the thought of development, this need not, however, be explained by the fact that our practical interests are attached to the ascending lines. It need not be any interest different from the thought-work itself that hereby makes itself felt. It can quite simply lie in the fact that dissolution presupposes development, since it presupposes wholes that can be dissolved, while development does not necessarily presuppose dissolution, since the objects and elements that cooperate could belong to existence's hitherto unused funds. And thought itself, the reflection sprung up on the basis of sensation, intuiting, and involuntary association, is itself "a higher unity," a development in a third stage; it is then natural that it is most occupied with tracing its own analoga.

There has, to be sure, among philosophers been an inclination to conceive all development that reaches the third stage as eo ipso valuable. In *Hegel* and *Spencer* this inclination shows itself distinctly. But thereby it is overlooked that the valueless and the value-adverse can just as well develop according to the indicated schema as the valuable. Development is in itself a neutral concept—and dissolution therefore really also.

It has, however, now in the most recent time been expressly maintained that the concept of development indicates a value-standpoint; only thereby is it supposed to be distinguishable from the concept of change. If we will speak purely theoretically, says *Rickert*[126], we ought to speak of a series of changes. Even in the most elementary development-processes, e.g. the formation of a cloud, we nonetheless presuppose a goal (Ziel) that is reached, and thus introduce a teleological moment. The concept of mere becoming (Genesis) leads us into the unlimited, but the concept of becoming directed toward a definite goal (Orthogenesis) makes the changes into an articulated and unified chain, and alone deserves the name development.

To this it must be remarked that while development in the neutral sense used above is of great methodical significance, in that it contains a direction to trace different stages in a series of changes, each stage denoted by its characteristic marks, it is impossible to say when a "goal" has been reached. It is arbitrary where we set the boundary, where the goal is to be reached. When a boil develops, is there a goal? Or when moral wickedness develops, is there then a movement toward a goal?

On the other hand, we cannot think at all without provisionally fixed points. It shows itself with time and place, with rationality and causality; also in the application of these categories a goal is presupposed. But we then also become conscious that we more or less arbitrarily hold fast certain members in the series as beginning- and end-points. Already the simplest formation of judgment must have an initial representation and a final representation. And this is sufficient to form an articulated and unified series. When the single act of cognition is completed, one can pass over to another—if, that is, thought itself is not placed in dissolution.

It must, however, be conceded that the categories whole and development have a closer relation to the concept of value than causality and than any of the formal categories. But it is appropriate to distinguish the value-standpoint from other standpoints. Hitherto we have only presupposed intellectual values and interests, because they are most intimately attached to the thought-work itself. We now extend the consideration to values in general and discuss the basis and significance of the value-concepts.

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## E. Ideal Categories

100. Our thinking treats its objects according to the forms that are the natural ones for it, and that are developed by the relation to the objects. But during this work an interest constantly makes itself felt that can stand in a more or less intimate relation to thought itself. It can be a purely outer motive, so that the thought-work itself is only a means,

[126] *Die Grenzen der naturwissenschaftlichen Begriffsbildung*. p. 451–463.

and it can be most intimately attached to the thought-activity itself, following it during its course, rising and falling according as thought succeeds in mastering a rich fullness of objects, or as this striving is hindered by resistance or by lack of objects. Such a purely intellectual interest we have in the foregoing for simplicity's sake presupposed (6). Now we bring out the fact that there are other interests and values than the intellectual, and investigate how far it is possible, on the presupposition of them, to undertake a thought-work, and under what special forms this must take place. In advance it is probable that the forms of valuing thought must present likeness, at least analogy, with the previously considered forms of thought.

The three concepts we here for clarity's sake distinguish between hang in reality closely together. Both forms and objects have their values; this is already presupposed in the foregoing. And value is itself an object that thought can treat and try to form.

Value has everything that satisfies an urge and thereby awakens a feeling of pleasure or averts a feeling of displeasure. The presupposition is thus an urge, a striving or longing in a certain direction. The concept of direction here again shows itself in its significance, and that in a significance not wholly different from the one brought out earlier (93), which led us to the concepts whole and development. An urge presupposes a whole that has a tendency to assert itself, to assert its inner connection and its independence in relation to everything else. Value then gets, from such a whole's point of view, everything that favors this tendency, while what hinders it becomes of the evil. Feelings of pleasure and displeasure are, psychologically and biologically regarded, symptoms of progress or regress for an individual whole's striving after subsistence and development. They contain in themselves the germ of valuing judgments, in simplest form exclamatory judgments (30). They appear as elementary forms of psychic energy, in that they make known how far the whole that a conscious life or an organism constantly constitutes is threatened with dissolution at one point or another. It is an old thought that a being that constituted an absolute unity would not be able to suffer: dissolution would then not be possible[127]. There is hereby indicated a presupposition that alone can make coherent reflection about values possible. For insofar as it holds, an investigation can be made of the conditions for the subsistence and development of the whole in question, and thereby the involuntary valuation can get its confirmation or its correction. Every immediate feeling of pleasure or displeasure can from this point of view be regarded as a hypothesis that is to stand its test. They are immediate qualities like the sense-qualities. Value-qualities we can call them. There is, as with all qualities, the question how far it

[127]Melissus of Samos (*Diels*: *Vorsokratiker*[1] p. 150 f.).

is possible to form rational series, so that grounded judgments and inferences can be set up. Such a possibility can precisely only be found if the value-quality corresponds to the relation between a given whole (an organism, a personality, a society) and the conditions for its subsistence. It is really such a possibility that was already discussed in the foregoing, in the relation between form and object. The whole that was here to be built up was a cognition, a thought-world, and the matter was what relation must be present between thought and its objects, if such a world was to subsist. Concepts like truth and reality are intellectual value-qualities that we have sought to give rational form. (See especially 36–38; 85–88.) We now seek to treat other value-qualities in a similar way.

Our category-series gets, as remarked several times, a narrower extension and a richer content the further we come forward within it. Of values we can, according to the given definition, speak only where a formation of a whole has taken place. And within the wholes it is directly only the three named—the organism, the personality, and society—that condition valuations. There can be a biological, an individual, and a social valuation. Yet we can very well by analogy apply a valuing train of thought to all wholes; we can ask about the conditions for a mountain's, a planet's, a world-system's subsistence and development. One will, e.g., be able to ask when a star stands at the culmination of its development—whether it is when it has developed from nebular mass to sun, or only when its temperature has reached so low a degree that organic life can become possible? In the latter case the presupposition will lie at the basis that it is life that gives everything else value. By this way one comes to a kind of cosmic valuation, so that existence gets a different stamp for us according as it shows itself to favor the development of life or not.

The valuation, the value-quality, becomes another according to the whole one lays at the basis. Within the same whole it can in turn be different parts or properties whose fate especially determines the value-quality. For every series of values there must therefore be laid a definite value at the basis, determined by the whole, and within it in turn by the part or property whose conditions of life are discussed. All valuation goes back to a fundamental value. Thereby is demonstrated the justification for setting up the value-categories as a peculiar group. Value can only be measured by value. Here is a circle one cannot get out of. It is the fundamental value that determines the standard. The special values are measured by their relation to the fundamental value.

It goes here as with time and space: every time- and place-determination must be undertaken from a definite standpoint. Time cannot be determined in relation to the timeless, or place in relation to the non-spatial. Thus value cannot be determined except in relation to other value. It becomes especially clear when we here, as with the other categories, start from the category's primitive, practical appearance. That something has value for me I show practically, perhaps quite involuntarily, by holding fast or acquiring it, doing a work for it. That something has more value for me than something else I show by preferring it to the latter. The work is released only by a difference, and in the concept "prefer" the difference is expressed still more distinctly. We can continue the series downward: a value is preferred to the indifferent (neutral), if there is any such, and still more to the "negative" values that make themselves known as displeasure and pain.

From the possibility of different, perhaps mutually conflicting fundamental values springs, as we shall see in a later section, the problem of valuation. In particular the religious problem springs from the relation between the human values and the vast existence that does not gather itself for us into a whole. We feel in our value-qualities the pulse of existence as it beats within the wholes that we ourselves are or are parts of; but have we thereby the right to infer a heart that beats in the whole of existence? Only by the boldest analogies can our most content-rich and therefore least comprehensive categories be extended to hold for the vast horizons that lead beyond the limits of our experience. About the justification of these analogies stands the great struggle between the views of life. But when this struggle is to be waged with the weapons of thought, it must use the means of thought that the categories can afford.

101. Provisionally we have merely to do with the categories themselves, not with the problems. The value-qualities can be divided from a formal and from a real point of view, and each of the groups thus formed in turn presents differences.

## 1) Formal Differences

a) *Elementary values* presuppose a sensuous present as object for enjoyment and appropriation, as spur and as support. *Ideal values* presuppose memory or expectation, reflection or imagination. Since they are not supported by sensuous presence, they presuppose greater psychic energy in order to be held fast. They have at the same time the advantage of being independent of outer conditions, so that they can be at one's disposal as soon as the inner conditions are present. Thereby they stand above the elementary values.

b) *Immediate values* presuppose no grounding or derivation from other values. They make themselves felt by their own power. The fundamental values are from the first such immediate values, and precisely therefore it is easily overlooked that they are determinative. Immediate values can be both ideal and elementary; their peculiarity consists in their standing as self-evident. Of *mediate values* there is a question only when an immediate value cannot without further ado, quite involuntarily, make itself felt or fall to our lot. They are means, ways, preparations for immediate values, must be derived from these and borrow their light from these. When such preparation is necessary, hence when there takes place a transition from the involuntary to the voluntary, the immediate value becomes a *purpose.* It does not thereby lose its immediacy (in the indicated sense); when the purpose is attained, the value makes itself immediately felt, does not necessarily point beyond itself. The immediate value is purpose only while the interval is being traversed. By displacement, what at first had only mediate value can become an immediate value; the means can become purpose, and the preparation become the best thing. The hunt can be better than the catch. It has been mentioned in the foregoing that intellectual value often arises in this way, in that it develops out of practical interests that needed reflection as a means. And within the intellectual interest the same process can then repeat itself, so that trains of thought that at first had value only as ways to the understanding of an object, later got independent value, formed a little thought-world for themselves. Most energetically this is expressed in Lessing's proposition that the constant striving after truth is to be preferred to the possession of truth (86).

We see here again the significance of the interval. Already the ideal values presuppose it, and the difference between immediate and mediate values could not come forth without it. The immediate values themselves could not become conscious to us without it.—As with the categories of time, place, and cause, the relation of distance also plays, in the category of value, an essential role for the development of thought. And it was precisely the distance between urge and goal that conditioned the development of concepts like time, place, and cause. The value-categories, which on account of their more special character form the last group, have thus constantly cooperated in the development-history of the preceding categories.

By the relation between immediate and mediate value the concept *Norm* is conditioned, which expresses a rule for the means and ways that are necessary for the attainment of the purpose. All different kinds of values have their norms, conditioned by the relation between the purposes and the experiential conditions under which they are to be reached. The value of the norm is mediate.

c) A third formal difference between values rests on the fact that a value can need a release, a finding-out, or a producing, before it can appear as value, while in other cases it appears in full activity. There can then be distinguished between *potential and actual values*, if one will, slumbering and waking values. This difference can make itself felt both as regards elementary and ideal, immediate and mediate values. There can arise something new in the world of values, as in the world of objects in general. There is the possibility of purposes no one has yet thought of, because the immediate value in question has not yet been able to make itself felt. It can be processes of displacement that are not yet concluded, or it can be natural processes that have not yet unfolded. Mediate values too can be potential; there can slumber, both in us and outside us, forces that we could take into the service of our actual values, if we could bring them forth to the light.

## 2) Real Differences

The formal differences mentioned in the foregoing can make themselves felt, of whatever kind the values are. When we call them formal, it is not thereby excluded that they condition decisive differences in the mental states. We feel differently when we enjoy than when we wait, long, or work, differently when we live on mere possibilities than when we move among actual values. So far all value-differences are real. But real differences in the narrower sense are those conditioned by the peculiarity of the objects to which the values are attached—by that which "has" value. Here two groups can be distinguished, of which the one has most to do with *the content of the values*, the other most with *their extent*.

a) It can be the purely organic subsistence, mere self-preservation, that conditions the value. It holds here literally and elementarily: to be or not to be. It is the organic feeling of life that here lies at the basis, but to it can in turn attach themselves ideal, mediate, and potential values; especially the feeling of power, conditioned by the consciousness of dominion over the conditions of life, stands near as a source of derived values that by displacement can become immediate. In a certain opposition to this kind of values—the *biological*—the *intellectual, aesthetic, and ethical values* can step. The *religious values* develop out of these different kinds of values' relation to existence as a whole.

b) With respect to extent there holds a difference between *individual, social, and cosmic values*. Whatever content-values man knows, it makes a difference whether it is only a matter to him that the values fall to his own lot, in conscious or unconscious opposition to others' lot, or whether he sees them as goods that are common to larger circles, and are for a part determined in their valuation thereby. Between individual and social values there will be a constant reciprocal action, now a harmonious, now a disharmonious one. From this arise derived categories, and from this arise special problems.—Of cosmic values there can be a question when we by way of analogy make a valuation of the conditions of existence of the most comprehensive wholes.—

102. The possibility of an irrational, if not an anti-rational, relation between categories lies nearer in the value-categories than in the other classes of categories. Throughout, human consciousness has hitherto relied on the assumption that there will constantly be application for value-categories as well as for the other kinds of categories. Thought's last work begins when there shows itself doubt about the forms of thought's mutual agreement as well as about their applicability to the objects that make themselves known to us. Such doubts lie at the basis of the problems of philosophy, to which we now pass over. Naturally it is also a problem to describe the development-history of thought and give a survey of its forms. But herein we have already presupposed the validity of the thought-work, as it happens to be constituted. It is this presupposition whose justification is now to be tested.

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# IV. The Tasks of Thought (Problems)

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## Introduction

103. The problem has in an earlier connection been discussed from the formal side, as a form of psychic energy (27). Energy is required both to grasp and hold fast the conflict between objects and to feel the lack of intermediate members between objects—both for liberation-problems and for completion-problems. It is only through memory and comparison that problems come to be. As long as continuity prevails, it is not discovered that the later states are different from the preceding; for that there is required a holding-together of members that perhaps lie far from one another in the series. The problem then becomes how continuity is compatible with the qualitative differences. And when consciousness, where discontinuous objects are present, lets go the one object when it passes over to the other, no lack of intermediate members can be felt, no urge can arise to transform discontinuity into continuity. To these two chief forms all problems can directly or indirectly be traced back (59). There arises in both cases a tension that sets a work going, until a solution is won, a provisional or a definitive, an individual or a universally valid solution. Such relations of tension will more easily set in the more experience is extended; but on the other hand tensions can also fall away by extended experience. Problems therefore do not, any more than the categories, continue to be the same at all times. In a natural history of problems, such as *Richard Avenarius* has given, the special conditions and differences in the arising of problems are treated. There arises, as he aptly remarks, a feeling of strangeness, of lack of security, when a spiritual equilibrium is abolished; every genuine problem is a kind of homesickness. But the home can be sought and found by different ways. One can seek back to the old home, let the oppositions fall, glide again along the stream of continuity, or cling to a single one of the members in the discontinuity-relation and forget the others. If one gives up comparison, the problem falls away. One returns to the state before the doubt. But one can also press

forward through the doubt and form a new spiritual home, which can then be either a permanent dwelling or a hut for provisional shelter.

The philosophical problems hang together with the nature of thought in such a way that they must follow thinking on its way and will constantly be set anew, even if it happens in new forms. The type for them is given with the psychological antinomy that springs from the very concept of psychic energy (4). Thought itself carries its enigma with it. It cannot leap over its own shadow. Just as little as the personality can be thought fully concluded, just as little can thinking. It goes from problem to problem. The urge for scientific inquiry is a special form of the urge for agreement with oneself under all manifold and changing experiences. It expresses itself both in the formal science and in the real—both as an urge to form trains of thought in which the one member with inner necessity develops out of the other, and as an urge to connect different objects in such a way that as comprehensive and as exact a continuity as possible is attained. Personality consists first and foremost in the inner connection between all representations, feelings, and endeavors. It is then no sacrifice that the personality brings when it stakes its life on inquiry. It produces in science—as in art—an objective counter-image of itself. Only thereby is understood the intimate character that the joy of cognition can take on, the amor intellectualis that the inquirer feels in working his way forward to a new standpoint from which the single object can be seen as a member in a great connection. The holy fire that constantly anew sets thought in motion, despite everything that can dampen and hinder it, is understood only when one has this in view. And the personality on its side needs to be purified by bowing to thought's objective connection, the firm order of things, within which it itself has its definite place. Freedom is won through truth. It belongs to the nobility of the personality that it can by its own power find the great laws that condition its subsistence and determine the wishes that arise in it, as well as the conditions for these wishes' being able to be realized.

104. If it is rightly said above that the philosophical problems hang together, more than other, more special problems, with thought's own nature, then we must in the preceding presentation of thought's growth and forms have a guide to finding what the philosophical problems are.

In every act of thought, at whatever stage of development it takes place, we can distinguish three different elements. Thought's psychic energy expresses itself in certain forms, which we can by abstraction distinguish from that to which they are applied, what in the foregoing has been called the objects. The concept object too is due to abstraction. No object is wholly unformed; synthesis comprises it from its moment of birth in consciousness. Only through analogy with an artist's relation to the material he works on, and that we can know before he works on it, can we speak of objects wholly independent of all form of thought or all forming by thought. Here we do not know the marble block before the hewing has begun. To the object we reckon everything that can stand as an object of reflection. Besides form and object there always cooperate motives, interests, values. They can vary in kind and degree, from the bitter need that requires reflection and inventiveness in order to keep life up, to the purely intellectual urge whose satisfaction is immediately given with the clear and grounded connection that reflection is able to bring about between as many objects as possible. In the doctrine of categories this triad appeared in the difference between the formal, the real, and the ideal categories.

It showed itself in the doctrine of categories that the forms took on ever more special character by reason of the objects that lay before us. Here lies the first philosophical problem, determined by the relation between form and object. Can the forms be derived from the objects, or conversely? With what right are the forms applied to the objects, if they cannot be derived from these? Is it possible to reach a perfect and exhaustive appropriation of the objects by the help of thought's forms, as these happen to be?—Such questions belong under *the problem of cognition*. The third element, the motive or the value, does indeed here too make itself felt. There could be raised the question whether the choice of the objects that are treated, and of the forms that are applied, is not due to a regard for value, a motive that can come more or less to consciousness. This question too must be discussed under the problem of cognition, even though it can otherwise be neutralized by our provisionally merely presupposing the intellectual interest that is immediately directed at clarity about the relation between forms and objects.

Another problem arises with the relation between the objects among themselves. The different sciences have each their object in a certain group of objects. But when the question is raised which object or group of objects is to be regarded as fundamental, as the object in whose light we regard all other objects, *the problem of existence* (the metaphysical or cosmological problem) is set up. It is the question how far a general world-view is possible, and what character it must take on. According to the group of objects that is laid at the basis, the world-view will get a different timbre.—Here too the value-element will be able to make itself felt; but it will be appropriate, as far as possible, to eliminate it, in order that it may be seen to what results the purely intellectual interest can here lead. It is necessary for clarity's sake to distinguish between the different points of view.

If we then finally bring out the value-standpoint, which, according to the investigation of the ideal categories, hangs closely together with the concept of whole, it gives occasion to two problems that can be summed up as *the problem of valuation.* There can, in the first place, be asked about the different values' mutual relation and about the possibility of tracing them back to a fundamental value, so that the relation to this determines their mutual sequence. This is the ethical problem. There can, in the second place, be asked about the relation between the values on the one side and the cognition that the forms and objects make possible. It is here a matter of the relation between the world of values and that of reality. Is there a harmony or a disharmony between these two worlds? Hereby arises the religious problem.—Under this falls also the question whether it is possible to develop a well-grounded world-view out of value-concepts, so that the validity of the forms and objects is made dependent on their relation to the values.—

If one misses *the aesthetic problem* in this arrangement, I will point to its analogy with the ethical problem. Aesthetic value rests on an object's appearing as a characteristic whole and being intuited and held fast merely as such, apart from intellectual and practical interest. Aesthetically we regard reality as image and the image as reality; it is enough for us when the object—be it now reality itself or its image—rounds itself off into a peculiar whole. The ugly and the disharmonious too can fall under this; the aesthetic problem would then concern the condition for such images being able to come to be and become the object of work and appropriation. In ethics, on the other hand, it is a matter of the work for the real whole itself, which stands as the fundamental value. Here the difference between image and reality becomes of decisive significance. It is the category of whole that lies at the basis of all valuation. We can conceive the whole as image (aesthetic problem), as basis for the judging of actions and endeavors (ethical problem), and as that whose fate determines our feeling of the value of life and of existence (religious problem). For a complete presentation of the philosophical problems it would be required that the aesthetic problem got a peculiar place. But the typical in our relation to the problems will hopefully be able to come forth, even if I keep to the problems I feel best prepared to treat. For that matter I must refer to my Psychology (V B, 12; VI C, 9) and my Ethics (XXX, 1–2).

105. That so definite a distinction can be made between the different problems is due to a spiritual division of labor that has slowly developed. The distinction steps forth more distinctly in Platonism than in Animism, and more distinctly in Positivism than in Platonism, on the whole more distinctly in modern than in ancient science. Cognition's forms and principles, existence's innermost nature, the rank and fate of values—these

were three points of view that at earlier standpoints were regarded as one. Epistemology, cosmology (metaphysics), and ethics went involuntarily together. The difference was in part acknowledged in post-Platonic Greek philosophy, especially in the division of philosophy into logic, physics (= metaphysics), and ethics. But it was only the eighteenth century's empirical and critical philosophy that carried it through. The romantic reaction in the first half of the nineteenth century was able only transiently to press it back.

But the division of labor must not lead to a splitting. It would conflict with thought's character as synthesis if the treatment of the different problems should lead to mutually contradictory results, or to results that did not fill out each other or show a mutual kinship. It would give occasion to a new problem. An identical solution of all problems, in the sense that a single principle contained the answer to the problem of cognition, of existence, and of valuation, will be an insoluble task. But there must be assumed to be an analogy between the different problems; this follows already from the fact that they hold for one and the same working thought, whose enigmas are determined by its nature, even if they do not exclusively have their origin in this.

Both the problems themselves and their solutions are determined in kind and direction by thought itself. Common to all problems is the great opposition between continuity and discontinuity—between the unhindered advance of the synthesis, of the psychic work, and its stoppage at continuities within which no distinguishing is possible, and at differences that cannot be equalized. Both in the world of thought, in existence as the aggregate of the objects we know, and in the world of values, the point is to determine differences within the continuity and to demonstrate transitions between discontinuous objects. Of course continuity and discontinuity get a different character in the different domains, just as there is, strictly speaking, on the whole only analogy between the different examples of one and the same concept.—

The difference of the problems makes itself known in the history of thinking by the fact that it is different motives that lead to philosophizing, and the analogy makes itself known by the fact that one is led from the one problem to the other. It can be an ethical or religious interest that leads to philosophizing, and one then begins with the problem of valuation. But it then shows itself that for its treatment one needs a conception of the nature and limits of cognition, as well as of the existence within which the ethical and religious ideals are to be realized. It leads in any case to an untenable dogmatism when one, out of one's conviction about the rank and fate of values, will give laws for cognition and construct a world-view. The division of labor came forth precisely because thought, experience, and the values themselves required the overthrow of dogmatism.

Another motive for philosophizing can lie in the urge to gather one's experience and one's knowledge into a comprehensive world-view. One then goes straight to the problem of existence, but will soon discover that the possibility and validity of a world-view presuppose the validity of our cognition in general, and that its nature and limits must be investigated. Likewise it will show itself to be necessary to investigate the influence that our interest, our life of feeling and will, has on the shaping of our world-view, and this will lead to a special treatment of the problem of valuation. Finally, one can begin with the urge to distinguish between what one knows and what one does not know, hence with the problem of cognition, and from there one will naturally come over to the two other problems, which each presuppose the former. It is no wonder that the problem of cognition makes itself known more and more as the fundamental problem.—The different philosophies get a different timbre according to the problem that in them forms the starting-point and determines the points of view.

Yet one more problem it will be justified to reckon among the philosophical ones. All three problems named have a relation to psychology. And it is often psychological interest, an urge for observation and investigation of the mental life, that forms the starting-point for a philosophizing. One then goes from the investigation of the psychic energy to the investigation of the tasks it sets itself, and of its capacity to treat these tasks. The question whether psychology is a part of philosophy, or the whole of philosophy, or perhaps not at all a philosophical discipline, must, according to what has been developed in the first section of our presentation, be answered thus, that every philosophy needs a historical introduction, an investigation of the psychic energy that makes itself known in the life of thought as well as in the life of feeling and will. So far psychology is a part of philosophy. It leads us to the point where the three problems spring forth. Psychology describes the place and the presuppositions from which we orient ourselves in existence. The special branches into which psychology divides, and the special methods it applies to the individual objects[128], have their independence in relation to philosophy, and the concept of the nature of the mental life that the latter lays at the basis must constantly be brought into as great agreement as possible with them. The subjective, descriptive, and analytic psychology forms an intermediate member between philosophy and special psychology. Besides, psychology itself, as regards presuppositions and methods, is, just like all other science, itself the object of epistemological investigation. There is a constant reciprocal action between epistemology and psychology: the forms that the former sets up must be psychologically possible, and the conception of the nature of the mental life

[128] See on this my *Psykologi*. Kap. I.

that psychology maintains itself builds on the application of such forms. It holds of the mental world as it holds of the world in general: man is understood by the world; but all understanding of the world is determined by the nature of man. It is no circle, for there can take place, and there in fact takes place, a rhythmic alternation of the two points of view.—In accordance with this consideration, the psychological problem has been discussed as an introduction to our whole presentation.

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# A. Cognition

## a. Forms and Principles

106. The categories express the ways in which thought operates in reciprocal relation with the objects. They acquaint us with our intellectual organization. What we understand, and whether we understand anything at all, depends not only on the constitution of the objects, but also on the nature of our mind, just as it depends not only on the outer things, but on the constitution of our organ of sight, which colors we see, and whether we see colors at all. It was the urge and the capacity for unity and continuity that denoted the pervading character of the forms of thought. This urge and capacity can express themselves in other ways than such as lead to what we nowadays in science call understanding. Of all the wealth of thought that the human mind has at its disposal, only a narrow and firmly closed series of thoughts can be used when it is a matter of such understanding. Here Wallenstein's words hold: Eng ist die Welt, und das Gehirn ist weit. Through Animism and Platonism the way in the history of thought leads to Positivism. It is not traversed once and for all; often individuals must each for themselves go through it anew, just as in embryonic development an analogy to the development of the species shows itself. The psychological possibilities are more comprehensive than the epistemological ones. What the understanding of the objects requires must of course be psychologically possible, accord with the general laws of conscious life; but the presuppositions of understanding are of so special a kind that they can only be known through an investigation of the very work of gaining understanding. Science's principles and presuppositions express the intellectual habitus that poses the questions and criticizes the answers in a stricter and more exact way than where the involuntary intuiting and connection of representations prevails, and where our wishes and drives rule without being subjected to the discipline of experience and reflection.

To the different groups of categories correspond just as many different groups of principles.—As a fundamental principle appears the demand that *every object shall be considered in as many relations as possible*, combined with as many other objects as possible, in order that it may become as perfectly determined as possible. It is a demand or a guiding thought that springs from the very nature and mode of operation of psychic energy; it formulates a condition for perfect determination.—In a certain sense it is derived from experience: we experience that only in this way can we cognize, and that it is also really possible to cognize by this way. But it is an ideal we set up, an anticipation of every possible cognition that is to be worthy of its name. Purely factually regarded, it is a hypothesis that everything stands in relations (or, with the ambiguous expression, that everything is relative). And it is a hypothesis that has shown itself to hold, everywhere where we can exactly test the conditions. We meet nothing relationless on our way. So far we can here speak of a law.—To the formal categories corresponds *the principle of consistency*, which requires strict agreement between all the thoughts we form about the objects. A subject in a judgment cannot have predicates that contradict it; and an inference cannot conflict with the premises. It is again here an instinctive tendency that lies at the basis: the urge for agreement with ourselves, for not disowning ourselves when we pass over to new thoughts or objects. Either the new must be brought into agreement with the old, or the old with the new. Against this urge there can in fact arise another urge: to be absorbed in the moment's incitements and connections of representations, without these being tested for their agreement with what other moments bring. The principle of consistency too (which is more closely formed in the special logical principles) can be called a hypothesis, insofar as we start from the assumption that all objects must be subject to its demand. But it is a hypothesis we rely on to such a degree that where strict inference from the objects at hand leads us to self-contradictory results, we sooner assume our experience (the aggregate of our objects) to be incomplete than we give up the principle of consistency.—To the real categories corresponds *the causal principle*, according to which, when an object emerges, we look around for another object to which we can attach it in analogy with the way in which we purely logically attach consequence to ground (conclusion to premise). This principle too sets a demand that springs from the nature of psychic energy. Of course the demand would long ago have fallen away had it not to a certain extent and degree been honored by the objects' mutual relations. The causal proposition can so far be called a hypothesis or a law. But the chief point of view is and remains that it states a thought under whose leadership we can, in the world of experience, satisfy our urge for unity and continuity[129]. The two other real categories too have their corresponding principles: as we pursue the directions of the causal series, there arises *the question of formations of wholes and of stages of development*, and for a perfect understanding there is required an attempt to apply these points of view, the most special of all, and the most hypothetical. That they are nonetheless set up as principles is justified by the fact that every object presents itself to us, already at its first emergence, as a whole, and the understanding of it can then only be reached when it is investigated through what stages this whole has come to be. In reality all cognition begins with the question about this, at least as soon as the object is recognized and classified.—As for the ideal categories, the value-categories, the principles corresponding to them will show themselves to belong to the named classes: relation, consistency, causality, and totality. Valuing reflection has no other principles of cognition than understanding reflection. When a value-thought leads to the setting-up of a purpose, I cannot, without breach of the principles of relation and consistency, hence without coming into conflict with myself, refrain from seeking and grasping the means that are necessary for the attainment of the purpose. And the relation between means and purpose is a causal relation, hence falls under the principles corresponding to the real categories. That peculiar value-categories and a peculiar problem of valuation are set up comes especially from the fact that the relations that make themselves felt in valuation present special properties, and that the difference between the different values among themselves too requires special investigations.

107. That a principle is true means that one can work with it, and when the talk is, as here, of the principles of cognition, it means that by their help one can get forward in understanding, in firm and intimate ordering and connection of the objects. Truth can be designated as a work-value, because it comes to be for us only through spiritual work. When therefore principles are designated as demands, these demands are directed not only at the objects, but are also directed at ourselves. For if the principles of cognition really for an essential part proceed from our nature, then it is our duty—the duty of intellectual honesty—that we follow and apply them, as far as we are able.

Already the sense-qualities are working hypotheses (cf. 12). And still more distinctly it is so with the fundamental propositions of reflection. But it would nonetheless be a misunderstanding if one therefore regarded them as wholly arbitrary. Every working-tool must correspond to the hand that is to use it, and that has from the first produced it. And in a still more intimate relation than the mechanical tools stand to the hand,

[129]More fully I have developed the conception of the causal principle here intimated in my *Psykologi*, V D., 1–4, where I have especially shown that a complete experiential proof of it is impossible.

thought's principles stand to its own essence. One will become clear about this when one goes back from the principles that have in fact guided and guide human inquiry to the categories, and from these to the psychic energy at its different stages: it is this way that we have here traversed, only in the opposite direction.

Truth can then not be defined as our thoughts' agreement with existence. We know existence, after all, only through constant work at bringing the objects under our forms of thought. Reality, the truth of the objects, consists already for sound common sense practically in a close connection between as many exactly apprehended objects as possible (cf. 36–38; 68).

With this it hangs together that we understand movement better than rest. Just as both sensation and reflection must be awakened by a change, so it is through the laws of changes that we come to know the nature of objects. Rest we therefore conceive as stopped movement, as an equilibrium of tendencies. If all our objects were in constant equilibrium, cognition could consist only in recognition and ordering. Even if the objects did not move, thought would nonetheless move back and forth between them—if, that is, it could be the only movable element in a resting world.

There is no absolute opposition between thought and object. Every object that emerges in consciousness shows itself already to have undergone an elaboration that presents analogy with the one reflection can undertake (11). And on the other hand thought can make itself its object; a later thought can have its object in earlier thoughts, test and compare them. *Kant* committed the fateful error of assuming "the matter" (what I have called the objects) as absolutely passively received. But the relation between activity and passivity has infinitely many degrees. An absolutely passive, a mere matter, a pure sensation cannot be demonstrated. It would, as *Maimon* aptly remarked[130], be an "Idea" in the Kantian sense: we can, by abstraction from the thought-work and from the reciprocal action of the sense-elements, approach something such, but not reach it. Thus it would also require an inconcludable thought-work (abstraction is, after all, a work) to reach a chaotic difference-series (62), which would hold for the pure matter. Everywhere in our cognition we have different elements operating together. We can therefore never come to compare either our objects in themselves, or our forms of thought in themselves, with an existence subsisting in and for itself. And in any case such a comparison would itself be a thought-activity, determined for an essential part by our thought's way of operating—so that the question of "agreement" would constantly have to repeat itself.

The static concept of truth, according to which truth would consist in cognition's

[130] *Den nyere Filosofis Historie*[2] II. p. 121; 123.

agreement with existence, must therefore be given up—partly because it is self-contradictory, partly because we can in our reflection never reach a point where we have, on the one side, our own cognition with its forms, on the other side the absolute objects. The only possible concept of truth becomes the dynamic one, according to which truth makes itself known in the firm and close connection between our objects brought about by thought-work. The dynamic concept of truth is symbolic: even if we think of an existence subsisting in itself, cognition's relation to it can, as regards neither objects nor forms, be congruence or qualitative likeness, but can only be analogy.

108. So far is such a concept of truth from leading to skepticism that precisely the opposite holds. Since we can never have "existence itself" before us as an object, so that we could compare it with our thoughts and with "other" objects, there will, according to the static concept of truth, constantly be doubt possible about the validity of our cognition, and a doubt that could never ever be corrected. According to the dynamic concept of truth, on the other hand, we are at once set to work—to find new objects, and to find closer connection between the objects we know. And this work is precisely of the kind that psychic energy exercises. It is then no foreign norm we bow to.

The dynamic concept of truth is a consequence of the critical philosophy, for which the truth of our fundamental thoughts consisted in their making strict experience possible. At the same time this philosophy found the fundamental presupposition of cognition by analysis of such an experience. A comparison between cognition and an absolute order of things is impossible; we can only compare thoughts and objects among themselves. This consequence *Kant* himself did not see as clearly as some of his disciples (Maimon, Fichte, Fries). The master himself was still caught in dogmatism's static concept of truth, as is seen from his doctrine of the "Ding an sich" as absolutely existing apart from all cognition. But when he designates the causal law as an "analogy of experience" and thereby means that the events in time relate analogously as ground and consequence in our thinking, he approaches conceiving it as a working hypothesis determined by the nature of our thought.

Natural science embraced, right up to the most recent time, a static concept of truth. It shows itself especially in Newton's doctrine of space and time, in the physicists' and chemists' atomic theory, and in the natural historians' species-concepts. Just as Pythagoras projected the numbers, Plato the Ideas, Spinoza the Substance, so space, time, and atoms were projected as the expression of existence's absolute essence, although all these concepts are formed during the thought-work and for the use of it. It was from such standpoints a defect that *we* could not reach the pure world of the numbers, the

Ideas, or the atoms, or the absolute Substance or the absolute time and the absolute space. The critical philosophy, conversely, regards all such concepts as means and forms that come to our aid during our work at reaching the perfect connection we call reality. It is not "reality" that has fallen away from the pure world of "Ideas"; it is the Ideas that, through the thought-work they call forth, lead us forward toward more perfect concepts of reality.

109. There shows itself, however, at every stage a difference between something that is taken as given and the work we more or less consciously exercise on it. The dynamic concept of truth does not efface the difference between object and form. On the contrary, it impresses that the way in which the objects have hitherto been elaborated in science cannot be proved to be the only possible one. It has been possible to set up principles and hypotheses from which the objects can be brought into firm and close connection; but we cannot prove that the principles and hypotheses we set up are the only possible ones, the only ways by which such a result could be reached. The success that has followed science is no proof of its full truth, as long as it cannot be proved that no other presuppositions could have led to the same result. There is thus no absolute relation of identity, only a partial relation of identity (67; 69) present between the connection of the objects and the presuppositions from which it is derived. Ground and consequence cannot here change places.

Already *Leibniz* remarked this, without, however, letting it get a decisive influence on his doctrine of cognition. A hypothesis's perfect confirmation, he states, is reached only when one can from the conclusion return to the premise: cela donne *un parfait retour*, suffisant à démontrer la vérité de l'hypothèse. Mathematically, he remarks further, this is often possible, so that synthesis and analysis correspond to each other *à rebours*; but in the astronomical or physical hypotheses no such return can take place. Success does not prove the truth of the hypothesis. True conclusions can be drawn from false premises. It is, however, probable that the premises are correct when they are simple, and when from them one can ground many truths.[131]

This is a consideration that is of great significance for seeing human science in the right light. Kant brought it out in a more general form than Leibniz (in the utterance cited), in that he emphasized that all our cognition is determined by the human organization. Think of another organization—and another world-picture might then be the consequence!—When Leibniz's monads each apprehended the universe "suivant son point de vue," Kant's theory did not lie far off.—

[131] *Leibniz*: *Nouveaux Essais.* IV., 12, 1; 17, 5. (Opera phil. ed. Erdmann p. 381 a; 397 b.) Cf. already *Hobbes*: *De corpore* XXV, I.

110. What in the individual case is designated as object depends on the point of view. A sense-, memory-, or imagination-intuition can be regarded as object for reflection, while from another point of view it is itself formed on the basis of still simpler objects. The difference always rests on the fact that the moment of activity is more pronounced in the form than in the object. What cannot be derived from the very nature of our thought is present in higher degree in the object than in the form, at whatever stage we seek out this opposition, which never quite vanishes. Whatever ground this underivable-from-our-own-activity may have, it nonetheless testifies that there subsists something *besides* us. This is what we mean when we speak of existence as more comprehensive than thought. We must here reverse Wallenstein's words and say: Gross ist die Welt, und das Gehirn ist klein. When we distinguish between cognition and existence, it is because our cognition is always an open system. New objects can always emerge that can be derived neither from earlier objects nor from our forms. It is by reason of this constant possibility that we speak of an existence of which we ourselves are members. Both our objects and our forms point out to it when we ask about their origin. Our forms can, after all, themselves be regarded as objects.

Only most indirectly does existence make itself known to us, namely through the work that the emerging objects compel us to undertake. It is not only the objects, it is also the forms that must have their ground in a larger connection.

*Kant*, who otherwise so energetically holds fast the opposition between our cognition and "the thing in itself," and who as a rule merely lets "the matter" spring from this, nonetheless intimates in several places—what for the most part is not heeded—that our "forms" too have their ground in "Ding an sich." If one keeps to these intimations, it shows itself, as I have shown elsewhere[132], that Kant comes to stand nearer to the philosophy of experience than he himself thinks. For his epistemology comes, when the forms too, the constant in our cognition, are to spring from "Ding an sich," to rest on the presupposition that the unknown basis that contains the cause of the forms of our cognition (of that which makes existence intelligible to us) operates in a constant and law-governed way. The whole system of categories and fundamental propositions rests on this presupposition, which Kant has not brought out, but which makes his whole philosophy hypothetical and deprives it of the strict rationalistic character Kant has given it. He flattered himself that nothing hypothetical was to be found in his doctrine. Such a thing is found, as mentioned in an earlier connection, in the psychological basis for his

[132] *Om Kontinuiteten i Kants filosofiske Udviklingsgang*. (Vid. Selsk. Skr. 6. Række.) p. 33–35.—To the places there cited may still be added *Kritik der reinen Vernunft.*[2] p. 145 f and letter to Reinhold, 12 May 1789.—Cf. with respect to the question itself my *Mindre Arbejder*. II. p. 13 f.

derivation of the categories; but it is also found in his intimations about the last origin of the forms. And when "matter" and "form" can have a common origin, there shows itself the possibility of a greater harmony between cognition and existence than Kant assumed—while at the same time the a priori, the absolutely experience-independent in our cognition, is limited to a high degree. When we apply our forms of thought to understand existence, it is, after all, in a certain way existence that thinks itself in us—to use *Spinoza*'s and *Hegel*'s mystical-speculative language. Thought is a part of existence that, from its presuppositions, seeks to form an image of the existence in which it is a member.

That there is an open place in our cognition—that this is not and can never become a closed system—in this Kant was right, and at bottom far more right than he was conscious of. He was also right that this changes not the least in our thinking's procedure. We must follow the ways to understanding that are pointed out to us by the nature of our mind and our position in existence. For *Copernicus*, with whom Kant liked to compare himself, it was possible to sketch an image of how the phenomena we regard from the earth's standpoint appear from a wholly different standpoint. But something corresponding is not possible for the philosopher. The philosopher can become clear that our thought's forms and principles cannot be proved to be the only possible ones; but he cannot construct a cognition from a standpoint other than the human. Fortunately there is also enough to do of spiritual work from this one.

111. During the constant reciprocal action and breaking between object and form, the third element, the motive or the value, always operates along with them. The attention to, or the discovery of, an object is often due to a definite motive, and the same holds of the way the object is treated. Such motives can come from the organic urge for self-preservation, or from the urge for hope or for consolation. There must in any case be nothing in the whole feeling-state and sphere of interest that hinders the acknowledgment of the object and its methodical treatment. For there can be a relation of opposition between the three elements that leads to unrest and strife, not only in the race or society, but in the individual's own interior, so that the "head" leads him one way, the "heart" another. But objects could also emerge with so immediate a power, and thought's unfolding through the forms that are natural to it can take place so involuntarily, that all misgivings from the side of the third element are blown away. In this way objects and forms react back on motives and interests. Of significance especially is, as stated several times above, that a purely intellectual interest is formed that is immediately attached to the life of thought. The struggle between "head" and "heart" stands in reality between

two interests. As already stated, it is to the good fortune of all sides of the mental life that thought gets self-governance—that it is not regarded merely as a means, still less as an enemy. Only then is there tolerable prospect of getting all objects brought out, and all forms applied, that are necessary for gaining understanding, e.g. for applying the criterion of reality in an individual case. There will always be more to discover both in the world of objects and in the pure world of thought than any special practical motive can lead one to bring out. It is not only philosophy or other science that could need the reminder that there is more in heaven and on earth than it thinks; it is also, and surely to a far higher degree, sound common sense and the faith built on tradition and inner experiences. Existence is more comprehensive, and its connection far more intimate, than practical motives lead one to assume. The greatest enigmas for the view of life arise precisely at the great horizons that go far beyond the sphere of human life-interests, although thought, guided by intellectual interest, is able to discover them and seek to measure them out.

112. The differences between epistemological views are determined by the different relation between form, object, and motive that they maintain. It is the irrational in this relation that conditions the different directions in the philosophy of cognition. Could object, form, and motive be derived from one another, or did merely a definite character of one of these elements correspond to a definite character of the others, the mutual divergences would not be so great.—The different epistemological views can be traced back to three groups, of which the one ascribes to the form, the second to the object, the third to the value (the motive) the decisive significance.

I take, in the following discussion, account only of modern theories. The problem of cognition is essentially a modern problem, although it of course already made itself felt among the Greeks. For *Plato* the great enigma was how the Ideas, despite their opposition to the phenomena, could nonetheless be called forth in us by these. *Aristotle* required a special organ, "reason" (νοῦς), for the apprehension of the principles (ἀρχαί) that lie at the basis of all proof and therefore cannot themselves be proved (Analyt. post. II, 19). *Grote*, with whom *Gomperz* agrees, holds that "reason" for Aristotle only completes and concludes the induction that is to lead up to the principles, but he concedes that Aristotle emphasizes the opposition between induction and "reason" in a way that recalls Plato[133]. Rightly, no doubt, *Teichmüller*[134] lays the chief weight on the agreement between Plato and Aristotle on this point. Induction is, he holds, for Aristotle only a means to grasp the principles; but it does not prove them; it only gives occasion for reason, which stems

[133] *Aristotle.* II. p. 237 f; 275; 375.

[134] *Studien zur Geschichte der Begriffe* p. 409–426.

from a wholly different source, to draw out the general propositions. In the two great thinkers of antiquity, then, the irrational steps forth clearly. It was this that called forth the interesting skeptical directions in antiquity's later time.

The modern epistemological directions we can group under the three names Formalism, Empiricism, and Pragmatism.—

α. It is *Kant*'s great merit to have seen that one must first take stock of which forms scientific inquiry has at its disposal and in fact applies, before one can decide how they relate to the objects. It is in his footsteps that I have developed the doctrine of categories, before I passed over to the treatment of the proper problem of cognition. Kant is right that these forms must be determined by the nature of our faculty of cognition, as this happens to be. But in his attempt to give proofs for the fundamental propositions that spring from the forms, the leaning toward dogmatizing, which despite all his criticism he did not free himself from, comes clearly forth. He holds that the conditions without which we cannot have strict experience can without further ado be formulated as necessary propositions, and overlooks the great question in what extent and to what degree we really have such strict experience. His "proof" for the causal proposition, e.g., presupposes that the objects are given completely, and that their essential properties will continue to be the same. But this is an unjustified assumption, since experience is never concluded, and new objects can constantly emerge. Furthermore Kant presupposed that the basis or source of all objects and forms operates constantly, a presupposition whose necessity likewise cannot be demonstrated (see 110). It is no accident that his nearest successor struck out in a speculative direction; Kant's rationalism on the one side, and on the other side the danger of empiricism that certain points in his philosophy could contain, had to lead to attempts to let thought and its forms become sole ruler. "Reason does not beg, it commands"—this proud word of Kant's united with the age's religious and poetic currents into the peculiar phenomenon in the history of philosophy that German romantic speculation is. Romanticism's philosophy issued commands according to which all objects had to conform. One overlooked that all commanders are nonetheless at least compelled to direct their commands according to the capacity to obey that can be assumed to be present.—

It is by way of analysis that the critical philosopher seeks to find thought's forms and the fundamental propositions corresponding to them. But since one can never know whether the analysis is complete, and since it in any case must always stop at a definite point, there becomes something arbitrary in at which point one sets the beginning of one's philosophizing. It is then a choice, an act, that concludes the analysis and proceeds to the setting-up of the principles. This point of view is with the greatest energy asserted by *Thomas Hobbes*. “It is,” he says, “we ourselves who make it that the first principles of reason are true (rationis prima principia vera esse facimus nosmet ipsi). The first principles are not objects of science, but of art or construction.” Yet Hobbes concedes that the principles we thus establish we find only by analysis, and he sets up as a special demand that they must not contradict each other[135]. A similar position on this point was taken by *Fichte*: the first and unconditioned fundamental proposition of human knowledge—the principle of identity—is set up by a free act. Our science rests not on a “Thatsache,” but on a “Thathandlung,” namely the thinking consciousness’s resolve constantly to remain in agreement with itself. Fichte too concedes that a work of reflection and abstraction (analysis) precedes that act, and that when different fundamental propositions are set up alongside one another, it must be possible to unite them[136]. *Kierkegaard* saw no other means of getting reflection stopped, so that a systematic development can begin, than a resolve, a leap. The beginning of philosophy cannot be rationalized, and therefore there is no presuppositionless philosophy[137]. While the three named thinkers expressly acknowledge that the act of will by which the principles are established presupposes an analysis (reflection, abstraction) by which it first becomes clear which principles must be established, *Kroman* sets up thinking’s principles by a pure postulate, a postulate that is, however, motivated by the fear of self-dissolution through the assumption of contradictory assertions and by the hope of finding stability in the course of things[138].

What is not sufficiently emphasized by the kind of critical philosophy that one could call the arbitrariness-theory is that there can in each individual case be a rational ground for stopping reflection and beginning to apply what it has found. For when our thinking has a definite task, e.g. to find the basis for a cognition’s validity, we stop reflection when we have found what is necessary for solving such a task. It is thus no purely arbitrary act. And in Fichte and Kroman it is moreover expressly conceded that in the first principles of cognition our own, the human nature must lay itself bare, since it is the endeavor to preserve the unity of our being that is decisive. It is the fundamental law of synthesis that is acknowledged. But this fundamental law is itself discovered by reflection on our mental nature, as this in fact shows itself to be; and likewise it will be impossible to find the principles for the understanding of the objects if we do not investigate these objects’

[135] Computatio sive Logica (1655) III, 8–9.
[136] *Grundlage der gesammten Wissenschaftslehre* § 1–3.
[137] *Uvidenskabelig Efterskrift* (Saml. Værker VII) p. 84.
[138] *Vor Naturerkendelse*. p. 270–274. By the emphasis on the motive Kroman approaches Pragmatism.

factual constitution. Nor can this form of criticism, then, free itself from a relation to experience.

β. Empiricism, which, when it takes account not merely of the individual's but of the whole race's experiences, becomes evolutionism, regards the objects as sufficient ground for cognition and especially for the setting-up of the first fundamental propositions. General principles can have arisen through the constant action of the objects' constitution. The objects deposit their general properties in the individual's or the race's nature, or rather in their habits. And then it is a matter of course that no principle has validity beyond the circle of objects that has imprinted it. Only as results do the principles have significance, while for Kant they were grounded in the nature of consciousness, and for Hobbes and his successor were essentially postulates. *Hume* and *Stuart Mill* have developed empiricism in its most radical form, in which all objects appear in consciousness as constituting an incoherent chaos, so that they enter into such firm connections that can operate as principles only through external association and habit. *Herbert Spencer* extended this way of considering from the individual to the race by virtue of the theory of development; those firm connections could, according to him, not form themselves in the course of the individual's life, but only in the course of generations.

Empiricism overlooks that we know the objects only in constant reciprocal action with the forms, just as, conversely, we know these only in constant reciprocal action with the former. Thereby it is understood that there is a constant moment of expectation, anticipation, seeking, and questioning in our cognition. One cannot demonstrate any absolute passivity in any living being. There would, as we have seen in the discussion of the chaotic difference-series, be needed an infinite analysis if its correctness were to be demonstrated. At every stage of psychic development we know, there are always certain inner conditions present, on which it depends both how the objects that are accessible appear in consciousness and how they are elaborated. This holds for each individual and each generation.

γ. Pragmatism lays the chief weight on the fact that the objects determine the forms' application and specialization. Cognition is a constant adaptation; the point is, as *Avenarius* has expressed it, to think the objects with as little expenditure of force as possible. Here, then, a psychic energy is presupposed, without its being more closely investigated whether it does not by its nature operate in certain definite ways that can in the individual cases be formed according to the objects' peculiarity. One keeps, in pragmatism, to the fact that we form the concepts and principles we need, and no others, and therefore we also seek to limit the formation of concepts and principles to the least possible. The principle of simplicity, which has played so great a role in the setting-up of new points of view and hypotheses, is derived from the economy of consciousness. We wish, through cognition, to make ourselves master of things, but with as few costs as possible. *Ernst Mach* has, on the basis of the history of natural science, developed this theory. *Avenarius* has developed it in connection with an investigation of the nature and history of problems in general. From a somewhat different side pragmatism is developed by *William James*, who especially emphasizes that we leave only for practical reasons the involuntary continuous stream that our mental life from the first forms. (Cf. 3.) We set up, for practical reasons, definitions and fundamental propositions, and the weight must be laid on the motives, not on the forms. While Mach and Avenarius keep mostly to the theoretical motives, James and his successors emphasize mostly the practical motives, although they protest against being supposed to overlook the theoretical interests.

The chief question in the face of pragmatism becomes whether cognition's form and motive can be distinguished in as external a way as it holds, and in such a way that the forms are essentially means. Whatever motives may guide us, they can nonetheless accomplish nothing when the capacities and forces are not there. There can never have been a stage of development where the motives operated independently of the forms. In the involuntary and immediate life, which James has so masterfully described, there is no difference between motive, form, and object; they operate together in an inseparable way, as in instinct. If it is an abstraction to isolate the object, as empiricism does, or the form, as criticism does, so it is also an abstraction to isolate the motive, as pragmatism does. In instinct an object at once releases an urge and an operation. Urge and capacity are here one. And they are so much one that there can be an urge to use the capacities or forces for their own sake, apart from every special motive. The intellectual interest, which we have in the foregoing so often mentioned, is an example of this kind of operation; yet it can also, by displacement of motive, develop out of other interests.

The truth in pragmatism consists in something that the preceding presentation of the doctrine of categories has presumably let come into its full right, namely that the special development of the forms is determined by the tasks set to thought. Therefore a distinction was made between fundamental categories on the one side, formal, real, and ideal categories on the other side. But already the fact that tasks can be set, and that they are set in a certain definite way, presupposes that besides urge and motive there is also the possibility of definite modes of operation present.

The economy according to which the history of thinking operates is not merely the fruit of a peculiar urge for thrift. It is understood not merely by the motive, but also by the form of thought's operation, as this lays itself bare in all domains. The principle of simplicity and the urge for unity has its last ground in the essence of consciousness as synthesis. The combined is always simpler than the unconnected, the chaotic. All thinking is so far in its nature economical, and biologically regarded one will only thereby be able to understand that thinking beings have been able to subsist in the struggle for existence. Thereby the external relation between form and motive falls away in the last instance. The intellectual feeling of displeasure, or perhaps even feeling of pain, that can arise from lack of connection or from self-contradiction and doubt, is surely due not merely to a feeling of impotence or vexation over waste of force, but is ultimately to be traced back to the fact that the very innermost essence of consciousness is violated[139].

Only through work do the forms of thought develop in the particular. And this work will succeed best when the intellectual feeling, determined by the urge for unity, continuity, and connection, is the decisive motive. Only from the moment such an interest is awakened does thought exist as an independent mental power, becomes more than a means. And even where it is to operate as a means for other interests than the intellectual, it will operate best when this latter provisionally takes the helm. Resignation is required in order to cognize, and the straight way is not always the best. In epistemology we in any case presuppose a stage where the intellectual interest prevails. Every task requires, after all, not only certain forces suited to its solution, but also motives that are as closely attached to the task itself as possible. If there are then other motives and interests that demand satisfaction, they must try to find this under the conditions brought about by the solution of that task. The spiritual division of labor cannot be abolished, however many difficulties it brings with it. There must be sought other ways to remedy these difficulties than that of making the thought-work dependent on principles other than its own.

113. Motive, form, and object operate in every act of cognition together with a timbre that can vary without any of the three elements ever quite vanishing. The difference between what one calls theoretical and practical thinking is only a difference in the timbre, respectively in the relation between the three elements by which this is determined. And thus too the difference between the three epistemological directions discussed is a

[139]The principle of simplicity has been conceived now as a maxim of thrift (frustra fit per plura quod fieri potest per pauciora): thus in *Will. Occam* (in the 14th century), who appeals to it,—now as a metaphysical proposition, holding for existence; *Copernicus* and *Newton* were thus convinced that nature followed the simplest ways. *Kant* found it, on the other hand, grounded in the very nature of our cognition and found it operative every time we form the simplest judgment. Already long before "Kritik der reinen Vernunft" he explained in this way "favor ille unitatis, philosophico ingenio proprius, a quo pervulgatus iste canon profluxit: principia non esse temere multiplicanda præter summam necessitatem" (*De mundi sensibilis atque intelligibilis forma et principiis*, § 30.) Cf. *Kritik der reinen Vernunft.*[2] p. 680.

difference in the timbre; more than that it should in any case not be.

But there is one moment at which criticism will always retain a prominent place in the epistemological discussion. It is the weight it lays on the analytic or regressive method of finding cognition's forms and principles. An investigation of this kind is, as shown in the foregoing, the presupposition both for dogmatism, arbitrariness-theory (postulate-theory), empiricism, and pragmatism, because it concerns *what* is to be set up as dogma, as postulate, as experience, or as good economy. Here is something every epistemology must provide. And here is something that can be separated out from the historical forms in which criticism has from the first appeared—both from Kant's dogmatizing rationalism and from the postulate-theory's emphasis on the leap. Thereby it will be able to let both motive and object come into the right they really have.

The concept of working hypothesis can very well accord with this consideration, when one lays weight on the fact that the way in which human thought works must be determined by its nature. And its nature we come to know by the study of the involuntary mental life, as psychology by the help of its different methods investigates it, as well as by the study of the history of thinking and of science. In the latter respect it shows itself that there is a certain type that constantly appears in the scientific principles and hypotheses, indeed even in mythological representations, a type that leads back to an urge for unity and continuity. All investigation of human thought is an investigation of the nature of this urge and its struggles to gain satisfaction.—Also in the more special assumptions that one as a rule first thinks of at the word working hypothesis, that stamp will be found. Such assumptions are merely more limited and adapted forms of the general type.

## b. Forms and Objects

114. We now go more closely into investigating the irrational relation between form and object that has already provisionally shown itself to be present.

Our highest, most transparent principle of thought, the principle of principles, the principle of identity, which makes it possible to put one concept in place of another and thereby abolish the differences that the objects present, meets resistance where we must stop at a plurality of concepts that can perhaps indeed be ordered in series, but not in such as make substitution possible. It was a romantic ideal that required all cognition to be derived from a single principle, which thus would have to be applicable at every single point in the series of objects and at the objects' transition into one another. The world, the world in which we are to think, is and remains a multiplicity, not a world of resting identities. And on this it rests that it is a living world. Nevertheless the principle of identity will constantly stand as a direction to work, namely to bring about as favorable conditions for substitution as possible—to find ways by which one can in each case conceive the differences as disguised identities.

The points of view from which different objects can thus be regarded as identical cannot, as stated several times, exhaust their whole nature. There will always make themselves felt oppositions that are determined by the limit of identity. The consideration of the most important relations of opposition of this kind will serve to illuminate the nature and conditions of our cognition.

### *α*. Quality and Quantity

115. As we have already seen in the doctrine of categories, science, in the face of irreducible differences, resorts to the expedient of putting, by way of analogy, quantitative differences in place of qualitative ones. It is the qualities time, number, degree, and place that can most easily be reshaped in such a way that the principle of identity can hold in the series they form, so that substitution can become possible. And in analogy with these series the other quality-series are ordered, so that they—as far as the analogy holds—can come under the principle of identity. These series can then be constructed by constant addition of identical members, just as we construct time-series, number-series, scales of degree, and geometrical figures.—It is first and foremost natural science that in more recent time has proceeded by this way. In psychology a similar method can be applied only in an imperfect degree (see 12–13).

That we can, in natural science, put quantitative changes in place of qualitative ones comes from the fact that we have several different senses, and that in some of these differences of magnitude are more clearly prominent than in others. Thus touch- and movement-sensations have a more quantitative character than sight and hearing—that is: the qualities of the former can more easily be interpreted as having arisen through the joining-together of identical elements. The qualitative differences in touching a larger and a smaller surface are easily seen to hang together with the difference of magnitude between the two surfaces; and likewise the qualitative difference between the movement by which I touch my head and the one necessary to take my hat from a high stand can easily be seen to hang together with the smaller or greater extension of my arm. Movement-sensations are connected with most other sense-sensations, and thereby measurement of these can become possible. Where merely qualitative estimation leads to disputes, measuring-out by movement—not only our own movement but also movements in the surrounding world that hang together with the object the estimate concerns—can turn the scale.

Numbers and figures can be constructed, in that they arise by repetition of the same act of thought, the one by which a unit is represented. New combinations can, by virtue of the repetition of the identical, be found out, without one needing to look to experience. Thought is therefore more self-active in the case of quantitative differences than when it stands before qualities. These it must wait for; those it can anticipate, merely by repeating, according to certain rules, the same acts of thought. *Kant* has clearly expressed this (in the remarkable letter to Herz of 21 Feb. 1772): “Only thereby do things become magnitudes for us, can they be represented as magnitudes, that we can produce the representation of them by repeating one thing a number of times. Therefore the concepts of magnitude can be self-active, and their fundamental propositions can be found a priori. But in the case of qualities the question how my understanding should, wholly a priori, be able to form concepts of things always leaves an obscurity with respect to how our faculty of understanding should be able to agree with the things.”

Already *Aristotle* taught that movement in space (φορά) lay at the basis of all other change (μεταβολή), whether this consisted in change of quality or in increase and decrease (αὔξησις καὶ φθορά). But he nonetheless combated *Democritus*’s and *Plato*’s attempt to trace all change back to the movement of small parts. Partly he maintained the originality of the qualitative differences, partly he found the teleology in nature violated by the atomistic mechanics. Only modern natural science made earnest of laying the quantitative properties at the basis. *Galileo* stands as the great founder of what one (in the narrowest sense) usually calls the mechanical conception of nature. Besides the simplicity that is attained by having merely to do with differences of magnitude—and Galileo was sure that nature follows the simplest ways—there was also gained the advantage that one could attain greater exactness in one’s descriptions and therefore greater certainty in one’s inferences. Therefore Galileo wanted everything measured that was measurable, and to make measurable what was not. He even considered the quantitative properties to be the things’ “first and real properties,” while the qualities were supposed to be only appearance. And this was in the following time the usual conception among the representatives of the mechanical conception of nature. The principles according to which natural science worked were conceived as the expression of absolute realities. One made the analogy between quantities and qualities into identity and saw in movement and the elements that moved a world that was the true material reality. This world was, according to *Descartes* and *Newton*, immediately created by God, while materialistic authors like *Holbach* held that it had subsisted from eternity as the only true reality. The physicists were on this point metaphysicians right up to the most recent time.

116. From various sides it can be shown that a complete solution of our problems is not reached by laying the mechanical conception of nature at the basis in a dogmatic way.

Even if we thought that everything in material nature could be explained according to this conception's principles, the qualities nonetheless remain standing as immediate facts, and there is given no explanation of how the purely quantitative differences can appear to our senses as qualitative differences. The chemical product's properties cannot be derived from the elements' properties, and when one kind of physical energy gets its equivalent in another kind of physical energy, the equivalent nonetheless has other properties than the first energy-form. The sensing subject cannot, after all, create these qualitative differences out of nothing. In any case one would thereby put insoluble psychological difficulties in place of the chemical and physical difficulties one pushes aside.

It must moreover not be forgotten that there is only a difference of degree between differences of magnitude and other differences. The former appear immediately just as much in a qualitative way as the latter. Extension and movement are qualities that can just as well need an explanation as the specially so-called sense-qualities. This point was especially emphasized by *Leibniz*, *Berkeley*, and *Kant* against "the mechanical mythology." We have no ground to believe that we, when we have come to the quantitative properties, are at the end, have the innermost essence of things before us. The purely methodical significance of the mechanical conception is very well compatible with only dynamic and symbolic truth being ascribed to this conception. Its significance consists in its laying at the basis a circle of symbols that can be applied with full consistency—in its having found an analogy by whose help inferences can be drawn that are confirmed by new experiences, which can again be expressed by help of the same symbols. But the possibility is never excluded that another circle of symbols or models could just as well, if not better, express the experiences and form the basis for exact inferences. There will never be able to be given a proof that any circle of symbols is the right one or the only possible one.

### β. Succession and Causality

117. The real ground of the grand attempt at conversion of quality into quantity must be sought in the fact that the causal relation is not clear and transparent so long as there continues to be a difference of quality between the event that stands as cause and the one that stands as effect. When thought seeks after a cause, there are two things it wants clarity about: why an event occurs precisely in this definite moment, and why it appears with this definite qualitative peculiarity. In respect both of time and of quality, the single event is sought to be determined in relation to other events, so that it can be inferred from these. The difficulties one has found in the concept of cause concern precisely the differences of quality and of time. With what right do we assume that, because the event A occurs in this moment, the event B must therefore occur in a later moment? How can events of different character and from different times be so connected that they cannot be separated?

The most important answer will be the counter-question: with what right does one assume that there are absolute differences of time and of quality? Should it not be possible to order the events in such series that from the earlier members one can infer the following ones?—This was the way *Kant* went, in that he pointed out that there was indeed no identity, but there could yet be analogy, between the series of events in time and the series of grounds and consequents (premises and conclusions) in logical and mathematical thinking. The real categories could thus be conceived in analogy with the formal ones. And—Kant adds—only because this analogy holds can we distinguish between dream and reality. If Kant had here said "in so far as" instead of "because," he would have been right. He moreover proceeds dogmatically from the assumption that this criterion of reality can be applied everywhere, in all changes in our objects. He presupposes, in other words, the validity of the causal principle at the very moment he wants to prove it. But the question will pose itself anew at every new change. That was surely also what *Hume* would have answered Kant. Even if the very same situation came again, one would not, he held, have the right to expect the same consequences; for it was not certain that the same things were now in possession of the same power as when the situation was present the previous time. And here no matter how many repetitions will not bring us further: "there will still be occasion for the same question, and that to infinity."[140]

Kant has, without more ado, made a principle or a hypothesis into a law. His apt connection of the causal problem with the criterion of reality (where he moreover has

[140]Cf. *Hume*: *Treatise on Human Nature.* I, 3, 6. (ed. Selby-Bigge p. 91.)

all more recent philosophers, right from Descartes on, as predecessors, see 37) does not justify going further than to maintain that we here have a means of thought and a form of thought that we must continually try to apply, already for the reason that we could otherwise not orient ourselves in our surrounding world. But the concept of reality is and remains an ideal, to which only approximation is possible. And therefore the concept of cause too designates an ideal of thought that can never find exhaustive conditions for its application. A complete confirmation the causal principle cannot get, and when it is called a law, this can strictly speaking hold only for the cases in which it has shown itself to be sufficient guidance for us. Its most important significance is constantly to lead us on to new tasks. And where we cannot solve these tasks, we have only the right to say that we have not found the cause, not that there is none. The possibility was not excluded that we could indeed apply the criterion of reality to establish the truth of a series of events each for itself, but could not find any connection among the events mutually (so that these form only a series of kind (2) (see 63), when the chronological order is assumed determined). It is such a situation that Hume seems to have imagined, in that he does not presuppose doubt about the reality of the single events each for itself.

Even if the application of the criterion of reality is forced upon us by practical grounds, and even if we must also apply the concept of cause in practice by seeking means to our ends, still causality is not a means of thought in a purely external sense. Neither is there here any purely external relation between form, need, and object. Our thought's peculiar forms it will have a spontaneous need to apply, without its needing to be conscious that it is setting up a hypothesis and carrying out an experiment. To Hume's question: Why do you apply the causal principle to new events?—the answer could be: Why not! Let us try our powers, try how far we can get with our thought's forms and means. We cannot behave like the man who would not go into the water before he could swim. And when the objects show themselves capable of being ordered in such series as the causal principle requires,—namely in series that are ordered analogically with a series of grounds and consequents,—then we also get courage to continue with our attempts.

Kant's discovery of the "analogies of experience" will be science's last word on the significance of the causal principle, although he himself believed that this last word contained more than it does. The experiences of daily life are here connected with science's highest investigations under one and the same type of thought, which hangs closely together with the inner organization of our thought. We get through this point of view a constant summons to investigate the objects as closely as possible, so that the analogy can be carried through as far as possible. In the logical relation between ground and consequent, differences of time and of quality play no role, and for the analogy to be carried through one must therefore reduce such differences between the objects to the least possible. Thought seeks to shape the objects into likeness with itself—thereby is attained both the greatest intellectual satisfaction, and there is won a more perfect means of orientation.

118. The relation between analogy and identity nevertheless brings with it still more questions with respect to the causal problem. From the empirical side it has been maintained that it has absolutely no significance whether the changes that are to stand in causal relation are mutually dissimilar. *Stuart Mill*[141] thus quite rejected the old proposition that like produces like. He regards it as an expression of superstition and points to the many cases in which one has placed things together as causes and effects merely because they had a certain outward likeness to each other—as when yellow flowers were supposed to be able to cure jaundice!—Of course the likeness must be such that the analogy with ground and consequent really fits. The more recent science aims at approaching the analogy to identity in as high a degree as possible. It is more than superstition, also more than a mere effect of experience, when thought sees in every difference of quality between events a task set for it; it is because thought is indeed awakened by the differences, but thereafter seeks out beyond the tension and wonder called forth by them. Wonder is only the beginning of wisdom. If one establishes that a causal relation can be wholly clear and comprehensible even though the members are ever so different, this will lead to one's coming to rest too early. One has then pulled out the sting that spurs to progress.

It was *Comte*'s and *Mill*'s mission in the philosophy of the 19th century to maintain the differences of the objects and their right over against speculation's zeal to find "the unity in all." *William James* continues their work in the most recent time. But in this, nevertheless, *Schelling* and *Hegel* were not mistaken, that it is unity we seek. Their error was that they too rashly believed themselves to have found it, and next, that they proclaimed the possibility of deriving the multiplicity of objects from the unity. But the doctrine of plurality (pluralism) contains just as much a danger as the doctrine of unity; it too easily stops too early. The life of thought develops only under the constant conflict between the tendency to multiplicity and the tendency to unity, and the highest fundamental principles for cognition must be such as contain directions for a labor under this condition.

*William Hamilton*, who conceived all categories as expressions of our imperfection,

[141] *System of Logic*. V., 3, 8. (10th ed. p. 336–344.)

because they show that we cognize only by setting the one thing in relation to the other, conceived it consistently also as an imperfection that we seek after causes. It was supposed to witness that we lacked the capacity to apprehend change as an increase of reality. It is in accordance with this imperfection that we seek to apprehend the new as a conversion of the old into a new form. There is this truth in it, that when such a conversion has taken place, the difference is evened out for us; there is now needed less energy to hold the objects together; there is, after all, won a point of view from which they can be regarded as identical. And here it is then indeed possible to end and to come to rest. But it is also possible that the equilibrium won and the energy saved can be applied to new tasks, so long as new objects present themselves. A rhythmic alternation of awakening and evening-out is characteristic of the life of thought.

119. From quite other points of view than the one that led Hume to pose his problem, one has in the most recent time criticized the concept of cause and demanded its elimination. It is the time-relation that this concept, on the usual conception, contains, that has called forth this criticism. It is indeed especially the time-relation that makes the causal relation different from the relation between ground and consequent and offers only an analogy with this.

From the speculative side it has been urged that so long as the time-relation is determining for our conception of existence, our conception is imperfect, and an absolute cognition is not reached. And our cognition works itself—one further maintains—more and more away from the time-relation: the more clearly we understand something, the smaller a role the differences of time play,—the more the real knowledge merges into the formal, the cause into the ground. What we at the beginning called cause and effect comes, with cognition's advance, to stand as members in a whole, a whole of thought, and the relation between them becomes purely logical, so that the time-sequence is inessential. That we first observe the one member, then the other, has now no significance; for in our cognition of the law-governed connection the whole process stands before us as a unity. Not the time-relation, but the unity behind and in spite of the time-relation, binds what we call cause to what we call effect. And there is no interval between the end of the event we call cause and the beginning of the one we call effect.[142]

From another point of view one has come to a similar result. The interest here goes to sorting out all the accompanying representations and auxiliary concepts with which the objects, what is given for thought, are involuntarily or on methodical grounds supplemented. Cognition is to approach as soon as possible a pure description of a

[142] See especially F. H. *Bradley*: *Appearance and Reality* (1893). Chap. 4–7; 18. — Similar train of thought already in Plotinus (Drews: *Plotinos*. p. 181.)

continuous process, within which the differences are reduced to a minimum. This is attained the more quantitative determinations replace qualitative ones, whereby it becomes possible to demonstrate relations of equivalence between the objects that replace one another in experience. It is then no longer the task of strict science to "explain"; whoever wants "explanations" is referred to mythology or metaphysics. Science's goal is to give an exact, methodical description of all relations and transitions in experience. As now especially regards the causal relation, the objects' relation of dependence is expressed better by help of the mathematical concept of function than by the indefinite expressions "cause and effect." Moreover there is this unfortunate feature in every application of the concepts cause and effect, that two objects are arbitrarily taken out of the whole continuous connection in which they occur, and set in opposition to each other. And the time-difference between "cause" and "effect" has no significance of principle when the relation is formulated as a mathematical relation of function, expressed in an equation. That time cannot be "reversed" really means merely that the changes of the physical magnitudes take place in a definite direction.[143] —

The epistemological views that thus, with different motivation, would eliminate the concept of cause and—in connection therewith—deny the significance of principle of time, can be characterized by the fact that they put an ideal of thought in place of the actual cognition we can possess at any given time. Every investigation must begin at a certain point—in so far, with a leap. But it is a leap that is made precisely at the point where the problem presents itself, and where wonder is awakened. That elimination really brings with it the abolition of all problems and of all wonder. The accidental element in the point of departure, "the leap," is really present in every act of thought. In an inference it is arbitrary that we begin precisely with these definite premises and seek what follows from them, and likewise it is arbitrary that we stop at a certain proposition and seek precisely its presupposition. All thinking presupposes limitation. Our cognition develops historically, because attention is awakened under certain definite conditions and cannot be directed all at once toward all objects in their successive appearance. Naturally it is of importance that one becomes conscious of the accidental or subjective element in the point of departure or the section, and if the labor of thought has progress, the section too will of course become worked into a great continuous connection or whole; only thereby is the full understanding won.

The historical character of our cognition becomes especially clear when we reflect that we always stand between the experiences of the past and the possibilities of the

[143] *Richard Avenarius*: *Kritik der reinen Erfahrung.*[1] II., p. 332–335. — *Ernst Mach*: *Zur Analyse der Empfindungen.* p. 168. *Die Mechanik in ihrer Entwickelung.*[6] p. 259 f.

future. In each single moment we must distinguish between what lies given and what we expect, and the expectation can never become more than a hypothesis. The single moment, in which toward the one side a “no longer” holds, toward the other side a “not yet,” poses the problem to us in its whole tension, which only the power of habit often diminishes. With new problems the tension cannot in any case fail to appear—and thought seeks and constantly finds new problems, so long as experience is continued. The full connection lies before us only afterward, and until it lies before us, the assumption of it stands as a hypothesis. We shall therefore never be able to do without concepts like force, energy, cause, possibility, which with different nuances express one and the same relation: the relation of the earlier-appearing objects to the later, apprehended in analogy with the relation between ground and consequent.

The matter is not changed by the relation between ground and consequent being expressed in the form of a mathematical equation. The equation is, after all, only a special kind of logical judgment. It expresses, like this, a timeless relation. For there is no time-difference between subject and predicate, or between ground and consequent. Between the logical and mathematical relations on the one side, the time-relations in the objects’ appearance on the other side, there can therefore always be only analogy, not identity. But this analogy nevertheless makes the empirical science we have possible, even though it leaves great questions standing behind.

120. One can therefore in no way say that Hume’s problem has been solved by the discovery of the conservation of energy, which makes it possible to formulate the relation between the different forms of energy as equivalence. The relation of equivalence excludes neither difference of time nor difference of quality, just as little as the relation of identity needs to do so (see 69 and 73). When A and B are equivalent, it is in a certain sense indifferent whether I pass over from B to A or from A to B. And yet in each single actual case it is either A that passes over to B, or B that passes over to A,—and this difference can be a difference between life and death. In the Carnot theorem (the entropy theorem) this comes clearly to the fore; here the direction is precisely decisive. Heat and motion are equivalents; but it is in fact more difficult to concentrate the heat in such a way that it can again be converted into quite the same quantum of motion than to carry out the reverse conversion. It is a great abstraction to disregard this difference. The Carnot theorem does not conflict with the theorem of the conservation of energy; it is precisely one of its predecessors. But it leads over to a more concrete problem, leads us into nature’s real economy, where precisely the directions of the causal series become of significance. An example from the human domain, corresponding to the Carnot one from

the physical domain, is offered by the relation between genius, exemplariness, new fresh contributions within human development, and the dominion of imitation and repetition. Between discovery and imitation one could imagine a relation similar to that between heat and motion, and here too there could lie a danger in the tendency to evening-out and distribution.[144]

*Ernst Mach*, who otherwise does not lay great weight on time's not being able to be "reversed," because it is "only" the direction that becomes different, nevertheless occasionally declares that the natural processes have a historical character, and that in this there lies a great problem. "When we say that time flows in a definite direction (Sinn), this means that the physical (and thus also the physiological) processes are completed only in a definite direction (Sinn). All differences of temperature, electrical differences, differences of level in general, are diminished,—not enlarged,—*when they are left to themselves*. If we consider two *bodies left to themselves*, touching each other, of different temperature, memory will offer greater differences of temperature, observation smaller, not the reverse. *In all this there expresses itself throughout only a peculiar, deep-lying connection in things*. But to want to demand complete clarity here already now is the same as, in the manner of speculative philosophy, to want to anticipate all future special research's results, thus to demand a completed natural science."[145] Here it is thus acknowledged that not all riddles are solved by the demonstration of equivalence. There is a historical element that one cannot get away from. The great question is to what degree the direction of the causal processes is left to itself. Just as the energy theorem can only be demonstrated within connections that are thought as closed, so that no supply or departure of energy can take place, so it goes also with the entropy theorem. By both ways it shows itself that it is an abstraction when we hold to the causal processes without taking account of the real whole within which they take place, and which can contain tendencies of opposite direction.

121. Although I have sought to show that the historical element cannot disappear from causal research, the immediately preceding consideration shows that it is first the concepts of whole and development that properly emphasize the historical. When research stands before objects that lead to the application of these categories, it stands before its most difficult tasks, and at the same time before its last tasks. Concerning the questions that here arise, there has partly already been discussion in the doctrine

[144]Cf. G. *Tarde*: *Les lois de l'imitation*. p. 412–415.

[145]*Die Mechanik in ihrer Entwickelung*.[6] p. 239 f. — [The emphases are mine.] — In his learned and acute work *Identité et Réalité* 1908 *Meyerson* has set the entropy theorem in sharp opposition to the principle of equivalence: the former witnesses to time's reality, in that the members cannot be exchanged, while the latter tends to abolish time and finally all difference.

of categories (93–99), and in the section on world-view we will come back to them. Here—under the problem of cognition—the matter concerns especially the principles for cognition that those categories lead to asserting, and this side of the matter we can investigate in close connection with the causal problem.

The analogy on which the causal principle is built becomes the more difficult to carry through, the more concrete and individual the object is that is to be explained, and that is to say: the more it constitutes a whole and runs through a development. There is here a whole scale with rising complications from simple locomotion through the physical and chemical processes to the organic, psychical, and social phenomena. The further we come forward in this series, the more the causal principle takes on the character of an ideal demand that we can satisfy only with a slight degree of approximation. It becomes ever more difficult to resolve the composite processes into simple causal series,—ever more difficult to resolve the single complicated objects into elements, each of which for itself could be regarded as a point of departure. Ever more difficult therefore it also becomes to establish that it is quite the same objects and the same states we have before us when a process seems to repeat itself. The old proposition that nature is always in agreement with itself (sibi consona, as Newton said) and makes no leaps, seems often to be contradicted by observation. Precisely the most recent time's research yields examples of this. In radium there was discovered a substance whose energy-discharges could seem to conflict with the theorem of the conservation of energy. In opposition to the older Darwinism one maintains, on the ground of observation, the sudden arising of new types (by mutation), which for the present cannot be established to be the result of a successive process. (Cf. however 92.) A master in descriptive psychology like *William James* maintains that there can occur new contributions in the mental development which could not be derived from the preceding processes, and prepares us on the whole for the fact that we must often content ourselves with a purely external connection of objects, a mere "conjunction" without rational connection, an "and" without a "therefore."

Such cases nevertheless do not lead to giving up the causal principle. They pose problems—precisely by virtue of the causal principle. Even if, besides differences of quality and of time, types of whole and series of development offer ever so many riddles, still thought has after all no other means than the causal principle with which to illuminate them. Yet there are some special fundamental principles of a methodical kind, which hang together with the categories whole and development, though they will in the particular show themselves at bottom to be applications of the causal principle.

α. Where a process seems to appear isolated, "left to itself," it must nevertheless always be pursued to see whether its direction does not lead to combination with other processes, which together with it could make possible a development of a whole. It must in general be set up as a fundamental principle that no element and no process is isolated. Isolation is due to an abstraction, an "accidental point of view." There will naturally also be occasion to investigate whether the apparently isolated and simple process is not due to a separating-out from a whole that has perhaps entered into dissolution.

It has been necessary to resolve the natural connection in order to find the laws that hold it together. These laws could not have been found except by, as far as possible, isolating single objects, elements, or processes. But this abstraction cannot be cognition's last word. It is not merely Romanticism, it is clear reflection itself upon science, its presuppositions and its method, that leads to demanding that the problem of the whole be heeded, whether it can now be solved or not.

β. If, conversely, it is a whole that is given, the task will be to see it as a "samfoldighed" of laws and processes. Where in the former case a synthesis was required, here an analysis is required. To let the whole explain itself, by presupposing mystical unities and thoughts of purpose, gives no explanation—save on the animistic standpoint. To explain an organism's coming-to-be, or a single organic phenomenon, by appealing to a "vital force," gives only an imagined explanation, just as when one explains special mental utterances by the intervention of "the soul," or in sociology lets "society" appear as a kind of substance different from and hovering over the single individuals.[146] When one sets the whole in opposition to the elements (e.g. the organism in opposition to its parts, the soul in opposition to its thoughts, feelings, or strivings, society in opposition to the single individuals), this can only mean that beyond the activity we in the single cases find in the single part, there is still a world of possibilities, a capital of potential energy, which under certain conditions can be released and alter the single part's activity. And there is here a constant reciprocal action. For the single part (cell, organ,—thought and striving,—individual and class) can itself contribute to continued formation of possibilities, just as it will itself show itself to have developed by help from a world of possibility that was more comprehensive than itself.

γ. A third task consists in finding the different steps through which the whole has reached its present form. Here shows itself quite especially all research's historical character—not merely in the sense that research itself has a history, a historical course,

[146]Cf. with respect to the organism my lecture on Vitalism (*Mindre Arbejder.* I.), — with respect to the soul my *Psykologi.* I, 1 and 7 as well as the present work § 10, — with respect to society my *Etik* VIII, 4; XIII, 5 and the present work § 40. — The analogy that is between the three concepts comes to light in the fact that one often uses the word organism about society, and the word society about the organism, as well as both words about the soul, and the word ensoulment about life and about society.

but also in the sense that its highest and last object is history, finally "the world's" history, when by the world we understand the most comprehensive whole of objects we are able to think. "The universe's" biography comprehends and presupposes all science—and points constantly out beyond every attainable science. Only art, not science, can finally give us pictures of a whole. Thought ends always with an open horizon,—with a question or an exclamation, never (when it understands itself) with a full stop. The "meaning" is never "all the way out."

122. There is thus enough for thought to do before it gets its last word said. And this last word itself must be said by virtue of the causal principle. It sounds perhaps as though no more causes can be found, but can never sound as though there are none.[147] As I believe I have sufficiently shown, no proof can be given for the causal principle's absolute validity. It is a working hypothesis—in the sense in which I above (113; 117) have taken this word. Should there now be objects in the face of which we must give up applying this working hypothesis?

One has thought that the human will was such an object, and one has—remarkably enough—grounded this opinion in the interest of maintaining the human personality's peculiarity. One did not see that one precisely violated this by denying the possibility of a law-governed connection within the personality. For personality precisely presupposes unity and continuity in a higher sense and in a higher degree than any other totality we know. It therefore presupposes the causal principle, even though its special application in the face of so intimate a whole is bound up with special difficulties. There is no ground why we should not here too meet with our working hypothesis and apply it as well as we can. It is in this sense that I maintain, and have long maintained, determinism. I do not dogmatically assert that everything, including the human volitions, is subject to the causal principle; but I assert that neither ethical nor psychological grounds speak against it. The ethical ones we have not here to do with;[148] on the other hand there is here, under the problem of cognition, occasion to dwell a little on the methodical side of the question, which will lead to bringing forward some psychological considerations.

If by determinism one understands a conception that sets motive and action, motive and personality, in a purely external relation to each other,—roughly like the weights and the balance-pan,—then it is the psychological absurdity that has already been rejected in an earlier consideration (10). Motives are not something one "has" or "gets"; there can here be no distinction between possessor and property, between receiving and giving.

[147] On the expedient of stopping at something that is "cause of itself," cf. my *Religionsfilosofi.* § 10. (Second edition p. 37–38.)

[148] See on this my *Etik.* Chap. V.

The will works throughout our whole life from the simplest utterances of instinct and drive to the highest resolutions; it stretches through all motives and actions as the inner binding band. The agent is himself in his motives and in his actions, when he really is a personality. There can here be distinguished between a central element (which I usually call the real self) and a peripheral, with all possible intermediate degrees; but out of the circle we never come. Here precisely the concept of the whole comes to light in its whole peculiarity. This is the main thought of determinism, and it is in trust in it that we seek to unravel as much of the inner world's connection as possible, which can happen under the guidance of the causal principle. Under this endeavor we must then often provisionally regard a single side, a single element of the personal life in isolation; we must take single members out of the objects' continuous stream. But this is only a means to finding a so much the deeper and firmer connection. He who denies the causal principle's validity for the will, and thus embraces indeterminism, must on the contrary once and for all abolish or deny the inner connection that makes the will, the personality, into a whole.

*Henri Bergson* stands in this question on a peculiar footing. He maintains, as we know, that the immediately given in mental life is a continuous stream, which one can only in an artificial way resolve into elements. And freedom consists precisely in a streaming-on, through which the past's I glides over into the future. Here there are no external differences, and no analysis is possible. Both determinism and indeterminism analyze, separate elements and set them in external relation to each other,—the former in a relation of cause and effect, the latter as wholly different from and independent of each other. It is an external relation, both when one separates elements as cause and effect, and when one sets up certain elements as having arisen without any connection whatever with other elements. Bergson rejects both indeterminism and determinism; but he warns against defining what "freedom" is, since this will always lead back to determinism, because one can only define "freedom" by making the act of will dependent on the personality. One can then, according to Bergson, only maintain "freedom" when one does not say what it is! And yet he has himself defined freedom, when he finds the continuity between past and future to be its essential mark.[149] He is then himself a determinist, not merely because he defines "freedom," but because he defines it in the way he does. Bergson's merit is precisely to have maintained mental life's continuity

[149] *Les données immédiates de la conscience.*[4] p. 166–168; 174 f. — A related but peculiar standpoint is taken by W. *James* in the following utterance: *The only "free will" I have ever thought of defending is the character of novelty in fresh activity-situations. The experience of activity* (The Psychological Review. January 1905. p. 15.)

over against such analyzing as confuses its elements, formed by the art of abstraction, with the very drops in life's stream.

When we so often stop in our search for causes in the psychological domain, it is especially on account of the fact that it is here far more difficult than in other domains to demonstrate identical relations and complete equivalence. But that this is more than a difference of degree, no one has the right to assert.

123. Often one appeals to an immediate consciousness that one could have acted otherwise than one did. Psychologically regarded, it must be said to this that if such a consciousness is given, the task must be to show how it has arisen. And there present themselves here several different possibilities.

The more a mental state lays claim to our interest and attention, the less we heed the preceding and conditioning circumstances, and the more easily we forget them. The whole state comes to stand for us as arisen "of itself," and we are disinclined to enter upon a genetic explanation of it. Most of all does this hold in the case of a serious decision of the will. When, after a long inner debate, the decisive resolution thrusts itself forward and in a moment chases away all opposing misgivings, then there seems to lie before us a break in the continuity. The new state, in which our whole mind is concentrated in the thought of the action, stands in such opposition to the deliberation's waste and strife that even though we retain a memory of this, we can nevertheless not easily imagine an inner connection between the two states. Achilles' sudden and decisive self-mastery in the face of Agamemnon (in the Iliad's first canto) Homer could only explain to himself by Athena's coming down from Olympus and—visible only to the wrathful hero—pulling him by his locks and holding his sword back in the sheath. The ancient Greek consciousness appealed in such cases to Olympian intervention; the modern indeterminism to "the free will."

The decision of the will itself, the resolution, moreover brings with it a peculiar sensation of unhamperedness and force by reason of the concentration upon a single thought. All inner resistance and constraint has fallen away. What hitherto perhaps rested hidden in us, held down by superficial and momentary promptings, now comes to light; there occurs a breakthrough of our real self; we have a sensation as at a whole and full breath; it is as if we now first "came to ourselves." No wonder that we cannot in one and the same moment be in such a state and at the same time clearly hold fast that it has developed out of preceding states.

If now nevertheless the memory of the state of deliberation can make itself felt, it will be able to lead to assuming the objective possibility of another resolution than the one actually made. There presented themselves during the deliberation different possible actions before our gaze. We entered into each of them, perhaps so vividly that each for itself almost already stood as a reality. Therefore such a rejected possibility can stand as more than merely thought, as a way that really existed side by side with the way we struck into in the moment of resolution, just as real as this. The two ways seem to have crossed each other, and from the one we actually walk along we cast a glance over to the other way, which also lies there, but which we did not go. And thus it will especially present itself when we, in the moment of repentance, command motives that now have a force they lacked in the moment of decision, and which we involuntarily lay back into the representation of that moment, forgetting that it is precisely a fruit of the repentance.

A similar illusion constantly makes itself felt when one, with strong interest and in tension, thinks of the future. The future will indeed become different, according as I choose the one or the other of two actions. It therefore stands for me in a double light: partly as it will become when the one action is carried out, partly as it will become when the other action is carried out. Neither of the two pictures of the future fixes itself, so long as the deliberation lasts, and it then looks as if the future itself were undetermined, wavering. Our own indeterminacy is involuntarily confused with an uncertainty in existence itself. In antiquity this assumption of the future's indeterminacy made itself felt not merely in the face of the part of existence that is determined by human action, but in the face of existence itself, as regards the sublunary world, the world of changes. *Aristotle* taught that to our problematic judgments there correspond objective possibilities in nature, just as to our apodictic judgments there correspond objective necessities in nature. This doctrine was due to a naive transference of the forms in which our labor of thought proceeds onto existence itself. A remnant of it is what has held itself in the assumption that we should immediately be able to know that we, at an earlier time, could have made a wholly different resolution than the one we really made. "The truth of an event," says *Hobbes* in his critique of the Aristotelian doctrine, "does not depend on our knowledge, but rests on what works prior to the event."[150] And it can be very difficult to discover all that works in us prior to our resolution.

### γ. Subject and Object

124. In the investigation of the relation between quality and quantity and of the relation between succession and causality we had to do with the relation between object and form. We now direct our reflection toward both of them together, in that we raise the

[150] *Philosophia prima.* IX, 5.

question how far they are merely to be set to the account of the cognizing subject or possess an objective character. The relation between subject and object does not without more ado coincide with the relation between form and object.

There is an involuntary tendency to regard both forms and objects as immediate revelations. There is ascribed to them a reality "outside" or "besides" consciousness. Thus it goes with the sense-qualities, with number, time, and space, with the atoms, with causality, and so forth. They are fixed as absolute existences. The question then becomes what is subjective and what is objective, and how a boundary can be staked out.

Since our criterion of reality is the fixed and law-governed connection between the objects, and since the differences of quality designate a limit for complete demonstration of such a connection, there will naturally arise a tendency in the thinking consciousness to set them to the account of the subject, while the objective was supposed to be the quality-less atoms and their motions. In this direction went the mechanical natural science. The speculative philosophy went still further and set also differences of degree, of time, and of space (intensive, protensive, and extensive differences) to the account of the subject; especially it was the difference of quality that for the speculation of all times stood as the irrational that had to be sorted out; had this happened, the pure logical thinking could unfold itself with inner necessity. Both the mechanical natural science and the speculative philosophy relied on the assumption that what was the condition for understanding and made cognition possible must be the most objective of all.

In opposition to this the critical philosophy urges that a distinction must indeed be made between the demands, in accordance with which by our thought's nature we arrange the objects for ourselves in order to be able to win what for us is understanding, —and the character of existence itself. On the ground of our thought's nature we work unceasingly and involuntarily at reshaping the objects in such a way that there can appear as great a continuity between them as possible. But the more our own mind's nature and mode of working gain influence on cognition's results, the less can we maintain these results as pure objects.—With criticism pragmatism here agrees, only that it regards the forms as mere means, not as expressions of the subject's nature.

From this last point of view it then becomes not discontinuity, but continuity, that is set to the account of the subject. The objects, on the contrary, become that which must have another origin than the subjective.

125. Just as object and form are abstractions, since in every act of cognition we have a connection of them, so it stands also with subject and object. They are two points of view that can be taken up in different ways, but never separated. And they hold each for

itself both for the forms and for the objects.

We can make our own subject into an object for us, namely when we study it psychologically, e.g. in order to find out the forms according to which it works in its cognition. These forms, which are systematized in the categories, must be taken as facts. They are made into objects when reflection is set going over them. Every cognition takes place from a definite standpoint, which it can have its significance to establish. We then objectify the subject. And this happens especially when we investigate how the cognitive standpoint in question, the subject in question, has developed to the step at which it now stands, and has gotten the forms it now works with. It is thus not merely the objects, but also the forms, that can appear as objects. —On the other hand no object is given or thinkable except in relation to a subject, and as it appears for such a one. In every description of an objective order of things we therefore presuppose, consciously or unconsciously, an apprehending subject, either ourselves or a subject we construct in analogy with ourselves. The subject is the Archimedean point in epistemology,—the point from which existence can be—not moved—but apprehended. And such a point must always be presupposed. More than what shows itself from such a point is not known, and otherwise than it shows itself from it, it is not known. This too holds both as regards objects and as regards forms.

When we regard something as object, we must state the character of the subject in relation to which it makes itself felt. And when we regard something as subject, we must partly seek the objective connection that determines its character and thereby the objects and forms that stand at its disposal, partly heed that we by this investigation make it itself into an object for a subject (if it is ourselves, then for ourselves in a somewhat other state, in any case in another moment, than before). We never have a pure subject ($S$), but always an objectively determined or at least an objectified subject ($S_o$). And we never have a pure object ($O$), but always a subjectified object ($O_s$). $S$ and $O$ are mere abstractions. What we can have before us is always $S_o$ and $O_s$.

In accordance with this general consideration we can take up three different views with respect to the relation between subject and object.

*α. Psychological view.* I here seek to explain the cognizing subject's coming-to-be, with its organization, its capacity of sensation and thought, and so forth, by the way of natural science and history. Every subject is a product of certain conditions, which I can seek, even if not always find. What is seen depends on the eyes that see, and I here seek to find out how the eyes have gotten their present character. So too with the thought-forms, which must have developed under physical and social conditions. I then derive, as well as I can, the subject ($S_1$) from the object ($O_1$) or from the aggregate of objects. This way go empiricism and evolutionism. But they overlook that the inner conditions are a constant presupposition for the outer influences. There are always presupposed earlier steps of what later appears as $S_1$. That which $O_1$ works upon we can designate as $S_o$, since it in each case forms an analogy to $S_1$. (Cf. 9.) Beyond the constant reciprocal action of inner and outer conditions we do not come. There is here possibility of an infinite series:

$$S_1 < O_1 < S_2 < O_2 \ldots,$$

The sign < here means "presupposes." The inner and the outer conditions presuppose each other to infinity. In our investigations we move in each single case at a definite place in such a series, but never come out beyond it. It is an identically varying difference-series (65).

*β. Psychological-epistemological view.* If I investigate the subject $S_1$ and find its mode of cognition conditioned by the object $O_1$, then I must by virtue of the constant belonging-together of $S$ and $O$ presuppose or construct a subject $S_2$, for which $O_1$ is there. $S_2$ will not be able quite to coincide with $S_1$. The actual character of $S_2$ is then perhaps conditioned by $O_2$, which presupposes $S_3$, and so forth. This series too will formally always be able to be continued. It comes down to whether we have sufficient objects. The problem poses itself ever anew, alternating in psychological and in epistemological form. It is an interplay of causal explanation and of interpretation. Every age, every people, every human being has its $O_s$ and its $S_o$, which it is a matter of determining, if they will understand themselves and the world that exists for them. But predominantly the explanation will be brought about by later generations, who must then set themselves back to that state of things which conditioned the earlier generations' apprehension and understanding, and learn to see with their eyes.

If we let, as before, the sign < mean "presupposes," and let the sign { mean "objectified by," we get the following series:

$$S_1 < O_1 \{ S_2 < O_2 \{ S_3 < O_3 \ldots.$$

It is inconvertible and intransitive and differs from an identically varying series only in that the relation is rhythmically changed. One could call it alternately varying.

*γ. Epistemological view.* It stands in our power to form a series that formally could constantly be continued, merely by constantly directing reflection toward our own thinking or sensing. The thought $S_1$ can become object for the thought $S_2$, this for $S_3$, and so forth. The act of self-consciousness will always be able to be exercised anew, though it will of course get ever slighter content, the more it removes itself from the original concrete and object-rich state. It is a self-producing series, always continued according to the same law.[151]

If we again let the sign { designate "objectified by," we get the identically varying difference-series:

$$S_1 \{ S_2 \{ S_3 \ldots .$$

The interest of this series is that it is produced by reflection itself, independently of special objects or conditions. But the interest is predominantly formal, and the alternately varying difference-series is richer in references to changing situations and tasks. Only apparently is the concept of object pushed out in the series $S_1 \{ S_2 \{ S_3 \ldots$. It is here $S$ itself that is constantly made into $O$. A more special form, which offers peculiar interest, these series get when we take account of the fact that both the psychological explanation and the epistemological objectification can again be made into an object of objectification. The relation between $S_1$ and $O_1$ is then in both respects object for $S_2$, and when this is again treated both psychologically and epistemologically, the series is continued further:

$$\left.\begin{matrix} \left.\begin{matrix} S_1 \\ O_1 \end{matrix}\right\} & \begin{matrix} S_2 \\ O_2 \end{matrix} \end{matrix}\right\} S_3 \ldots .$$

It is the advance through "higher unities" of a purely formal kind, which the speculative philosophy tried to give a real significance. The interest of such a series is that it expresses the law of synthesis, the constant combining of applied views and won results. Could this series really be continued—which would require constantly new fullness of objects and constantly wakeful psychical energy—then idealism's highest aspirations would be satisfied. In any case the series contains a significant Excelsior!

126. From all this it follows that cognition's advance consists now in making the objective subjective, that is, sorting out in cognition's objects what is conditioned by the character of the cognizing subject, now in making the subjective objective, that is, making the very nature of the cognizing subject into an object of research. We cognize the world through the human being, but we also cognize the human being through the world. Romanticism emphasized the first point of view and was inclined to overlook the second; positivism emphasized the second point of view and was inclined to overlook the first. There is a constant interplay between the two points of view; neither of them

[151] Cf. on such a series *Dedekind*: Was sind und was bedeuten die Zahlen? § 60 (English trans. Chicago 1901. p. 60.) — Already *Fichte* operates with series of this kind.

can be eliminated, and the one cannot be derived from the other. In changing forms the relation continues to make itself felt, so long as cognition is in development. In each single case we must determine the $S_o$ and the $O_s$ we have to do with (whether $S_o$ is ourselves or another),—in analogy with the fact that one can only describe a motion or apply the law of inertia from a definite given point.[152]

It would be a misunderstanding to assume that subjectivity in itself is a hindrance to cognition's constant advance. On the contrary—the more richly equipped the subject is, the more forms and means it commands, and the more and the more many-sided its objects can therefore be. $S$ and $O$ do not necessarily stand in an inverse relation to each other; of this biology already witnesses through the nervous system's and the sense-organs' increasing differentiation. Cognition is not a passive receiving or taking-in,—and even if it were merely that, a rich equipment of capacities and organs would be an advantage. But it is a labor, which can be carried out the better, the more many-sided the powers that are commanded. Of course it comes down to whether precisely the capacities are present that the definite objects presuppose. The subject can be richly equipped in its life of feeling and fantasy; it will then perhaps not be able to perform the labor required to develop strictly exact observations and trains of thought, but in return it can then be other sides of existence that become accessible: *Shakespeare* understood how human beings are disposed under fate's heavy course and in severe conflicts, and in great religious personalities there has been a feeling of life's oppositions, an inwardness and a loftiness in the life of the spirit, which too can lay claim to bearing witness to what it is to exist. "Only they know the sea's secret who defy its dangers."

Conversely, a fuller development of objectivity will also make a more perfect development of the subject possible. Cognition is, after all, no closed system. The alternately varying difference-series $S_1 < O_1 \{ S_2 < O_2 \{ S_3 \ldots$ expresses the possibility that $S$ and $O$ can rise with each other. They also fall with each other; the slighter the subjectivity is, the poorer will also the corresponding objectivity be. But when the cognitive subject merely uses the energy it commands in its own independent way, there will thereby nonetheless always be won the objective that can come to be at this stage. It becomes perhaps only "a personal truth," but it becomes a truth. *Poul Møller* maintained with right, in so far, that an assumption due to serious self-labor could never be wholly false.—If $S$ wholly goes out, all truths and all riddles go away with it.

Cognition is constantly placed in this relation between $S$ and $O$. This necessary relation *Kant* overlooked, when he set up the "Ding an sich" as an absolute object. G.

[152] On theologizing Hegelians' attempt to cut the knot by postulates about an "ontological subjectivity" see my remarks on the occasion of *Rasmus Nielsen*'s "Grundideernes Logik" in *Danske Filosofer.* p. 189 f.

E. *Schultze*, and later *Albert Lange*, have with right remarked that Kant consistently ought to have granted that the distinction between the cognizable and the uncognizable "Ding an sich," just as well as all our cognition, could only hold for human subjectivity. There would then have to be posited a new "Ding an sich" as ground for the first, and so forth.—What Kant has had in view has nevertheless especially been precisely the irrational, inexhaustible relation between subject and object (and from a somewhat other point of view between form and object), and "Ding an sich" can in so far serve as symbol for this inexhaustibility, which requires ever-renewed labor.

If we look at the different systems or world-pictures that human thought has formed in the course of its historical development—let us call them $O_x$ $O_y$ $O_z$ …—then they cannot by a purely logical way be derived one from the other. The transition from $O_x$ to $O_y$ (e.g. from Ptolemy to Copernicus or from Aristotle to Kant) presupposes the emergence of new objects, which were not there for the $S_x$ corresponding to $O_x$. There must always be taken up new premises, in order that we can come from the one system to the other. Cognition, which is to present and explain existence for us, is itself always a part of existence; therefore there can always emerge new objects for it and, consequently, new objectivities. We have no knowledge beyond experience; but at no point have we the right to regard experience as concluded. Cognition gives us, even where it reaches highest, only a section of existence. Every reality we find is always itself again part of a greater reality.

127. The problem of cognition can only be posed from the human standpoint. It is thus a special $S$ that lies at the ground. We can call it $S_m$. The human subject has certain qualitatively marked peculiarities in its sensation and in its thinking, which we must take as pure facts (125 α and β). Their necessity and absolute validity cannot be established. By the way of the thought-experiment we can imagine an $S_n$ and $S_p$ and so forth—subjective, psychical energies with other special forms, objects, and motives than $S_m$—in analogy with the mathematicians' hypotheses about other spaces than the Euclidean. Unfortunately we cannot construct these other subjectivities, but we use the thought-experiment to impress upon ourselves the purely factual presuppositions that are given for a cognition with $S_m$. One could well imagine that to the same objective differences of refraction there corresponded other visual qualities than ours, and one could imagine that the logical principles that stand for us as self-evident were derived from more universal principles, just as the laws of inertia and of energy are perhaps special cases of more general propositions.

In the face of a consideration of this kind one has declared that the truth is only one, whether it is human beings, sub-human beings, or super-human beings that grasp it. A proposition must, after all, either be true or false![153] —But to this it must be answered that what is really true for $S_m$ does not necessarily become false for $S_n$, when $S_n$ from sure presuppositions (as forms and objects) can see that the truth had to appear for $S_m$ as it did. There could perhaps be taken up a law according to which what appears in a certain way for $S_n$ must appear in a certain definite other way for $S_m$ and in a third way for $S_p$. This law would then be the real truth, if it could be extended to all thinkable $S$. One of Leibniz's monads could perhaps discover (and the monad that Leibniz himself was believed itself to have discovered) that existence is mirrored differently in the infinitely different existing beings, without this abolishing its unity. There is indeed not much material for such a comparative epistemology, which of course would have to be written from the standpoint $S_m$. But $S_m$ is itself constantly placed in development, and the history of the sciences, of philosophy, of art, and of religion now shows us a series $S_\alpha$ $S_\beta$ $S_\gamma$ … belonging under $S_m$. The most deep-going task for the historian is not to describe mere events, doctrines, thoughts, pictures, and dogmas, but to bring forward the subjectivities for which all these things had to stand as they did, as regards content, validity, and value. This side of historical research is only in its beginning. Quite predominantly we have hitherto read history from the most modern $S_m$.

Between contemporary subjectivities too the same holds. At essentially the same stage of development different types will appear. It shows itself already as regards sensation, e.g. in the difference between the color-blind and those who apprehend all colors, or between those in whom one or another sense is especially developed. As regards the life of memory and representation, it shows itself in the difference between visual, auditory, and motor individuals. In some the forms, in others the objects, in others again the values, can within the life of cognition play the greatest role. And the more the life of thought takes on a personal character, the more types will appear.[154]

When truth is defined as the fixed and exact connection between our objects produced by the labor of thought, then to the objects must also be reckoned the different standpoints from which truth can be sought. The full truth must also contain the explanation of the errors and of the fact that they could stand as truths. If there is conflict between different attempts to find the truth, then the very definition of truth contains a measure for which attempt stands highest: namely that in which the greatest multiplicity of objects is

[153] *Husserl*: *Logische Untersuchungen.* I, 117; 151 f. — II, 669. (The polemic is directed against Benno Erdmann.)

[154] Cf. on religious types my *Religionsfilosofi.* § 35–43; 94–99. Cf. *Mindre Arbejder.* Second series. p. 103–140.

connected in the most intimate way. That the truth is only one shows itself through its history, in which unity works as a constantly onward-driving force, leading now to seeking out new objects, now to finding a closer connection between those already found. Every seeker of truth has his place in this history, of which we surely know only a slight part.

## B. World-View

### a. Science and World-View

128. The problem of cognition arose with the question how far and how existence could find expression in and for a single part of it. It showed itself that here one must constantly struggle with difficulties; thought was now too big, now too small in relation to existence, since this now demanded a single narrow way followed, now set us before immeasurable expanses. Many thoughts must be pushed aside when the concept of reality is to be formed; but the objects that set thought to work constantly demand new thoughts in place of the old.

We now regard the relation between cognition and existence from another side, in that we raise the question whether any of the objects that stand at our disposal is suited to be laid at the ground of a concluding characterization of existence, so far as this is accessible to us. Which object shall determine the timbre of the tone in which existence makes itself known to us?—There is, as we have seen, an involuntary need to combine the objects' multiplicity into unity, not merely in such a way that they are seen in their law-governed relation to each other, but in such a way that a coherent picture forms itself. From reflection's and analysis's going-through of the single objects we wish to ascend to an intuiting, in which all the found and compared particulars find their place. From the discursive we long up toward the intuitive. It is an intuiting that differs from the one that formed the presupposition for reflection by the fact that it has gone through reflection's purgatory. It is thus a concluding synthesis that is sought. If one will use the old word metaphysics of an attempt in this direction, one can say that in all there is a metaphysical need. Involuntarily the different objects that are known thrust themselves together, and one of them will become determining for the whole's character. Even in those for whom the word metaphysics is the expression of all evil in the world of thought, one will, where closer investigation is possible, find such an involuntary thrusting-together, a system-formation that has taken place without reflection having expressly been directed toward it. Here too works what *Kant* has called "the hidden art in the interior of the human being." Observation and association of representations, life of mood and interest, work here in an interplay that can be difficult to unravel, but is not therefore less significant. In folk-religions as well as in individual views of life this interplay deposits its results. "Metaphysics" is in so far a fact, and there is then here room for a comparative investigation of the origin, motives, and character of such involuntary thrustings-together. Philosophical interest such metaphysics will have only when the thrusting-together has taken place around a significant object, and when a firm bearing rules throughout the whole train of thought. It is then a peculiar expression of psychical energy, which can be typical of the human thought's position in existence. Otherwise where the satisfaction of the metaphysical need leads to a sinking into a single side of life,—and where it is merely the need for security, or perhaps even spiritual comfort, that makes itself felt. The eyes then close to essential sides. It was this that *Goethe* reproached his pious friends (Lavater, Jacobi, Claudius) with: "Dem Lehrbegierigen bleibt wenig Trost bei ihnen." Such involuntary metaphysics gets a dogmatic character. It does not know its own sources and often believes itself to be the expression of "sound common sense," or to be the voice of nature or of the deity.

Already on account of the involuntary tendency to metaphysical representations it will be of significance to set up a special problem here. As a French philosopher has said: Hélas, il est plus difficile qu'on ne l'imagine de ne point faire de métaphysique; la pire de toutes est celle qui s'ignore.[155] Philosophy must test the spiritual powers that utter themselves in such involuntary thought-activity, and seek to determine the possibilities that by thought's nature and conditions must be said to be present. It is philosophy's honor that it has distinguished between metaphysics and science, before the special sciences, which up to the most recent time have confounded the two, have gotten an eye for the difference. That philosophy cannot without more ado push metaphysics out comes from the fact that the treatment of the problem of cognition establishes that the whole human science, the most exact as the purely empirical, rests on presuppositions whose absolute necessity cannot be grounded. Here is and remains an open place.

It will nevertheless be expedient to divide the consideration, so that we provisionally investigate how far a world-view is possible that presupposes no other motives than the purely intellectual interest that is all science's psychological motive, and that attaches itself closely around thought's own laws, will only know what their consequence leads to. In so far as a world-view is determined by the need to keep hope and courage up, or to get air for the fundamental moods that the course of life awakens, it is most correct to

[155]G. *Séailles*: *Les affirmations de la conscience moderne* p. 229.

regard it under the problem of valuation.

If one will have another word than the for many frightening "metaphysics," I propose the word cosmology. This expression is used in older time partly in opposition to psychology, partly in opposition to theology. The old metaphysics, as it was set in system by Chr. *Wolff*, comprehended, besides ontology (which corresponds to the doctrine of categories and epistemology), the three parts: cosmology, psychology, and theology.[156] Here I will use the word cosmology in the most comprehensive sense: world-view, which can also etymologically be defended, since Kosmos means world and Logos thought or doctrine.

The main difficulty for such a cosmology will be that the cosmological principles, that is, the fundamental thoughts that condition and bear the world-view, cannot without more ado be fetched from any single domain of experience, any single group of objects, since it should precisely be a matter of connecting all domains, all objects, into a whole and giving a characterization of this.

An attempt to treat the problem of existence will have partly a formal, partly a real character. Partly it must be asked what demands the concept that is to express existence must satisfy, and whether it is possible to satisfy them. Partly it must be asked what content one can give such a concept.

129. Formally the cosmological concepts spring out of the need for combining, in which we have found mental life's nature, and which especially gets practical significance through the criterion of reality. The completed combining would give us a concluded world-picture, in which all objects find their place. But already the discussion of the problem of cognition has taught us that an absolutely concluded world-picture will be impossible. New objects constantly emerge, and new tasks with them. And in any case our world-picture, if it were to be an object of complete cognition, would have to be held together and compared with something different from itself; only then could it get its full determinacy and its exhaustive characterization,—but *if* there were something such, our world-picture would not rightly bear its name: it would not be an absolute whole-picture. —It is here indifferent whether one distinguishes between cosmology and theology, or not. The contradictions become the same, whether we speak of "the world" or of "God."[157] The irrational announces itself here just as in the problem of cognition. And perhaps there is here an inner connection between the two problems. When cognition constantly ends in new seeking and cannot be concluded, and when it is itself a part of existence,

[156] Such a *Metaphysics* we have in Danish by *Jens Kraft* (Copenhagen 1751–1753). See on this *Danske Filosofer.* p. 19–20.

[157] Cf. my *Religionsfilosofi.* § 17–18.

then already therein lies that existence cannot be a concluded whole. And on the other side, what is irrational for cognition, which indeed especially hangs together with the time-relation, could stem from the fact that existence itself is not finished, is constantly placed in development, and therefore does not offer such clear and rational forms as our research must presuppose. The philosophical systems have been too sure that existence in itself was a closed system, and that it was only our will and thought that constantly had to struggle to subsist and to win through to harmony. There is moreover the possibility that existence has properties or sides that do not reveal themselves along the lines along which alone our thought can move. According to *Spinoza* there are, of existence's infinitely many fundamental forms ("attributes"), only two (spirit and matter) that are accessible to us. *Wundt* has here applied the comparison between the so-called imaginary numbers' [sideways] series and the positive and negative numbers' [forward and backward] series.[158] The formula $a + b\sqrt{-1}$ could be existence's formula, but in such a way that in our experience $b$ was always 0. Within the dimension of existence in which our thought can move, it strikes on the irrational, the inexhaustible; but the other dimensions it would not strike on, however rich a fullness of objects it commanded, and however consistently it worked through these objects. In one of the more special problems we will come back to this possibility.

130. A real motive for speculation lies in the prominent role an object or a group of objects plays or seems to play within our experience. There can be a purely intellectual interest in getting connection into our world-view by laying an object at the ground for the understanding of all the others, and by this way the need for formal unity can then be sought satisfied. The different cosmological systems are just so many attempts to sound existence's depth by testing how far it is possible to understand it from a single side in our experience. It is a series of thought-experiments by which our most important thoughts' reach has been tested.

Every such attempt must, as *Leibniz* first has seen, go the way of analogy. We are indeed here, by help of the given objects, to form for ourselves a conception of the whole to which they all belong. Even if we now disregard the whole-concept's irrationality, the difficulty will arise that the whole itself is never given us as an object. It must thus be a single object (or group of objects) that is to express the whole's character, and to which all other objects would be referred. It is, as history shows, not merely philosophical systems, but all religions have gone this way, whether they themselves are clear about it or not.

[158]Spinoza: *Ethica*, I, 11; II Ax. 3. On Wundt see my book *Moderne Filosofer* p. 18.

Analogy is now, as we have seen (74–75), a necessary and justified means of thought in science. Science's history shows us how constantly one domain of experience is used to illuminate another. But when an object is used to illuminate another, the analogy can always be tested through the fruitfulness and durability of the working hypothesis it leads to. One can compare the analogy's consequences with experience and find it confirmed or contested by the emergence of new objects. But cosmological analogies cannot be tested in this way, since the whole that is to be expressed by a single object being laid at the ground never itself appears as an object, as little as it ever lies before us concluded. We are quite otherwise placed than in an analogy between parts of our experience mutually.

A world-view's character and significance rests partly on how clearly and consistently the underlying analogy is carried through, partly on the object that is laid at the ground. This object we can, with an expression fetched from *Goethe*, call the primal phenomenon. Yet Goethe understood by "primal phenomenon" not merely an object that could serve to illuminate other objects for us. It means for him a fact that is at the same time law, so that one merely needs to utter it in order to have the explanation of it. It must moreover not be regarded as composite; it is a "limit for our beholding"; it awakens "the great wonder," but designates at the same time our cognition's limit.[159] It is known how fateful a role this concept played in Goethe's doctrine of colors (where it found its expression in "Gesetz der Trübe"); the primal phenomenon—shadows that on dark ground showed themselves blue, on light ground yellow-red—could and must not be more closely investigated and explained! But so dogmatically the concept primal phenomenon need not be taken. It can be used to designate the point in our experience from which we try to orient ourselves in all directions, but which therefore may very well itself be an object of further-going investigation and explanation. In epistemology we have seen that although the cognizing subject stands as a presupposition that must be understood in any cognition whatever, it can therefore nevertheless very well itself be an object of psychological investigation. There is not any absolutely fixed point in the world of spirit, as little as in the world of matter. Whatever significance a cosmological attempt may otherwise have, it is the first demand that must be made of it, that it not appear dogmatically by pushing all

[159] *Goethe*'s *Farbenlehre* § 124; 175–177. *Gespräche mit Eckermann*, 13. and 18. Feb. 1829; 21. Dec. 1831. Goethe applied this concept not merely in the doctrine of colors, but also in the doctrine of magnetism and in biology. *Sprüche in Prosa (über Naturwissenschaft). Gespräche mit Eckermann.* 27. Jan. 1830. *Unterhaltungen mit Kanzler v. Müller.* p. 39. "Ein Urphänomen muss man nicht weiter erklären wollen; Gott selbst weiss nicht mehr davon als ich." — *Schopenhauer* applied the Goethean expression to every interpretation of "the thing in itself," every attempt to reach out beyond the world of phenomena. He himself saw in the will (this concept taken in the widest sense) the primal phenomenon, "the most immediate manifestation" of the thing in itself. Cf. *Den nyere Filosofis Historie.*[2] II. p. 220–222.

epistemological principles aside.

131. Every attempt to develop a world-view must rest on clarity about its moving on science's boundary. It must not intervene in science's daily and special work, in that of the mental sciences as in that of natural science. The primal phenomenon must itself share all phenomena's, all objects' lot. The very fact that existence, as we know it, is to a great extent accessible to understanding, shows itself subject to general laws, must in every world-view become of essential significance, whether this very fact is made into primal phenomenon or not. Even if we would regard all principles and laws as mere working hypotheses, it nevertheless serves to characterize existence that one, as regards understanding of it and dominion over it, can only get at it along certain definite ways. Every, even the most special branch of natural science or mental science, thereby gets cosmological significance. It is indeed only most indirectly that existence makes itself known in the working hypotheses; but a connection there must in any case be, and a cosmology that would quite scorn the special sciences' views and results would contradict itself. While *Schelling* and *Hegel* conceived the special sciences as imperfect approaches to cosmology and boldly would set their systems in place of those sciences' imperfect philosophy, their modern adherents often grasp the expedient of proclaiming the special sciences' independence, but at the same time declaring them to be, for philosophy as such, quite indifferent.[160] But the difficulty for every attempt lies precisely in the fact that a whole-conception can only be built up on the ground of points of view and tendencies that already make themselves known within the special sciences.

The work on a world-view takes place on science's boundary. But even there one must work in science's spirit. A significant world-view is now only formed when at the ground there lies a great enthusiasm or at least being-seized by the thought of existence regarded as a whole, especially by a single object within it. Here it is then the task to unite being-seized and critique. It is a matter at once of holding fast the interest for the ideas by whose help one tries to cast light over existence, and of applying them with full consciousness of their problematic character. There belongs great psychical energy to working on these conditions, great elasticity in mind and thought to swing between the deep conviction a great idea can call forth, and the firm conviction that continued discussion is constantly necessary. Fantasy, with its capacity to form whole-pictures, has

[160]Thus *Bradley* (see *Moderne Filosofer.* p. 48,) and *Benedetto Croce.* Croce reproaches Hegel with wanting to continue Schelling's philosophy of nature instead of proclaiming "l'indifferenza filosofica delle discipline naturale matematiche e la loro piena autonomia" and referring these sciences "al non — teoretico, e cioè al pratico." (*Ciò che è vivo e ciò che è morto nella filosofia del Hegel.* 1907. p. 149; 155).

its great significance in science as in art. But in science such whole-pictures—whether they are to hold only for single object-domains or span all—can only stand as hypotheses, as stations on a way that can always be continued. In art, on the contrary, a definitive goal is reached when a full and characteristic picture has been created.[161]

When I in my time have spoken of philosophy as art,[162] I thereby thought of that union of opposite properties which every attempt at a scientific world-view requires: a world-picture is to be brought forth with exertion of energy and in enthusiasm—and yet constantly stand as a hypothesis. From other points of view the artistic element in philosophy's concluding ideas has earlier been brought forward. *Schopenhauer*[163] thought most of the involuntary coming-to-be of a concluding thought-construction, whereby art's "What?" relieves science's "Why?" *Albert Lange*[164] thought nearest of the idealizing tendency, the need to find in ideal pictures expression for the highest reality.

The transition from philosophy's empirical and critical parts (psychology and epistemology) to cosmology offers an analogy to the transition from historical critique to history-writing. The history-writer unites the critically treated material into a whole-picture of events and characters. There is thereby carried out a rounding-off, in which the researcher's personality involuntarily plays a role along. Thus the philosopher tries to work together the critically worked objects into a whole, within which a single object determines the timbre. The point of view (the primal phenomenon) and its carrying-through here unavoidably get a personal stamp. A great philosophical system is a work of art, a drama, in which the single lines and scenes join together, now more with logical, now more with aesthetic harmony. *Plato*'s, *Spinoza*'s, and *Schopenhauer*'s systems especially offer this stamp.

In any case an attempt at a world-view stands constantly between science and art. If such attempts have a future, there will perhaps here develop itself a type of spirit that on account of the hitherto ruling dogmatism has not yet been able to unfold itself properly. *Georg Simmel* has from this point of view denied that metaphysics (cosmology) should be called art: "So poor in independent fundamental forms our soul is not, that every picture of things that is not science should be only art."[165] I would oppose this "only"; it could indeed be that the artistic form was the highest, when it rested on a great experience and a serious labor of thought. But the main thing is that the endeavors that constantly must be made in cosmological direction get freedom to stir themselves, in

[161] Cf. my *Psykologi.* V B, 12.
[162] *Mindre Arbejder*. First series. p. 184–186.
[163] *Den nyere Filosofis Historie.*[2] II. p. 215 f.
[164] Op. cit. p. 550.
[165] *Schopenhauer und Nietzsche — Ein Vortragscyclus.* (1907) p. 128 f.

order that their significance for life's different domains can become established. Such a significance they will moreover always have, when they really provide intellectual satisfaction for highly developed personalities.

132. In a very sharp and interesting way the cosmological problem is posed in the now-living philosopher who most directly has taken aim at a solution of this problem, *Henri Bergson*. It has already earlier (3) been mentioned how sharp an opposition this thinker assumes between the immediately given and reflection or analysis. Science's work is essentially analytic. Out of the immediately given's continuity are sorted the elements, parts, and properties, which later can only in an external way be put together again. It is the practical need, technique, and language, that necessitate these distinctions. And then especially language, which predominantly applies space-pictures and therefore really can only express purely external relations. What in immediate experience is connected and at once appears as unity and multiplicity, is now made into a mosaic without real inner unity. If a metaphysics is to be possible, we must by an exertion of will return from analysis to intuition, look at the objects, not with "the understanding's" eyes (les yeux de la seule intelligence), which only discover the outer, but with "the spirit" (l'esprit), which is one with intuition and with life itself. "The spirit" is the will's bending toward itself instead of the bending outward toward action in the surrounding world. Here then existence (l'Etre, l'Absolu) is grasped directly, not by the detour of logical and mathematical abstractions. Through the intellectual sympathy that is one with intuition, the moving force in reality is grasped and followed through all its turns. Only thereby is time's reality maintained over against the mechanical crystallization of the moments. The mechanical causal relations are seen as sensuous symptoms of the inner stream that is the true real, and it then becomes possible to follow life under its continuous development, follow it under its constant striving, in spite of the material elements' resistance, to break itself a way to higher forms that cannot be foreseen. The spirit or intuition, as it appears in limited form in the human being, is a stage in the great process. It is not a product of outer causes, but an unfolding of life's and existence's inner process (l'élan de la vie), which only at a certain definite step becomes able to grasp itself in a beholding, and thereupon by "intellectual sympathy" interprets all the other currents in existence, all other striving under, around, and over us.

Bergson laments that metaphysics must use the understanding's language, because only the understanding has a language. From this is understood the application of the many pictures in his brilliant presentations. When the question is raised what means of thought metaphysics, on Bergson's conception, really commands, we get no clear answer. This is understandable, since the understanding has laid claim to language, and since philosophy (metaphysics) according to Bergson only comes to be by our going in the opposite direction of thought's accustomed movement: Philosopher consiste à invertir la direction habituelle de la vie de la pensée — remonter la pente naturelle de l'intelligence!

By this definition of philosophy[166] Bergson recalls *Carlyle*, for whom philosophy also was a struggle against habit. It has its justification over against dogmatic application of special sciences' methods to solve existence's fundamental problems. But one thing is that we can see that our usual means of thought here do not suffice; another is what means of thought we then command with the last great problem. Already above I found the difficulty of the cosmological problem in the fact that it was here the objects' aggregate, existence, so far as it could be regarded as whole, that was to be cognized;—but existence is not given as object, and we must therefore use a single object to illuminate all others by the way of analogy. If this conception is right, it will be of interest to see whether the spirited French thinker would be able to attach himself to it.

Bergson applies so great a labor to critique of language's and technique's external and mechanical representations and expressions, that he does not come to give us the new logic or doctrine of method that his denial of "the understanding" and his appeal to "intuition" or "the spirit" really require. It stands especially not clear what relation there is between the intuition to which we—when we with exertion of will have worked ourselves out of the strong habit of parceling-out—are to return, and the analysis in which the scientific work consists. And yet he tries in his last work (*Évolution créatrice*) to develop a kind of speculative vitalism, in that he will show how the idea of life as a self-working-forward upswing (élan) is able to gather series of facts about itself. But whence comes this idea, and on what does it ground itself, since it is not to be won by the way of analysis?

In seeking the answer to this question in Bergson's presentation I find the hypothesis of all metaphysics as grounded on analogy confirmed. In his *Introduction à la Métaphysique* (p. 25 f.) it runs, namely: La conscience que nous avons de notre propre personne, dans son continuel écoulement, nous introduit à l'intérieur d'une réalité *sur le modèle de laquelle* vous devons vous représenter les autres. Toute réalité est donc tendance.—Although the word analogy is not used, the concept is nevertheless clearly applied. Likewise there is here clearly indicated the object that lies at the ground for analogizing. From the psychological domain a bridge is struck, through the analogy,

[166] *Introduction à la Métaphysique.* (Revue de Métaphysique et de Morale 1903). p. 27. — *Évolution créatrice* (1907). p. 32. — On Carlyle cf. *Den nyere Filosofis Historie.*[2] II. p. 378. (Sartor Resartus I, 10.)

which leads in to all objects' interior. Only by this way can Bergson come to the result that "the absolute is of psychological, not of logical or mathematical nature" (*Evol. créatr.* p. 323). Like all idealism, Bergson's too is grounded on the analogy with the human being's mental life.

The artistic trait in Bergson's presentation lies especially in the way in which he brings "the immediately given" and analysis into the sharpest opposition to each other and thereby forces us to the leap with which the metaphysical speculation, according to his theory, must begin. The whole way that leads to the catastrophe is depicted in the most spirited way. There is only the transition or leap itself that remains standing as a riddle. He has avoided the schematic, especially the striving after symmetry in the arrangement of concepts, which according to Poul Møller is affectation. He points energetically to the conditions under which we can reach out beyond all the piecemeal in the scientific conception of existence. But even after we have followed this pointing-finger, the opposition to the scientific analysis nevertheless remains standing for us, and there is opened no way out either to do without it or to use it in the service of a concluding world-view. —

After having found our hypothesis about cosmology's nature and mode of procedure confirmed by the last attempt made in this direction, we now go over to a closer consideration of the different primal phenomena that can be laid at the ground.

## b. Unity and Multiplicity

133. For the philosopher it lies near to take the point of departure for his world-view in the very fact that it is possible to orient oneself in existence by cognition's help. The primal phenomenon then becomes intelligibility itself, as it is from thought's side anticipated in the categories of rationality and of causality. There cannot subsist any absolute relation of indifference between our thought's mode of working and the source of the objects that are worked by it. When we can recognize objects,—when hypotheses formed on the ground of the past's objects show themselves to hold for later-appearing objects,—and when with advancing experience there shows itself an ever greater continuity in the world of qualities and of changes,—then in this lies a witness to existence's, not merely to our thought's, nature. Our very mark of reality consists indeed in the unity and connection that shows itself between our observations, those now lying before us and those earlier made. It is by reason of the applicability of this mark that we can distinguish between what are mere thoughts, and what is a reality we cannot get around, even if we push our thoughts aside, but which we can perhaps master when we know its laws. The

concept of reality arises partly by a limitation, partly by an extension of our thoughts (110; 126).

The connection between thought and existence is, according to the dynamic concept of truth, only indirect. But nevertheless it can, according to what has just been developed, with right be said that existence reveals itself through thought's work. Way and goal cannot here fall quite outside each other, especially when one remembers that detours and errors can often be more fruitful in truth's service than the straight ways. The criterion of reality teaches that it is not merely a matter of gathering as many objects as possible, but also of thinking them through with consistency. Error rests on a limited thing, a part, being made into a whole; but when one then merely consistently thinks through this limited thing, one will come to see where its limit lies, just as one will soonest find one's way out of a forest when one follows one definite direction instead of constantly changing direction. Intellectio fictionem terminat (Spinoza). This, that existence's nature can be present in the midst of error, shows perhaps most clearly of all that here is a kinship, however near or far it may be. He who seeks rightly has already found. In so far *Lessing*'s alternative (seeking of truth—and possession of truth) does not hold, and *Spinoza* and *Hegel* were right in their mystically sounding proposition that in that we think existence, existence thinks itself in us. (Cf. 110.)

The innermost nerve in the work thought has to do is the application of the relation between ground and consequent to the changing and manifold objects. Causality becomes primal phenomenon. Its applicability—in how great or small an extent it holds—is a fact of the greatest significance for every world-view that will in earnest distinguish between dream and reality. It is no wonder that the essence and validity of the concept of cause has played so great a role in the more recent philosophy's discussions.

Mythologies and theologies too build on the concept of cause. But they apply it chiefly to make an ascent to a "first cause"—or to several first causes, which are then gradually ordered in a hierarchy. One acknowledges perhaps subordinate causes (causæ secundæ), but yearns out beyond all finite causes and seeks at any price to conclude the causal series and to hang the whole chain of causes and effects up at an absolute point of departure or end-point, on a secure hook. The causal principle does not permit this. It demands to know in which wall the hook sits. It requires a cause for the first member in the causal series as well as for each single one of the others. If we are forced to conclude, we end with an unanswered question. How far we can actually follow the causal series back is a question that at each given time is answered by empirical science. For the philosopher it is not this endless series that is the main thing, however much interest it may in itself offer; the essential thing is for him the thought according to which there must constantly be sought, and can be found, new members in the causal series. He finds here a world-order expressed, which encloses all that exists in the most intimate way,—a unity-power that connects all objects, however peculiar and independent they each for themselves may be, so as to stand as members in a great, immeasurable connection. It was this power that *Spinoza* in a mythological way conceived as a being ("an eternal and infinite being") and called substance, that which subsists. What subsists is the great unity in existence, which makes itself known in the law-governed connection. This idea is merely a carrying-further and carrying-through of the simple thought all involuntarily apply in daily life, when they seek to distinguish between the apparent and the real. The "sound common sense's" daily practice here meets with the highest metaphysics.

134. There are constantly members lacking within the causal series, and this can be thought continued in both its directions. The unity-picture it is possible to form stands constantly as a great hypothesis, as a resting-point where the synthesis has provisionally done its work, and where there are perhaps means lacking to come further. New objects can arise or be discovered, so that new series-formations become necessary. Ever anew thought must ascend from provisionally chaotic difference-series to rational series. The word "absolute" has its scientific significance within the single rational series; it expresses the relation between premise and conclusion, ground and consequent, but cannot be used to designate a conclusion of all thought-connection. "The absolute" is a mythological remnant, in *Hegel* as in *Spinoza*, a sign that metaphysicians too can become tired and feel a need to rest out in a last thought.

Even if conclusion is not possible, the world-view can nevertheless bear unity's stamp, and every world-view must, whether it itself lays weight on it or not. Monism is therefore invincible, even though it constantly must struggle with the advancing and manifold experience and therefore must be designated as *critical monism*. It is by virtue of synthesis's own law that every unity-view we can form for ourselves of existence points out beyond itself: every object is set in relation to something else, something different from itself, since only thereby can it get its full characterization. (Cf. 58–62.) And experience then also constantly gives occasion for satisfaction of this need, in that new objects emerge and pose their demands, which often require reshaping or extension of the hitherto developed view. It is no imperfection in our thought that it thus constantly seeks to set one object in relation to another. It is precisely a perfection that we cannot stop at any point. We constantly seek anew to get out beyond the limitation (both in content and in extent) that every finished world-picture offers. While one has hastened to form concluding "world" or "God" as secure knots on the thought-thread,[167] it can precisely be called a spark of divinity in the human thought, that no such finished concept can satisfy it.

There shows itself here, within the problem of existence, an antinomy that forms an analogy to the antinomies we in different forms have met in thought's psychology, in its history, in the doctrine of categories, and in the problem of cognition. It is the conflicting relation between continuity and discontinuity that under the different forms makes itself felt. For our world-view the consequence becomes a constant conflict of unity and multiplicity, without a complete harmony's being here possible. It is no use, with Hegel,[168] to distinguish between the seeking cognition and the true cognition. In the seeking cognition, says Hegel, there is difference between subject, method, and object; but in the true cognition these differences fall away. The question is precisely whether thought can ever cease to be seeking. Hegel's own system has fallen to the dialectic he himself constantly demanded as truth's medium, and which utters itself in that every thought-content is set over against another and thereby leads out beyond itself. Thus it has gone with Hegel's system in the face of new objects and new points of view (also in the face of certain old objects and points of view!). We should, according to the Hegelian dialectic, by this way constantly come to "higher unities." Of this we now cannot be so sure (cf. 99). We could indeed provisionally move in a downward direction; the development of truth can follow winding ways, and we must be glad if we can hold fast the analogy with the spiral, which bends only in order to rise: inclinata resurget!

135. But when the lack of conclusion, which every world-picture thus must offer, has its ground in the fact that the relation between unity and multiplicity constantly makes itself felt in new forms,—with what right is then greater weight laid on unity than on multiplicity? Why has one the right to sympathize more with Spinoza's one "substance" than with Leibniz's countless "monads"? When there are two sides in our cognition that cannot be done without either in our science or in our world-view, why then regard the one of them as the more significant and as the one that shall determine the timbre in our world-view? Should it not be just as justified to embrace a pluralism as to embrace a monism?

*Charles Renouvier* and *William James* have in our days with energy taken up the word for a pluralistic world-view. Renouvier protests, on the ground of his neocriticism, against what he calls "la superstition de l'unité"; James, on the contrary, will build his pluralism on empirical ground.

[167] Cf. *Religionsfilosofi.* § 17–18, cf. § 52–55.
[168] *Wissenschaft der Logik.* (1812–16.) II. p. 375 f.

Pluralism, that is, the world-view for which the objects' multiplicity and difference is the primal phenomenon, can seek its justification in the fact that difference, opposition, and change is the condition for all sensation and thought. (Cf. 11; 21.) There is surely also, under philosophy's present condition, good ground to make pluralistic points of view felt. There is still constantly too much speculative metaphysics in the air, whether it now appears in philosophical or in theological form. It is of significance that we not forget what positivism and criticism have taught us. The old English school had the mission of holding philosophers' attention fixed on experience, and it awakened the great movement against dogmatism in the two last centuries. It is no accident that the great poser-of-problems *David Hume* belonged to this school. Pluralism makes the world new for us, and it forces us, where it appears with authority, to revise our categories, our principles, and our methods. We are too inclined to dogmatic sleep, to believing that we possess thoughts in which all existence can find expression.

A "radical" pluralism is nevertheless untenable. If multiplicity were sole ruler, no understanding would be possible. For to understand is to find an inner connection between different objects, find a unity-principle for them. We must indeed very often content ourselves with establishing the reality of single, isolated objects each for itself. But in that case a problem arises by reason of the isolation itself. That the isolation is unconditioned, observation can in any case not teach us. As several times remarked, the assertion that a chaotic difference-series lies before us requires a long investigation. Pluralism can just as well appear dogmatically as monism, though it does not offer such rest as the mystical unity has yielded to many a tired thought.

The single objects are constantly determined and characterized through their connection with other objects within a comprehensive world-order. The single object's place in time and in space is determined by its relation to other objects within the world of space and time that is accessible to us. Every property we ascribe to the single object expresses the way it affects other objects or is affected by them. The force or energy the single object is said to possess is determined by a law according to which its changes bring with them changes in other objects. The single object's individuality is expressed in the law according to which its states and its properties change.

We do not know the objects outside a connection of one of these kinds. It is constantly due to an analysis, a separating and distinguishing, when we speak of different objects; we have then already taken them out of "the immediately given's" continuous connection. Every analysis rests on an original, primitive, involuntary synthesis, which conditions the whole that by analysis can be separated into its elements. Where does the one object end, and where does the other begin? Even if the connection is only "conjunctive" (to use James's expression), and can only be designated by the word "and," it nevertheless encloses problems: have we drawn the boundary-line rightly? and even if we have, what degree and kind of separation does then this "and" mean? Analogously it stands when judgments are formed on the ground of intuitions, and inferences of judgments. Every act of cognition means a section of a greater whole. Every conceptual determination is a limitation, every judgment rests on presuppositions that point out beyond the judgment-act itself; every inference presupposes several premises fetched from a greater, ramified connection. We move in our cognition constantly within a whole that comprehends more than the objects we think over, and that contains the conditions for the results to which our reflection comes.

For these grounds monism must be said to have greater right than pluralism to characterize our world-view. It supports itself on that in existence which makes it possible for our reflection at all to be directed in a fruitful way toward the objects it contains. As we have seen, monism subsists only under a constant struggle with multiplicity. But our world-view rightly bears its name after that one of the struggling parties on whose side we place ourselves. *Renouvier*, whose main category was relation (see 56), was—in his maintenance of pluralism—really only concerned to hold fast the constant relation of unity and multiplicity (la correlation de l'un et du plusieurs), which thought cannot do without. For *James* "the radical pluralism" hung together from the first with "the philosophy of pure experience,"[169] while he later has declared that pragmatism, which is to be a radical expression of the philosophy of experience, must in equal degree reject the absolute monism and the absolute pluralism, and he now designates his standpoint as "noëtic Pluralism," that is, a critical doctrine of plurality that lays weight on the fact that there are many objects between which no necessary connection can be demonstrated.[170] This conception seems not to be in principle different from the critical monism presented above; only center of gravity or emphasis are different—but it is indeed a difference that can become decisive for the world-view's timbre. When James calls his pluralism a method, he must thereby mean that it especially aims at maintaining and securing the single objects in their peculiarity and independence over against the tendency to let them disappear in a mystical unity, which easily becomes the night in which all cats are gray. This is an endeavor with which the critical monism can fully sympathize.

Monism and pluralism are in no way necessary oppositions, when the unity-principle

[169] *A World of Pure Experience.* (Journal of Psychology, Philosophy and Scientific Method. Oct. 1904. p. 570).

[170] *Pragmatism. A new name for some old ways of thinking.* (1907.) p. 51; 155; 166.

on which monism lays the weight is conceived in such a way that it precisely reveals itself through the single objects (individuals, events, relations) as well as through the law-governed world-order that bears and conditions them. And in this direction went *Spinoza*'s, the most pronounced modern monist's, thoughts. For him it stood thus, that the unity in existence could only be expressed through an infinity of single beings; in the single being's capacity and need to remain in its state and maintain its existence he saw a peculiar form of the unity-principle in existence. There is thus for him no external opposition between the single thing and the whole: in the single individual existence's innermost direction and striving there made itself known precisely existence's fundamental principle. Spinoza's real fundamental thought surely appears at this point. When his system does not manage to bring this point into full and consistent connection with other points, the main ground lies in the fact that he believed he could get all the irrational out of the way and with right found its last source in the time-relation. If the last principle in our world-view is timeless, the single existences, which develop successively, cannot find a natural place within it. If, on the contrary, one makes earnest of Spinoza's proposition that the more we understand the single beings (res singulares), the more we understand existence's fundamental principle,[171] then we must also take the consequence, which would have been to him an offense,—that existence's innermost essence, so far as we can at all form a hypothesis about it, must be thought to unfold itself in time. But in the same moment our monism becomes not absolute or dogmatic, but critical.

136. Monism's strength is the fact of understanding. Its weak side rests on the fact that there are constantly limits, hitherto unsolved tasks. Empiricists and skeptics will constantly be able to stop the monist by pointing to these barriers. Not even the criterion of reality can we apply everywhere, thus not carry through a strict opposition between dream and reality. The world-views will perhaps constantly conflict at this point. But it will not be any unfruitful conflict, since it will sharpen the interest in finding unity and connection as regards as many objects as possible. The monist's cause stands and falls with the possibility of cognition's advance. And he will perhaps with irony leave to the pluralist the chaos that at each time seems to remain, with the reservation that it cannot be established whether it is a real chaos, but that its time for clearing-up will surely come—then perhaps to be relieved by new objects' apparent chaos.

But the possibility is not excluded that the ground why full understanding is not possible is to be sought not merely in the fact that the cognizing subjects are placed

[171] Qvo magis res singulares intelligimus eo magis Deum intelligimus. *Ethica.* V, 24.

in development, but finally in the fact that existence itself is not finished, but placed in becoming and development, just as cognition is. Concluded cognition would only be possible if existence were concluded. The time-form, the source of all irrationality, might be not merely one of our categories, but an existence-form that we have no right to think away.

If now something wholly new arises in existence, so that the later real moments are not quite as the earlier,—in Kant's language: if "Ding an sich" does not work constantly,—then it is no wonder that our cognition constantly has to struggle with difficulties and perhaps finally with insoluble riddles. And there is in any case a point in existence where something new arises, namely there where understanding develops out of ignorance, error, or doubt. Existence indeed comprehends cognition too; both forms, objects, and values belong to it;—it cannot then be finished so long as the worlds of forms, of objects, and of values are concluded and brought into equilibrium mutually. There is a cosmic development that takes place through the human thought,—perhaps also, as will later be discussed, through human will. It was the great rationalistic systems' (Plato's, Spinoza's, Hegel's) wonderful contradiction that even though they could explain everything else, they nevertheless could not explain the very striving and working thought whose works the systems themselves were.

The critical monism, which holds fast the time-relation's validity and thereby existence's unconcludability as well as cognition's, nevertheless finds a primal phenomenon in the fact that more and more objects are brought under the causal principle,—the analogy between ground and consequent on the one side, the series of changes on the other side, constantly anew shows itself to have real applicability. Thereby is grounded a belief in an advancing unity-power in existence, which leads out beyond the sporadic and conflicting. The labor of thought is seen in cosmic illumination. There occurs a cosmic advance when thought advances—however it may then stand with other objects within the immeasurable existence.

## c. Spirit and Matter

137. The first object we investigated in the capacity of primal phenomenon was accessibility to thought. This investigation led to seeing the possibility of a critical monism. Should there now not be objects of a more concrete kind that could lay claim to being regarded as primal phenomena? The world-view would thereby get a more definite character.

Reflection could then turn toward the fact that all we experience is either of a spiritual or of a material kind. All objects fall into two great main groups—the one comprehending sensations, representations, feelings of pleasure and displeasure, striving and willing, the other comprehending all that is extended and movable. It is two fundamental qualities that here stand over against each other. Experience shows them to us at once in their difference and in their close belonging-together. The question becomes whether one of them, and if so which, is suited to be primal phenomenon and thus can determine our world-view's timbre. Since our experience offers no more main kinds of objects than these two, there seems to be a bound choice between them.

In the section on thought's psychology the relation between the spiritual and the material was already discussed, but essentially from a psychological, empirical, and methodological point of view. It was especially a matter of finding a view that at once expressed what we really know about this relation, and what was suited to be laid at the ground as a working hypothesis in continued research. What we really know is that there subsists such a relation between the two kinds of objects that inferences can be drawn from the one to the other,—that there thus subsists a relation of function in the mathematical sense between them. On the other hand it was doubtful whether there also subsists a time-relation between them, with conversions from physical energy to psychical, and conversely. But that relation of function is now also what first and foremost is of use in continued research, so that it is suited to express the relation between the objects psychology occupies itself with, and those physiology occupies itself with.

When we now return to this question, it is to discuss how far our world-view is forced to stop at this provisional result, or whether it is possible to go a step further in the one or the other direction. Whether we methodically and empirically are dualists, materialists, or Spinozists, it can be asked whether in either of the two groups of objects we find existence's innermost essence expressed, in such a way that the other stands only as symptom or symbol for it.

Every attempt to answer this question must support itself on analogy, when it is given that one can neither derive the spiritual from the material, nor conversely. The question then becomes whether we conceive the spiritual in analogy with the material, or the material in analogy with the spiritual,—or whether perhaps neither of these analogies can be carried through.

138. If I lay the analogy with what is extended and movable in space at the ground, my cosmology, not merely my method, becomes materialistic. Materialism is the human being's natural philosophy. It recommends itself by its perspicuity: space is indeed

the clearest form of apprehension we have, wherefore there is (especially in visual individuals) a need to convert all thoughts into space-pictures. Language witnesses to this, in that all expressions for spiritual objects are transferred from the material, often by a very characteristic symbolism. Communications from one human being to another about mental experiences get a clearer character when they can make use of space-pictures, express mental life's "inner" suffering and working through pictures of positions and movements in space. Even time we make perspicuous only by space's help. Confusions of identity and analogy lie then here extraordinarily near, and it is on such a confusion that the cosmological materialism as a rule rests.

Even after the difference between the spiritual and the material has been acknowledged, there can be grounds that speak in favor of regarding the material as the primal phenomenon and thus letting it occupy the fundamental place in the world-view.—The spiritual shows itself indeed, when we strictly hold to experience, only to appear in some earthly organisms, while the material world spreads itself in immeasurable regions. Can then so vanishing an object with right stand as expression for existence's innermost essence—in opposition to the vast masses?—And if we agree to regard the material as the primal phenomenon, we attain a greater continuity in our world-view than we would otherwise be able to get. The spiritual arises sporadically—in different individuals and often also in the same individual: there is no medium for the different consciousnesses mutually, such as there is for the different bodies mutually,—and in the same individual the life of consciousness is often interrupted by unconscious intervals. On the other hand the material objects offer, the more we penetrate into them, a great continuity. Space's form makes it possible to follow the material changes down into the vanishingly small, carry the same laws down through what for observation stands as an interruption. The world of spirit stands as a fragment, or as a collection of fragments, while the material world offers connections to infinity.

Is there not all ground to conceive the non-perspicuous in analogy with the perspicuous, the small in analogy with the great, the fragmentary in analogy with the connected?

However energetically such grounds are made felt, they will nevertheless again and again be outweighed by the thought that is all idealism's fundamental thought, that we know matter only through spiritual activity, through the energy that utters itself in sensation and thinking, and which therefore naturally becomes our last refuge. The very concept matter, as natural science determines it in ever more "ethereal" forms,[172] is a

[172] *Paul Langevin* (*Physics of the Electron.* Congress of Arts and Science. 1904. IV. p. 137.) holds that the ether must now be regarded as ground for the concept of matter. The electronic theory of matter, in which

thought-product, built on observations and experiments. And even if one then again would assert that the thought that has produced this concept is itself again a product of matter, then there follows from idealism's side as rejoinder that this very hypothesis about thought's arising (be it now tenable or not) is also a thought-product. We meet here the earlier (125) treated relation between subject and object in a special form (for there are other objects than the material). If $T$ means thought, $M$ matter, then the series $T < M \{ T \ldots$ is a special case of the series $S_1 < O \{ S_2 \ldots$. The single combat can in a certain sense not get an end. But when we ask about the last ground for all assumptions, it is nevertheless thought, the only object in and for which grounds at all exist, that we come back to. It was therefore that the last proposition in my "Psychology" came to run thus: "Further back than to what for the human being stands as a thought-necessity we cannot come; the thought that explains everything becomes its last and constant problem." Even if what is extended and movable in space occupies a predominant place among our objects, both by reason of its extent, its continuity, and its perspicuity, it can nevertheless only in and for thought make itself felt. It is an object that constantly bears the stamp of the subject's work (an $O_s$).

But thought has other objects than the material. It works also in the world of spirit, in the study of mental life in the thinking subject itself and in other subjects, which it conceives in analogy with itself. The mental science makes its claim felt with the same right in the world-view's coming-to-be as does natural science. That its objects do not fill space in the perspicuous world-picture, as natural science's do, is not the decisive thing. There could indeed in a corner of the universe have arisen something that was of far greater significance for the deepest understanding of existence than all the much that moves in space. And if one lays the weight on the fact that thought seems to appear at a very late step in the world-development, then this could indeed be assumed to indicate that the most essential qualities need the longest time to unfold themselves properly.

By space—says *Pascal*—the universe comprehends (comprend) me as a vanishing point; but by thought I comprehend (comprends) the universe.

139. Instead of conceiving the spiritual as a form of the material, idealism conversely conceives the material as the form under which existence's content, which in itself is of a spiritual kind, presents itself to the sensuous intuition. The justification or necessity of taking one's point of departure from the spiritual or making this into primal phenomenon is grounded in different ways.

When materialism emphasizes the continuity the material world offers, it is answered

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matter is partly synonymous with electricity in motion, explains an ever-rising number of phenomena. — See also *Christiansen*'s Program (1905). passim.

to this that this continuity is external and mechanical, as space's form brings with it. A true continuity is only possible in thought, which connects the single elements in its unity. The most intimate unity that can at all be thought is the one that comes to light in that thought advances from ground to consequent. In the material world there are found only outer copies of this inner unity, by whose help we understand everything. We cannot connect any meaning with an existence that does not bear such a unity's stamp. If existence is to constitute a whole, we can only think it as a thought-whole. This was *Hegel*'s solution of the world-riddle. Thought and existence are for him identical, because thought's form is the only one in which existence can be conceived as a world.—That it is a great analogy he here operates with,—an extension of a single object's peculiarity to being expression for all existence,—this Hegel does not admit.[173]

*Leibniz*, the metaphysical idealism's real founder, was on the other hand fully conscious that he built on analogy. It was by this's help that he conceived existence as a continuum of monads, in that these represent infinitely many different degrees of the same thing that in the human being appears as mental life. The fragmentary character of the world of spirit is only apparent. Nature makes no leaps. Unconsciousness and consciousness are only gradually different, and thus it stands also with the many different individual consciousnesses. To these psychical differences of degree correspond the physical differences of degree between rest and motion, and between different motions mutually.—In a similar direction as Leibniz go, on this point, Schopenhauer, Beneke, Lotze, and in the most recent time Bergson (see 132).

The metaphysical idealism would only be able to lead to a concluding solution if it could provide means for a positive determination of the different degrees and nuances of spiritual existence which according to it constitute existence's innermost essence. But there are no means for this, and this rests on the fact that the concept potential psychical energy, whose application is necessary for the carrying-through of the idea of psychical continuity, only designates a problem or a postulate (see 8). Already for this ground the metaphysical idealism can never become anything other than a belief. And when one tries to formulate idealism in a single idea, it unavoidably gets more the character of the picture than of the thought.[174] We stand at the boundary between thinking and poetry.

140. There is no bound choice between idealism and materialism, so that he who rejects the one solution would necessarily have to assume the other. For it cannot

[173] Cf. also *Brøchner*'s standpoint. *Danske Filosofer.* p. 202–204. — *Wundt* too stands on Hegel's standpoint here. Neither does he admit that he analogizes. He maintains that he—led by thought's need for unity in the world-view—merely continues and supplements what empirical cognition gives. *Moderne Filosofer.* p. 21–24.

[174] See on this *Religionsfilosofi.* § 21–22.

be proved that what is not spiritual must necessarily be material, and conversely. The division into spiritual and material objects is purely empirical, and it cannot be established that existence's content should be exhausted by this division. On this point too existence could be inexhaustible. All our knowledge develops through a series of predicate-determinations, conclusion-representations; but whether this series is ever complete, so that all that "lies in" the ground, in the indeterminate point-of-departure representation, comes to appear as member in the series (cf. 35), cannot be decided, and still less whether all sides of existence make themselves known in the object-ground on which our knowledge thus builds. An Either—Or would only lie before us if it could be proved that all existence's fundamental traits made themselves known in our experience, and that this experience was exhaustively characterized by the twofold division into spiritual and material objects.

Spirit and matter stand (cf. 129) not to each other as Yes to No. There is a relation of opposition or of difference between them, but there need not be a relation of contradiction between them. It is possible that they stand to each other as two color-qualities stand to each other; the one is not what the other is, but therefore it is not said that what does not have the one quality must have the other. If we knew only two colors, e.g. yellow and blue, we would nevertheless obviously not be justified in saying that what is not yellow must necessarily be blue. The necessity would in any case have to be limited to our experience. Our color-scale has a purely empirical character, an upper and a lower limit, which hangs together with more deep-lying physical and physiological relations. When there is more in existence than we think of in our philosophy, the ground need not be that we have not philosophized rightly; it can lie in the fact that this more does not make itself known as object for us. Again here we strike on the irrational.

According to this consideration we cannot, as regards the here-lying part of the problem of existence, conclude our thinking at a primal phenomenon, since none of the objects is suited to it. If one would hold that one could take them both together and say that existence was "psychophysical," one would thereby only have gotten a sounding word, no solution (cf. 9).

According to *Goethe* "the highest a human being can reach is wonder,"—"and," he adds, "when the primal phenomenon sets him in wonder, let him be content; a higher it cannot yield him, and further he shall not seek behind it; here is the limit." Goethe's need for poetic beholding here comes to light. But even though we find no primal phenomenon,—and precisely because we find none,—there is room for the great wonder. Out beyond all objects and all reflection over them there shows itself an order of things, within which our experience and our thinking are like islands in the ocean.—It is of course a belief, when we here speak of an *order* of things; it is a belief that supports itself on the ground on which the critical monism (134) builds.

When a feeling of loneliness can arise at the apparently vanishing and unassuming position the spiritual occupies in relation to the material (138), then a feeling of loneliness of a deeper kind will arise when we become clear that our whole world of spiritual and material objects, together with the knowledge and the problems it conditions, is again only a single section of a far more comprehensive existence. Wonder has here natural ground; it stands at our wisdom's end, not merely at its beginning. Whether it is to become admiration is a question that must be discussed in another connection.

What speaks for seeing the matter in this way is that there are such great difficulties for thought in reaching a satisfactory conception of the relation between the spiritual and the material. These two objects cannot be reduced to each other, and we know no third from which we could derive them as special cases. It was then possible that these difficulties hung together with the fact that there are sides of existence that do not appear for us, but that contain the ground for the twofold division within our experience. Our cognition's history shows us many examples of extended experience's being able to solve problems or get them out of the way. We have indeed no prospect of an extension of our experience that could here help us; but we have the right to conceive our position here in analogy with such cases where a problem has shown itself to rest on limited experience.

Existence would offer great simplicity if there were only two fundamental forms under which it stepped forth. But what stands for us as simple and easily surveyable can very well be due to a very intricate connection. In our simplest sense-sensations we have examples of this, when we compare them with the physical and physiological processes to which they correspond. It can be "economical" or "pragmatic" to operate with the principle of simplicity. But the attempt to develop a world-view has significance only when earnest is made of investigating all possibilities. It is here not a matter of our convenience, but of a last orientation concluding for us. Both idealists and materialists as a rule come too quickly to their last word. It can also, in a single, relatively limited domain, be very dangerous to proceed from the assumption that only two possibilities, an Either—Or, lie before us, because experience provisionally offers no others. *Charles Darwin* experienced this in his youth, when he in a geological treatise had concluded that the parallel lines on the Scottish coast must be due to the sea, since no other explanation seemed tenable. Agassiz's glacier theory showed a new possibility. And Darwin learned from this that in empirical science one should be careful with applying the principle of the excluded middle.[175]

Among thought's masters *Spinoza* has here seen most clearly, when he taught that the difference between the spiritual and the material was purely empirical, and that existence in itself unfolded itself under infinitely many fundamental forms (attributes). In one of his earlier writings he seems to have imagined the possibility that we could come to conceive more attributes than those two; but in his main work (Ethica) he has given up this thought. The ground why there must be infinitely many attributes lies for Spinoza in the fact that existence's whole fullness can only thereby find expression. It is Spinoza's belief that here comes to light. He could simply have supported himself on the purely empirical character of the difference between the spiritual and the material; but he would then have had to express himself somewhat less dogmatically. Neither can we follow him in the assumption that all the different attributes are logically coordinate or "parallel." The possibility cannot be rejected that one or more of the attributes inaccessible to us contain the ground for the peculiar difference and the peculiar "parallel" relation that our experience leads us to assume between spirit and matter.

Spinoza's fundamental thought preserves its significance even when the dogmatic element in his doctrinal structure is corrected. It shows us out to a greater horizon than the one within which idealism and materialism carry on their strife.

### d. Subsistence and Development

## C. Valuation

### a. The Ethical Problem

#### α. *Ethical Work.* (Formal Ethics)

#### β. *The Rationality of Ethical Valuation.* (Real Ethics)

### b. The Religious Problem

#### α. *The Psychological Place of Religion*

#### β. *The Historical and the Philosophical Consideration*

[175] *Charles Darwin*: *Life and Letters.* I. p. 69. 291.